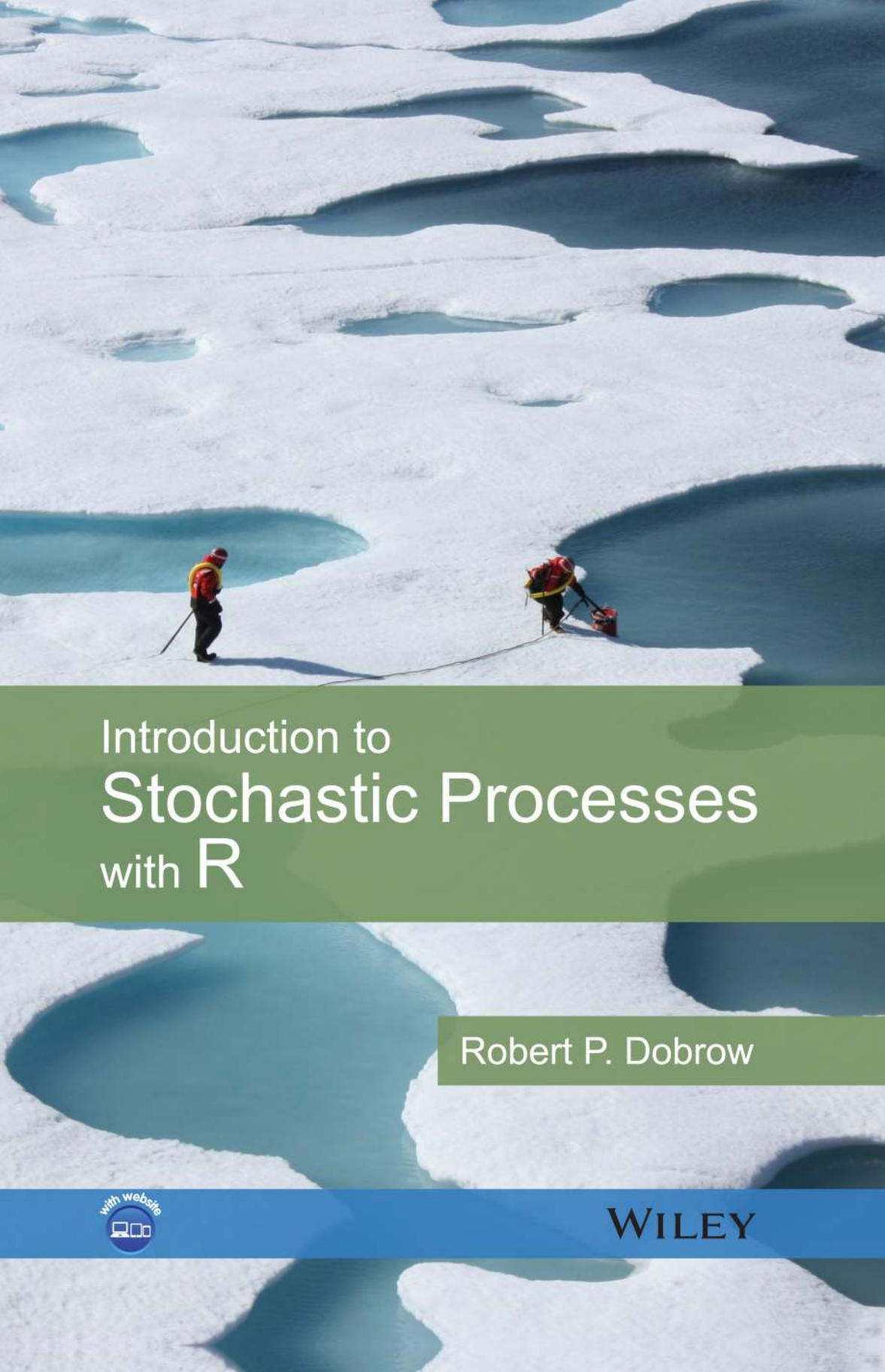




Introduction to Stochastic Processes with R

Robert P. Dobrow



WILEY

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ROBERT P. DOBROW

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To my family

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PREFACE

The last thing one discovers in composing a work is what to put first.

—Blaise Pascal

The intended audience for this book are students who like probability. With that prerequisite, I am confident that you will love stochastic processes.

Stochastic, or random, processes is the dynamic side of probability. What differential equations is to calculus, stochastic processes is to probability. The material appeals to those who like applications and to those who like theory. It is both excellent preparation for future study, as well as a *terminal* course, in the sense that we do not have to tell students to wait until the next class or the next year before seeing the good stuff. This *is* the good stuff! Stochastic processes, as a branch of probability, speaks in the language of rolling dice, flipping coins, and gambling games, but in the service of applications as varied as the spread of infectious diseases, the evolution of genetic sequences, models for climate change, and the growth of the World Wide Web.

The book assumes that the reader has taken a calculus-based probability course and is familiar with matrix algebra. Conditional probability and conditional expectation, which are essential tools, are offered in the introductory chapter, but may be skimmed over depending upon students' background. Some topics assume a greater knowledge of linear algebra than basic matrices (such as eigenvalues and eigenvectors) but these are optional, and relevant sections are starred. The book does not assume background in combinatorics, differential equations, or real analysis. Necessary mathematics is introduced as needed.

A focus of this book is the use of simulation. I have chosen to use the popular statistical freeware R, which is an accessible interactive computing environment. The use

of simulation, important in its own right for applied work and mathematical research, is a powerful pedagogical tool for making theoretical concepts come alive with practical, hands-on demonstrations. It is not necessary to use R in order to use this book; code and script files are supplemental. However, the software is easy—and fun—to learn, and there is a tutorial and exercises in an appendix for bringing students up to speed.

The book contains more than enough material for a standard one-semester course. Several topics may lend themselves to individual or group projects, such as card shuffling, perfect sampling (coupling from the past), queueing theory, stochastic calculus, martingales, and stochastic differential equations. Such specialized material is contained in starred sections.

An undergraduate textbook poses many challenges. I have struggled with trying to find the right balance between theory and application, between conceptual understanding and formal proof. There are, of course, some things that cannot be said. Continuous-time processes, in particular, require advanced mathematics based on measure theory to be made precise. Where these subjects are presented I have emphasized intuition over rigor.

Following is a synopsis of the book's nine chapters.

Chapter 1 introduces stochastic and deterministic models, the generic features of stochastic processes, and simulation. This is essential material. The second part of the chapter treats conditional probability and conditional expectation, which can be reviewed at a fast pace.

The main features of discrete-time Markov chains are covered in Chapters 2 and 3. Many examples of Markov chains are introduced, and some of them are referenced throughout the book. Numerical and simulation-based methods motivate the discussion of limiting behavior. In addition to basic computations, topics include stationary distributions, ergodic and absorbing chains, time reversibility, and the strong Markov property. Several important limit theorems are discussed in detail, with proofs given at the end of the chapter. Instructors may choose to limit how much time is spent on proofs.

Branching processes are the topic of Chapter 4. Although branching processes are Markov chains, the methods of analysis are different enough to warrant a separate chapter. Probability-generating functions are introduced, and do not assume prior exposure.

The focus of Chapter 5 is Markov chain Monte Carlo, a relatively new topic but one with exponentially growing application. Instructors will find many subjects to pick and choose. Several case studies make for excellent classroom material, in particular (i) a randomized method for decoding text, from Diaconis (2009), and (ii) an application that combines ecology and counting matrices with fixed row and column totals, based on Cobb and Chen (2003). Other topics include coupling from the past, card shuffling, and rates of convergence of Markov chains.

Chapter 6 is devoted to the Poisson process. The approach emphasizes three alternate definitions and characterizations, based on the (i) counting process, (ii) arrival process, and (iii) infinitesimal description. Additional topics are spatial processes, nonhomogeneous Poisson processes, embedding, and arrival time paradoxes.

Continuous-time Markov chains are discussed in Chapter 7. For continuous-time stochastic processes, here and in Chapter 8, there is an emphasis on intuition, examples, and applications. In addition to basic material, there are sections on queueing theory (with Little's formula), absorbing processes, and Poisson subordination.

Brownian motion is the topic of Chapter 8. The material is more challenging. Topics include the invariance principle, transformations, Gaussian processes, martingales, and the optional stopping theorem. Examples include scoring in basketball and animal tracking. The Black–Scholes options pricing formula is derived.

Chapter 9 is a gentle introduction to stochastic calculus. *Gentle* means no measure theory, sigma fields, or filtrations, but an emphasis on examples and applications. I decided to include this material because of its growing popularity and application. Stochastic differential equations are introduced. Simulation and numerical methods help make the topic accessible.

Book appendices include (i) getting started with R, with exercises, (ii) probability review, with short sections on the main discrete and continuous probability distributions, (iii) summary table of common probability distributions, and (iv) matrix algebra review. Resources for students include a suite of R functions and script files for generating many of the processes from the book.

The book contains more than 200 examples, and about 600 end-of-chapter exercises. Short solutions to most odd-numbered exercises are given at the end of the book. A web site www.people.carleton.edu/rdobrow/stochbook is established. It contains errata and relevant files. All the R code and script files used in the book are available at this site. A solutions manual with detailed solutions to all exercises is available for instructors.

Much of this book is a reflection of my experience teaching the course over the past 10 years. Here is a suggested one-semester syllabus, which I have used.

1. Introduction and review—1.1, 1.2, 1.3 (quickly skim 1.4 and 1.5)
2. One-day introduction to R—Appendix A
3. Markov chains—All of chapters 2 and 3
4. Branching processes—Chapter 4
5. MCMC—5.1, 5.2
6. Poisson process—6.1, 6.2, 6.4, 6.5, 6.8
7. Continuous-time Markov chains—7.1, 7.2, 7.3, 7.4
8. Brownian motion—8.1, 8.2, 8.3, 8.4, 8.5, 8.7

If instructors have questions on syllabus, homework assignments, exams, or projects, I am happy to share resources and experiences teaching this most rewarding course.

Stochastic Processes is a great mathematical adventure. Bon voyage!

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LIST OF SYMBOLS AND NOTATION

Such is the advantage of a well constructed language that its simplified notation often becomes the source of profound theories.

—Pierre-Simon Laplace

$\emptyset$	empty set
$A \cup B$	union of sets A and B
$A \cap B$	intersection of sets A and B
$A \subseteq B$	A is a subset of B
$x \in A$	x is an element of set A
$\binom{n}{k}$	binomial coefficient $\frac{n!}{k!(n-k)!}$
$\lfloor x \rfloor$	floor of x
$\lceil x \rceil$	ceiling of x
$\ \mathbf{x} \ $	length of vector $\mathbf{x}$
A^{-1}	matrix inverse of A
A^T	matrix transpose of A
e^A	matrix exponential of A
B_t	standard Brownian motion
$\text{Corr}(X, Y)$	correlation of X and Y
$\text{Cov}(X, Y)$	covariance of X and Y
$d(i)$	period of state i
$\deg(v)$	degree of vertex v in a graph
e	extinction probability (for branching process)

$E(X)$	expectation of X
$E(X^k)$	k th moment of X
$E(X Y)$	conditional expectation of X given Y
f_j	probability of eventual return to j
$f_X(x)$	probability density function of X
$f(x, y)$	joint probability density function of X and Y
$f_{X Y}(x y)$	conditional density function of X given $Y = y$
$G_X(s)$	probability generating function of X
I	identity matrix
I_A	indicator function of the event A
$k(s, t)$	covariance function
$K_t(x, y)$	Markov transition kernel
λ_*	second largest eigenvalue
$m_X(t)$	moment generating function of X
N_t	Poisson process
N_A	spatial Poisson process ($A \subseteq \mathbb{R}^d$)
ω	simple outcome of a sample space
Ω	sample space
π	stationary distribution of Markov chain
P	transition matrix for Markov chain
$P(t)$	transition function for continuous-time Markov chain
Q	generator matrix
$\mathbb{R}$	real numbers
$\mathbb{R}^2$	two-dimensional Euclidean space
$\mathbb{R}^n$	n -dimensional Euclidean space
S	state space for Markov chain
$SD(X)$	standard deviation of X
T_a	first hitting time of a
$U_{(1)}, \dots, U_{(n)}$	order statistics
$v(t)$	total variation distance
$Var(X)$	variance of X
$Var(X Y)$	conditional variance of X given Y

NOTATION CONVENTIONS

Matrices are represented by bold, capital letters, for example, $\mathbf{M}$, $\mathbf{P}$.

Vectors are represented by bold, lowercase letters, for example, $\boldsymbol{\alpha}$, $\boldsymbol{\lambda}$, $\boldsymbol{\pi}$. When vectors are used to represent discrete probability distributions they are row vectors.

ABBREVIATIONS

cdf	cumulative distribution function
gcd	greatest common divisor
i.i.d.	independent and identically distributed
MCMC	Markov chain Monte Carlo
mgf	moment-generating function
pdf	probability density function
pgf	probability generating function
pmf	probability mass function
sde	stochastic differential equation
slln	strong law of large numbers
wlln	weak law of large numbers

ABOUT THE COMPANION WEBSITE

This book is accompanied by a companion website:
<http://www.people.carleton.edu/~rdobrow/stochbook/>

The website includes:

- Solutions manual available to instructors.
- R script files
- Errata

1

INTRODUCTION AND REVIEW

We demand rigidly defined areas of doubt and uncertainty!

—Douglas Adams, *The Hitchhiker's Guide to the Galaxy*

1.1 DETERMINISTIC AND STOCHASTIC MODELS

Probability theory, the mathematical science of uncertainty, plays an ever growing role in how we understand the world around us—whether it is the climate of the planet, the spread of an infectious disease, or the results of the latest news poll.

The word “stochastic” comes from the Greek *stokhazesthai*, which means to aim at, or guess at. A stochastic process, also called a random process, is simply one in which outcomes are uncertain. By contrast, in a deterministic system there is no randomness. In a deterministic system, the same output is always produced from a given input.

Functions and differential equations are typically used to describe deterministic processes. Random variables and probability distributions are the building blocks for stochastic systems.

Consider a simple exponential growth model. The number of bacteria that grows in a culture until its food source is exhausted exhibits exponential growth. A common

deterministic growth model is to assert that the population of bacteria grows at a fixed rate, say 20% per minute. Let $y(t)$ denote the number of bacteria present after t minutes. As the growth rate is proportional to population size, the model is described by the differential equation

$$\frac{dy}{dt} = (0.20)y.$$

The equation is solved by the exponential function

$$y(t) = y_0 e^{(0.20)t},$$

where $y_0 = y(0)$ is the initial size of the population.

As the model is deterministic, bacteria growth is described by a function, and no randomness is involved. For instance, if there are four bacteria present initially, then after 15 minutes, the model asserts that the number of bacteria present is

$$y(15) = 4e^{(0.20)15} = 80.3421 \approx 80.$$

The deterministic model does not address the uncertainty present in the reproduction rate of individual organisms. Such uncertainty can be captured by using a stochastic framework where the times until bacteria reproduce are modeled by random variables. A simple stochastic growth model is to assume that the times until individual bacteria reproduce are independent exponential random variables, in this case with rate parameter 0.20. In many biological processes, the exponential distribution is a common choice for modeling the times of *births* and *deaths*.

In the deterministic model, when the population size is n , the number of bacteria increases by $(0.20)n$ in 1 minute. Similarly, for the stochastic model, after n bacteria arise the time until the next bacteria reproduces has an exponential probability distribution with rate $(0.20)n$ per minute. (The stochastic process here is called a *birth process*, which is introduced in Chapter 7.)

While the outcome of a deterministic system is fixed, the outcome of a stochastic process is uncertain. See Figure 1.1 to compare the graph of the deterministic exponential growth function with several possible outcomes of the stochastic process.

The dynamics of a stochastic process are described by random variables and probability distributions. In the deterministic growth model, one can say with certainty how many bacteria are present after t minutes. For the stochastic model, questions of interest might include:

- What is the *average* number of bacteria present at time t ?
- What is the *probability* that the number of bacteria will exceed some threshold after t minutes?
- What is the *distribution* of the time it takes for the number of bacteria to double in size?

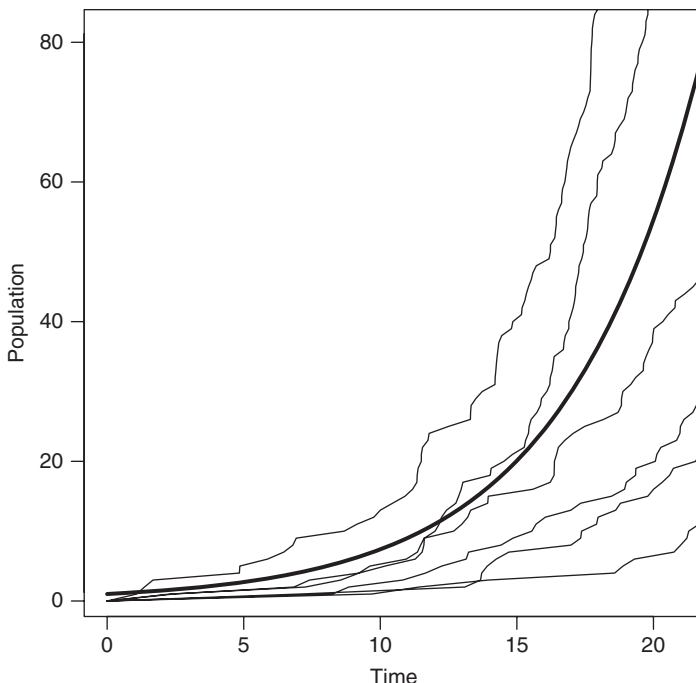


Figure 1.1 Growth of a bacteria population. The deterministic exponential growth curve (dark line) is plotted against six realizations of the stochastic process.

In more sophisticated stochastic growth models, which allow for births and deaths, one might be interested in the likelihood that the population goes extinct, or reaches a long-term equilibrium.

In all cases, conclusions are framed using probability with the goal of quantifying the uncertainty in the system.

Example 1.1 (PageRank) The power of internet search engines lies in their ability to respond to a user's query with an ordered list of web sites ranked by importance and relevance. The heart of Google's search engine is the PageRank algorithm, which assigns an *importance value* to each web page, called its *page rank*. The algorithm is remarkable given the massiveness of the web with over one trillion web pages, and is an impressive achievement of mathematics, particularly linear algebra.

Although the actual PageRank algorithm is complex with many technical (and secret) details, the page rank of a particular web page is easily described by means of a stochastic model. Consider a hypothetical web surfer who travels across the internet moving from page to page at random. When the surfer is on a particular web page, they pick one of the available hypertext links on that page uniformly at random and then move to that page.

The model can be described as a random walk by the web surfer on a giant graph called the *webgraph*. In the webgraph, vertices (nodes) are web pages. Vertex x is joined to vertex y by a directed edge if there is a hypertext link on page x that leads to page y . When the surfer is at vertex x , they choose an edge leading away from x uniformly at random from the set of available edges, and move to the vertex which that edge points to.

The random surfer model is an example of a more general stochastic process called *random walk on a graph*.

Imagine that the web surfer has been randomly walking across the web for a long, long time. What is the probability that the surfer will be at, say, page x ? To make this more precise, let p_x^k denote the probability that the surfer is at page x after k steps. The long-term probability of being at page x is defined as $\lim_{k \rightarrow \infty} p_x^k$.

This long-term probability is precisely the page rank of page x . Intuitively, the long-term probability of being at a particular page will tend to be higher for pages with more incoming links and smaller for pages with few links, and is a measure of the importance, or popularity, of a page. The PageRank algorithm can be understood as an assignment of probabilities to each site on the web.

Figure 1.2 shows a simplified network of five pages. The numbers under each vertex label are the long-term probabilities of reaching that vertex, and thus the page rank assigned to that page.

Many stochastic processes can be expressed as random walks on graphs in discrete time, or as the limit of such walks in continuous time. These models will play a central role in this book. ■

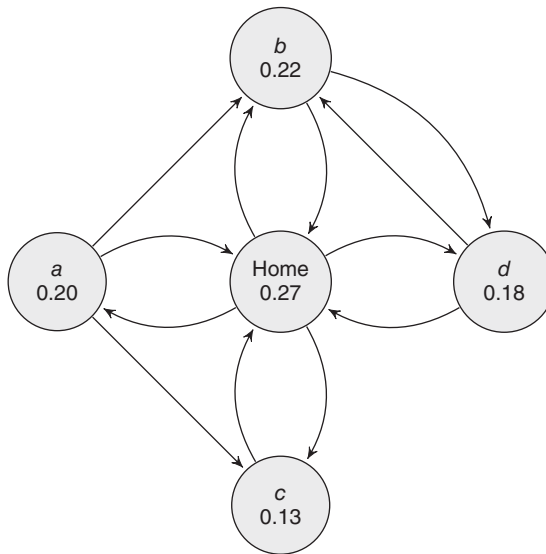


Figure 1.2 Five-page webgraph. Vertex labels show long-term probabilities of reaching each page.

■ **Example 1.2 (Spread of infectious diseases)** Models for the spread of infectious diseases and the development of epidemics are of interest to health scientists, epidemiologists, biologists, and public health officials. Stochastic models are relevant because of the randomness inherent in person-to-person contacts and population fluctuations.

The SIR (Susceptible–Infected–Removed) model is a basic framework, which has been applied to the spread of measles and other childhood diseases. At time t , let S_t represent the number of people susceptible to a disease, I_t the number infected, and R_t the number recovered and henceforth immune from infection. Individuals in the population transition from being susceptible to possibly infected to recovered ($S \rightarrow I \rightarrow R$).

The deterministic SIR model is derived by a system of three nonlinear differential equations, which model interactions and the rate of change in each subgroup.

A stochastic SIR model in discrete time was introduced in the 1920s by medical researchers Lowell Reed and Wade Frost from Johns Hopkins University. In the Reed–Frost model, when a susceptible individual comes in contact with someone who is infected there is a fixed probability z that they will be infected.

Assume that each susceptible person is in contact with all those who are infected. Let p be the probability that a susceptible individual is infected at time t . This is equal to 1 minus the probability that the person is not infected at time t , which occurs if they are not infected by any of the already infected persons, of which there are I_t . This gives

$$p = 1 - (1 - z)^{I_t}.$$

Disease evolution is modeled in discrete time, where one time unit is the incubation period—also the recovery time—of the disease.

The model can be described with a coin-flipping analogy. To find I_{t+1} , the number of individuals infected at time $t + 1$, flip S_t coins (one for each susceptible), where the probability of heads for each coin is the infection probability p . Then, the number of newly infected individuals is the number of coins that land heads.

The number of heads in n independent coin flips with heads probability p has a binomial distribution with parameters n and p . In other words, I_{t+1} has a binomial distribution with $n = S_t$ and $p = 1 - (1 - z)^{I_t}$.

Having found the number of infected individuals at time $t + 1$, the number of susceptible persons decreases by the number of those infected. That is,

$$S_{t+1} = S_t - I_{t+1}.$$

Although the Reed–Frost model is not easy to analyze exactly, it is straightforward to simulate on a computer. The graphs in Figure 1.3 were obtained by simulating the process assuming an initial population of 3 infected and 400 susceptible individuals, with individual infection probability $z = 0.004$. The number of those infected is plotted over 20 time units. ■

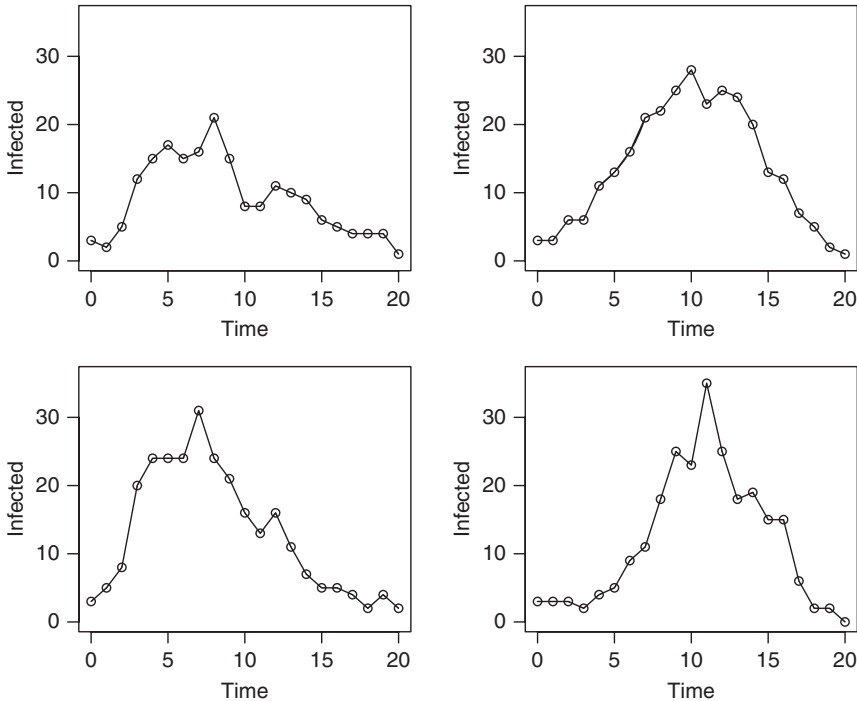


Figure 1.3 Four outcomes of the Reed–Frost epidemic model.

1.2 WHAT IS A STOCHASTIC PROCESS?

In its most general expression, a stochastic process is simply a collection of random variables $\{X_t, t \in I\}$. The index t often represents time, and the set I is the *index set* of the process. The most common index sets are $I = \{0, 1, 2, \dots\}$, representing discrete time, and $I = [0, \infty)$, representing continuous time. Discrete-time stochastic processes are sequences of random variables. Continuous-time processes are uncountable collections of random variables.

The random variables of a stochastic process take values in a common *state space* $\mathcal{S}$, either discrete or continuous. A stochastic process is specified by its index and state spaces, and by the dependency relations among its random variables.

Stochastic Process

A stochastic process is a collection of random variables $\{X_t, t \in I\}$. The set I is the *index set* of the process. The random variables are defined on a common *state space* $\mathcal{S}$.

■ **Example 1.3 (Monopoly)** The popular board game *Monopoly* can be modeled as a stochastic process. Let $X_0, X_1, X_2, \dots$ represent the successive board positions of an individual player. That is, X_k is the player's board position after k plays.

The state space is $\{1, \dots, 40\}$ denoting the 40 squares of a Monopoly board—from Go to Boardwalk. The index set is $\{0, 1, 2, \dots\}$. Both the index set and state space are discrete.

An interesting study is to rank the squares of the board in increasing order of probability. Which squares are most likely to be landed on?

Using Markov chain methods (discussed in Chapter 2), Stewart (1996) shows that the most landed-on square is Jail. The next most frequented square is Illinois Avenue, followed by Go, whereas the least frequented location on the board is the third Chance square from Go. ■

■ **Example 1.4 (Discrete time, continuous state space)** An air-monitoring station in southern California records oxidant concentration levels every hour in order to monitor smog pollution. If it is assumed that hourly concentration levels are governed by some random mechanism, then the station's data can be considered a realization of a stochastic process $X_0, X_1, X_2, \dots$, where X_k is the oxidant concentration level at the k th hour. The time variable is discrete. Since concentration levels take a continuum of values, the state space is continuous. ■

■ **Example 1.5 (Continuous time, discrete state space)** Danny receives text messages at random times day and night. Let X_t be the number of texts he receives up to time t . Then, $\{X_t, t \in [0, \infty)\}$ is a continuous-time stochastic process with discrete state space $\{0, 1, 2, \dots\}$.

This is an example of an *arrival process*. If we assume that the times between Danny's texts are independent and identically distributed (i.i.d.) exponential random variables, we obtain a *Poisson process*. The Poisson process arises in myriad settings involving random *arrivals*. Examples include the number of births each day on a maternity ward, the decay of a radioactive substance, and the occurrences of oil spills in a harbor. ■

■ **Example 1.6 (Random walk and gambler's ruin)** A random walker starts at the origin on the integer line. At each discrete unit of time the walker moves either right or left, with respective probabilities p and $1 - p$. This describes a *simple random walk* in one dimension.

A stochastic process is built as follows. Let $X_1, X_2, \dots$ be a sequence of i.i.d. random variables with

$$X_k = \begin{cases} +1, & \text{with probability } p, \\ -1, & \text{with probability } 1 - p, \end{cases}$$

for $k \geq 1$. Set

$$S_n = X_1 + \dots + X_n, \text{ for } n \geq 1,$$

with $S_0 = 0$. Then, S_n is the random walk's position after n steps. The sequence $S_0, S_1, S_2, \dots$ is a discrete-time stochastic process whose state space is $\mathbb{Z}$, the set of all integers.

Consider a gambler who has an initial stake of k dollars, and repeatedly wagers \$1 on a game for which the probability of winning is p and the probability of losing is $1 - p$. The gambler's successive fortunes is a simple random walk started at k .

Assume that the gambler decides to stop when their fortune reaches $\$n$ ($n > k$), or drops to 0, whichever comes first. What is the probability that the gambler is eventually ruined? This is the classic gambler's ruin problem, first discussed by mathematicians Blaise Pascal and Pierre Fermat in 1656.

See Figure 1.4 for simulations of gambler's ruin with $k = 20$, $n = 60$, and $p = 1/2$. Observe that four of the nine outcomes result in the gambler's ruin before 1,000 plays. In the next section, it is shown that the probability of eventual ruin is $(n - k)/n = (60 - 20)/60 = 2/3$. ■

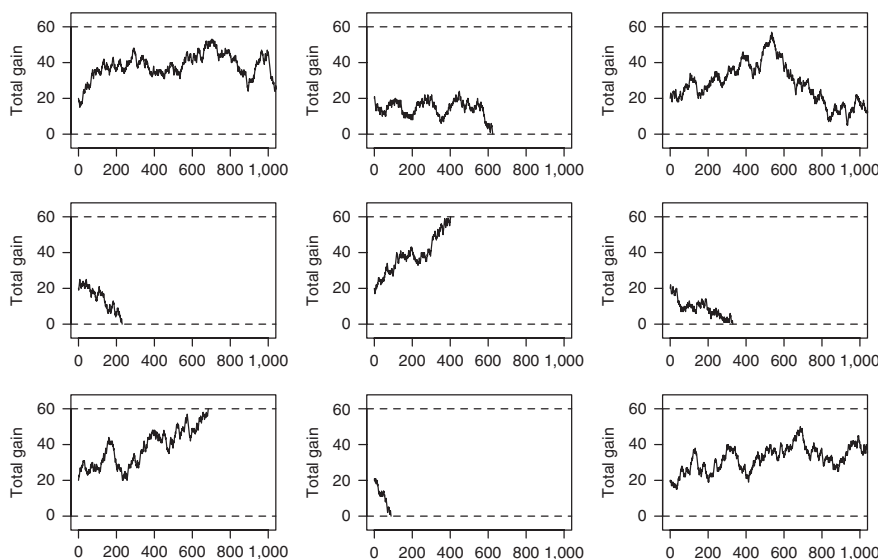


Figure 1.4 Random walk and gambler's ruin.

■ **Example 1.7 (Brownian motion)** Brownian motion is a continuous-time, continuous state space stochastic process. The name also refers to a physical process, first studied by the botanist Robert Brown in 1827. Brown observed the seemingly erratic, zigzag motion of tiny particles ejected from pollen grains suspended in water. He gave a detailed study of the phenomenon but could not explain its cause. In 1905, Albert Einstein showed that the motion was the result of water molecules bombarding the particles.

The mathematical process known as Brownian motion arises as the *limiting process* of a discrete-time random walk. This is obtained by *speeding up* the walk, letting

the interval between discrete steps tend to 0. The process is used as a model for many phenomena that exhibit “erratic, zigzag motion,” such as stock prices, the growth of crystals, and signal noise.

Brownian motion has remarkable properties, which are explored in Chapter 8. Paths of the process are continuous everywhere, yet differentiable nowhere. Figure 1.5 shows simulations of two-dimensional Brownian motion. For this case, the index set is $[0, \infty)$ and the state space is $\mathbb{R}^2$. ■

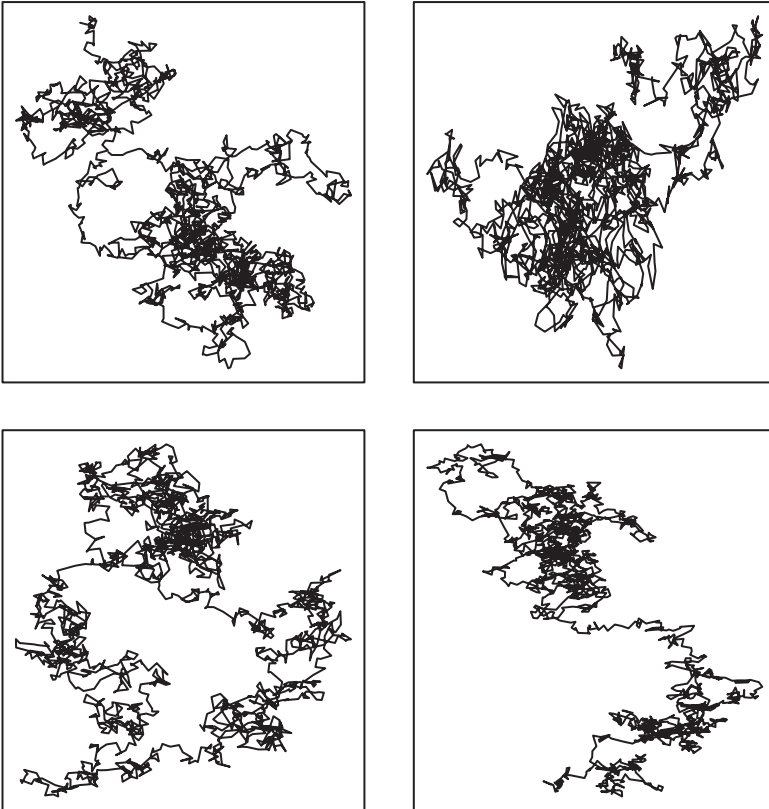


Figure 1.5 Simulations of two-dimensional Brownian motion.

1.3 MONTE CARLO SIMULATION

Advancements in modern computing have revolutionized the study of stochastic systems, allowing for the visualization and simulation of increasingly complex models.

At the heart of the many simulation techniques developed to generate random variables and stochastic processes lies the Monte Carlo method. Given a random experiment and event A , a Monte Carlo estimate of $P(A)$ is obtained by repeating the

random experiment many times and taking the proportion of trials in which A occurs as an approximation for $P(A)$.

The name Monte Carlo evidently has its origins in the fact that the mathematician Stanislaw Ulam, who developed the method in 1946, had an uncle who regularly gambled at the Monte Carlo casino in Monaco.

Monte Carlo simulation is intuitive and matches up with our sense of how probabilities *should* behave. The relative frequency interpretation of probability says that the probability of an event is the long-term proportion of times that the event occurs in repeated trials. It is justified theoretically by the strong law of large numbers.

Consider repeated independent trials of a random experiment. Define the sequence $X_1, X_2, \dots$, where

$$X_k = \begin{cases} 1, & \text{if } A \text{ occurs on the } k\text{th trial,} \\ 0, & \text{if } A \text{ does not occur on the } k\text{th trial,} \end{cases}$$

for $k \geq 1$. Then, $(X_1 + \dots + X_n)/n$ is the proportion of n trials in which A occurs. The X_k are identically distributed with common mean $E(X_k) = P(A)$.

By the strong law of large numbers,

$$\lim_{n \rightarrow \infty} \frac{X_1 + \dots + X_n}{n} = P(A), \text{ with probability 1.} \quad (1.1)$$

For large n , the Monte Carlo estimate of $P(A)$ is

$$P(A) \approx \frac{X_1 + \dots + X_n}{n}.$$

In this book, we use the software package R for simulation. R is a flexible and interactive environment. We often use R to illustrate the result of an exact, theoretical calculation with numerical verification. The easy-to-learn software allows the user to see the impact of varying parameters and assumptions of the model. For example, in the Reed–Frost epidemic model of Example 1.2, it is interesting to see how small changes in the infection probability affect the duration and intensity of the epidemic. See the R script file **ReedFrost.R** and Exercise 1.36 to explore this question.

If you have not used R before, work through the exercises in the introductory tutorial in Appendix A: Getting Started with R.

1.4 CONDITIONAL PROBABILITY

The simplest stochastic process is a sequence of i.i.d. random variables. Such sequences are often used to model random samples in statistics. However, most real-world systems exhibit some type of dependency between variables, and an independent sequence is often an unrealistic model.

Thus, the study of stochastic processes really begins with conditional probability—conditional distributions and conditional expectation. These will become essential tools for all that follows.

Starting with a random experiment, the sample space Ω is the set of all possible outcomes. An *event* is a subset of the sample space. For events A and B , the *conditional probability of A given B* is

$$P(A|B) = \frac{P(A \cap B)}{P(B)},$$

defined for $P(B) > 0$. Events A and B are *independent* if $P(A|B) = P(A)$. Equivalently, A and B are independent if

$$P(A \cap B) = P(A)P(B).$$

Events that are not independent are said to be *dependent*.

For many problems where the goal is to find $P(A)$, partial information and dependencies between events in the sample space are brought to bear. If the sample space can be partitioned into a collection of disjoint events $B_1, \dots, B_k$, then A can be expressed as the disjoint union

$$A = (A \cap B_1) \cup \dots \cup (A \cap B_k).$$

If conditional probabilities of the form $P(A|B_i)$ are known, then the law of total probability can be used to find $P(A)$.

Law of Total Probability

Let $B_1, \dots, B_k$ be a sequence of events that partition the sample space. That is, the B_i are mutually exclusive (disjoint) and their union is equal to Ω . Then, for any event A ,

$$P(A) = \sum_{i=1}^k P(A \cap B_i) = \sum_{i=1}^k P(A|B_i)P(B_i).$$

Example 1.8 According to the Howard Hughes Medical Institute, about 7% of men and 0.4% of women are colorblind—either cannot distinguish red from green or see red and green differently from most people. In the United States, about 49% of the population is male and 51% female. A person is selected at random. What is the probability they are colorblind?

Solution Let C , M , and F denote the events that a random person is colorblind, male, and female, respectively. By the law of total probability,

$$\begin{aligned} P(C) &= P(C|M)P(M) + P(C|F)P(F) \\ &= (0.07)(0.49) + (0.004)(0.51) = 0.03634. \end{aligned}$$

Using the law of total probability in this way is called *conditioning*. Here, we find the *total probability* of being colorblind by conditioning on sex.

■ **Example 1.9** In a standard deck of cards, the probability that the suit of a random card is hearts is $13/52 = 1/4$. Assume that a standard deck has one card missing. A card is picked from the deck. Find the probability that it is a heart.

Solution Assume that the missing card can be any of the 52 cards picked uniformly at random. Let M denote the event that the missing card is a heart, with the complement M^c the event that the missing card is not a heart. Let H denote the event that the card that is picked from the deck is a heart. By the law of total probability,

$$\begin{aligned} P(H) &= P(H|M)P(M) + P(H|M^c)P(M^c) \\ &= \left(\frac{12}{51}\right) \frac{1}{4} + \left(\frac{13}{51}\right) \frac{3}{4} = \frac{1}{4}. \end{aligned}$$

The result can also be obtained by appealing to symmetry. Since all cards are equally likely, and all four suits are equally likely, the argument by symmetry gives that the desired probability is $1/4$. ■

■ **Example 1.10 (Gambler's ruin)** The gambler's ruin problem was introduced in Example 1.6. A gambler starts with k dollars. On each play a fair coin is tossed and the gambler wins \$1 if heads occurs, or loses \$1 if tails occurs. The gambler stops when he reaches \$ n ($n > k$) or loses all his money. Find the probability that the gambler will eventually lose.

Solution We make two observations, which are made more precise in later chapters. First, the gambler will eventually stop playing, either by reaching n or by reaching 0. One might argue that the gambler could play forever. However, it can be shown that that event occurs with probability 0. Second, assume that after, say, 100 wagers, the gambler's capital returns to \$ k . Then, the probability of eventually winning \$ n is the same as it was initially. The memoryless character of the process means that the probability of winning \$ n or losing all his money only depends on how much capital the gambler has, and not on how many previous wagers the gambler made.

Let p_k denote the probability of reaching n when the gambler's fortune is k . What is the gambler's status if heads is tossed? Their fortune increases to $k + 1$ and the probability of winning is the same as it would be if the gambler had started the game with $k + 1$. Similarly, if tails is tossed and the gambler's fortune decreases to $k - 1$. Hence,

$$p_k = p_{k+1} \left(\frac{1}{2}\right) + p_{k-1} \left(\frac{1}{2}\right),$$

or

$$p_{k+1} - p_k = p_k - p_{k-1}, \quad \text{for } k = 1, \dots, n-1, \quad (1.2)$$

with $p_0 = 0$ and $p_n = 1$. Unwinding the recurrence gives

$$p_k - p_{k-1} = p_{k-1} - p_{k-2} = p_{k-2} - p_{k-3} = \cdots = p_1 - p_0 = p_1,$$

for $k = 1, \dots, n$. We have that $p_2 - p_1 = p_1$, giving $p_2 = 2p_1$. Also, $p_3 - p_2 = p_3 - 2p_1 = p_1$, giving $p_3 = 3p_1$. More generally, $p_k = kp_1$, for $k = 1, \dots, n$.

Sum Equation (1.2) over suitable k to obtain

$$\sum_{k=1}^{n-1} (p_{k+1} - p_k) = \sum_{k=1}^{n-1} (p_k - p_{k-1}).$$

Both sums telescope to

$$p_n - p_1 = p_{n-1} - p_0,$$

which gives $1 - p_1 = p_{n-1} = (n-1)p_1$, so $p_1 = 1/n$. Thus,

$$p_k = kp_1 = \frac{k}{n}, \text{ for } k = 0, \dots, n.$$

The probability that the gambler eventually wins $\$n$ is k/n . Hence, the probability of the gambler's ruin is $(n-k)/n$. ■

R : Simulating Gambler's Ruin

The file **gamblersruin.R** contains the function `gamble(k, n, p)`, which simulates the gambler's ruin process. At each wager, the gambler wins with probability p , and loses with probability $1-p$. The gambler's initial stake is $\$k$. The function `gamble` returns 1, if the gambler is eventually ruined, or 0, if the gambler gains $\$n$.

In the simulation the function is called 1,000 times, creating a list of 1,000 ruins and wins, which are represented by 1s and 0s. The mean of the list gives the proportion of 1s, which estimates the probability of eventual ruin.

```
> k <- 20
> n <- 60
> p <- 1/2
> trials <- 1000
> simlist <- replicate(trials, gamble(k,n,p))
> mean(simlist) # Estimate of probability of ruin
[1] 0.664
# Exact probability of ruin is 2/3
```

Sometimes, we need to find a conditional probability of the form $P(B|A)$, but what is given in the problem are *reverse* probabilities of the form $P(A|B)$ and $P(A|B^c)$. Bayes' rule provides a method for *inverting* the conditional probability.

Bayes' Rule

For events A and B ,

$$P(B|A) = \frac{P(A|B)P(B)}{P(A|B)P(B) + P(A|B^c)P(B^c)}.$$

Bayes' rule is a consequence of the law of total probability and the definition of conditional probability, as

$$P(B|A) = \frac{P(A \cap B)}{P(A)} = \frac{P(A|B)P(B)}{P(A)}.$$

For any event B , the events B and B^c partition the sample space. Given a countable sequence of events $B_1, B_2, \dots$, which partition the sample space, a more general form of Bayes' rule is

$$P(B_i|A) = \frac{P(A|B_i)P(B_i)}{\sum_j P(A|B_j)P(B_j)}, \quad \text{for } i = 1, 2, \dots$$

 **Example 1.11** The use of polygraphs (lie detectors) is controversial, and many scientists feel that they should be banned. On the contrary, some polygraph advocates claim that they are mostly accurate. In 1998, the Supreme Court (*United States v. Sheffer*) supported the right of state and federal governments to bar polygraph evidence in court.

Assume that one person in a company of 100 employees is a thief. To find the thief the company will administer a polygraph test to all its employees. The lie detector has the property that if a subject is a liar, there is a 95% probability that the polygraph will detect that they are lying. However, if the subject is telling the truth, there is a 10% chance the polygraph will report a *false positive* and assert that the subject is lying.

Assume that a random employee is given the polygraph test and asked whether they are the thief. The employee says “no,” and the lie detector reports that they are lying. Find the probability that the employee is in fact lying.

Solution Let L denote the event that the employee is a liar. Let D denote the event that the lie detector reports that the employee is a liar. The desired probability is $P(L|D)$. By Bayes' rule,

$$\begin{aligned} P(L|D) &= \frac{P(D|L)P(L)}{P(D|L)P(L) + P(D|L^c)P(L^c)} \\ &= \frac{(0.95)(0.01)}{(0.95)(0.01) + (0.10)(0.99)} = 0.088. \end{aligned}$$

There is less than a 10% chance that the employee is in fact the thief!

Many people, when first given this problem and asked to guess the answer, choose a probability closer to 90%. The mistake is a consequence of confusing the conditional probabilities $P(L|D)$ and $P(D|L)$, the probability that the individual is a liar, given that the polygraph says they are, with the probability that the polygraph says they are lying, given that they are a liar. Since the population of truth tellers is relatively big, the number of false positives—truth tellers whom the lie detector falsely records as being a liar—is also significant. In this case, about 10% of 99, or about 10 employees will be false positives. Assuming that the lie detector correctly identifies the thief as lying, there will be about 11 employees who are identified as liars by the polygraph. The probability that one of them chosen at random is in fact the thief is only about $1/11$. ■

Conditional Distribution

The *distribution* of a random variable X refers to the set of values of X and their corresponding probabilities. The distribution of a random variable is specified with either a probability mass function (pmf), if X is discrete, or a probability density function (pdf), if X is continuous.

For more than one random variable, there is a joint distribution, specified by either a joint pmf or a joint pdf.

If X and Y are discrete random variables, their joint pmf is $P(X = x, Y = y)$, considered a function of x and y . If X and Y are continuous, the joint density function $f(x, y)$ satisfies

$$P(X \leq x, Y \leq y) = \int_{-\infty}^x \int_{-\infty}^y f(s, t) dt ds,$$

for all $x, y \in \mathbb{R}$.

For jointly distributed random variables X and Y , the *conditional distribution of Y given $X = x$* is specified by either a conditional pmf or a conditional pdf.

Discrete Case

The *conditional pmf of Y given $X = x$* is

$$P(Y = y|X = x) = \frac{P(X = x, Y = y)}{P(X = x)},$$

defined when $P(X = x) > 0$. The conditional pmf is a function of y , where x is treated as fixed.

■ **Example 1.12** Max chooses an integer X uniformly at random between 1 and 100. If $X = x$, Mary then chooses an integer Y uniformly at random between 1 and x . Find the conditional pmf of Y given $X = x$.

Solution By the structure of this two-stage random experiment, the conditional distribution of Y given $X = x$ is uniform on $\{1, \dots, x\}$. Thus, the conditional pmf is

$$P(Y = y|X = x) = \frac{1}{x}, \text{ for } y = 1, \dots, x. \quad \blacksquare$$

Note that the conditional pmf is a probability function. For fixed x , the probabilities $P(Y = y|X = x)$ are nonnegative and sum to 1, as

$$\sum_y P(Y = y|X = x) = \sum_y \frac{P(X = x, Y = y)}{P(X = x)} = \frac{P(X = x)}{P(X = x)} = 1.$$

■ **Example 1.13** The joint pmf of X and Y is

$$P(X = x, Y = y) = \frac{x+y}{18}, \text{ for } x, y = 0, 1, 2.$$

Find the conditional pmf of Y given $X = x$.

Solution The marginal distribution of X is

$$P(X = x) = \sum_{y=0}^2 P(X = x, Y = y) = \frac{x}{18} + \frac{x+1}{18} + \frac{x+2}{18} = \frac{x+1}{6},$$

for $x = 0, 1, 2$. The conditional pmf is

$$P(Y = y|X = x) = \frac{P(X = x, Y = y)}{P(X = x)} = \frac{(x+y)/18}{(x+1)/6} = \frac{x+y}{3(x+1)},$$

for $y = 0, 1, 2$. ■

■ **Example 1.14** A bag contains 2 red, 3 blue, and 4 white balls. Three balls are picked from the bag (sampling without replacement). Let B be the number of blue balls picked. Let R be the number of red balls picked. Find the conditional pmf of B given $R = 1$.

Solution We have

$$\begin{aligned} P(B = b|R = 1) &= \frac{P(B = b, R = 1)}{P(R = 1)} \\ &= \frac{\binom{3}{b} \binom{2}{1} \binom{4}{3-b-1} / \binom{9}{3}}{\binom{2}{1} \binom{7}{2} / \binom{9}{3}} = \frac{2 \binom{3}{b} \binom{4}{2-b}}{42} \end{aligned}$$

$$= \frac{\binom{3}{b} \binom{4}{2-b}}{21} = \begin{cases} 2/7, & \text{if } b = 0, \\ 4/7, & \text{if } b = 1, \\ 1/7, & \text{if } b = 2. \end{cases}$$

■

Continuous Case

For continuous random variables X and Y , the *conditional density function of Y given $X = x$* is

$$f_{Y|X}(y|x) = \frac{f(x, y)}{f_X(x)},$$

where f_X is the marginal density function of X . The conditional density is a function of y , where x is treated as fixed.

Conditional densities are used to compute conditional probabilities. For $R \subseteq \mathbb{R}$,

$$P(Y \in R | X = x) = \int_R f_{Y|X}(y|x) dy.$$

■ **Example 1.15** Random variables X and Y have joint density

$$f(x, y) = e^{-x}, \text{ for } 0 < y < x < \infty.$$

Find $P(Y < 2 | X = 5)$.

Solution The desired probability is

$$P(Y < 2 | X = 5) = \int_0^2 f_{Y|X}(y|5) dy.$$

To find the conditional density function $f_{Y|X}(y|x)$, find the marginal density

$$f_X(x) = \int_{-\infty}^{\infty} f(x, y) dy = \int_0^x e^{-x} dy = xe^{-x}, \quad \text{for } x > 0.$$

This gives

$$f_{Y|X}(y|x) = \frac{f(x, y)}{f_X(x)} = \frac{e^{-x}}{xe^{-x}} = \frac{1}{x}, \quad \text{for } 0 < y < x.$$

The conditional distribution of Y given $X = x$ is uniform on $(0, x)$. Hence,

$$P(Y < 2 | X = 5) = \int_0^2 f_{Y|X}(y|5) dy = \int_0^2 \frac{1}{5} dy = \frac{2}{5}.$$

■

■ **Example 1.16** Tom picks a real number X uniformly distributed on $(0, 1)$. Tom shows his number x to Marisa who then picks a number Y uniformly distributed on $(0, x)$. Find (i) the conditional distribution of Y given $X = x$; (ii) the joint distribution of X and Y ; and (iii) the marginal density of Y .

Solution

- (i) The conditional distribution of Y given $X = x$ is given directly in the statement of the problem. The distribution is uniform on $(0, x)$. Thus,

$$f_{Y|X}(y|x) = \frac{1}{x}, \quad \text{for } 0 < y < x.$$

- (ii) For the joint density,

$$f(x, y) = f_{Y|X}(y|x)f_X(x) = \frac{1}{x}(1) = \frac{1}{x}, \quad \text{for } 0 < y < x < 1.$$

- (iii) To find the marginal density of Y , integrate out the x variable in the joint density function. This gives

$$f_Y(y) = \int_{-\infty}^{\infty} f(x, y) dx = \int_y^1 \frac{1}{x} dx = -\ln y, \quad \text{for } 0 < y < 1. \quad \blacksquare$$

1.5 CONDITIONAL EXPECTATION

A conditional expectation is an expectation computed with respect to a conditional probability distribution. Write $E(Y|X = x)$ for the *conditional expectation of Y given $X = x$* .

Conditional Expectation of Y given $X = x$

$$E(Y|X = x) = \begin{cases} \sum_y yP(Y = y|X = x), & \text{discrete,} \\ \int_{-\infty}^{\infty} yf_{Y|X}(y|x) dy, & \text{continuous.} \end{cases}$$

Most important is that $E(Y|X = x)$ is a function of x .

■ **Example 1.17** A school cafeteria has two registers. Let X and Y denote the number of students in line at the first and second registers, respectively. The joint pmf of X and Y is specified by the following joint distribution table.

		Y				
		0	1	2	3	4
X	0	0.15	0.14	0.03	0	0
	1	0.14	0.12	0.06	0.01	0
	2	0.03	0.06	0.10	0.03	0.02
	3	0	0.01	0.03	0.02	0.01
	4	0	0	0.02	0.01	0.01

Find the expected number of people in line at the second register if there is one person at the first register.

Solution The problem asks for $E(Y|X = 1)$. We have

$$E(Y|X = 1) = \sum_{y=0}^4 yP(Y = y|X = 1) = \sum_{y=0}^4 y \frac{P(X = 1, Y = y)}{P(X = 1)}.$$

The marginal probability $P(X = 1)$ is obtained by summing over the $X = 1$ row of the table. That is, $P(X = 1) = 0.14 + 0.12 + 0.06 + 0.01 + 0 = 0.33$. Hence,

$$\begin{aligned} E(Y|X = 1) &= \frac{1}{0.33} \sum_{y=0}^4 yP(X = 1, Y = y) \\ &= \frac{1}{0.33} [0(0.14) + 1(0.12) + 2(0.06) + 3(0.01) + 4(0)] \\ &= 0.818. \end{aligned}$$

■

■ **Example 1.18** Let X and Y be independent Poisson random variables with respective parameters λ and μ . Find the conditional expectation of Y given $X + Y = n$.

Solution First find the conditional pmf of Y given $X + Y = n$. We use the fact that the sum of independent Poisson random variables has a Poisson distribution whose parameter is the sum of the individual parameters. That is, $X + Y$ has a Poisson distribution with parameter $\lambda + \mu$. This gives

$$\begin{aligned} P(Y = y|X + Y = n) &= \frac{P(Y = y, X + Y = n)}{P(X + Y = n)} = \frac{P(Y = y, X = n - y)}{P(X + Y = n)} \\ &= \frac{P(Y = y)P(X = n - y)}{P(X + Y = n)} \\ &= \frac{(e^{-\mu}\mu^y/y!)(e^{-\lambda}\lambda^{n-y}/(n-y)!)}{e^{-(\lambda+\mu)}(\lambda + \mu)^n/n!} \end{aligned}$$

$$\begin{aligned}
&= \frac{n!}{y!(n-y)!} \left(\frac{\mu^y \lambda^{n-y}}{(\lambda + \mu)^n} \right) \\
&= \binom{n}{y} \left(\frac{\mu}{\lambda + \mu} \right)^y \left(\frac{\lambda}{\lambda + \mu} \right)^{n-y},
\end{aligned}$$

for $y = 0, \dots, n$. The form of the conditional pmf shows that the conditional distribution is binomial with parameters n and $p = \mu/(\lambda + \mu)$. The desired conditional expectation is the mean of this binomial distribution. That is,

$$E(Y|X + Y = n) = np = \frac{n\mu}{\lambda + \mu}. \quad \blacksquare$$

Example 1.19 Assume that X and Y have joint density

$$f(x, y) = \frac{2}{xy}, \text{ for } 1 < y < x < e.$$

Find $E(Y|X = x)$.

Solution The marginal density of X is

$$f_X(x) = \int_1^x \frac{2}{xy} dy = \frac{2 \ln x}{x}, \text{ for } 1 < x < e.$$

The conditional density of Y given $X = x$ is

$$f_{Y|X}(y|x) = \frac{f(x, y)}{f_X(x)} = \frac{2/(xy)}{2 \ln x/x} = \frac{1}{y \ln x}, \text{ for } 1 < y < x,$$

with conditional expectation

$$E(Y|X = x) = \int_1^x y f_{Y|X}(y|x) dy = \int_1^x \frac{y}{y \ln x} dy = \frac{x-1}{\ln x}. \quad \blacksquare$$

Key properties of conditional expectation follow.

Properties of Conditional Expectation

1. (Linearity) For constants a and b and random variables X , Y , and Z ,

$$E(aY + bZ|X = x) = aE(Y|X = x) + bE(Z|X = x).$$

2. If g is a function,

$$E(g(Y)|X = x) = \begin{cases} \sum_y g(y)P(Y = y|X = x), & \text{discrete,} \\ \int_{-\infty}^{\infty} g(y)f_{Y|X}(y|x) dy, & \text{continuous.} \end{cases}$$

3. (Independence) If X and Y are independent,

$$E(Y|X = x) = E(Y).$$

4. If $Y = g(X)$ is a function of X ,

$$E(Y|X = x) = g(x).$$

Proof. Properties 1 and 2 are consequences of the fact that conditional expectation is an expectation and thus retains all the properties, such as linearity, of the regular expectation. For a proof of property 2, which is sometimes called *the law of the unconscious statistician*, see Dobrow (2013).

For the independence property in the discrete case, if X and Y are independent,

$$E(Y|X = x) = \sum_y yP(Y = y|X = x) = \sum_y yP(Y = y) = E(Y).$$

The continuous case is similar. For property 4,

$$E(Y|X = x) = E(g(X)|X = x) = E(g(x)|X = x) = g(x). \quad \blacksquare$$

Example 1.20 Consider random variables X , Y , and U , where U is uniformly distributed on $(0, 1)$. Find the conditional expectation

$$E(UX^2 + (1 - U)Y^2|U = u).$$

Solution By linearity of conditional expectation,

$$\begin{aligned} E(UX^2 + (1 - U)Y^2|U = u) &= E(uX^2 + (1 - u)Y^2|U = u) \\ &= uE(X^2|U = u) + (1 - u)E(Y^2|U = u). \end{aligned}$$

If X and Y are also independent of U , the latter expression reduces to

$$uE(X^2) + (1 - u)E(Y^2). \quad \blacksquare$$

Extending the scope of conditional expectation, we show how to condition on a general event. Given an event A , the *indicator* for A is the 0–1 random variable defined as

$$I_A = \begin{cases} 1, & \text{if } A \text{ occurs,} \\ 0, & \text{if } A \text{ does not occur.} \end{cases}$$

Conditional Expectation Given an Event

Let A be an event such that $P(A) > 0$. The *conditional expectation of Y given A* is

$$E(Y|A) = \frac{E(YI_A)}{P(A)}.$$

For the discrete case, the formula gives

$$E(Y|A) = \frac{1}{P(A)} \sum_y yP(\{Y = y\} \cap A) = \sum_y yP(Y = y|A).$$

Let $A_1, \dots, A_k$ be a sequence of events that partition the sample space. Observe that

$$I_{A_1} + \dots + I_{A_k} = 1,$$

since every outcome $\omega \in \Omega$ is contained in exactly one of the A_i s. It follows that

$$Y = \sum_{i=1}^k YI_{A_i}.$$

Taking expectations gives

$$E(Y) = \sum_{i=1}^k E(YI_{A_i}) = \sum_{i=1}^k \left(\frac{E(YI_{A_i})}{P(A_i)} \right) P(A_i) = \sum_{i=1}^k E(Y|A_i)P(A_i).$$

The result is known as the law of total expectation. Note the similarity with the law of total probability.

The law of total expectation is often used with partitioning events $\{X = x\}$. This gives

$$E(Y) = \sum_x E(Y|X = x)P(X = x).$$

In summary, here are two forms of the law of total expectation.

Law of Total Expectation

Let Y be a random variable. Let $A_1, \dots, A_k$ be a sequence of events that partition the sample space. Then,

$$E(Y) = \sum_{i=1}^k E(Y|A_i)P(A_i).$$

If X and Y are jointly distributed random variables,

$$E(Y) = \sum_x E(Y|X = x)P(X = x).$$

■ **Example 1.21** A fair coin is flipped repeatedly. Find the expected number of flips needed to get two heads in a row.

Solution Let Y be the number of flips needed. Consider three events: (i) T , the first flip is tails; (ii) HT , the first flip is heads and the second flip is tails; and (iii) HH , the first two flips are heads. The events T, HT, HH partition the sample space. By the law of total expectation,

$$\begin{aligned} E(Y) &= E(Y|T)P(T) + E(Y|HT)P(HT) + E(Y|HH)P(HH) \\ &= E(Y|T)\frac{1}{2} + E(Y|HT)\frac{1}{4} + (2)\frac{1}{4}. \end{aligned}$$

Consider $E(Y|T)$. Assume that the first flip is tails. Since successive flips are independent, after the first tails we *start over* waiting for two heads in a row. Since one flip has been used, it follows that $E(Y|T) = 1 + E(Y)$. Similarly, after first heads and then tails we start over again, having used up two coin tosses. Thus, $E(Y|HT) = 2 + E(Y)$. This gives

$$E(Y) = (1 + E(Y))\frac{1}{2} + (2 + E(Y))\frac{1}{4} + (2)\frac{1}{4} = E(Y)\frac{3}{4} + \frac{3}{2}.$$

Solving for $E(Y)$ gives $E(Y)(1/4) = 3/2$, or $E(Y) = 6$. ■

■ **Example 1.22** Every day Bob goes to the pizza shop, orders a slice of pizza, and picks a topping—pepper, pepperoni, pineapple, prawns, or prosciutto—uniformly at random. On the day that Bob first picks pineapple, find the expected number of prior days in which he picked pepperoni.

Solution Let Y be the number of days, before the day Bob first picked pineapple, in which he picks pepperoni. Let X be the number of days needed for Bob to first pick pineapple. Then, X has a geometric distribution with parameter $1/5$.

If $X = x$, then on the first $x - 1$ days pineapple was not picked. And for each of these days, given that pineapple was not picked, there was a $1/4$ chance of picking pepperoni. The conditional distribution of Y given $X = x$ is binomial with parameters $x - 1$ and $1/4$. Thus, $E[Y|X = x] = (x - 1)/4$, and

$$E(Y) = \sum_{x=1}^{\infty} E(Y|X = x)P(X = x)$$

$$\begin{aligned}
&= \sum_{x=1}^{\infty} \left(\frac{x-1}{4} \right) \left(\frac{4}{5} \right)^{x-1} \frac{1}{5} \\
&= \left(\frac{1}{4} \sum_{x=1}^{\infty} x \left(\frac{4}{5} \right)^{x-1} \frac{1}{5} \right) - \left(\frac{1}{4} \sum_{x=1}^{\infty} \left(\frac{4}{5} \right)^{x-1} \frac{1}{5} \right) \\
&= \frac{1}{4} E(X) - \frac{1}{4} = \frac{5}{4} - \frac{1}{4} = 1.
\end{aligned}$$

■

R : Simulation of Bob's Pizza Probability

```

> trials <- 10000 # simulation repeated 10,000 times
> simlist <- numeric(trials)
> toppings <- c("pepper", "pepperoni", "pineapple",
  "prawns", "prosciutto")
> for (i in 1:trials) {
>   pineapple <- 0
>   pepperoni <- 0 # counts pepperonis before pineapple
>   while (pineapple == 0) {
      # pick toppings until pineapple is selected
      pick <- sample(toppings, 1)
      if (pick == "pepperoni") pepperoni <- pepperoni + 1
      if (pick == "pineapple") pineapple <- 1
    }
>   simlist[i] <- pepperoni
>   mean(simlist)
[1] 0.9966

```

The next example illustrates conditional expectation given an event in the continuous case.

■ **Example 1.23** The time that Joe spends talking on the phone is exponentially distributed with mean 5 minutes. What is the expected length of his phone call if he talks for more than 2 minutes?

Solution Let Y be the length of Joe's phone call. With $A = \{Y > 2\}$, the desired conditional expectation is

$$\begin{aligned}
E(Y|A) &= E(Y|Y > 2) = \frac{1}{P(Y > 2)} \int_2^{\infty} y \frac{1}{5} e^{-y/5} dy \\
&= \left(\frac{1}{e^{-2/5}} \right) 7e^{-2/5} = 7 \text{ minutes.}
\end{aligned}$$

Note that the solution can be obtained using the memoryless property of the exponential distribution. The conditional distribution of Y given $Y > 2$ is equal to the

distribution of $2 + Z$, where Z has the same distribution as Y . Thus,

$$E(Y|Y > 2) = E(2 + Z) = 2 + E(Z) = 2 + E(Y) = 2 + 5 = 7.$$

■

Conditioning on a Random Variable

From conditioning on an event, we introduce the notion of conditioning on a random variable, a powerful tool for computing conditional expectations and probabilities.

Recall that if X is a random variable and g is a function, then $Y = g(X)$ is a random variable, which is a function of X . The conditional expectation $E(Y|X = x)$ is a function of x . We apply this function to the random variable X and define a new random variable called the *conditional expectation of Y given X* , written $E(Y|X)$. The defining properties of $E(Y|X)$ are given here.

Conditional Expectation of Y given X

The conditional expectation $E(Y|X)$ has three defining properties.

1. $E(Y|X)$ is a random variable.
2. $E(Y|X)$ is a function of X .
3. $E(Y|X)$ is equal to $E(Y|X = x)$ whenever $X = x$. That is, if

$$E(Y|X = x) = g(x), \text{ for all } x,$$

then $E(Y|X) = g(X)$.

■ **Example 1.24** Let X be uniformly distributed on $(0, 1)$. If $X = x$, a second number Y is picked uniformly on $(0, x)$. Find $E(Y|X)$.

Solution For this two-stage random experiment, the conditional distribution of Y given $X = x$ is uniform on $(0, x)$, for $0 < x < 1$. It follows that $E(Y|X = x) = x/2$. Since this is true for all x , $E(Y|X) = X/2$. ■

It may seem that the difference between $E(Y|X)$ and $E(Y|X = x)$ is just a matter of notation, with capital letters replacing lowercase letters. However, as much as they look the same, there is a fundamental difference. The conditional expectation $E(Y|X = x)$ is a function of x . Its domain is a set of real numbers. The deterministic function can be evaluated and graphed. For instance, in the last example $E(Y|X = x) = x/2$ is a linear function of x with slope $1/2$.

On the contrary, $E(Y|X)$ is a random variable. As such, it has a probability distribution. Since $E(Y|X)$ is a random variable with some probability distribution, it makes sense to take *its* expectation with respect to that distribution. The expectation

of a conditional expectation may seem pretty far out. But it leads to one of the most important results in probability.

Law of Total Expectation

For random variables X and Y ,

$$E(Y) = E(E(Y|X)).$$

We prove this result for the discrete case, and leave the continuous case as an exercise for the reader.

Proof.

$$\begin{aligned} E(E(Y|X)) &= \sum_x E(Y|X=x)P(X=x) \\ &= \sum_x \left(\sum_y yP(Y=y|X=x) \right) P(X=x) \\ &= \sum_y y \sum_x P(Y=y|X=x)P(X=x) \\ &= \sum_y y \sum_x P(X=x, Y=y) \\ &= \sum_y yP(Y=y) = E(Y). \end{aligned}$$

■

■ **Example 1.25** Angel will harvest N tomatoes in her garden, where N has a Poisson distribution with parameter λ . Each tomato is checked for defects. The chance that a tomato has defects is p . Defects are independent from tomato to tomato. Find the expected number of tomatoes with defects.

Solution Let X be the number of tomatoes with defects. The conditional distribution of X given $N = n$ is binomial with parameters n and p . This gives $E(X|N=n) = np$. Since this is true for all n , $E(X|N) = Np$. By the law of total expectation,

$$E(X) = E(E(X|N)) = E(Np) = pE(N) = p\lambda.$$

■

■ **Example 1.26** Ellen's insurance will pay for a medical expense subject to a \$100 deductible. Assume that the amount of the expense is exponentially distributed with mean \$500. Find the expectation and standard deviation of the payout.

Solution Let M be the amount of the medical expense and let X be the insurance company's payout. Then,

$$X = \begin{cases} M - 100, & \text{if } M > 100, \\ 0, & \text{if } M \leq 100, \end{cases}$$

where M is exponentially distributed with parameter $1/500$. To find the expected payment, apply the law of total expectation, giving

$$\begin{aligned} E(X) &= E(E(X|M)) = \int_0^{\infty} E(X|M=m) \lambda e^{-\lambda m} dm \\ &= \int_{100}^{\infty} E(M-100|M=m) \frac{1}{500} e^{-m/500} dm \\ &= \int_{100}^{\infty} (m-100) \frac{1}{500} e^{-m/500} dm \\ &= 500e^{-100/500} = \$409.365. \end{aligned}$$

For the standard deviation, first find

$$\begin{aligned} E(X^2) &= E(E(X^2|M)) = \int_0^{\infty} E(X^2|M=m) \lambda e^{-\lambda m} dm \\ &= \int_{100}^{\infty} E((M-100)^2|M=m) \frac{1}{500} e^{-m/500} dm \\ &= \int_{100}^{\infty} (m-100)^2 \frac{1}{500} e^{-m/500} dm \\ &= 500000e^{-1/5} = 409365. \end{aligned}$$

This gives

$$\begin{aligned} SD(X) &= \sqrt{\text{Var}(X)} = \sqrt{E(X^2) - E(X)^2} \\ &= \sqrt{409365 - (409.365)^2} = \$491.72. \end{aligned}$$

■

R : Simulation of Ellen's Payout

```
> trials <- 100000
> simlist <- numeric(trials)
> for (i in 1:trials) {
>   expense <- rexp(1,1/500)
>   payout <- max(0, expense-100)
```

```

> simlist[i] <- payout}
> mean(simlist)
[1] 410.0308
> sd(simlist)
[1] 493.5457

```

■ **Example 1.27 (Random sum of random variables)** A stochastic model for the cost of damage from traffic accidents is given by Van der Lann and Louter (1986). Let X_k be the amount of damage from an individual's k th traffic accident. Assume $X_1, X_2, \dots$ is an i.i.d. sequence with mean μ . The number of accidents N for an individual driver is a random variable with mean λ . It is assumed that the number of accidents is independent of the amount of damages for each accident. That is, N is independent of the X_k . For an individual driver, find the total cost of damages.

Solution Let T be the total cost of damages. Then,

$$T = X_1 + \cdots + X_N = \sum_{k=1}^N X_k.$$

The number of summands is random. The random variable T is a random sum of random variables. By the law of total expectation $E(T) = E(E(T|N))$. To find $E(T|N)$, consider

$$\begin{aligned} E(T|N = n) &= E\left(\sum_{k=1}^N X_k | N = n\right) = E\left(\sum_{k=1}^n X_k | N = n\right) \\ &= E\left(\sum_{k=1}^n X_k\right) = \sum_{k=1}^n E(X_k) = n\mu, \end{aligned}$$

where the third equality is because N is independent of the X_k . Since the final equality holds for all n , $E(T|N) = N\mu$. By the law of total expectation,

$$E(T) = E(E(T|N)) = E(N\mu) = \mu E(N) = \mu\lambda.$$

The result is intuitive. The expected total cost is the product of the expected number of accidents and the expected cost per accident.

Note that it would have been incorrect to write

$$E\left(\sum_{k=1}^N X_k\right) = \sum_{k=1}^N E(X_k).$$

Linearity of expectation does not apply here because the number of summands is random, not fixed. Indeed, this equation does not even make sense as the left-hand

side is a fixed number (the expectation of a random variable), while the right-hand side is a random variable. ■

Computing Probabilities by Conditioning

For an event A , let I_A be the indicator for A . Then,

$$E(I_A) = (1)P(A) + (0)P(A^c) = P(A).$$

From this simple fact, one sees that probabilities can always be expressed as expectations. As such, the law of total expectation can be used when computing probabilities. In particular, if X is a discrete random variable,

$$\begin{aligned} P(A) &= E(I_A) = E(E(I_A|X)) \\ &= \sum_x E(I_A|X=x)P(X=x) \\ &= \sum_x [(1)P(I_A=1|X=x)P(X=x) + (0)P(I_A=0|X=x)P(X=x)] \\ &= \sum_x P(A|X=x)P(X=x). \end{aligned}$$

We have recaptured the law of total probability, where the conditioning events are $\{X=x\}$ for all x .

If X is continuous with density function f_X ,

$$P(A) = \int_{-\infty}^{\infty} E(I_A|X=x)f_X(x) dx = \int_{-\infty}^{\infty} P(A|X=x)f_X(x) dx,$$

which gives the continuous version of the law of total probability.

■ **Example 1.28** Max arrives to class at time X . Mary arrives at time Y . Assume that X and Y have exponential distributions with respective parameters λ and μ . If arrival times are independent, find the probability that Mary arrives before Max.

Solution Let $A = \{Y < X\}$ be the event that Mary arrives to class before Max. By conditioning on Max's arrival time,

$$\begin{aligned} P(A) &= P(Y < X) = \int_{-\infty}^{\infty} P(Y < X|X=x)f_X(x) dx \\ &= \int_0^{\infty} P(Y < x|X=x)\lambda e^{-\lambda x} dx \\ &= \int_0^{\infty} P(Y < x)\lambda e^{-\lambda x} dx \end{aligned}$$

$$\begin{aligned}
&= \int_0^{\infty} (1 - e^{-\mu x}) \lambda e^{-\lambda x} dx \\
&= 1 - \int_0^{\infty} \lambda e^{-(\lambda+\mu)x} dx \\
&= 1 - \frac{\lambda}{\lambda + \mu} = \frac{\mu}{\lambda + \mu}.
\end{aligned}$$

The fourth equality is by independence of X and Y . ■

■ **Example 1.29 (Sums of independent random variables)** Assume that X and Y are independent continuous random variables with density functions f_X and f_Y , respectively. (i) Find the density function of $X + Y$. (ii) Use part (i) to find the density of the sum of two independent standard normal random variables.

Solution

(i) Conditioning on Y ,

$$\begin{aligned}
P(X + Y \leq t) &= \int_{-\infty}^{\infty} P(X + Y \leq t | Y = y) f_Y(y) dy \\
&= \int_{-\infty}^{\infty} P(X \leq t - y | Y = y) f_Y(y) dy \\
&= \int_{-\infty}^{\infty} P(X \leq t - y) f_Y(y) dy.
\end{aligned}$$

Differentiating with respect to t gives

$$f_{X+Y}(t) = \int_{-\infty}^{\infty} f_X(t - y) f_Y(y) dy. \quad (1.3)$$

Equation (1.3) is known as a *convolution* formula.

(ii) For X and Y independent standard normal random variables, by Equation (1.3), $X + Y$ has density

$$\begin{aligned}
f_{X+Y}(t) &= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-(t-y)^2/2} \frac{1}{\sqrt{2\pi}} e^{-y^2/2} dy \\
&= \frac{1}{\sqrt{4\pi}} e^{-t^2/4} \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}(1/2)} e^{-(y-t/2)^2/2(1/2)} dy \\
&= \frac{1}{\sqrt{4\pi}} e^{-t^2/4},
\end{aligned} \quad (1.4)$$

which is the density of a normal distribution with mean 0 and variance 2. The last equality is because the integrand in Equation (1.4) is the density of a normal distribution with mean $t/2$ and variance $1/2$, and thus integrates to 1. ■

Conditional Variance

Analogous to conditional expectation, the conditional variance is a variance taken with respect to a conditional distribution. Given random variables X and Y , let $\mu_x = E(Y|X = x)$. Then, the conditional variance $\text{Var}(Y|X = x)$ is defined as

$$\begin{aligned} \text{Var}(Y|X = x) &= E((Y - \mu_x)^2 | X = x) \\ &= \begin{cases} \sum_y (y - \mu_x)^2 P(Y = y | X = x), & \text{discrete,} \\ \int_{-\infty}^{\infty} (y - \mu_x)^2 f_{Y|X}(y|x) dy, & \text{continuous.} \end{cases} \end{aligned}$$

Compare with the regular variance formula

$$\text{Var}(Y) = E((Y - \mu)^2),$$

where $\mu = E(Y)$. Observe that the conditional expectation $E(Y|X = x)$ takes the place of the unconditional expectation $E(Y)$ in the regular variance formula.

■ **Example 1.30** Let N be a positive, integer-valued random variable. If $N = n$, flip n coins, each of which has heads probability p . Let Y be the number of coins which come up heads. Find $\text{Var}(Y|N = n)$.

Solution The conditional distribution of Y given $N = n$ is binomial with parameters n and p . From the properties of the binomial distribution,

$$\text{Var}(Y|N = n) = np(1 - p). \quad \blacksquare$$

Properties of the variance transfer to the conditional variance.

Properties of Conditional Variance

1.

$$\text{Var}(Y|X = x) = E(Y^2|X = x) - (E(Y|X = x))^2.$$

2. For constants a and b ,

$$\text{Var}(aY + b|X = x) = a^2 \text{Var}(Y|X = x).$$

As with the development of conditional expectation, define the *conditional variance* $\text{Var}(Y|X)$ as the random variable which is a function of X and which takes the value $\text{Var}(Y|X = x)$ when $X = x$.

The law of total variance shows how to find the variance of a random variable by conditioning.

Law of Total Variance

$$\text{Var}(Y) = E(\text{Var}(Y|X)) + \text{Var}(E(Y|X)).$$

The proof is easier than you might think. We have that

$$\begin{aligned} E(\text{Var}(Y|X)) &= E(E(Y^2|X) - (E(Y|X))^2) \\ &= E(E(Y^2|X)) - E((E(Y|X))^2) \\ &= E(Y^2) - E((E(Y|X))^2). \end{aligned}$$

And

$$\begin{aligned} \text{Var}(E(Y|X)) &= E((E(Y|X))^2) - (E(E(Y|X)))^2 \\ &= E((E(Y|X))^2) - (E(Y))^2. \end{aligned}$$

Thus,

$$\begin{aligned} &E(\text{Var}(Y|X)) + \text{Var}(E(Y|X)) \\ &= (E(Y^2) - E((E(Y|X))^2)) + (E((E(Y|X))^2) - (E(Y))^2) \\ &= E(Y^2) - (E(Y))^2 = \text{Var}(Y). \end{aligned}$$

Example 1.31 A number X is uniformly distributed on $(0, 1)$. If $X = x$, then Y is picked uniformly on $(0, x)$. Find the variance of Y .

Solution The conditional distribution of Y given $X = x$ is uniform on $(0, x)$. From properties of the uniform distribution,

$$E(Y|X = x) = \frac{x}{2} \quad \text{and} \quad \text{Var}(Y|X = x) = \frac{x^2}{12}.$$

This gives $E(Y|X) = X/2$ and $\text{Var}(Y|X) = X^2/12$. By the law of total variance,

$$\text{Var}(Y) = E(\text{Var}(Y|X)) + \text{Var}(E(Y|X)) = E\left(\frac{X^2}{12}\right) + \text{Var}\left(\frac{X}{2}\right)$$

$$\begin{aligned}
 &= \frac{1}{12} E(X^2) + \frac{1}{4} \text{Var}(X) = \frac{1}{12} \left(\frac{1}{3} \right) + \frac{1}{4} \left(\frac{1}{12} \right) \\
 &= \frac{7}{144} = 0.04861.
 \end{aligned}$$

■

R : Simulation of $\text{Var}(Y)$

The structure of this two-stage random experiment makes it especially easy to simulate in R .

```

> trials <- 100000
> simlist <- replicate(trials, runif(1, 0, runif(1)))
> var(simlist)
[1] 0.04840338

```

■ **Example 1.32 (Variance of a random sum of random variables)** Assume that $X_1, X_2, \dots$ is an i.i.d. sequence with common mean μ_X and variance σ_X^2 . Let N be a positive, integer-valued random variable that is independent of the X_i with mean μ_N and variance σ_N^2 . Let $T = X_1 + \dots + X_N$. Find the variance of T .

Solution Random sums of random variables were introduced in Example 1.27 where the expectation $E(T) = \mu_X \mu_N$ was found using the law of total expectation. By the law of total variance,

$$\text{Var}(T) = \text{Var} \left(\sum_{k=1}^N X_k \right) = E \left(\text{Var} \left(\sum_{k=1}^N X_k | N \right) \right) + \text{Var} \left(E \left(\sum_{k=1}^N X_k | N \right) \right).$$

We have that

$$\begin{aligned}
 \text{Var} \left(\sum_{k=1}^N X_k | N = n \right) &= \text{Var} \left(\sum_{k=1}^n X_k | N = n \right) \\
 &= \text{Var} \left(\sum_{k=1}^n X_k \right) = \sum_{k=1}^n \text{Var}(X_k) \\
 &= n \sigma_X^2.
 \end{aligned}$$

The second equality is because N is independent of the X_k . The third equality is because the X_k are independent. This gives

$$\text{Var} \left(\sum_{k=1}^N X_k | N \right) = N \sigma_X^2.$$

From results for conditional expectation,

$$E\left(\sum_{k=1}^N X_k | N\right) = NE(X_1) = N\mu_X.$$

This gives

$$\begin{aligned} \text{Var}(T) &= E\left(\text{Var}\left(\sum_{k=1}^N X_k | N\right)\right) + \text{Var}\left(E\left(\sum_{k=1}^N X_k | N\right)\right) \\ &= E(N\sigma_X^2) + \text{Var}(N\mu_X) \\ &= \sigma_X^2 E(N) + \mu_X^2 \text{Var}(N) \\ &= \sigma_X^2 \mu_N + \mu_X^2 \sigma_N^2. \end{aligned}$$

■

Random sums of independent random variables will arise in several contexts. Results are summarized here.

Random Sums of Random Variables

Let $X_1, X_2, \dots$ be an i.i.d. sequence of random variables with common mean μ_X and variance σ_X^2 . Let N be a positive, integer-valued random variable independent of the X_i with $E(N) = \mu_N$ and $\text{Var}(N) = \sigma_N^2$. Let $T = \sum_{i=1}^N X_i$. Then,

$$E(T) = \mu_X \mu_N \text{ and } \text{Var}(T) = \sigma_X^2 \mu_N + \sigma_N^2 \mu_X^2.$$

EXERCISES

1.1 For the following scenarios identify a stochastic process $\{X_t, t \in I\}$, describing (i) X_t in context, (ii) state space, and (iii) index set. State whether the state space and index set are discrete or continuous.

- (a) From day to day the weather in International Falls, Minnesota is either rain, clear, or snow.

Solution: Let X_t denote the weather on day t . Discrete state space is $S = \{\text{Rain, Clear, Snow}\}$. Discrete index set is $I = \{0, 1, 2, \dots\}$.

- (b) At the end of each year, a 4-year college student either advances in grade, repeats their grade, or drops out.

- (c) Seismologists record daily earthquake magnitudes in Chile. The largest recorded earthquake in history was the Valdivia, Chile earthquake on May 22, 1960, which had a magnitude of 9.5 on the Richter scale.
 - (d) Data are kept on the circumferences of trees in an arboretum. The arboretum covers a two square-mile area.
 - (e) Starting Monday morning at 9 a.m., as students arrive to class, the teacher records student arrival times. The class has 30 students and lasts for 60 minutes.
 - (f) A card player shuffles a standard deck of cards by the following method: the top card of the deck is placed somewhere in the deck at random. The player does this 100 times to mix up the deck.
- 1.2** A regional insurance company insures homeowners against flood damage. Half of their policyholders are in Florida, 30% in Louisiana, and 20% in Texas. Company actuaries give the estimates in Table 1.1 for the probability that a policyholder will file a claim for flood damage over the next year.
- (a) Find the probability that a random policyholder will file a claim for flood damage next year.
 - (b) A claim was filed. Find the probability that the policyholder is from Texas.

TABLE 1.1 Probability of Claim for Flood Damage

Florida	Louisiana	Texas
0.03	0.015	0.02

- 1.3** Let $B_1, \dots, B_k$ be a partition of the sample space. For events A and C , prove the *law of total probability for conditional probability*

$$P(A|C) = \sum_{i=1}^k P(A|B_i \cap C)P(B_i|C).$$

- 1.4** See Exercise 1.2. Among all policyholders who live within five miles of the Atlantic Ocean, 75% live in Florida, 20% live in Louisiana, and 5% live in Texas. For those who live close to the ocean the probabilities of filing a claim increase, as given in Table 1.2. Assume that a policyholder lives within five miles of the Atlantic coast. Use the law of total probability for conditional probability in Exercise 1.3 to find the chance they will file a claim for flood damage next year.
- 1.5** Two fair, six-sided dice are rolled. Let X_1, X_2 be the outcomes of the first and second die, respectively.

TABLE 1.2 Probability of Claim for Those Within Five Miles of Atlantic Coast

Florida	Louisiana	Texas
0.10	0.06	0.06

- (a) Find the conditional distribution of X_2 given that $X_1 + X_2 = 7$.
- (b) Find the conditional distribution of X_2 given that $X_1 + X_2 = 8$.
- 1.6** Bob has n coins in his pocket. One is two-headed, the rest are fair. A coin is picked at random, flipped, and shows heads. Find the probability that the coin is two-headed.
- 1.7** A rat is trapped in a maze with three doors and some hidden cheese. If the rat takes door one, he will wander around the maze for 2 minutes and return to where he started. If he takes door two, he will wander around the maze for 3 minutes and return to where he started. If he takes door three, he will find the cheese after 1 minute. If the rat returns to where he started he immediately picks a door to pass through. The rat picks each door uniformly at random. How long, on average, will the rat wander before finding the cheese?
- 1.8** A bag contains 1 red, 3 green, and 5 yellow balls. A sample of four balls is picked. Let G be the number of green balls in the sample. Let Y be the number of yellow balls in the sample.
- (a) Find the conditional probability mass function of G given $Y = 2$ assuming the sample is picked without replacement.
- (b) Find the conditional probability mass function of G given $Y = 2$ assuming the sample is picked with replacement.
- 1.9** Assume that X is uniformly distributed on $\{1, 2, 3, 4\}$. If $X = x$, then Y is uniformly distributed on $\{1, \dots, x\}$. Find
- (a) $P(Y = 2|X = 2)$
- (b) $P(Y = 2)$
- (c) $P(X = 2|Y = 2)$
- (d) $P(X = 2)$
- (e) $P(X = 2, Y = 2)$.
- 1.10** A die is rolled until a 3 occurs. By conditioning on the outcome of the first roll, find the probability that an even number of rolls is needed.
- 1.11** Consider the gambler's ruin process where at each wager, the gambler wins with probability p and loses with probability $q = 1 - p$. The gambler stops when reaching $\$n$ or losing all their money. If the gambler starts with $\$k$, with $0 < k < n$, find the probability of eventual ruin. See Example 1.10.

1.12 In n rolls of a fair die, let X be the number of times 1 is rolled, and Y the number of times 2 is rolled. Find the conditional distribution of X given $Y = y$.

1.13 Random variables X and Y have joint density

$$f(x, y) = 3y, \quad \text{for } 0 < x < y < 1.$$

- (a) Find the conditional density of Y given $X = x$.
- (b) Find the conditional density of X given $Y = y$. Describe the conditional distribution.

1.14 Random variables X and Y have joint density function

$$f(x, y) = 4e^{-2x}, \quad \text{for } 0 < y < x < \infty.$$

- (a) Find the conditional density of X given $Y = y$.
- (b) Find the conditional density of Y given $X = x$. Describe the conditional distribution.

1.15 Let X and Y be uniformly distributed on the disk of radius 1 centered at the origin. Find the conditional distribution of Y given $X = x$.

1.16 A poker hand consists of five cards drawn from a standard 52-card deck. Find the expected number of aces in a poker hand given that the first card drawn is an ace.

1.17 Let X be a Poisson random variable with $\lambda = 3$. Find $E(X|X > 2)$.

1.18 From the definition of conditional expectation given an event, show that

$$E(I_B|A) = P(B|A).$$

1.19 See Example 1.21. Find the variance of the number of flips needed to get two heads in a row.

1.20 A fair coin is flipped repeatedly.

- (a) Find the expected number of flips needed to get three heads in a row.
- (b) Find the expected number of flips needed to get k heads in a row.

1.21 Let T be a nonnegative, continuous random variable. Show

$$E(T) = \int_0^{\infty} P(T > t) dt.$$

1.22 Find $E(Y|X)$ when (X, Y) is uniformly distributed on the following regions.

- (a) The rectangle $[a, b] \times [c, d]$.
- (b) The triangle with vertices $(0, 0)$, $(1, 0)$, $(1, 1)$.
- (c) The disc of radius 1 centered at the origin.

- 1.23** Let $X_1, X_2, \dots$ be an i.i.d. sequence of random variables with common mean μ . Let $S_n = X_1 + \dots + X_n$, for $n \geq 1$.
- Find $E(S_m|S_n)$, for $m \leq n$.
 - Find $E(S_m|S_n)$ for $m > n$.
- 1.24** Prove the law of total expectation $E(Y) = E(E(Y|X))$ for the continuous case.
- 1.25** Let X and Y be independent exponential random variables with respective parameters 1 and 2. Find $P(X/Y < 3)$ by conditioning.
- 1.26** The density of X is $f(x) = xe^{-x}$, for $x > 0$. Given $X = x$, Y is uniformly distributed on $(0, x)$. Find $P(Y < 2)$ by conditioning on X .
- 1.27** A restaurant receives N customers per day, where N is a random variable with mean 200 and standard deviation 40. The amount spent by each customer is normally distributed with mean \$15 and standard deviation \$3. The amounts that customers spend are independent of each other and independent of N . Find the mean and standard deviation of the total amount spent at the restaurant per day.
- 1.28** On any day, the number of accidents on the highway has a Poisson distribution with parameter Λ . The parameter Λ varies from day to day and is itself a random variable. Find the mean and variance of the number of accidents per day when Λ is uniformly distributed on $(0, 3)$.
- 1.29** If X and Y are independent, does $\text{Var}(Y|X) = \text{Var}(Y)$?
- 1.30** Assume that $Y = g(X)$ is a function of X . Find simple expressions for
- $E(Y|X)$.
 - $\text{Var}(Y|X)$.
- 1.31** Consider a sequence of i.i.d. Bernoulli trials with success parameter p . Let X be the number of trials needed until the first success occurs. Then, X has a geometric distribution with parameter p . Find the variance of X by conditioning on the first trial.
- 1.32** R: Simulate flipping three fair coins and counting the number of heads X .
- Use your simulation to estimate $P(X = 1)$ and $E(X)$.
 - Modify the above to allow for a biased coin where $P(\text{Heads}) = 3/4$.
- 1.33** R: Cards are drawn from a standard deck, with replacement, until an ace appears. Simulate the mean and variance of the number of cards required.
- 1.34** R: The time until a bus arrives has an exponential distribution with mean 30 minutes.
- Use the command `rexp()` to simulate the probability that the bus arrives in the first 20 minutes.
 - Use the command `pexp()` to compare with the exact probability.

- 1.35** R: See the script file **gamblersruin.R**. A gambler starts with a \$60 initial stake.
- (a) The gambler wins, and loses, each round with probability $p = 0.50$. Simulate the probability the gambler wins \$100 before he loses everything.
 - (b) The gambler wins each round with probability $p = 0.51$. Simulate the probability the gambler wins \$100 before he loses everything.
- 1.36** R: See Example 1.2 and the script file **ReedFrost.R**. Observe the effect on the course of the disease by changing the initial values for the number of people susceptible and infected. How does increasing the number of infected people affect the duration of the disease?
- 1.37** R: Simulate the results of Exercise 1.28. Estimate the mean and variance of the number of accidents per day.

2

MARKOV CHAINS: FIRST STEPS

Let us finish the article and the whole book with a good example of dependent trials, which approximately can be considered as a simple chain.

—Andrei Andreyevich Markov

2.1 INTRODUCTION

Consider a game with a playing board consisting of squares numbered 1–10 arranged in a circle. (Think of miniature Monopoly.) A player starts at square 1. At each turn, the player rolls a die and moves around the board by the number of spaces shown on the face of the die. The player keeps moving around and around the board according to the roll of the die. (Granted, this is not a very exciting game.)

Let X_k be the number of squares the player lands on after k moves, with $X_0 = 1$. Assume that the player successively rolls 2, 1, and 4. The first four positions are

$$(X_0, X_1, X_2, X_3) = (1, 3, 4, 8).$$

Given this information, what can be said about the player's next location X_4 ? Even though we know the player's full past history of moves, the only information relevant

for predicting their future position is their most recent location X_3 . Since $X_3 = 8$, then necessarily $X_4 \in \{9, 10, 1, 2, 3, 4\}$, with equal probability. Formally,

$$P(X_4 = j | X_0 = 1, X_1 = 3, X_2 = 4, X_3 = 8) = P(X_4 = j | X_3 = 8) = \frac{1}{6},$$

for $j = 9, 10, 1, 2, 3, 4$. Given the player's most recent location X_3 , their future position X_4 is independent of past history X_0, X_1, X_2 .

The sequence of player's locations $X_0, X_1, \dots$ is a stochastic process called a *Markov chain*. The game illustrates the essential property of a Markov chain: the future, given the present, is independent of the past.

Markov Chain

Let S be a discrete set. A Markov chain is a sequence of random variables $X_0, X_1, \dots$ taking values in S with the property that

$$\begin{aligned} P(X_{n+1} = j | X_0 = x_0, \dots, X_{n-1} = x_{n-1}, X_n = i) \\ = P(X_{n+1} = j | X_n = i), \end{aligned} \quad (2.1)$$

for all $x_0, \dots, x_{n-1}, i, j \in S$, and $n \geq 0$. The set S is the *state space* of the Markov chain.

We often use descriptive language to describe the evolution of a Markov chain. For instance, if $X_n = i$, we say that the chain *visits* state i , or *hits* i , at *time* n .

A Markov chain is *time-homogeneous* if the probabilities in Equation (2.1) do not depend on n . That is,

$$P(X_{n+1} = j | X_n = i) = P(X_1 = j | X_0 = i), \quad (2.2)$$

for all $n \geq 0$. Unless stated otherwise, the Markov chains in this book are time-homogeneous.

Since the probabilities in Equation (2.2) only depend on i and j , they can be arranged in a matrix P , whose ij th entry is $P_{ij} = P(X_1 = j | X_0 = i)$. This is the *transition matrix*, or *Markov matrix*, which contains the one-step transition probabilities of moving from state to state.

If the state space has k elements, then the transition matrix is a square $k \times k$ matrix. If the state space is countably infinite, the transition matrix is infinite.

For the simple board game Markov chain, the sample space is

$$S = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\},$$

with transition matrix

$$P = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \\ 6 \\ 7 \\ 8 \\ 9 \\ 10 \end{matrix} & \left(\begin{array}{cccccccccc} 0 & 1/6 & 1/6 & 1/6 & 1/6 & 1/6 & 1/6 & 0 & 0 & 0 \\ 0 & 0 & 1/6 & 1/6 & 1/6 & 1/6 & 1/6 & 1/6 & 0 & 0 \\ 0 & 0 & 0 & 1/6 & 1/6 & 1/6 & 1/6 & 1/6 & 1/6 & 0 \\ 0 & 0 & 0 & 0 & 1/6 & 1/6 & 1/6 & 1/6 & 1/6 & 1/6 \\ 1/6 & 0 & 0 & 0 & 0 & 1/6 & 1/6 & 1/6 & 1/6 & 1/6 \\ 1/6 & 1/6 & 0 & 0 & 0 & 0 & 1/6 & 1/6 & 1/6 & 1/6 \\ 1/6 & 1/6 & 1/6 & 0 & 0 & 0 & 0 & 1/6 & 1/6 & 1/6 \\ 1/6 & 1/6 & 1/6 & 1/6 & 0 & 0 & 0 & 0 & 1/6 & 1/6 \\ 1/6 & 1/6 & 1/6 & 1/6 & 1/6 & 0 & 0 & 0 & 0 & 1/6 \\ 1/6 & 1/6 & 1/6 & 1/6 & 1/6 & 1/6 & 0 & 0 & 0 & 0 \end{array} \right) \end{matrix}.$$

The entries of every Markov transition matrix P are nonnegative, and each row sums to 1, as

$$\sum_j P_{ij} = \sum_j P(X_1 = j | X_0 = i) = \sum_j \frac{P(X_1 = j, X_0 = i)}{P(X_0 = i)} = \frac{P(X_0 = i)}{P(X_0 = i)} = 1,$$

for all rows i . A nonnegative matrix whose rows sum to 1 is called a *stochastic matrix*.

Stochastic Matrix

A stochastic matrix is a square matrix P , which satisfies

1. $P_{ij} \geq 0$ for all i, j .
2. For each row i ,

$$\sum_j P_{ij} = 1.$$

2.2 MARKOV CHAIN CORNUCOPIA

Following is a taste of the wide range of applications of Markov chains. Many of these examples are referenced throughout the book.

■ **Example 2.1 (Heads you win)** Successive coin flips are the very model of independent events. Yet in a fascinating study of how people actually flip coins, Diaconis (2007) shows that vigorously flipped coins are ever so slightly biased to come up the same way they started. “For natural flips,” Diaconis asserts, “the chance of coming up as started is about 0.51.”

In other words, successive coin flips are not independent. They can be described by a Markov chain with transition matrix

$$P = \begin{matrix} & \begin{matrix} H & T \end{matrix} \\ \begin{matrix} H \\ T \end{matrix} & \begin{pmatrix} 0.51 & 0.49 \\ 0.49 & 0.51 \end{pmatrix} \end{matrix}.$$

Note on Notation

In Example 2.1, the state space is $S = \{H, T\}$, and the transition matrix is labeled with row and column identifiers. For many Markov chains, such as this, random variables take values in a discrete state space whose elements are not necessarily numbers.

For matrix entries, we can use suitable labels to identify states. For instance, for this example $P_{HH} = P_{TT} = 0.51$ and $P_{HT} = P_{TH} = 0.49$.

■ **Example 2.2 (Poetry and dependent sequences)** Andrei Andreyevich Markov, the Russian mathematician who introduced Markov chains over 100 years ago, first applied them in the analysis of the poem *Eugene Onegin* by Alexander Pushkin. In the first 20,000 letters of the poem, Markov counted (by hand!) 8,638 vowels and 11,362 consonants. He also tallied pairs of successive letters. Of the 8,638 pairs that start with vowels, 1,104 pairs are vowel–vowel. Of the 11,362 pairs that start with consonants, 3,827 are consonant–consonant. Markov treated the succession of letter types as a random sequence. The resulting transition matrix is

$$P = \begin{matrix} & \begin{matrix} \text{vowel} & \text{consonant} \end{matrix} \\ \begin{matrix} \text{v} \\ \text{c} \end{matrix} & \begin{pmatrix} 1,104/8,638 & 7,534/8,638 \\ 7,535/11,362 & 3,827/11,362 \end{pmatrix} = \begin{pmatrix} 0.175 & 0.825 \\ 0.526 & 0.474 \end{pmatrix} \end{matrix}.$$

Markov showed that the succession of letter types was not an independent sequence. For instance, if letter types were independent, the probability of two successive consonants would be $(11,362/20,000)^2 = 0.323$, whereas from Pushkin's poem the probability is $P_{cc} = 0.474$.

Markov's work was a polemic against a now obscure mathematician who argued that the law of large numbers only applied to independent sequences. Markov disproved the claim by showing that the Pushkin letter sequence was a dependent sequence for which the law of large numbers applied. ■

■ **Example 2.3 (Chained to the weather)** Some winter days in Minnesota it seems like the snow will never stop. A Minnesotan's view of winter might be described by the following transition matrix for a weather Markov chain, where r , s , and c denote

rain, snow, and clear, respectively.

$$P = \begin{matrix} & \begin{matrix} r & s & c \end{matrix} \\ \begin{matrix} r \\ s \\ c \end{matrix} & \begin{pmatrix} 0.2 & 0.6 & 0.2 \\ 0.1 & 0.8 & 0.1 \\ 0.1 & 0.6 & 0.3 \end{pmatrix} \end{matrix}.$$

For this model, no matter what the weather today, there is always at least a 60% chance that it will snow tomorrow. ■

■ **Example 2.4 (I.i.d. sequence)** An independent and identically distributed sequence of random variables is trivially a Markov chain. Assume that $X_0, X_1, \dots$ is an i.i.d. sequence that takes values in $\{1, \dots, k\}$ with

$$P(X_n = j) = p_j, \text{ for } j = 1, \dots, k, \text{ and } n \geq 0,$$

where $p_1 + \dots + p_k = 1$. By independence,

$$P(X_1 = j | X_0 = i) = P(X_1 = j) = p_j.$$

The transition matrix is

$$P = \begin{matrix} & \begin{matrix} 1 & 2 & \dots & k \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ \vdots \\ k \end{matrix} & \begin{pmatrix} p_1 & p_2 & \dots & p_k \\ p_1 & p_2 & \dots & p_k \\ \vdots & \vdots & \ddots & \vdots \\ p_1 & p_2 & \dots & p_k \end{pmatrix} \end{matrix}.$$

The matrix for an i.i.d. sequence has identical rows as the next state of the chain is independent of, and has the same distribution as, the present state. ■

■ **Example 2.5 (Gambler's ruin)** Gambler's ruin was introduced in Chapter 1. In each round of a gambling game a player either wins \$1, with probability p , or loses \$1, with probability $1 - p$. The gambler starts with \$ k . The game stops when the player either loses all their money, or gains a total of \$ n ($n > k$).

The gambler's successive fortunes form a Markov chain on $\{0, 1, \dots, n\}$ with $X_0 = k$ and transition matrix given by

$$P_{ij} = \begin{cases} p, & \text{if } j = i + 1, \quad 0 < i < n, \\ 1 - p, & \text{if } j = i - 1, \quad 0 < i < n, \\ 1, & \text{if } i = j = 0, \text{ or } i = j = n, \\ 0, & \text{otherwise.} \end{cases}$$

Here is the transition matrix with $n = 6$ and $p = 1/3$.

$$P = \begin{matrix} & \begin{matrix} 0 & 1 & 2 & 3 & 4 & 5 & 6 \end{matrix} \\ \begin{matrix} 0 \\ 1 \\ 2 \\ 3 \\ 4 \\ 5 \\ 6 \end{matrix} & \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 2/3 & 0 & 1/3 & 0 & 0 & 0 & 0 \\ 0 & 2/3 & 0 & 1/3 & 0 & 0 & 0 \\ 0 & 0 & 2/3 & 0 & 1/3 & 0 & 0 \\ 0 & 0 & 0 & 2/3 & 0 & 1/3 & 0 \\ 0 & 0 & 0 & 0 & 2/3 & 0 & 1/3 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix} \end{matrix}.$$

Gambler's ruin is an example of *simple random walk with absorbing boundaries*. Since $P_{00} = P_{nn} = 1$, when the chain reaches 0 or n , it stays there forever. ■

■ **Example 2.6 (Wright–Fisher model)** The Wright–Fisher model describes the evolution of a fixed population of k genes. Genes can be one of two types, called alleles: A or a . Let X_n denote the number of A alleles in the population at time n , where time is measured by generations. Under the model, the number of A alleles at time $n + 1$ is obtained by drawing with replacement from the gene population at time n . Thus, conditional on there being i alleles of type A at time n , the number of A alleles at time $n + 1$ has a binomial distribution with parameters k and $p = i/k$. This gives a Markov chain with transition matrix defined by

$$P_{ij} = \binom{k}{j} \left(\frac{i}{k}\right)^j \left(1 - \frac{i}{k}\right)^{k-j}, \quad \text{for } 0 \leq i, j \leq k.$$

Observe that $P_{00} = P_{kk} = 1$. As the chain progresses, the population is eventually made up of all a alleles (state 0) or all A alleles (state k). A question of interest is what is the probability that the population evolves to the all- A state? ■

■ **Example 2.7 (Squash)** Squash is a popular racket sport played by two or four players in a four-walled court. In the international scoring system, the first player to score nine points is the winner. However, if the game is tied at 8-8, the player who reaches 8 points first has two options: (i) to play to 9 points (set one) or to play to 10 points (set two). Set one means that the next player to score wins the game. Set two means that the first player to score two points wins the game. Points are only scored by the player who is serving. A player who wins a rally serves the next rally. Thus, if the game is tied 8-8, the player who is not serving decides. Should they choose set one or set two? This endgame play is modeled by a Markov chain in Broadie and Joneja (1993). The two players are called A and B and a score of xy means that A has scored x points and B has scored y points. The states of the chain are defined by the score and the server. The authors let p be the probability that A wins a rally given that A is serving, and q be the probability that A wins a rally given that B is serving. Assumptions are that p and q are constant over time and independent of the current

score. Following is the transition matrix of the Markov chain for the set two options. The chain is used to solve for the optimal strategy.

$$P = \begin{matrix} & \begin{matrix} 88B & 88A & 89B & 89A & 98B & 98A & 99B & 99A & A \text{ loses} & A \text{ wins} \end{matrix} \\ \begin{matrix} 88B \\ 88A \\ 89B \\ 89A \\ 98B \\ 98A \\ 99B \\ 99A \\ A \text{ loses} \\ A \text{ wins} \end{matrix} & \begin{pmatrix} 0 & q & 1-q & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1-p & 0 & 0 & 0 & 0 & p & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & q & 0 & 0 & 0 & 0 & 1-q & 0 \\ 0 & 0 & 1-p & 0 & 0 & 0 & 0 & p & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & q & 1-q & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1-p & 0 & 0 & 0 & 0 & p \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & q & 1-q & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1-p & 0 & 0 & p \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix} \end{matrix}.$$

■ **Example 2.8 (Random walk on a graph)** A *graph* is a set of vertices and a set of edges. Two vertices are *neighbors* if there is an edge joining them. The *degree* of vertex v is the number of neighbors of v . For the graph in Figure 2.1, $\deg(a) = 1$, $\deg(b) = \deg(c) = \deg(d) = 4$, $\deg(e) = 3$, and $\deg(f) = 2$.

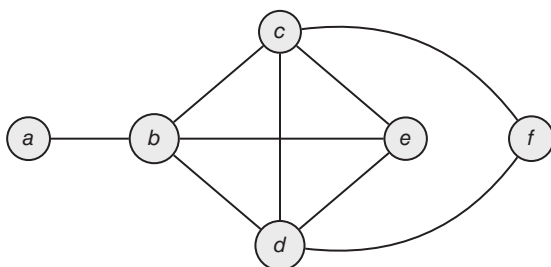


Figure 2.1 Graph on six vertices.

Imagine the vertices as lily pads on a pond. A frog is sitting on one lily pad. At each discrete unit of time, the frog hops to a neighboring lily pad chosen uniformly at random. For instance, if the frog is on lily pad c , it jumps to b , d , e , or f with probability $1/4$ each. If the frog is on f , it jumps to c or d with probability $1/2$ each. If the frog is on a , it always jumps to b .

Let X_n be the frog's location after n hops. The sequence $X_0, X_1, \dots$ is a Markov chain. Given a graph G such a process is called *simple random walk on G* .

For vertices i and j , write $i \sim j$ if i and j are neighbors. The one-step transition probabilities are

$$P(X_1 = j | X_0 = i) = \begin{cases} \frac{1}{\deg(i)}, & \text{if } i \sim j, \\ 0, & \text{otherwise.} \end{cases}$$

The transition matrix for simple random walk on the graph in Figure 2.1 is

$$P = \begin{matrix} & \begin{matrix} a & b & c & d & e & f \end{matrix} \\ \begin{matrix} a \\ b \\ c \\ d \\ e \\ f \end{matrix} & \begin{pmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 1/4 & 0 & 1/4 & 1/4 & 1/4 & 0 \\ 0 & 1/4 & 0 & 1/4 & 1/4 & 1/4 \\ 0 & 1/4 & 1/4 & 0 & 1/4 & 1/4 \\ 0 & 1/3 & 1/3 & 1/3 & 0 & 0 \\ 0 & 0 & 1/2 & 1/2 & 0 & 0 \end{pmatrix} \end{matrix}.$$

Of particular interest is the long-term behavior of the random walk. What can be said of the frog's position after it has been hopping for a long time?

- (a) The *cycle graph* on nine vertices is shown in Figure 2.2. Simple random walk on the cycle moves left or right with probability $1/2$. Each vertex has degree two. The transition matrix is defined using clock arithmetic. For a cycle with k vertices,

$$P_{ij} = \begin{cases} 1/2, & \text{if } j \equiv i \pm 1 \pmod{k}, \\ 0, & \text{otherwise.} \end{cases}$$

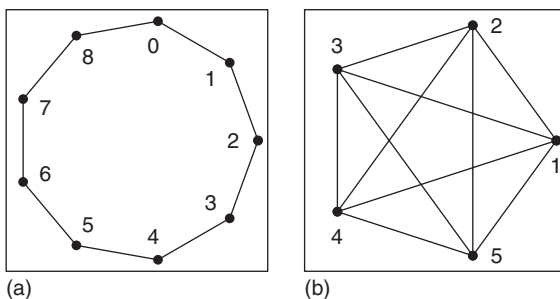


Figure 2.2 (a) Cycle graph on nine vertices. (b) Complete graph on five vertices.

- (b) In the *complete graph*, every pair of vertices is joined by an edge. The complete graph on five vertices is shown in Figure 2.2. The complete graph on k vertices has $\binom{k}{2}$ edges. Each vertex has degree $k - 1$. The entries of the transition matrix are

$$P_{ij} = \begin{cases} 1/(k - 1), & \text{if } i \neq j \\ 0, & \text{if } i = j. \end{cases}$$

- (c) The *k-hypercube graph* has vertex set consisting of all k -element sequences of 0s and 1s. Two vertices (sequences) are connected by an edge if they differ in exactly one coordinate. The graph has 2^k vertices and $k2^{k-1}$ edges. Each vertex has degree k . See Figure 2.3.

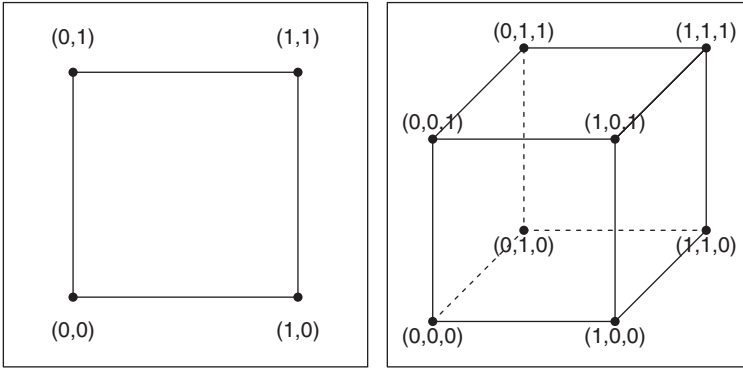


Figure 2.3 The k -hypercube graphs for $k = 2$ and $k = 3$.

Random walk on the k -hypercube can be described as follows. Assume that the walk is at a particular k -element 0–1 sequence. The next sequence of the walk is found by picking one of the k coordinates uniformly at random and *flipping the bit* at that coordinate. That is, switch from 0 to 1, or from 1 to 0. Here is the transition matrix for random walk on the 3-hypercube graph.

$$P = \begin{matrix} & \begin{matrix} 000 & 100 & 010 & 110 & 001 & 101 & 011 & 111 \end{matrix} \\ \begin{matrix} 000 \\ 100 \\ 010 \\ 110 \\ 001 \\ 101 \\ 011 \\ 111 \end{matrix} & \begin{pmatrix} 0 & 1/3 & 1/3 & 0 & 1/3 & 0 & 0 & 0 \\ 1/3 & 0 & 0 & 1/3 & 0 & 1/3 & 0 & 0 \\ 1/3 & 0 & 0 & 1/3 & 0 & 0 & 1/3 & 0 \\ 0 & 1/3 & 1/3 & 0 & 0 & 0 & 0 & 1/3 \\ 1/3 & 0 & 0 & 0 & 0 & 1/3 & 1/3 & 0 \\ 0 & 1/3 & 0 & 0 & 1/3 & 0 & 0 & 1/3 \\ 0 & 0 & 1/3 & 0 & 1/3 & 0 & 0 & 1/3 \\ 0 & 0 & 0 & 1/3 & 0 & 1/3 & 1/3 & 0 \end{pmatrix} \end{matrix}.$$

■

■ **Example 2.9 (Birth-and-death chain)** A birth-and-death Markov chain is a process with countably infinite state space and two types of transitions: *births* from i to $i + 1$ and *deaths* from i to $i - 1$. Define transition probabilities

$$P_{ij} = \begin{cases} q_i, & \text{if } j = i - 1, \\ p_i, & \text{if } j = i + 1, \\ 1 - p_i - q_i, & \text{if } j = i, \\ 0, & \text{otherwise,} \end{cases}$$

for $0 \leq p_i, q_i$ and $p_i + q_i \leq 1$. The infinite, tri-diagonal transition matrix is

$$P = \begin{matrix} & \begin{matrix} 0 & 1 & 2 & 3 & \dots \end{matrix} \\ \begin{matrix} 0 \\ 1 \\ 2 \\ 3 \\ \vdots \end{matrix} & \begin{pmatrix} 1 - p_0 & p_0 & 0 & 0 & \dots \\ q_1 & 1 - q_1 - p_1 & p_1 & 0 & \dots \\ 0 & q_2 & 1 - q_2 - p_2 & p_2 & \dots \\ 0 & 0 & q_3 & 1 - q_3 - p_3 & \dots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix} \end{matrix}.$$

Birth-and-death chains are used to model population size, spread of disease, and the number of customers in line at the supermarket. The case when $p_i = p$ and $q_i = q$ are constant gives random walk on the non-negative integers. ■

■ **Example 2.10 (The lazy librarian and move-to-front)** A library has k books and one very long bookshelf. Each book's popularity is measured by a probability. The chance that book i will be checked out is p_i , with $p_1 + \cdots + p_k = 1$. When patrons look for books they go to the bookshelf, start at the front (left end), and scan down the bookshelf, left to right, until they find the book they want.

The library wants to organize the bookshelf in such a way as to minimize patrons' search time, measured by how many books they need to scan before they find the one they want. If the probabilities p_i are known then the best way to organize the bookshelf is by order of popularity, with the most popular, high probability books at the front. However, the actual probabilities are not known.

A lazy librarian uses the following method for organizing the books. When a patron returns a book the librarian simply puts the book at the front of the shelf. All the other books move down the shelf to the right. For instance, assume that the library has six books labeled a to f . If the bookshelf is currently ordered (b, c, a, f, e, d) and book e is chosen, the new ordering of the shelf after the book is returned is (e, b, c, a, f, d) . We assume that one book is chosen and returned at a time.

Such a scheme has the advantage that over time, as books get taken out and returned, the most popular books will gravitate to the front of the bookshelf and the least popular books will gravitate to the back. The process is known as the move-to-front self-organizing scheme. Move-to-front is studied in computer science as a dynamic data structure for maintaining a linked list.

Let X_n be the order of the books after n steps. Then, the move-to-front process $X_0, X_1, \dots$ is a Markov chain whose state space is the set of all permutations (orderings) of k books. The transition matrix has dimension $k! \times k!$. Let $\sigma = (\sigma_1, \dots, \sigma_k)$ and $\tau = (\tau_1, \dots, \tau_k)$ denote permutations. Then,

$$P_{\sigma, \tau} = P(X_1 = \tau | X_0 = \sigma) = p_x,$$

if $\tau_1 = x$ and τ can be obtained from σ by moving item x to the front of σ . Here is the transition matrix for move-to-front for a $k = 3$ book library.

$$P = \begin{matrix} & \begin{matrix} abc & acb & bac & bca & cab & cba \end{matrix} \\ \begin{matrix} abc \\ acb \\ bac \\ bca \\ cab \\ cba \end{matrix} & \begin{pmatrix} p_a & 0 & p_b & 0 & p_c & 0 \\ 0 & p_a & p_b & 0 & p_c & 0 \\ p_a & 0 & p_b & 0 & 0 & p_c \\ p_a & 0 & 0 & p_b & 0 & p_c \\ 0 & p_a & 0 & p_b & p_c & 0 \\ 0 & p_a & 0 & p_b & 0 & p_c \end{pmatrix} \end{matrix}.$$

Move-to-front is related to a card-shuffling scheme known as *random-to-top*. Given a standard deck of cards, pick a card uniformly at random and move it to the top of the deck. Random-to-top is obtained from move-to-front by letting $k = 52$

and $p_i = 1/52$, for all i . Of interest is how many such shuffles will mix up the deck of cards. ■

■ **Example 2.11 (Weighted, directed graphs)** A *weighted graph* associates a positive number (weight) with every edge. An example is shown in Figure 2.4. The graph contains *loops* at vertices b , c , and f , which are edges joining a vertex to itself.

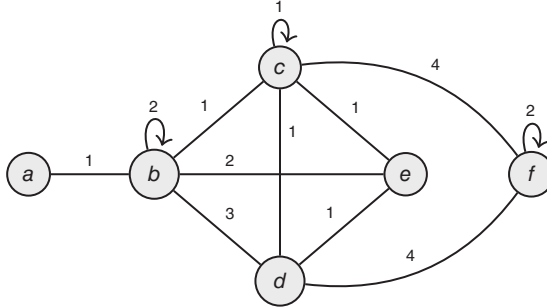


Figure 2.4 Weighted graph with loops.

For random walk on a weighted graph, transition probabilities are proportional to the sum of the weights. If vertices i and j are neighbors, let $w(i, j)$ denote the weight of the edge joining i and j . Let $w(i) = \sum_{i \sim k} w(i, k)$ be the sum of the weights on all edges joining i to its neighbors. The transition matrix is given by

$$P_{ij} = \begin{cases} \frac{w(i, j)}{w(i)}, & \text{if } i \sim j, \\ 0, & \text{otherwise.} \end{cases}$$

For the weighted graph in Figure 2.4, the transition matrix is

$$P = \begin{matrix} & \begin{matrix} a & b & c & d & e & f \end{matrix} \\ \begin{matrix} a \\ b \\ c \\ d \\ e \\ f \end{matrix} & \begin{pmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 1/9 & 2/9 & 1/9 & 3/9 & 2/9 & 0 \\ 0 & 1/8 & 1/8 & 1/8 & 1/8 & 4/8 \\ 0 & 3/9 & 1/9 & 0 & 1/9 & 4/9 \\ 0 & 2/4 & 1/4 & 1/4 & 0 & 0 \\ 0 & 0 & 4/10 & 4/10 & 0 & 2/10 \end{pmatrix} \end{matrix}.$$

A *directed graph* is a graph where edges have an associated direction. For every pair of vertices i and j one can have an edge from i to j and an edge from j to i . In a weighted, directed graph, there is a weight function $w(i, j)$ which gives the weight for the directed edge from i to j .

Every Markov chain can be described as a random walk on a weighted, directed graph whose vertex set is the state space of the chain. We call such a graph a *transition graph* for the Markov chain.

To create a transition graph from a transition matrix P , for each pair of vertices i and j such that $P_{ij} > 0$, put a directed edge between i and j with edge weight P_{ij} .

Conversely, given a weighted, directed graph with non-negative weight function $w(i, j)$, to obtain the corresponding transition matrix let

$$P_{ij} = \frac{w(i, j)}{\sum_{i \sim k} w(i, k)} = \frac{w(i, j)}{w(i)}, \quad \text{for all } i, j.$$

Observe that matrix entries are non-negative and rows sum to 1, as for each i ,

$$\sum_j P_{ij} = \sum_j \frac{w(i, j)}{w(i)} = \frac{1}{w(i)} \sum_{j \sim i} w(i, j) = \frac{w(i)}{w(i)} = 1.$$

For example, consider the transition matrix

$$P = \begin{matrix} & \begin{matrix} a & b & c \end{matrix} \\ \begin{matrix} a \\ b \\ c \end{matrix} & \begin{pmatrix} 0 & 1 & 0 \\ 0.1 & 0.2 & 0.7 \\ 0.4 & 0 & 0.6 \end{pmatrix} \end{matrix}.$$

Two versions of the transition graph are shown in Figure 2.5. Note that multiplying the weights in the transition graph by a constant does not change the resulting transition matrix. ■

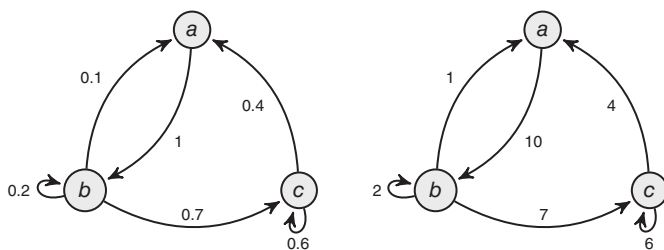


Figure 2.5 Markov transition graphs.

■ **Example 2.12** The metastatic progression of lung cancer throughout the body is modeled in Newton et al. (2012). The 50 possible metastatic locations for cancer spread are the state space for a Markov chain model. Matrix entries were estimated from autopsy data extracted from 3,827 patients. The progress of the disease is observed as a random walk on the weighted, directed graph in Figure 2.6. Site 23 represents the lung.

An important quantity associated with this model is the *mean first-passage time*, the average number of steps it takes to move from the lung to each of the other locations in the body. ■

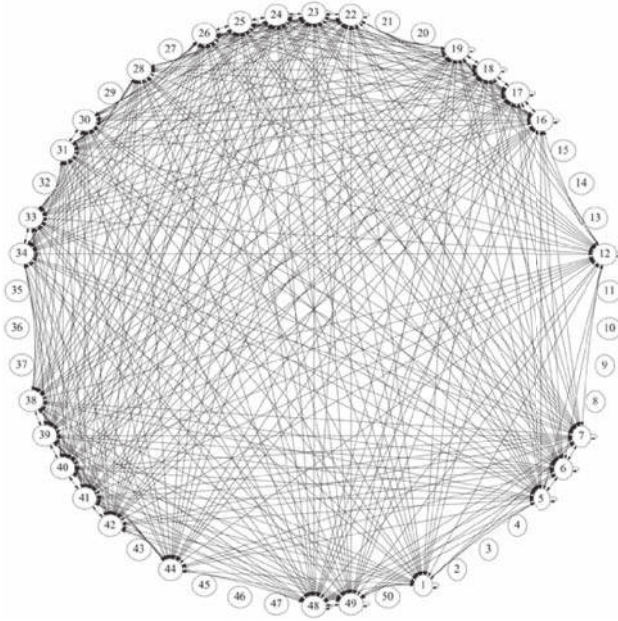


Figure 2.6 Lung cancer network as a weighted, directed graph (weights not shown). *Source:* Newton et al. (2012).

2.3 BASIC COMPUTATIONS

A powerful feature of Markov chains is the ability to use matrix algebra for computing probabilities. To use matrix methods, we consider probability distributions as vectors.

A *probability vector* is a row vector of non-negative numbers that sum to 1. Bold Greek letters, such as α , λ , and π , are used to denote such vectors.

Assume that X is a discrete random variable with $P(X = j) = \alpha_j$, for $j = 1, 2, \dots$. Then, $\alpha = (\alpha_1, \alpha_2, \dots)$ is a probability vector. We say that *the distribution of X is α* . For matrix computations we will identify discrete probability distributions with row vectors.

For a Markov chain $X_0, X_1, \dots$, the distribution of X_0 is called the *initial distribution* of the Markov chain. If α is the initial distribution, then $P(X_0 = j) = \alpha_j$, for all j .

n-Step Transition Probabilities

For states i and j , and $n \geq 1$, $P(X_n = j | X_0 = i)$ is the probability that the chain started in i hits j in n steps. The n -step transition probabilities can be arranged in a matrix. The matrix whose ij th entry is $P(X_n = j | X_0 = i)$ is the *n -step transition matrix* of the Markov chain. Of course, for $n = 1$, this is just the usual transition matrix P .

For $n \geq 1$, one of the central computational results for Markov chains is that the n -step transition matrix is precisely P^n , the n th matrix power of P . To show that

$P(X_n = j | X_0 = i) = (P^n)_{ij}$, condition on X_{n-1} , which gives

$$\begin{aligned} P(X_n = j | X_0 = i) &= \sum_k P(X_n = j | X_{n-1} = k, X_0 = i) P(X_{n-1} = k | X_0 = i) \\ &= \sum_k P(X_n = j | X_{n-1} = k) P(X_{n-1} = k | X_0 = i) \\ &= \sum_k P_{kj} P(X_{n-1} = k | X_0 = i), \end{aligned}$$

where the second equality is by the Markov property, and the third equality is by time-homogeneity.

For $n = 2$, this gives

$$P(X_2 = j | X_0 = i) = \sum_k P_{kj} P(X_1 = k | X_0 = i) = \sum_k P_{kj} P_{ik} = (P^2)_{ij}.$$

Hence, the two-step transition matrix is P^2 . Similarly, for $n = 3$,

$$P(X_3 = j | X_0 = i) = \sum_k P_{kj} P(X_2 = k | X_0 = i) = \sum_k P_{kj} P_{ik}^2 = (P^3)_{ij}.$$

The three-step transition matrix is P^3 . Induction establishes the general result. (See Section 2.6 for an introduction to mathematical induction.)

***n*-Step Transition Matrix**

Let $X_0, X_1, \dots$ be a Markov chain with transition matrix P . The matrix P^n is the n -step transition matrix of the chain. For $n \geq 0$,

$$P^n_{ij} = P(X_n = j | X_0 = i), \text{ for all } i, j.$$

Note that $P^n_{ij} = (P^n)_{ij}$. Do not confuse this with $(P_{ij})^n$, which is the number P_{ij} raised to the n th power. Also note that P^0 is the identity matrix. That is,

$$P^0_{ij} = P(X_0 = j | X_0 = i) = \begin{cases} 1, & \text{if } i = j, \\ 0, & \text{if } i \neq j. \end{cases}$$

 **Example 2.13** Consider random walk on the cycle graph consisting of five vertices $\{0, 1, 2, 3, 4\}$. Describe the six-step transition probabilities of the chain.

with

$$P^4 = \begin{matrix} & \begin{matrix} 0 & 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \end{matrix} \\ \begin{matrix} 0 \\ 1 \\ 2 \\ 3 \\ 4 \\ 5 \\ 6 \\ 7 \\ 8 \end{matrix} & \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0.496 & 0.115 & 0 & 0.259 & 0 & 0.130 & 0 & 0 & 0 \\ 0.237 & 0 & 0.288 & 0 & 0.346 & 0 & 0.130 & 0 & 0 \\ 0.064 & 0.115 & 0 & 0.346 & 0 & 0.346 & 0 & 0.130 & 0 \\ 0.026 & 0 & 0.154 & 0 & 0.346 & 0 & 0.346 & 0 & 0.130 \\ 0 & 0.026 & 0 & 0.154 & 0 & 0.346 & 0 & 0.259 & 0.216 \\ 0 & 0 & 0.026 & 0 & 0.154 & 0 & 0.288 & 0 & 0.533 \\ 0 & 0 & 0 & 0.026 & 0 & 0.115 & 0 & 0.115 & 0.744 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix} \end{matrix}.$$

The gambler's expected fortune is

$$\begin{aligned} E(X_4|X_0 = 3) &= \sum_{j=0}^8 jP(X_4 = j|X_0 = 3) = \sum_{j=0}^8 jP_{3,j}^4 \\ &= 0(0.064) + 1(0.115) + 3(0.346) + 5(0.346) + 7(0.130) \\ &= \$3.79. \end{aligned} \quad \blacksquare$$

Chapman–Kolmogorov Relationship

For $m, n \geq 0$, the matrix identity $P^{m+n} = P^m P^n$ gives

$$P_{ij}^{m+n} = \sum_k P_{ik}^m P_{kj}^n, \text{ for all } i, j.$$

By time-homogeneity, this gives

$$\begin{aligned} P(X_{n+m} = j|X_0 = i) &= \sum_k P(X_m = k|X_0 = i)P(X_n = j|X_0 = k) \\ &= \sum_k P(X_m = k|X_0 = i)P(X_{m+n} = j|X_m = k). \end{aligned}$$

The probabilistic interpretation is that transitioning from i to j in $m+n$ steps is equivalent to transitioning from i to some state k in m steps and then moving from that state to j in the remaining n steps. This is known as the *Chapman–Kolmogorov relationship*.

Distribution of X_n

In general, a Markov chain $X_0, X_1, \dots$ is not a sequence of identically distributed random variables. For $n \geq 1$, the marginal distribution of X_n depends on the n -step

transition matrix $\mathbf{P}^n$, as well as the initial distribution α . To obtain the probability mass function of X_n , condition on the initial state X_0 . For all j ,

$$P(X_n = j) = \sum_i P(X_n = j | X_0 = i) P(X_0 = i) = \sum_i P_{ij}^n \alpha_i. \quad (2.3)$$

The sum in the last expression of Equation (2.3) can be interpreted in terms of matrix operations on vectors. It is the dot product of the initial probability vector α with the j th column of $\mathbf{P}^n$. That is, it is the j th component of the vector–matrix product $\alpha \mathbf{P}^n$. (Remember: α is a *row* vector.)

Distribution of X_n

Let $X_0, X_1, \dots$ be a Markov chain with transition matrix $\mathbf{P}$ and initial distribution α . For all $n \geq 0$, the distribution of X_n is $\alpha \mathbf{P}^n$. That is,

$$P(X_n = j) = (\alpha \mathbf{P}^n)_j, \text{ for all } j.$$

■ **Example 2.15** Consider the weather chain introduced in Example 2.3. For tomorrow, the meteorologist predicts a 50% chance of snow and a 50% chance of rain. Find the probability that it will snow 2 days later.

Solution As the ordered states of the chain are rain, snow, and clear, the initial distribution is $\alpha = (0.5, 0.5, 0)$. We have

$$\mathbf{P} = \begin{matrix} & \begin{matrix} r & s & c \end{matrix} \\ \begin{matrix} r \\ s \\ c \end{matrix} & \begin{pmatrix} 0.2 & 0.6 & 0.2 \\ 0.1 & 0.8 & 0.1 \\ 0.1 & 0.6 & 0.3 \end{pmatrix} \end{matrix} \quad \text{and} \quad \mathbf{P}^2 = \begin{matrix} & \begin{matrix} r & s & c \end{matrix} \\ \begin{matrix} r \\ s \\ c \end{matrix} & \begin{pmatrix} 0.12 & 0.72 & 0.16 \\ 0.11 & 0.76 & 0.13 \\ 0.11 & 0.72 & 0.17 \end{pmatrix} \end{matrix}$$

This gives

$$\alpha \mathbf{P}^2 = (0.5, 0.5, 0) \begin{pmatrix} 0.12 & 0.72 & 0.16 \\ 0.11 & 0.76 & 0.13 \\ 0.11 & 0.72 & 0.17 \end{pmatrix} = (0.115, 0.74, 0.145).$$

The desired probability of snow is $P(X_2 = s) = (\alpha \mathbf{P}^2)_s = 0.74$. ■

Present, Future, and Most Recent Past

The Markov property says that past and future are independent given the present. It is also true that past and future are independent, given *the most recent* past.

Markov Property

Let $X_0, X_1, \dots$ be a Markov chain. Then, for all $m < n$,

$$\begin{aligned} P(X_{n+1} = j | X_0 = i_0, \dots, X_{n-m-1} = i_{n-m-1}, X_{n-m} = i) \\ &= P(X_{n+1} = j | X_{n-m} = i) \\ &= P(X_{m+1} = j | X_0 = i) = P_{ij}^{m+1}, \end{aligned} \quad (2.4)$$

for all $i, j, i_0, \dots, i_{n-m-1}$, and $n \geq 0$.

Proof. With $m = 0$, Equation (2.4) reduces to the defining Markov relationship as stated in Equation (2.1).

Let $m = 1$. By conditioning on X_n ,

$$\begin{aligned} P(X_{n+1} = j | X_0 = i_0, \dots, X_{n-1} = i) \\ &= \sum_k P(X_{n+1} = j | X_0 = i_0, \dots, X_{n-1} = i, X_n = k) \\ &\quad \times P(X_n = k | X_0 = i_0, \dots, X_{n-1} = i) \\ &= \sum_k P(X_{n+1} = j | X_n = k) P(X_n = k | X_{n-1} = i) \\ &= \sum_k P(X_1 = j | X_0 = k) P(X_1 = k | X_0 = i) \\ &= \sum_k P_{kj} P_{ik} = P_{ij}^2. \end{aligned}$$

The third equality is by time-homogeneity. For general m , induction gives the result by conditioning on X_{n-m+1} . ■

Joint Distribution

The marginal distributions of a Markov chain are determined by the initial distribution α and the transition matrix P . However, a much stronger result is true. In fact, α and P determine all the joint distributions of a Markov chain, that is, the joint distribution of any finite subset of $X_0, X_1, X_2, \dots$. In that sense, the initial distribution and transition matrix give a complete probabilistic description of a Markov chain.

To illustrate, consider an arbitrary joint probability, such as

$$P(X_5 = i, X_6 = j, X_9 = k, X_{17} = l), \text{ for some states } i, j, k, l.$$

For the underlying event, the chain moves to i in five steps, then to j in one step, then to k in three steps, and then to l in eight steps. With initial distribution α , intuition

suggests that

$$P(X_5 = i, X_6 = j, X_9 = k, X_{17} = l) = (\alpha P^5)_i P_{ij} P_{jk}^3 P_{kl}^8.$$

Indeed, conditional probability, the Markov property, and time-homogeneity give

$$\begin{aligned} P(X_5 = i, X_6 = j, X_9 = k, X_{17} = l) \\ &= P(X_{17} = l | X_5 = i, X_6 = j, X_9 = k) P(X_9 = k | X_5 = i, X_6 = j) \\ &\quad \times P(X_6 = j | X_5 = i) P(X_5 = i) \\ &= P(X_{17} = l | X_9 = k) P(X_9 = k | X_6 = j) P(X_6 = j | X_5 = i) P(X_5 = i) \\ &= P(X_8 = l | X_0 = k) P(X_3 = k | X_0 = j) P(X_1 = j | X_0 = i) P(X_5 = i) \\ &= P_{kl}^8 P_{jk}^3 P_{ij} (\alpha P^5)_i. \end{aligned}$$

The joint probability is obtained from just the initial distribution α and the transition matrix P . For completeness, here is the general formula.

Joint Distribution

Let $X_0, X_1, \dots$ be a Markov chain with transition matrix P and initial distribution α . For all $0 \leq n_1 < n_2 < \dots < n_{k-1} < n_k$ and states $i_1, i_2, \dots, i_{k-1}, i_k$,

$$\begin{aligned} P(X_{n_1} = i_1, X_{n_2} = i_2, \dots, X_{n_{k-1}} = i_{k-1}, X_{n_k} = i_k) \\ = (\alpha P^{n_1})_{i_1} (P^{n_2 - n_1})_{i_1 i_2} \dots (P^{n_k - n_{k-1}})_{i_{k-1} i_k}. \end{aligned} \quad (2.5)$$

 **Example 2.16** Danny's daily lunch choices are modeled by a Markov chain with transition matrix

$$P = \begin{array}{c} \begin{array}{l} \text{Burrito} \\ \text{Falafel} \\ \text{Pizza} \\ \text{Sushi} \end{array} \left(\begin{array}{cccc} \text{Burrito} & \text{Falafel} & \text{Pizza} & \text{Sushi} \\ \begin{array}{l} 0.0 \\ 0.5 \\ 0.4 \\ 0.0 \end{array} & \begin{array}{l} 0.5 \\ 0.0 \\ 0.0 \\ 0.2 \end{array} & \begin{array}{l} 0.5 \\ 0.5 \\ 0.0 \\ 0.6 \end{array} & \begin{array}{l} 0.0 \\ 0.0 \\ 0.6 \\ 0.2 \end{array} \end{array} \right).$$

On Sunday, Danny chooses lunch uniformly at random. Find the probability that he chooses sushi on the following Wednesday and Friday, and pizza on Saturday.

Solution Let b, f, p, s denote Danny's lunch choices, respectively. Let X_0 denote Danny's lunch choice on Sunday. The desired probability is

$$P(X_3 = s, X_5 = s, X_6 = p) = (\alpha P^3)_s P_{ss}^2 P_{sp},$$

where $\alpha = (1/4, 1/4, 1/4, 1/4)$. We have

$$P^2 = \begin{matrix} & \begin{matrix} b & f & p & s \end{matrix} \\ \begin{matrix} b \\ f \\ p \\ s \end{matrix} & \begin{pmatrix} 0.45 & 0.00 & 0.25 & 0.30 \\ 0.20 & 0.25 & 0.25 & 0.30 \\ 0.00 & 0.32 & 0.56 & 0.12 \\ 0.34 & 0.04 & 0.22 & 0.40 \end{pmatrix} \end{matrix}, \quad P^3 = \begin{matrix} & \begin{matrix} b & f & p & s \end{matrix} \\ \begin{matrix} b \\ f \\ p \\ s \end{matrix} & \begin{pmatrix} 0.100 & 0.285 & 0.405 & 0.210 \\ 0.225 & 0.160 & 0.405 & 0.210 \\ 0.384 & 0.024 & 0.232 & 0.360 \\ 0.108 & 0.250 & 0.430 & 0.212 \end{pmatrix} \end{matrix}.$$

The desired probability is

$$(\alpha P^3)_s P_{ss}^2 P_{sp} = (0.248)(0.40)(0.60) = 0.05952. \quad \blacksquare$$

2.4 LONG-TERM BEHAVIOR—THE NUMERICAL EVIDENCE

In any stochastic—or deterministic—process, the long-term behavior of the system is often of interest.

The Canadian Forest Fire Weather Index is widely used as a means to estimate the risk of wildfire. The Ontario Ministry of Natural Resources uses the index to classify each day's risk of forest fire as either nil, low, moderate, high, or extreme.

Martell (1999) gathered daily fire risk data over 26 years at 15 weather stations across Ontario to construct a five-state Markov chain model for the daily changes in the index. The transition matrix from one location for the early summer subseason is

$$P = \begin{matrix} & \begin{matrix} \text{Nil} & \text{Low} & \text{Moderate} & \text{High} & \text{Extreme} \end{matrix} \\ \begin{matrix} \text{Nil} \\ \text{Low} \\ \text{Moderate} \\ \text{High} \\ \text{Extreme} \end{matrix} & \begin{pmatrix} 0.575 & 0.118 & 0.172 & 0.109 & 0.026 \\ 0.453 & 0.243 & 0.148 & 0.123 & 0.033 \\ 0.104 & 0.343 & 0.367 & 0.167 & 0.019 \\ 0.015 & 0.066 & 0.318 & 0.505 & 0.096 \\ 0.000 & 0.060 & 0.149 & 0.567 & 0.224 \end{pmatrix} \end{matrix}.$$

Of interest to forest managers is the long-term probability distribution of the daily index. Regardless of the risk on any particular day, what is the long-term likelihood of risk for a typical day in the early summer?

Consider the n -step transition matrix for several increasing values of n .

$$P^2 = \begin{pmatrix} 0.404 & 0.164 & 0.218 & 0.176 & 0.038 \\ 0.388 & 0.173 & 0.212 & 0.185 & 0.042 \\ 0.256 & 0.234 & 0.259 & 0.210 & 0.041 \\ 0.079 & 0.166 & 0.304 & 0.372 & 0.079 \\ 0.051 & 0.117 & 0.277 & 0.446 & 0.109 \end{pmatrix},$$

$$P^3 = \begin{pmatrix} 0.332 & 0.176 & 0.235 & 0.211 & 0.046 \\ 0.326 & 0.175 & 0.235 & 0.216 & 0.047 \\ 0.283 & 0.192 & 0.247 & 0.229 & 0.049 \\ 0.158 & 0.183 & 0.280 & 0.312 & 0.067 \\ 0.118 & 0.165 & 0.286 & 0.353 & 0.078 \end{pmatrix},$$

$$\begin{aligned}
 \mathbf{P}^5 &= \begin{pmatrix} 0.282 & 0.180 & 0.248 & 0.239 & 0.051 \\ 0.279 & 0.180 & 0.248 & 0.241 & 0.052 \\ 0.273 & 0.181 & 0.250 & 0.244 & 0.052 \\ 0.235 & 0.183 & 0.259 & 0.266 & 0.057 \\ 0.217 & 0.183 & 0.264 & 0.277 & 0.060 \end{pmatrix}, \\
 \mathbf{P}^{10} &= \begin{pmatrix} 0.264 & 0.181 & 0.252 & 0.249 & 0.053 \\ 0.264 & 0.181 & 0.252 & 0.249 & 0.054 \\ 0.264 & 0.181 & 0.252 & 0.249 & 0.054 \\ 0.263 & 0.181 & 0.252 & 0.25 & 0.054 \\ 0.262 & 0.181 & 0.252 & 0.251 & 0.054 \end{pmatrix}, \\
 \mathbf{P}^{17} &= \begin{pmatrix} 0.264 & 0.181 & 0.252 & 0.249 & 0.054 \\ 0.264 & 0.181 & 0.252 & 0.249 & 0.054 \\ 0.264 & 0.181 & 0.252 & 0.249 & 0.054 \\ 0.264 & 0.181 & 0.252 & 0.249 & 0.054 \\ 0.264 & 0.181 & 0.252 & 0.249 & 0.054 \end{pmatrix}, \\
 \mathbf{P}^{18} &= \begin{pmatrix} 0.264 & 0.181 & 0.252 & 0.249 & 0.054 \\ 0.264 & 0.181 & 0.252 & 0.249 & 0.054 \\ 0.264 & 0.181 & 0.252 & 0.249 & 0.054 \\ 0.264 & 0.181 & 0.252 & 0.249 & 0.054 \\ 0.264 & 0.181 & 0.252 & 0.249 & 0.054 \end{pmatrix}.
 \end{aligned}$$

Numerical evidence suggests that matrix powers are converging to a limit. Furthermore, the rows of that limiting matrix are all the same. The fact that the rows of $\mathbf{P}^{17}$ are the same means that the probability of a particular fire index after 17 days does not depend on today's level of risk. After 17 days, the effect of the initial state has worn off, and no longer affects the distribution of the fire index.

Furthermore, $\mathbf{P}^{18} = \mathbf{P}^{17}$ (at least to three decimal places). In fact, $\mathbf{P}^n = \mathbf{P}^{17}$, for $n \geq 17$. The latter is intuitive since if the probability of hitting state j in 17 steps is independent of the initial state, then the probability of hitting j in 17 or more steps is also independent of the initial state. See also Exercise 2.16.

The long-term fire index distribution taken from the common row of $\mathbf{P}^{17}$ is

Nil	Low	Moderate	High	Extreme
0.264	0.181	0.252	0.249	0.054

R : Matrix Powers

The function `matrixpower(mat, n)` computes the n th power of a square matrix `mat`, for $n = 0, 1, 2, \dots$. The function is found in the R script file **utilities.R**, which includes several useful utility functions for working with Markov chains.

R : Fire Weather Index

> P

	Nil	Low	Moderate	High	Extreme
Nil	0.575	0.118	0.172	0.109	0.026
Low	0.453	0.243	0.148	0.123	0.033
Moderate	0.104	0.343	0.367	0.167	0.019
High	0.015	0.066	0.318	0.505	0.096
Extreme	0.000	0.060	0.149	0.567	0.224

> matrixpower(P,17)

	Nil	Low	Moderate	High	Extreme
Nil	0.263688	0.181273	0.251976	0.249484	0.0535768
Low	0.263688	0.181273	0.251976	0.249485	0.0535769
Moderate	0.263685	0.181273	0.251977	0.249486	0.0535772
High	0.263671	0.181273	0.251981	0.249495	0.0535789
Extreme	0.263663	0.181274	0.251982	0.249499	0.0535798

round entries to three decimal places

> round(matrixpower(P,17),3)

	Nil	Low	Moderate	High	Extreme
Nil	0.264	0.181	0.252	0.249	0.054
Low	0.264	0.181	0.252	0.249	0.054
Moderate	0.264	0.181	0.252	0.249	0.054
High	0.264	0.181	0.252	0.249	0.054
Extreme	0.264	0.181	0.252	0.249	0.054

 **Example 2.17** Changes in the distribution of wetlands in Yinchuan Plain, China are studied in Zhang et al. (2011). Wetlands are considered among the most important ecosystems on earth. A Markov model is developed to track yearly changes in wetland type. Based on imaging and satellite data from 1991, 1999, and 2006, researchers measured annual distributions of wetland type throughout the region and estimated the Markov transition matrix

$$P = \begin{matrix} & \begin{matrix} \text{River} & \text{Lake} & \text{Pond} & \text{Paddy} & \text{Non} \end{matrix} \\ \begin{matrix} \text{River} \\ \text{Lake} \\ \text{Pond} \\ \text{Paddy} \\ \text{Non} \end{matrix} & \begin{pmatrix} 0.342 & 0.005 & 0.001 & 0.020 & 0.632 \\ 0.001 & 0.252 & 0.107 & 0.005 & 0.635 \\ 0.000 & 0.043 & 0.508 & 0.015 & 0.434 \\ 0.001 & 0.002 & 0.004 & 0.665 & 0.328 \\ 0.007 & 0.007 & 0.007 & 0.025 & 0.954 \end{pmatrix} \end{pmatrix}.$$

The state *Non* refers to nonwetland regions. Based on their model, the scientists predict that “The wetland distribution will essentially be in a steady state in Yinchuan Plain in approximately 100 years.”

With technology one checks that $P^{100} = P^{101}$ has identical rows. The common row gives the predicted long-term, *steady-state* wetland distribution.

River	Lake	Pond	Paddy	Non
0.010	0.010	0.015	0.068	0.897

R : Limiting Distribution for Wetlands Type					
> P					
	River	Lake	Pond	Paddy	Non
River	0.342	0.005	0.001	0.020	0.632
Lake	0.001	0.252	0.107	0.005	0.635
Pond	0.003	0.043	0.507	0.014	0.433
Paddy	0.001	0.002	0.004	0.665	0.328
Non	0.007	0.007	0.007	0.025	0.954
> matrixpower(P,100)					
	River	Lake	Pond	Paddy	Non
River	0.01	0.01	0.015	0.068	0.897
Lake	0.01	0.01	0.015	0.068	0.897
Pond	0.01	0.01	0.015	0.068	0.897
Paddy	0.01	0.01	0.015	0.068	0.897
Non	0.01	0.01	0.015	0.068	0.897



Random Walk on Cycle

Assume that the hopping frog of Example 2.8 now finds itself on a 25-lily pad cycle graph. If the frog starts hopping from vertex 1, where is it likely to be after many hops?

After a small number of hops the frog will tend to be close to its starting position at the top of the pond. But after a large number of hops the frog’s position will tend to be randomly distributed about the cycle, that is, it will tend to be uniformly distributed on all the vertices. See Figure 2.7 and Table 2.1, where probabilities are shown for the frog’s position after n steps, for several values of n .

Vertices 24, 25, 1, 2, and 3 are closest to the frog’s starting position. We consider these vertices *near the top* of the cycle. Vertices 12, 13, 14, and 15 are the furthest away. We consider these *near the bottom* of the cycle. After just 12 hops, the frog is still relatively close to the starting vertex. The chance that the frog is near the top of the cycle is 0.61. After 25 steps, the probability of being near the top is eight times greater than the probability of being near the bottom—0.32 compared with 0.04. Even after 100 steps it is still almost twice as likely that the frog will be near the starting vertex as compared to being at the opposite side of the cycle.

After 400 steps, however, the frog’s position is *mixed up* throughout the cycle and is very close to being uniformly distributed on all the vertices. The dependency on the frog’s initial position has worn off and all vertices are essentially equally likely. Numerical evidence suggests that the long-term distribution of the frog’s position is uniform on the vertices and independent of starting state.

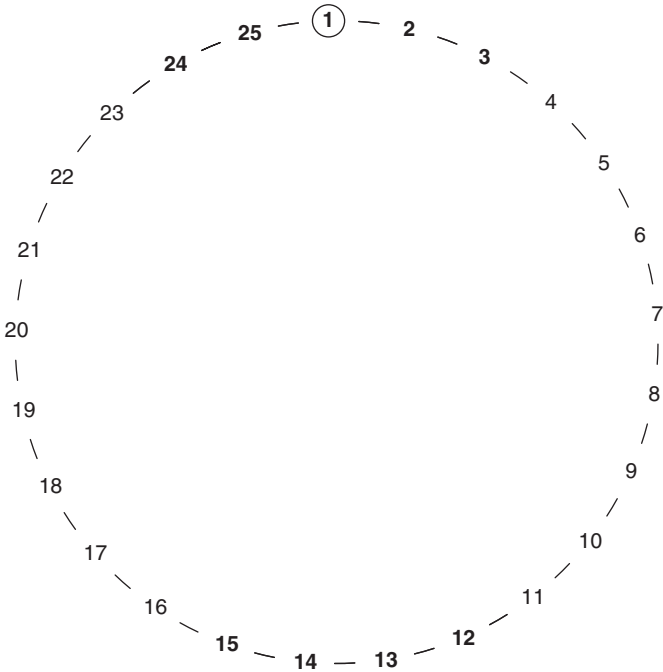


Figure 2.7 Frog starts hopping from vertex 1 of the 25-cycle graph.

TABLE 2.1 Probabilities, After n Steps, of the Frog’s Position Near the Top and Bottom of the 25-Cycle Graph

n	24	25	1	2	3	12	13	14	15
1	0	0.5	0	0.5	0	0	0	0	0
2	0.25	0	0.5	0	0.25	0	0	0	0
6	0.23	0	0.31	0	0.23	0	0	0	0
12	0.19	0	0.23	0	0.19	0	0.00	0.00	0
25	0.00	0.16	0.00	0.16	0.0	0.01	0.01	0.01	0.01
60	0.10	0.00	0.10	0.00	0.10	0.02	0.03	0.03	0.02
100	0.08	0.01	0.08	0.01	0.08	0.03	0.04	0.04	0.03
400	0.04	0.04	0.04	0.04	0.04	0.04	0.04	0.04	0.04
401	0.04	0.04	0.04	0.04	0.04	0.04	0.04	0.04	0.04
402	0.04	0.04	0.04	0.04	0.04	0.04	0.04	0.04	0.04

Not all Markov chains exhibit the kind of long-term limiting behavior seen in random walk on the 25-cycle graph or in the fire index chain. Consider random walk on the cycle graph with six vertices. Here are several powers of the transition matrix.

$$\begin{aligned}
P &= \begin{matrix} & \begin{matrix} 0 & 1 & 2 & 3 & 4 & 5 \end{matrix} \\ \begin{matrix} 0 \\ 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{matrix} & \begin{pmatrix} 0 & 0.5 & 0 & 0 & 0 & 0.5 \\ 0.5 & 0 & 0.5 & 0 & 0 & 0 \\ 0 & 0.5 & 0 & 0.5 & 0 & 0 \\ 0 & 0 & 0.5 & 0 & 0.5 & 0 \\ 0 & 0 & 0 & 0.5 & 0 & 0.5 \\ 0.5 & 0 & 0 & 0 & 0.5 & 0 \end{pmatrix} \end{matrix}, \\
P^2 &= \begin{pmatrix} 0.5 & 0 & 0.25 & 0 & 0.25 & 0 \\ 0 & 0.5 & 0 & 0.25 & 0 & 0.25 \\ 0.25 & 0 & 0.5 & 0 & 0.25 & 0 \\ 0 & 0.25 & 0 & 0.5 & 0 & 0.25 \\ 0.25 & 0 & 0.25 & 0 & 0.5 & 0 \\ 0 & 0.25 & 0 & 0.25 & 0 & 0.5 \end{pmatrix}, \\
P^3 &= \begin{pmatrix} 0 & 0.375 & 0 & 0.25 & 0 & 0.375 \\ 0.375 & 0 & 0.375 & 0 & 0.25 & 0 \\ 0 & 0.375 & 0 & 0.375 & 0 & 0.25 \\ 0.25 & 0 & 0.375 & 0 & 0.375 & 0 \\ 0 & 0.25 & 0 & 0.375 & 0 & 0.375 \\ 0.375 & 0 & 0.25 & 0 & 0.375 & 0 \end{pmatrix}, \\
P^8 &= \begin{pmatrix} 0.336 & 0 & 0.332 & 0 & 0.332 & 0 \\ 0 & 0.336 & 0 & 0.332 & 0 & 0.332 \\ 0.332 & 0 & 0.336 & 0 & 0.332 & 0 \\ 0 & 0.332 & 0 & 0.336 & 0 & 0.332 \\ 0.332 & 0 & 0.332 & 0 & 0.336 & 0 \\ 0. & 0.332 & 0 & 0.332 & 0 & 0.336 \end{pmatrix}, \\
P^{11} &= \begin{pmatrix} 0 & 0.333 & 0 & 0.333 & 0 & 0.333 \\ 0.336 & 0 & 0.336 & 0 & 0.333 & 0 \\ 0 & 0.333 & 0 & 0.333 & 0 & 0.333 \\ 0.333 & 0 & 0.333 & 0 & 0.333 & 0 \\ 0 & 0.333 & 0 & 0.333 & 0 & 0.333 \\ 0.333 & 0 & 0.333 & 0 & 0.333 & 0 \end{pmatrix}, \\
P^{12} &= \begin{pmatrix} 0.333 & 0 & 0.333 & 0 & 0.333 & 0 \\ 0 & 0.333 & 0 & 0.333 & 0 & 0.333 \\ 0.333 & 0 & 0.333 & 0 & 0.333 & 0 \\ 0 & 0.333 & 0 & 0.333 & 0 & 0.333 \\ 0.333 & 0 & 0.333 & 0 & 0.333 & 0 \\ 0. & 0.333 & 0 & 0.333 & 0 & 0.333 \end{pmatrix}.
\end{aligned}$$

The higher-order transition matrices flip-flop for odd and even powers. If the walk starts at an even vertex, it will always be on an even vertex after an even number of steps, and on an odd vertex after an odd number of steps. The parity of the position

of the walk matches the parity of the starting vertex after an even number of steps. The parity switches after an odd number of steps.

High powers of the transition matrix do not converge to a limiting matrix. The long-term behavior of the walk depends on the starting state and depends on how many steps are taken.

2.5 SIMULATION

Simulation is a powerful tool for studying Markov chains. For many chains that arise in applications, state spaces are huge and matrix methods may not be practical, or even possible, to implement.

For instance, the card-shuffling chain introduced in Example 2.10 has a state space of $k!$ elements. The transition matrix for a standard deck of cards is $52! \times 52!$, which has about 6.5×10^{135} entries.

Even for a moderately sized 50×50 matrix, as in the cancer study of Example 2.12, numerical matrix computations can be difficult to obtain. The researchers of that study found it easier to derive their results by simulation.

A Markov chain can be simulated from an initial distribution and transition matrix. To simulate a Markov sequence $X_0, X_1, \dots$, simulate each random variable sequentially conditional on the outcome of the previous variable. That is, first simulate X_0 according to the initial distribution. If $X_0 = i$, then simulate X_1 from the i th row of the transition matrix. If $X_1 = j$, then simulate X_2 from the j th row of the transition matrix, and so on.

Algorithm for Simulating a Markov Chain

Input: (i) initial distribution α , (ii) transition matrix P , (iii) number of steps n .

Output: $X_0, X_1, \dots, X_n$

Algorithm:

```

Generate  $X_0$  according to  $\alpha$ 
FOR  $i = 1, \dots, n$ 
  Assume that  $X_{i-1} = j$ 
  Set  $p = j$ th row of  $P$ 
  Generate  $X_i$  according to  $p$ 
END FOR

```

To simulate a finite Markov chain, the algorithm is implemented in R by the function `markov(init, mat, n)`, which is contained in the file **utilities.R**. The arguments of the function are `init`, the initial distribution, `mat`, the transition matrix, and `n`, the number of steps to simulate. A call to `markov(init, mat, n)` generates the $(n + 1)$ -element vector $(X_0, \dots, X_n)$. The `markov` function allows for an optional fourth argument `states`, which is the state space given as a vector. If the state space has k elements, the function assigns the default value to `states` of $(1, \dots, k)$.

- **Example 2.18 (Lung cancer study)** Medical researchers can use simulation to study the progression of lung cancer in the body, as described in Example 2.12. The 50×50 transition matrix is stored in an Excel spreadsheet and can be downloaded into R from the file **cancerstudy.R**. The initial distribution is a vector of all 0s with a 1 at position 23, corresponding to the lung. See the documentation in the script file for the 50-site numbering system. Common sites are 24 and 25 (lymph nodes) and 22 (liver). Following are several simulations of the process for eight steps.

R : Simulating Lung Cancer Growth

```
> mat <- read.csv("lungcancer.csv",header=TRUE)
> init <- c(rep(0,22),1,rep(0,27))
  # all 0s with 1 at site 23 (lung)
> n <- 8
> markov(init,mat,n)
[1] 23 17 24 23 44 6 1 24 28
> markov(init,mat,n)
[1] 23 25 25 19 25 1 24 22 22
> markov(init,mat,n)
[1] 23 18 44 7 22 23 30 24 33
> markov(init,mat,n)
[1] 23 22 24 24 30 24 24 23 24
```

Newton et al. (2012) use simulation to estimate the *mean first passage time*—the number of steps, on average, it takes for cancer to pass from the lung to each other location in the body, “something a static autopsy data set cannot give us directly.” The authors conclude that their study gives “important baseline quantitative insight into the structure of lung cancer progression networks.” ■

- **Example 2.19** University administrators have developed a Markov model to simulate graduation rates at their school. Students might drop out, repeat a year, or move on to the next year. Students have a 3% chance of repeating the year. First-years and sophomores have a 6% chance of dropping out. For juniors and seniors, the drop-out rate is 4%. The transition matrix for the model is

$$P = \begin{matrix} & \begin{matrix} \text{Drop} & \text{Fr} & \text{So} & \text{Jr} & \text{Sr} & \text{Grad} \end{matrix} \\ \begin{matrix} \text{Drop} \\ \text{Fr} \\ \text{So} \\ \text{Jr} \\ \text{Sr} \\ \text{Grad} \end{matrix} & \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0.06 & 0.03 & 0.91 & 0 & 0 & 0 \\ 0.06 & 0 & 0.03 & 0.91 & 0 & 0 \\ 0.04 & 0 & 0 & 0.03 & 0.93 & 0 \\ 0.04 & 0 & 0 & 0 & 0.03 & 0.93 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix} \end{matrix}.$$

Eventually, students will either drop out or graduate. See Figure 2.8 for the transition graph. To simulate the long-term probability that a new student graduates, the chain is run for 10 steps with initial distribution $\alpha = (0, 1, 0, 0, 0, 0)$, taking X_{10} as a *long-term* sample. (With high probability, a student will either drop out or graduate by 10 years.) The simulation is repeated 10,000 times, each time keeping track of whether a student graduates or drops out. The estimated long-term probability of graduating is 0.8037.

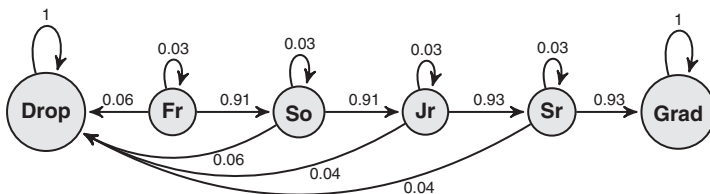


Figure 2.8 Transition graph of the graduation Markov chain.

R : Simulating Graduation, Drop-out Rate

```
# graduation.R
> init <- c(0,1,0,0,0,0) #Student starts as first-year
> P <- matrix(c(1,0,0,0,0,0,0.06,0.03,0.91,0,
  0,0,0.06,0,0,0.03,0.91,0,0,0.04,0,0,0.03,0.93,0,
  0.04,0,0,0,0.03,0.93,0,0,0,0,0,0,1),nrow=6,byrow=T)
> states <- c("Drop","Fr","So","Jr","Se","Grad")
> rownames(P) <- states
> colnames(P) <- states
> P
      Drop   Fr   So   Jr   Se Grad
Drop 1.00 0.00 0.00 0.00 0.00 0.00
Fr   0.06 0.03 0.91 0.00 0.00 0.00
So   0.06 0.00 0.03 0.91 0.00 0.00
Jr   0.04 0.00 0.00 0.03 0.93 0.00
Se   0.04 0.00 0.00 0.00 0.03 0.93
Grad 0.00 0.00 0.00 0.00 0.00 1.00
> markov(init,P,10,states)
[1] "Fr"   "So"   "Ju"   "Se"   "Grad" "Grad"
[7] "Grad" "Grad" "Grad" "Grad"
> sim <- replicate(10000,markov(init,P,10,states)[11])
> table(sim)/10000
      Drop   Grad
0.1963 0.8037
```

The graduation transition matrix is small enough so that it is possible to use technology to take high matrix powers. We find that

$$\mathbf{P}^{20} = \mathbf{P}^{21} = \begin{array}{c} \begin{array}{ccccc} & \text{Drop} & \text{Fr} & \text{So} & \text{Jr} & \text{Sr} & \text{Grad} \\ \text{Drop} & \left(\begin{array}{cccccc} 1 & 0 & 0 & 0 & 0 & 0 \\ 0.1910 & 0 & 0 & 0 & 0 & 0.8090 \\ 0.1376 & 0 & 0 & 0 & 0 & 0.8624 \\ 0.0808 & 0 & 0 & 0 & 0 & 0.9192 \\ 0.0412 & 0 & 0 & 0 & 0 & 0.9588 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{array} \right) \end{array} \end{array}.$$

The interpretation of this limiting matrix as representing long-term probabilities shows that the probability that a first-year student eventually graduates is 0.809. The last column of the matrix gives the probability of eventually graduating for each class year.

Note that although matrix powers $\mathbf{P}^n$ converge to a limiting matrix, as $n \rightarrow \infty$, unlike the forest fire and frog-hopping examples, the rows of this matrix are not identical. In this case, the long-term probability of hitting a particular state depends on the initial state. ■

2.6 MATHEMATICAL INDUCTION*

Mathematical induction is a technique for proving theorems, or properties, which hold for the natural numbers 1, 2, 3, ... An example of such a theorem is that the sum of the first n positive integers is

$$1 + 2 + \cdots + n = \binom{n+1}{2} = \frac{n(n+1)}{2}.$$

An example for Markov chains is given in Section 2.3. Let $X_0, X_1, \dots$ be a Markov chain with transition matrix $\mathbf{P}$ and initial distribution $\boldsymbol{\alpha}$. Then, for all $n \geq 1$, the distribution of X_n is $\boldsymbol{\alpha}\mathbf{P}^n$. That is,

$$P(X_n = j) = (\boldsymbol{\alpha}\mathbf{P}^n)_j = \sum_i \alpha_i P_{ij}^n \text{ for all states } j.$$

Both of these results can be proven using induction.

The *principle of mathematical induction* states that (i) if a statement is true for $n = 1$, and (ii) whenever the statement is true for a natural number $n = k$, it is also true for $n = k + 1$, then the statement will be true for all natural numbers.

Proving theorems by mathematical induction is a two-step process.

First, the *base case* is established by proving the result true for $n = 1$.

Second, one assumes the result true for a given n , and then shows that it is true for $n + 1$. Assuming that the result true for a given n is called the *induction hypothesis*.

To illustrate the proof technique, consider the claim that the sum of the first n integers is equal to $n(n+1)/2$.

For the base case, when $n = 1$, the sum of the first n integers is trivially equal to 1. And

$$n(n+1)/2 = 1(2)/2 = 1.$$

This establishes the base case.

Assume that the property true for a given n . We need to show that the sum of the first $n + 1$ integers is $(n + 1)(n + 2)/2$. The sum of the first $n + 1$ integers is

$$\sum_{k=1}^{n+1} k = \left(\sum_{k=1}^n k \right) + (n + 1).$$

By the induction hypothesis, the latter is equal to

$$\frac{n(n + 1)}{2} + (n + 1) = \frac{n(n + 1) + 2(n + 1)}{2} = \frac{n^2 + 3n + 2}{2} = \frac{(n + 2)(n + 1)}{2}.$$

This establishes the sum formula.

For another example of an induction proof, take the Markov chain result that the distribution of X_n is αP^n . The base case $n = 1$ is shown in Section 2.3. The distribution of X_1 is $\alpha P^1 = \alpha P$. Assume the result true for a given n . For the distribution of X_{n+1} , condition on X_n . This gives

$$\begin{aligned} P(X_{n+1} = j) &= \sum_i P(X_{n+1} = j | X_n = i) P(X_n = i) \\ &= \sum_i P_{ij} P(X_n = i) \\ &= \sum_i P_{ij} (\alpha P^n)_i \\ &= (\alpha P^{n+1})_j, \end{aligned}$$

where the next-to-last equality is by the induction hypothesis.

If the reader would like more practice applying induction, see Exercise 2.22.



Figure 2.9 Andrei Andreyevich Markov (1856–1922). *Source:* Wikimedia Commons, <https://commons.wikimedia.org/wiki/File:AAMarkov.jpg>. Public domain.

EXERCISES

2.1 A Markov chain has transition matrix

$$P = \begin{matrix} & \begin{matrix} 1 & 2 & 3 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \end{matrix} & \begin{pmatrix} 0.1 & 0.3 & 0.6 \\ 0 & 0.4 & 0.6 \\ 0.3 & 0.2 & 0.5 \end{pmatrix} \end{matrix}$$

with initial distribution $\alpha = (0.2, 0.3, 0.5)$. Find the following:

- (a) $P(X_7 = 3 | X_6 = 2)$
- (b) $P(X_9 = 2 | X_1 = 2, X_5 = 1, X_7 = 3)$
- (c) $P(X_0 = 3 | X_1 = 1)$
- (d) $E(X_2)$

2.2 Let $X_0, X_1, \dots$ be a Markov chain with transition matrix

$$P = \begin{matrix} & \begin{matrix} 1 & 2 & 3 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \end{matrix} & \begin{pmatrix} 0 & 1/2 & 1/2 \\ 1 & 0 & 0 \\ 1/3 & 1/3 & 1/3 \end{pmatrix} \end{matrix}$$

and initial distribution $\alpha = (1/2, 0, 1/2)$. Find the following:

- (a) $P(X_2 = 1 | X_1 = 3)$
- (b) $P(X_1 = 3, X_2 = 1)$
- (c) $P(X_1 = 3 | X_2 = 1)$
- (d) $P(X_9 = 1 | X_1 = 3, X_4 = 1, X_7 = 2)$

2.3 See Example 2.6. Consider the Wright–Fisher model with a population of $k = 3$ genes. If the population initially has one A allele, find the probability that there are no A alleles in three generations.

2.4 For the general two-state chain with transition matrix

$$P = \begin{matrix} & \begin{matrix} a & b \end{matrix} \\ \begin{matrix} a \\ b \end{matrix} & \begin{pmatrix} 1-p & p \\ q & 1-q \end{pmatrix} \end{matrix}$$

and initial distribution $\alpha = (\alpha_1, \alpha_2)$, find the following:

- (a) the two-step transition matrix
- (b) the distribution of X_1

- 2.5** Consider a random walk on $\{0, \dots, k\}$, which moves left and right with respective probabilities q and p . If the walk is at 0 it transitions to 1 on the next step. If the walk is at k it transitions to $k - 1$ on the next step. This is called *random walk with reflecting boundaries*. Assume that $k = 3$, $q = 1/4$, $p = 3/4$, and the initial distribution is uniform. For the following, use technology if needed.
- Exhibit the transition matrix.
 - Find $P(X_7 = 1 | X_0 = 3, X_2 = 2, X_4 = 2)$.
 - Find $P(X_3 = 1, X_5 = 3)$.
- 2.6** A tetrahedron die has four faces labeled 1, 2, 3, and 4. In repeated independent rolls of the die $R_0, R_1, \dots$, let $X_n = \max\{R_0, \dots, R_n\}$ be the maximum value after $n + 1$ rolls, for $n \geq 0$.
- Give an intuitive argument for why $X_0, X_1, \dots$ is a Markov chain, and exhibit the transition matrix.
 - Find $P(X_3 \geq 3)$.
- 2.7** Let $X_0, X_1, \dots$ be a Markov chain with transition matrix $\mathbf{P}$. Let $Y_n = X_{3n}$, for $n = 0, 1, 2, \dots$. Show that $Y_0, Y_1, \dots$ is a Markov chain and exhibit its transition matrix.
- 2.8** Give the Markov transition matrix for random walk on the weighted graph in Figure 2.10.
- 2.9** Give the transition matrix for the transition graph in Figure 2.11.

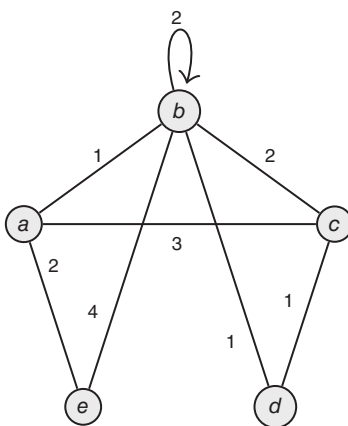


Figure 2.10

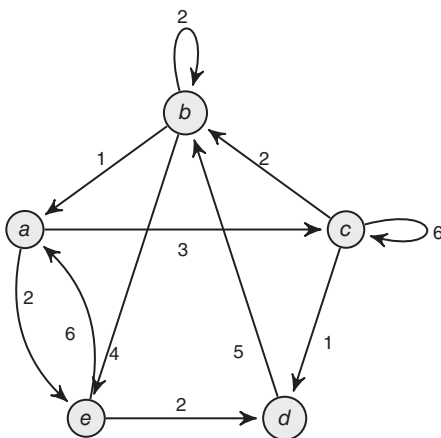


Figure 2.11

2.10 Consider a Markov chain with transition matrix

$$P = \begin{matrix} & \begin{matrix} a & b & c & d \end{matrix} \\ \begin{matrix} a \\ b \\ c \\ d \end{matrix} & \begin{pmatrix} 0 & 3/5 & 1/5 & 1/5 \\ 3/4 & 0 & 1/4 & 0 \\ 1/4 & 1/4 & 1/4 & 1/4 \\ 1/4 & 0 & 1/4 & 1/2 \end{pmatrix} \end{matrix}.$$

- (a) Exhibit the directed, weighted transition graph for the chain.
 - (b) The transition graph for this chain can be given as a weighted graph without directed edges. Exhibit the graph.
- 2.11** You start with five dice. Roll all the dice and put aside those dice that come up 6. Then, roll the remaining dice, putting aside those dice that come up 6. And so on. Let X_n be the number of dice that are sixes after n rolls.
- (a) Describe the transition matrix P for this Markov chain.
 - (b) Find the probability of getting all sixes by the third play.
 - (c) What do you expect P^{100} to look like? Use technology to confirm your answer.
- 2.12** Two urns contain k balls each. Initially, the balls in the left urn are all red and the balls in the right urn are all blue. At each step, pick a ball at random from each urn and exchange them. Let X_n be the number of blue balls in the left urn. (Note that necessarily $X_0 = 0$ and $X_1 = 1$.) Argue that the process is a Markov chain. Find the transition matrix. This model is called the Bernoulli–Laplace model of diffusion and was introduced by Daniel Bernoulli in 1769 as a model for the flow of two incompressible liquids between two containers.

- 2.13** See the move-to-front process in Example 2.10. Here is another way to organize the bookshelf. When a book is returned it is put back on the library shelf one position forward from where it was originally. If the book at the front of the shelf is returned it is put back at the front of the shelf. Thus, if the order of books is (a, b, c, d, e) and book d is picked, the new order is (a, b, d, c, e) . This reorganization method is called the *transposition*, or *move-ahead-1*, scheme. Give the transition matrix for the transposition scheme for a shelf with three books.
- 2.14** There are k songs on Mary's music player. The player is set to *shuffle* mode, which plays songs uniformly at random, sampling with replacement. Thus, repeats are possible. Let X_n denote the number of *unique* songs that have been heard after the n th play.
- (a) Show that $X_0, X_1, \dots$ is a Markov chain and give the transition matrix.
- (b) If Mary has four songs on her music player, find the probability that all songs are heard after six plays.
- 2.15** Assume that $X_0, X_1, \dots$ is a two-state Markov chain on $S = \{0, 1\}$ with transition matrix

$$P = \begin{matrix} & \begin{matrix} 0 & 1 \end{matrix} \\ \begin{matrix} 0 \\ 1 \end{matrix} & \begin{pmatrix} 1-p & p \\ q & 1-q \end{pmatrix} \end{matrix}.$$

The present state of the chain only depends on the previous state. One can model a bivariate process that looks back two time periods by the following construction. Let $Z_n = (X_{n-1}, X_n)$, for $n \geq 1$. The sequence $Z_1, Z_2, \dots$ is a Markov chain with state space $S \times S = \{(0, 0), (0, 1), (1, 0), (1, 1)\}$. Give the transition matrix of the new chain.

- 2.16** Assume that P is a stochastic matrix with equal rows. Show that $P^n = P$, for all $n \geq 1$.
- 2.17** Let P be a stochastic matrix. Show that $\lambda = 1$ is an eigenvalue of P . What is the associated eigenvector?
- 2.18** A stochastic matrix is called *doubly stochastic* if its columns sum to 1. Let $X_0, X_1, \dots$ be a Markov chain on $\{1, \dots, k\}$ with doubly stochastic transition matrix and initial distribution that is uniform on $\{1, \dots, k\}$. Show that the distribution of X_n is uniform on $\{1, \dots, k\}$, for all $n \geq 0$.
- 2.19** Let P be the transition matrix of a Markov chain on k states. Let $\mathbf{I}$ denote the $k \times k$ identity matrix. Consider the matrix

$$Q = (1 - p)\mathbf{I} + pP, \text{ for } 0 < p < 1.$$

Show that Q is a stochastic matrix. Give a probabilistic interpretation for the dynamics of a Markov chain governed by the Q matrix in terms of the original Markov chain.

2.20 Let $X_0, X_1, \dots$ be a Markov chain with transition matrix

$$P = \begin{matrix} & \begin{matrix} 1 & 2 & 3 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \end{matrix} & \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ p & 1-p & 0 \end{pmatrix} \end{matrix},$$

for $0 < p < 1$. Let g be a function defined by

$$g(x) = \begin{cases} 0, & \text{if } x = 1, \\ 1, & \text{if } x = 2, 3. \end{cases}$$

Let $Y_n = g(X_n)$, for $n \geq 0$. Show that $Y_0, Y_1, \dots$ is not a Markov chain.

2.21 Let P and Q be two transition matrices on the same state space. We define two processes, both started in some initial state i .

In process #1, a coin is flipped. If it lands heads, then the process unfolds according to the P matrix. If it lands tails, the process unfolds according to the Q matrix.

In process #2, at each step a coin is flipped. If it lands heads, the next state is chosen according to the P matrix. If it lands tails, the next state is chosen according to the Q matrix.

Thus, in #1, one coin is initially flipped, which governs the entire evolution of the process. And in #2, a coin is flipped at each step to decide the next step of the process.

Decide whether either of these processes is a Markov chain. If not, explain why, if yes, exhibit the transition matrix.

2.22 Prove the following using mathematical induction.

- (a) $1 + 3 + 5 + \dots + (2n - 1) = n^2$.
- (b) $1^2 + 2^2 + \dots + n^2 = n(n + 1)(2n + 1)/6$.
- (c) For all real $x > -1$, $(1 + x)^n \geq 1 + nx$.

2.23 R : Simulate the first 20 letters (vowel/consonant) of the Pushkin poem Markov chain of Example 2.2.

2.24 R : Simulate 50 steps of the random walk on the graph in Figure 2.1. Repeat the simulation 10 times. How many of your simulations end at vertex c ? Compare with the exact long-term probability the walk visits c .

2.25 R : The behavior of dolphins in the presence of tour boats in Patagonia, Argentina is studied in Dans et al. (2012). A Markov chain model is developed, with state space consisting of five primary dolphin activities (socializing,

traveling, milling, feeding, and resting). The following transition matrix is obtained.

$$P = \begin{matrix} & \begin{matrix} s & t & m & f & r \end{matrix} \\ \begin{matrix} s \\ t \\ m \\ f \\ r \end{matrix} & \begin{pmatrix} 0.84 & 0.11 & 0.01 & 0.04 & 0.00 \\ 0.03 & 0.80 & 0.04 & 0.10 & 0.03 \\ 0.01 & 0.15 & 0.70 & 0.07 & 0.07 \\ 0.03 & 0.19 & 0.02 & 0.75 & 0.01 \\ 0.03 & 0.09 & 0.05 & 0.00 & 0.83 \end{pmatrix} \end{matrix}.$$

Use technology to estimate the long-term distribution of dolphin activity.

- 2.26 R :** In computer security applications, a *honeypot* is a trap set on a network to detect and counteract computer hackers. Honeypot data are studied in Kimou et al. (2010) using Markov chains. The authors obtain honeypot data from a central database and observe attacks against four computer ports—80, 135, 139, and 445—over 1 year. The ports are the states of a Markov chain along with a state corresponding to no port is attacked. Weekly data are monitored, and the port most often attacked during the week is recorded. The estimated Markov transition matrix for weekly attacks is

$$P = \begin{matrix} & \begin{matrix} 80 & 135 & 139 & 445 & \text{No attack} \end{matrix} \\ \begin{matrix} 80 \\ 135 \\ 139 \\ 445 \\ \text{No} \end{matrix} & \begin{pmatrix} 0 & 0 & 0 & 0 & 1 \\ 0 & 8/13 & 3/13 & 1/13 & 1/13 \\ 1/16 & 3/16 & 3/8 & 1/4 & 1/8 \\ 0 & 1/11 & 4/11 & 5/11 & 1/11 \\ 0 & 1/8 & 1/2 & 1/8 & 1/4 \end{pmatrix} \end{matrix},$$

with initial distribution $\alpha = (0, 0, 0, 0, 1)$.

- Which are the least and most likely attacked ports after 2 weeks?
- Find the long-term distribution of attacked ports.

- 2.27 R :** See **gamblersruin.R**. Simulate gambler's ruin for a gambler with initial stake \$2, playing a fair game.

- Estimate the probability that the gambler is ruined before he wins \$5.
- Construct the transition matrix for the associated Markov chain. Estimate the desired probability in (a) by taking high matrix powers.
- Compare your results with the exact probability.

3

MARKOV CHAINS FOR THE LONG TERM

There exists everywhere a medium in things, determined by equilibrium.

—Dmitri Mendeleev

3.1 LIMITING DISTRIBUTION

In many cases, a Markov chain exhibits a long-term limiting behavior. The chain settles down to an equilibrium distribution, which is independent of its initial state.

Limiting Distribution

Let $X_0, X_1, \dots$ be a Markov chain with transition matrix P . A *limiting distribution* for the Markov chain is a probability distribution λ with the property that for all i and j ,

$$\lim_{n \rightarrow \infty} P_{ij}^n = \lambda_j.$$

The definition of limiting distribution is equivalent to each of the following:

- (i) For any initial distribution, and for all j ,

$$\lim_{n \rightarrow \infty} P(X_n = j) = \lambda_j.$$

(ii) For any initial distribution α ,

$$\lim_{n \rightarrow \infty} \alpha P^n = \lambda.$$

(iii)

$$\lim_{n \rightarrow \infty} P^n = \Lambda,$$

where Λ is a stochastic matrix all of whose rows are equal to λ .

We interpret λ_j as the long-term probability that the chain hits state j . By the uniqueness of limits, if a Markov chain has a limiting distribution, then that distribution is unique.

If a limiting distribution exists, a *quick and dirty* numerical method to find it is to take high matrix powers of the transition matrix until one obtains an obvious limiting matrix with equal rows. The common row is the limiting distribution. Examples of this approach have been given in Section 2.5.

Numerical methods, however, have their limits (no pun intended), and the emphasis in this chapter is on exact solutions and theoretical results.

For the general two-state Markov chain, matrix powers can be found exactly in order to compute the limiting distribution.

■ **Example 3.1 (Two-state Markov chain)** The transition matrix for a general two-state chain is

$$P = \frac{1}{2} \begin{pmatrix} 1-p & p \\ q & 1-q \end{pmatrix},$$

for $0 \leq p, q \leq 1$. If $p + q = 1$,

$$P = \begin{pmatrix} 1-p & p \\ 1-p & p \end{pmatrix},$$

and $P^n = P$ for all $n \geq 1$. Thus, $\lambda = (1-p, p)$ is the limiting distribution.

Assume $p + q \neq 1$. To find P^n , consider the entry P_{11}^n . As $P^n = P^{n-1}P$,

$$\begin{aligned} P_{11}^n &= (P^{n-1}P)_{11} = P_{11}^{n-1}P_{11} + P_{12}^{n-1}P_{21} \\ &= P_{11}^{n-1}(1-p) + P_{12}^{n-1}q \\ &= P_{11}^{n-1}(1-p) + (1 - P_{11}^{n-1})q \\ &= q + (1-p-q)P_{11}^{n-1}, \text{ for } n \geq 1. \end{aligned}$$

The next-to-last equality uses the fact that $P_{11}^{n-1} + P_{12}^{n-1} = 1$, since P^{n-1} is a stochastic matrix. Unwinding the recurrence gives

$$\begin{aligned}
 P_{11}^n &= q + (1 - p - q)P_{11}^{n-1} \\
 &= q + q(1 - p - q) + (1 - p - q)^2 P_{11}^{n-2} \\
 &= q + q(1 - p - q) + q(1 - p - q)^2 + (1 - p - q)^3 P_{11}^{n-3} \\
 &= \cdots = q \sum_{k=0}^{n-1} (1 - p - q)^k + (1 - p - q)^n \\
 &= (1 - p - q)^n + q \frac{1 - (1 - p - q)^n}{1 - (1 - p - q)} \\
 &= \frac{q}{p + q} + \frac{p}{p + q} (1 - p - q)^n.
 \end{aligned}$$

The matrix entry

$$P_{22}^n = \frac{p}{p + q} + \frac{q}{p + q} (1 - p - q)^n$$

is found similarly. Since the rows of P^n sum to 1,

$$P^n = \frac{1}{p + q} \begin{pmatrix} q + p(1 - p - q)^n & p - p(1 - p - q)^n \\ q - q(1 - p - q)^n & p + q(1 - p - q)^n \end{pmatrix}. \quad (3.1)$$

If p and q are not both 0, nor both 1, then $|1 - p - q| < 1$ and

$$\lim_{n \rightarrow \infty} P^n = \frac{1}{p + q} \begin{pmatrix} q & p \\ q & p \end{pmatrix}.$$

The limiting distribution is

$$\lambda = \left(\frac{q}{p + q}, \frac{p}{p + q} \right).$$

Observe that in addition to giving the limiting distribution, Equation (3.1) reveals the rate of convergence to that limit. The convergence is exponential and governed by the quantity $(1 - p - q)^n$. ■

Proportion of Time in Each State

The limiting distribution gives the long-term probability that a Markov chain hits each state. It can also be interpreted as the long-term proportion of time that the chain visits each state. To make this precise, let $X_0, X_1, \dots$ be a Markov chain with transition matrix P and limiting distribution λ . For state j , define indicator random variables

$$I_k = \begin{cases} 1, & \text{if } X_k = j, \\ 0, & \text{otherwise,} \end{cases}$$

for $k = 0, 1, \dots$. Then, $\sum_{k=0}^{n-1} I_k$ is the number of times the chain visits j in the first n steps (counting X_0 as the first step). From initial state i , the long-term expected proportion of time that the chain visits j is

$$\begin{aligned} \lim_{n \rightarrow \infty} E \left(\frac{1}{n} \sum_{k=0}^{n-1} I_k | X_0 = i \right) &= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=0}^{n-1} E(I_k | X_0 = i) \\ &= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=0}^{n-1} P(X_k = j | X_0 = i) \\ &= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=0}^{n-1} P_{ij}^k \\ &= \lim_{n \rightarrow \infty} P_{ij}^n = \lambda_j. \end{aligned}$$

The next-to-last equality applies a result in analysis known as Cesaro's lemma. The lemma states that if a sequence of numbers converges to a limit, then the sequence of partial averages also converges to that limit. That is, if $x_n \rightarrow x$, as $n \rightarrow \infty$, then $(x_1 + \dots + x_n)/n \rightarrow x$, as $n \rightarrow \infty$.

 **Example 3.2** After work, Angel goes to the gym and either does aerobics, weights, yoga, or gets a massage. Each day, Angel decides her workout routine based on what she did the previous day according to the Markov transition matrix

$$P = \begin{matrix} & \begin{matrix} \text{Aerobics} & \text{Massage} & \text{Weights} & \text{Yoga} \end{matrix} \\ \begin{matrix} \text{Aerobics} \\ \text{Massage} \\ \text{Weights} \\ \text{Yoga} \end{matrix} & \begin{pmatrix} 0.1 & 0.2 & 0.4 & 0.3 \\ 0.4 & 0.0 & 0.4 & 0.2 \\ 0.3 & 0.3 & 0.0 & 0.4 \\ 0.2 & 0.1 & 0.4 & 0.3 \end{pmatrix} \end{matrix}.$$

Taking high matrix powers gives the limiting distribution

Aerobics	Massage	Weight	Yoga
0.238	0.164	0.286	0.312

See the following R code, where Angel's gym visits are simulated for 100 days. During that time, Angel did aerobics on 26 days, got a massage 14 times, did weights on 31 days, and did yoga 29 times. The proportion of visits to each state is

Aerobics	Massage	Weighs	Yoga
0.26	0.14	0.31	0.29

These proportions are relatively close to the actual limiting distribution of the chain notwithstanding the fact that the estimates are based on just 100 steps. Compare the results to the million step simulation, also given in the R code. ■

R: Angel at the Gym

```
# gym.R
> P
      Aerobics  Massage  Weights  Yoga
Aerobics    0.1      0.2      0.4    0.3
Massage     0.4      0.0      0.4    0.2
Weights     0.3      0.3      0.0    0.4
Yoga        0.2      0.1      0.4    0.3
> init <- c(1/4,1/4,1/4,1/4) # initial distribution
> states <- c("a","m","w","y")
# simulate Markov chain for 100 steps
> simlist <- markov(init,P,100,states)
> mwyaaawyayawyayawymwywamwawawyawywywywyawyamwaway
> amamawymyayawmawymwywmwyaywaywywamwywymwawamaay
> table(simlist)/100
Aerobics  Massage  Weights      Yoga
   0.26      0.14      0.31      0.29
> steps <- 1000000 # one million steps
> simlist <- markov(init,P,steps,states)
> table(simlist)/steps
Aerobics  Massage  Weights      Yoga
0.237425  0.164388  0.285548  0.312640
```

3.2 STATIONARY DISTRIBUTION

It is interesting to consider what happens if we assign the limiting distribution of a Markov chain to be the initial distribution of the chain.

For the two-state chain, as in Example 3.1, the limiting distribution is

$$\lambda = \left(\frac{q}{p+q}, \frac{p}{p+q} \right).$$

Let λ be the initial distribution for such a chain. Then, the distribution of X_1 is

$$\begin{aligned} \lambda P &= \left(\frac{q}{p+q}, \frac{p}{p+q} \right) \begin{pmatrix} 1-p & p \\ q & 1-q \end{pmatrix} \\ &= \left(\frac{q(1-p) + pq}{p+q}, \frac{qp + p(1-q)}{p+q} \right) \end{aligned}$$

$$= \left(\frac{q}{p+q}, \frac{p}{p+q} \right) = \lambda.$$

That is, $\lambda P = \lambda$. A probability vector π that satisfies $\pi P = \pi$ plays a special role for Markov chains.

Stationary Distribution

Let $X_0, X_1, \dots$ be a Markov chain with transition matrix P . A *stationary distribution* is a probability distribution π , which satisfies

$$\pi = \pi P. \quad (3.2)$$

That is,

$$\pi_j = \sum_i \pi_i P_{ij}, \text{ for all } j.$$

If we assume that a stationary distribution π is the initial distribution, then Equation (3.2) says that the distribution of X_0 is the same as the distribution of X_1 . Since the chain started at $n = 1$ is also a Markov chain with transition matrix P , it follows that X_2 has the same distribution as X_1 . In fact, all of the X_n have the same distribution, as

$$\pi P^n = (\pi P)P^{n-1} = \pi P^{n-1} = (\pi P)P^{n-2} = \pi P^{n-2} = \dots = \pi P = \pi.$$

If the initial distribution is a stationary distribution, then $X_0, X_1, X_2, \dots$ is a sequence of identically distributed random variables.

The name *stationary* comes from the fact that if the chain starts in its stationary distribution, then it stays in that distribution. We refer to the *stationary Markov chain* or the Markov chain *in stationarity* for the chain started in its stationary distribution.

If $X_0, X_1, X_2, \dots$ is a stationary Markov chain, then for any $n > 0$, the sequence $X_n, X_{n+1}, X_{n+2}, \dots$ is also a stationary Markov chain with the same transition matrix and stationary distribution as the original chain.

(The fact that the stationary chain is a sequence of identically distributed random variables does not mean that the random variables are independent. On the contrary, the dependency structure between successive random variables in a Markov chain is governed by the transition matrix, regardless of the initial distribution.)

Other names for the stationary distribution are *invariant*, *steady-state*, and *equilibrium* distribution. The latter highlights the fact that there is an intimate connection between the stationary distribution and the limiting distribution. If a Markov chain has a limiting distribution then that distribution is a stationary distribution.

Limiting Distributions are Stationary Distributions

Lemma 3.1. Assume that π is the limiting distribution of a Markov chain with transition matrix P . Then, π is a stationary distribution.

Proof. Assume that π is the limiting distribution. We need to show that $\pi P = \pi$. For any initial distribution α ,

$$\pi = \lim_{n \rightarrow \infty} \alpha P^n = \lim_{n \rightarrow \infty} \alpha (P^{n-1} P) = \left(\lim_{n \rightarrow \infty} \alpha P^{n-1} \right) P = \pi P,$$

which uses the fact that if $\lim_{n \rightarrow \infty} x_n = x$, then $\lim_{n \rightarrow \infty} x_{n-1} = x$. ■

Unfortunately, the converse of Lemma 3.1 is not true—stationary distributions are not necessarily limiting distributions. For a counterexample, take the Markov chain with transition matrix

$$P = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Solving $\pi P = \pi$, or

$$(\pi_1, \pi_2) \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = (\pi_1, \pi_2),$$

gives $\pi_1 = \pi_2$. Since the stationary distribution is a probability vector, the unique solution is $\pi = (1/2, 1/2)$. The stationary distribution is uniform on each state. However, the chain has no limiting distribution. The process evolves by flip-flopping back and forth between states. As in the case of random walk on a cycle with an even number of vertices, the position of the walk after n steps depends on the starting vertex and the parity of n .

Another counterexample is the Markov chain with transition matrix

$$P = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

This process is rather boring—the chain simply stays forever in its starting state. The chain has no limiting distribution, as the long-term state of the chain depends upon the starting state. However, *every* probability vector is a stationary distribution since $xP = x$, for all vectors x .

Thus, there are Markov chains with more than one stationary distribution; there are Markov chains with unique stationary distributions that are not limiting distributions; and there are even Markov chains that do not have stationary distributions.

However, a large and important class of Markov chains has unique stationary distributions that are the limiting distribution of the chain. A goal of this chapter is to characterize such chains.

Regular Matrices

A matrix M is said to be *positive* if all the entries of M are positive. We write $M > 0$. Similarly, write $x > 0$ for a vector x with all positive entries.

Regular Transition Matrix

A transition matrix P is said to be *regular* if some power of P is positive. That is, $P^n > 0$, for some $n \geq 1$.

For example,

$$P = \begin{pmatrix} 0 & 1/2 & 1/2 \\ 1 & 0 & 0 \\ 1/2 & 1/2 & 0 \end{pmatrix}$$

is regular, since

$$P^4 = \begin{pmatrix} 9/16 & 5/16 & 1/8 \\ 1/4 & 3/8 & 3/8 \\ 1/2 & 5/16 & 3/16 \end{pmatrix}$$

is positive. However,

$$P = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}$$

is not regular, since the powers of P cycle through the matrices

$$P = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}, \quad P^2 = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \quad \text{and} \quad P^3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

If the transition matrix of a Markov chain is regular, then the chain has a limiting distribution, which is the unique stationary distribution of the chain.

Limit Theorem for Regular Markov Chains

Theorem 3.2. *A Markov chain whose transition matrix P is regular has a limiting distribution, which is the unique, positive, stationary distribution of the chain. That is, there exists a unique probability vector $\pi > 0$, such that*

$$\lim_{n \rightarrow \infty} P_{ij}^n = \pi_j,$$

for all i, j , where

$$\sum_i \pi_i P_{ij} = \pi_j.$$

Equivalently, there exists a positive stochastic matrix Π such that

$$\lim_{n \rightarrow \infty} P^n = \Pi,$$

where $\mathbf{\Pi}$ has equal rows with common row $\boldsymbol{\pi}$, and $\boldsymbol{\pi}$ is the unique probability vector, which satisfies

$$\boldsymbol{\pi}P = \boldsymbol{\pi}.$$

The proof of Theorem 3.2 is deferred to Section 3.10.

■ **Example 3.3** Assume that a Markov chain has transition matrix

$$P = \begin{pmatrix} 0 & 1-p & p \\ p & 0 & 1-p \\ 1-p & p & 0 \end{pmatrix},$$

for $0 < p < 1$. Find the limiting distribution.

Solution We find that

$$P^2 = \begin{pmatrix} 2p(1-p) & p^2 & (1-p)^2 \\ (1-p)^2 & 2p(1-p) & p^2 \\ p^2 & (1-p)^2 & 2p(1-p) \end{pmatrix}.$$

Since $0 < p < 1$, the matrix P^2 is positive. Thus, P is regular. By Theorem 3.2, the limiting distribution is the stationary distribution.

How to find the stationary distribution of a Markov chain is the topic of the next section. However, for now we give the reader a little help and urge them to try $\boldsymbol{\pi} = (1/3, 1/3, 1/3)$.

Indeed, $\boldsymbol{\pi}P = \boldsymbol{\pi}$, as

$$\left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right) \begin{pmatrix} 0 & 1-p & p \\ p & 0 & 1-p \\ 1-p & p & 0 \end{pmatrix} = \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right).$$

The uniform distribution $\boldsymbol{\pi}$ is the unique stationary distribution of the chain, and thus the desired limiting distribution.

The example is interesting because the limiting distribution is uniform for *all* choices of $0 < p < 1$. ■

Here is one way to tell if a stochastic matrix is not regular. If for some power n , all the 0s in P^n appear in the same locations as all the 0s in P^{n+1} , then they will appear in the same locations for all higher powers, and the matrix is not regular.

■ **Example 3.4** A Markov chain has transition matrix

$$\mathbf{P} = \begin{pmatrix} 0 & 0 & 0 & 0 & 1 \\ \frac{1}{4} & 0 & 0 & \frac{1}{2} & \frac{1}{4} \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{4} & \frac{1}{2} & 0 & 0 & \frac{1}{4} \end{pmatrix}.$$

Determine if the matrix is regular.

Solution We find that

$$\mathbf{P}^4 = \begin{pmatrix} \frac{9}{64} & \frac{7}{32} & \frac{1}{8} & \frac{3}{16} & \frac{21}{64} \\ \frac{21}{256} & \frac{11}{128} & \frac{15}{32} & \frac{11}{64} & \frac{49}{256} \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & \frac{15}{16} & \frac{1}{16} & 0 \\ \frac{35}{256} & \frac{21}{128} & \frac{7}{32} & \frac{13}{64} & \frac{71}{256} \end{pmatrix} \text{ and } \mathbf{P}^5 = \begin{pmatrix} \frac{35}{256} & \frac{21}{128} & \frac{7}{32} & \frac{13}{64} & \frac{71}{256} \\ \frac{71}{1024} & \frac{49}{512} & \frac{71}{128} & \frac{33}{256} & \frac{155}{1024} \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & \frac{31}{32} & \frac{1}{32} & 0 \\ \frac{113}{1024} & \frac{71}{512} & \frac{41}{128} & \frac{47}{256} & \frac{253}{1024} \end{pmatrix}.$$

Since the 0s are in the same locations for both matrices, we conclude that $\mathbf{P}$ is not regular. ■

Finding the Stationary Distribution

Assume that $\boldsymbol{\pi}$ is a stationary distribution for a Markov chain with transition matrix $\mathbf{P}$. Then,

$$\sum_i \pi_i P_{ij} = \pi_j, \text{ for all states } j,$$

which gives a system of linear equations. If $\mathbf{P}$ is a $k \times k$ matrix, the system has k equations and k unknowns. Since the rows of $\mathbf{P}$ sum to 1, the $k \times k$ system will contain a redundant equation.

For the general two-state chain, with

$$\mathbf{P} = \begin{pmatrix} 1-p & p \\ q & 1-q \end{pmatrix},$$

the equations are

$$(1-p)\pi_1 + q\pi_2 = \pi_1$$

$$p\pi_1 + (1-q)\pi_2 = \pi_2.$$

The equations are redundant and lead to $\pi_1 p = \pi_2 q$. If p and q are not both zero, then together with the condition $\pi_1 + \pi_2 = 1$, the unique solution is

$$\pi = \left(\frac{q}{p+q}, \frac{p}{p+q} \right).$$

■ **Example 3.5** Find the stationary distribution of the weather Markov chain of Example 2.3, with transition matrix

$$P = \begin{array}{c} \begin{array}{ccc} & \text{Rain} & \text{Snow} & \text{Clear} \\ \text{Rain} & 1/5 & 3/5 & 1/5 \\ \text{Snow} & 1/10 & 4/5 & 1/10 \\ \text{Clear} & 1/10 & 3/5 & 3/10 \end{array} \end{array}.$$

Solution The linear system to solve is

$$(1/5)\pi_1 + (1/10)\pi_2 + (1/10)\pi_3 = \pi_1$$

$$(3/5)\pi_1 + (4/5)\pi_2 + (3/5)\pi_3 = \pi_2$$

$$(1/5)\pi_1 + (1/10)\pi_2 + (3/10)\pi_3 = \pi_3$$

$$\pi_1 + \pi_2 + \pi_3 = 1.$$

One of the first three equations is redundant. The unique solution is

$$\pi = \left(\frac{1}{9}, \frac{3}{4}, \frac{5}{36} \right).$$

R : Finding the Stationary Distribution

The R function `stationary(P)` finds the stationary distribution of a Markov chain with transition matrix `P`. The function is contained in the **utilities.R** file.

```
> P
      Rain Snow Clear
Rain   0.2  0.6  0.2
Snow   0.1  0.8  0.1
Clear  0.1  0.6  0.3
> stationary(P)
[1] 0.1111 0.7500 0.1389
# check that this is a stationary distribution
> stationary(P) %*% P
      Rain   Snow   Clear
[1] 0.1111 0.75 0.1389
```

Here is a useful technique for finding the stationary distribution, which reduces by one the number of equations to solve. It makes use of the fact that if $\mathbf{x}$ is a vector, not necessarily a probability vector, which satisfies $\mathbf{xP} = \mathbf{x}$, then $(c\mathbf{x})\mathbf{P} = c\mathbf{x}$, for all constants c . It follows that if one can find a non-negative $\mathbf{x}$, which satisfies $\mathbf{xP} = \mathbf{x}$, then a unique probability vector solution $\boldsymbol{\pi} = c\mathbf{x}$ can be gotten by an appropriate choice of c so that the rows of $c\mathbf{x}$ sum to 1. In particular, let $c = 1/\sum_j x_j$, the reciprocal of the sum of the components of $\mathbf{x}$.

The linear system $\sum_i \pi_i P_{ij} = \pi_j$, without the constraint $\sum_i \pi_i = 1$, has one redundant equation. Our solution method consists of (i) eliminating a redundant equation and (ii) solving the resulting system for $\mathbf{x} = (1, x_2, x_3, \dots)$, where the first (or any) component of $\mathbf{x}$ is replaced by 1.

For a Markov chain with k states, this method reduces the problem to solving a $(k-1) \times (k-1)$ linear system. If the original chain has a unique stationary distribution, then the reduced linear system will have a unique solution, but one which is not necessarily a probability vector. To make it a probability vector whose components sum to 1, divide by the sum of the components. In other words, the unique stationary distribution is

$$\boldsymbol{\pi} = \frac{1}{1 + x_2 + \dots + x_k} (1, x_2, \dots, x_k).$$

To illustrate the method, consider the transition matrix

$$\mathbf{P} = \begin{pmatrix} 1/3 & 1/2 & 1/6 \\ 1/2 & 1/2 & 0 \\ 1/4 & 1/2 & 1/4 \end{pmatrix}.$$

To find the stationary distribution, first let $\mathbf{x} = (1, x_2, x_3)$. Then, $\mathbf{xP} = \mathbf{x}$ gives a 3×3 linear system. The first two equations are

$$(1/3)(1) + (1/2)x_2 + (1/4)x_3 = 1,$$

$$(1/2)(1) + (1/2)x_2 + (1/2)x_3 = x_2,$$

or

$$(1/2)x_2 + (1/4)x_3 = 2/3,$$

$$(-1/2)x_2 + (1/2)x_3 = -1/2,$$

with unique solution $\mathbf{x} = (1, 11/9, 2/9)$. The sum of the components is

$$1 + \frac{11}{9} + \frac{2}{9} = \frac{22}{9}.$$

The stationary distribution is

$$\boldsymbol{\pi} = \frac{9}{22} \left(1, \frac{11}{9}, \frac{2}{9} \right) = \left(\frac{9}{22}, \frac{11}{22}, \frac{2}{22} \right).$$

■ **Example 3.6** A Markov chain on $\{1, 2, 3, 4\}$ has transition matrix

$$P = \begin{pmatrix} p & 1-p & 0 & 0 \\ (1-p)/2 & p & (1-p)/2 & 0 \\ 0 & (1-p)/2 & p & (1-p)/2 \\ 0 & 0 & 1-p & p \end{pmatrix},$$

for $0 < p < 1$. Find the stationary distribution.

Solution Let $\mathbf{x} = (x_1, x_2, x_3, x_4)$, with $x_1 = 1$. Take

$$x_j = \sum_{i=1}^4 x_i P_{ij},$$

for $j = 1, 2$, and 4 . This gives

$$\begin{aligned} p + \left(\frac{1-p}{2}\right)x_2 &= 1, \\ 1-p + px_2 + \left(\frac{1-p}{2}\right)x_3 &= x_2, \\ \left(\frac{1-p}{2}\right)x_3 + px_4 &= x_4, \end{aligned}$$

with solution $x_2 = x_3 = 2$ and $x_1 = x_4 = 1$. The stationary distribution is

$$\pi = \frac{1}{1+2+2+1} (1, 2, 2, 1) = \left(\frac{1}{6}, \frac{2}{6}, \frac{2}{6}, \frac{1}{6}\right).$$

■

■ **Example 3.7 (The Ehrenfest dog–flea model)** The Ehrenfest dog–flea model was originally proposed by physicists Tatyana and Paul Ehrenfest to describe the diffusion of gases. Mathematician Mark Kac called it “one of the most instructive models in the whole of physics.”

Two dogs—Lisa and Cooper—share a population of N fleas. At each discrete unit of time, one of the fleas jumps from the dog it is on to the other dog. Let X_n denote the number of fleas on Lisa after n jumps. If there are i fleas on Lisa, then on the next jump the number of fleas on Lisa either goes up by one, if one of the $N - i$ fleas on Cooper jumps to Lisa, or goes down by one, if one of the i fleas on Lisa jumps to Cooper.

The process is a Markov chain on $\{0, 1, \dots, N\}$, with transition matrix

$$P_{ij} = \begin{cases} i/N, & \text{if } j = i - 1, \\ (N - i)/N, & \text{if } j = i + 1, \\ 0, & \text{otherwise.} \end{cases}$$

Here is the Ehrenfest transition matrix for $N = 5$ fleas:

$$P = \begin{matrix} & \begin{matrix} 0 & 1 & 2 & 3 & 4 & 5 \end{matrix} \\ \begin{matrix} 0 \\ 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{matrix} & \begin{pmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 1/5 & 0 & 4/5 & 0 & 0 & 0 \\ 0 & 2/5 & 0 & 3/5 & 0 & 0 \\ 0 & 0 & 3/5 & 0 & 2/5 & 0 \\ 0 & 0 & 0 & 4/5 & 0 & 1/5 \\ 0 & 0 & 0 & 0 & 1 & 0 \end{pmatrix} \end{matrix}.$$

To find the stationary distribution for the general Ehrenfest chain, let $\mathbf{x} = (x_0, x_1, \dots, x_N)$, with $x_0 = 1$. Set

$$x_j = \sum_{i=0}^N x_i P_{ij} = x_{j-1} \frac{N - (j - 1)}{N} + x_{j+1} \frac{j + 1}{N}, \quad (3.3)$$

for $j = 1, \dots, N - 1$. Also, $1 = (1/N)x_1$, so $x_1 = N$. Solving Equation (3.3) starting at $j = 1$ gives $x_2 = N(N - 1)/2$, then $x_3 = N(N - 1)(N - 2)/6$. The general term is

$$x_j = \frac{N(N - 1) \cdots (N - j + 1)}{j!} = \frac{N!}{j!(N - j)!} = \binom{N}{j}, \quad \text{for } j = 0, 1, \dots, N,$$

which can be derived by induction. The stationary distribution is

$$\pi_j = \frac{1}{\sum_{i=0}^N \binom{N}{i}} \binom{N}{j} = \binom{N}{j} \frac{1}{2^N}, \quad \text{for } j = 0, \dots, N.$$

The distribution is a binomial distribution with parameters N and $1/2$.

The Ehrenfest transition matrix is not regular, and the chain does not have a limiting distribution. However, we can interpret the stationary distribution as giving the long-term proportion of time spent in each state. See Example 3.19 for the description of a modified Ehrenfest scheme, which has a limiting distribution. ■

For the next example, rather than solve a linear system to find the stationary distribution, we take a *guess* at the distribution $\boldsymbol{\pi}$ based on our intuition for how the Markov chain evolves. The candidate distribution $\boldsymbol{\pi}$ is then checked to see if it satisfies $\boldsymbol{\pi} = \boldsymbol{\pi}P$.

Guessing before proving! Need I remind you that it is so that all important discoveries have been made?

—Henri Poincaré

■ **Example 3.8 (Random walk on a graph)** To find the stationary distribution for simple random walk on a graph, consider the interpretation of the distribution as the long-term fraction of time that the walk visits each vertex.

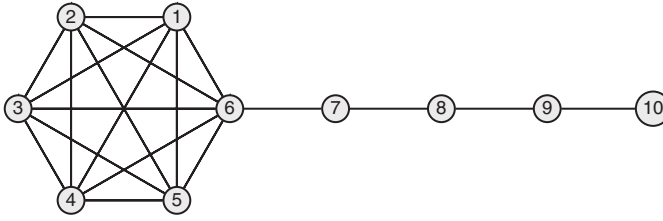


Figure 3.1 Lollipop graph.

For a concrete example, consider the *lollipop graph*, shown in Figure 3.1. One expects that in the long term a random walk on the graph is most likely to be on the *candy* at the leftmost end of the graph, and least likely to be at the right end of the stick. (The candy is a complete graph where all pairs of vertices are joined by edges.)

Intuition suggests that vertices that have more connections are more likely to be visited. That is, the time spent at vertex v is related to the degree of v .

This suggests considering a distribution on vertices that is related to the degree of the vertices. One possibility is a distribution that is proportional to the degree of the vertex. Let

$$\pi_v = \frac{\deg(v)}{\sum_w \deg(w)} = \frac{\deg(v)}{2e},$$

where e is the number of edges in the graph. The sum of the vertex degrees is equal to twice the number of edges since every edge contributes two vertices (its endpoints) to the sum of the vertex degrees.

It remains to check, for this choice of π , whether in fact $\pi = \pi P$. For vertex v ,

$$\begin{aligned} (\pi P)_v &= \sum_w \pi_w P_{wv} = \sum_{w \sim v} \left(\frac{\deg(w)}{2e} \right) \frac{1}{\deg(w)} \\ &= \frac{1}{2e} \sum_{w \sim v} 1 = \frac{\deg(v)}{2e} = \pi_v. \end{aligned}$$

Indeed, our candidate π is a stationary distribution.

For the lollipop graph in Figure 3.1, the sum of the vertex degrees is 38. Here is the stationary distribution.

Vertex	1	2	3	4	5	6	7	8	9	10
π_v	5/38	5/38	5/38	5/38	5/38	6/38	2/38	2/38	2/38	1/38

■

Simple random walk on a graph is a special case of random walk on a weighted graph, with edge weights all equal to 1. The stationary distribution for a weighted graph has a similar form as for an unweighted graph. If v is a vertex, say that an edge is *incident* to v if v is an endpoint of the edge.

Stationary Distribution for Random Walk on a Weighted Graph

Let G be a weighted graph with edge weight function $w(i, j)$. For random walk on G , the stationary distribution π is proportional to the sum of the edge weights incident to each vertex. That is,

$$\pi_v = \frac{w(v)}{\sum_z w(z)}, \text{ for all vertices } v, \quad (3.4)$$

where

$$w(v) = \sum_{z \sim v} w(v, z)$$

is the sum of the edge weights on all edges incident to v .

Stationary Distribution for Simple Random Walk on a Graph

For simple random walk on a nonweighted graph, set $w(i, j) = 1$, for all i, j . Then, $w(v) = \deg(v)$, which gives

$$\pi_v = \frac{\deg(v)}{\sum_z \deg(z)} = \frac{\deg(v)}{2e},$$

where e is the number of edges in the graph.

■ **Example 3.9** The two-state Markov chain with transition matrix

$$P = \frac{1}{2} \begin{pmatrix} 1-p & p \\ q & 1-q \end{pmatrix}$$

can be expressed as a random walk on the weighted graph in Figure 3.2. Then, $w(1) = q(1-p) + pq = q$ and $w(2) = pq + p(1-q) = p$. By Equation (3.4), the stationary distribution is

$$\pi = \left(\frac{q}{p+q}, \frac{p}{p+q} \right).$$

■

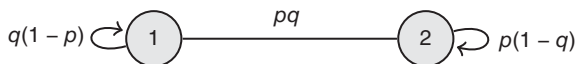


Figure 3.2 The general two-state Markov chain expressed as a weighted graph.

The reader may wonder how the weighted graph in Figure 3.2 is derived from the transition matrix $\mathbf{P}$. The algorithm will be explained in Section 3.7.

■ **Example 3.10** Find the stationary distribution for random walk on the hypercube.

Solution The k -hypercube graph, as described in Example 2.8, has 2^k vertices. Each vertex has degree k . The sum of the vertex degrees is $k2^k$, and the stationary distribution π is given by

$$\pi_v = \frac{k}{k2^k} = \frac{1}{2^k}, \text{ for all } v.$$

That is, the stationary distribution is uniform on the set of vertices. ■

The hypercube is an example of a *regular* graph. A graph is regular if all the vertex degrees are the same. For simple random walk on a regular graph, the stationary distribution π is uniform on the set of vertices, since π_j is constant for all j . In addition to the hypercube, examples of regular graphs with uniform stationary distributions include the cycle graph ($\deg(v) = 2$) and the complete graph on k vertices ($\deg(v) = k - 1$).

The Eigenvalue Connection*

The stationary distribution of a Markov chain is related to the eigenstructure of the transition matrix.

First, a reminder on notation. For a matrix $\mathbf{M}$, the *transpose* of $\mathbf{M}$ is denoted $\mathbf{M}^T$. In this book, vectors are considered as row vectors. If $\mathbf{x}$ is a vector, $\mathbf{x}^T$ is a column vector.

Recall that an eigenvector of $\mathbf{M}$ is a column vector $\mathbf{x}^T$ such that $\mathbf{M}\mathbf{x}^T = \lambda\mathbf{x}^T$, for some scalar λ . We call such a vector a *right eigenvector* of $\mathbf{M}$. A *left eigenvector* of $\mathbf{M}$ is a row vector $\mathbf{y}$, which satisfies $\mathbf{y}\mathbf{M} = \mu\mathbf{y}$, for some scalar μ . A left eigenvector of $\mathbf{M}$ is simply a right eigenvector of $\mathbf{M}^T$.

If π is the stationary distribution of a Markov chain and satisfies $\pi\mathbf{P} = \pi$, then π is a left eigenvector of $\mathbf{P}$ corresponding to eigenvalue $\lambda = 1$.

Let $\mathbf{1}$ denote the column vector of all 1s. Since the rows of a stochastic matrix sum to 1, $\mathbf{P}\mathbf{1} = \mathbf{1} = (1)\mathbf{1}$. That is, $\mathbf{1}$ is a right eigenvector of $\mathbf{P}$ corresponding to eigenvalue $\lambda = 1$.

A matrix and its transpose have the same set of eigenvalues, with possibly different eigenvectors. It follows that $\lambda = 1$ is an eigenvalue of $\mathbf{P}^T$ with some corresponding right eigenvector $\mathbf{y}^T$. Equivalently, $\mathbf{y}$ is a left eigenvector of $\mathbf{P}$. That is, there exists a row vector $\mathbf{y}$ such that $\mathbf{y}\mathbf{P} = \mathbf{y}$. If a multiple of $\mathbf{y}$ can be normalized so that its components are non-negative and sum to 1, then this gives a stationary distribution. However, some of the entries of $\mathbf{y}$ might be negative, or complex-valued, and the vector might not be able to be normalized to give a probability distribution.

If a Markov chain has a unique stationary distribution, then the distribution is an eigenvector of $\mathbf{P}^T$ corresponding to $\lambda = 1$.

R: Stationary Distribution and Eigenvectors

The R command `eigen(P)` returns the eigenvalues and eigenvectors of a square matrix P . These are given in a list with two components: `values` contains the eigenvalues, and `vectors` contains the corresponding eigenvectors stored as a matrix. If P is a stochastic matrix, an eigenvector corresponding to eigenvalue $\lambda = 1$ will be stored in the first column of the `vectors` matrix. The R command `t(P)` gives the transpose of P .

In the following, an eigenvector is found for P^T corresponding to $\lambda = 1$. The vector is then normalized so that components sum to 1 in order to compute the stationary distribution. We illustrate on the weather matrix with stationary distribution $\pi = (1/9, 3/4, 5/36) = (0.111, 0.750, 0.139)$.

```
> P
      Rain Snow Clear
Rain   0.2  0.6   0.2
Snow   0.1  0.8   0.1
Clear  0.1  0.6   0.3
> eigen(P)
$values
[1] 1.0 0.2 0.1
$vectors
      [,1]      [,2]      [,3]
[1,] 0.5773503 0.6882472 -0.9847319
[2,] 0.5773503 -0.2294157 0.1230915
[3,] 0.5773503 0.6882472 0.1230915
> eigen(P)$values # eigenvalues
[1] 1.0 0.2 0.1
# eigenvalues of P and its transpose are the same
> eigen(t(P))$values
[1] 1.0 0.2 0.1
# eigenvectors of P-transpose
> eigen(t(P))$vectors
      [,1]      [,2]      [,3]
[1,] 0.1441500 -2.008469e-16 7.071068e-01
[2,] 0.9730125 -7.071068e-01 3.604182e-16
[3,] 0.1801875 7.071068e-01 -7.071068e-01

# first column gives eigenvector for eigenvalue 1
> x <- eigen(t(P))$vectors[,1]
> x
[1] 0.1441500 0.9730125 0.1801875
# normalize so rows sum to 1
> x/sum(x)
```

```
[1] 0.1111111 0.7500000 0.1388889

# one-line command to find stationary distribution
> x <- eigen(t(P))$vectors[,1]; x/sum(x)
[1] 0.1111111 0.7500000 0.1388889
```

3.3 CAN YOU FIND THE WAY TO STATE a ?

The long-term behavior of a Markov chain is related to how often states are visited. Here, we look more closely at the relationship between states and how reachable, or accessible, groups of states are from each other.

Say that state j is *accessible* from state i , if $P_{ij}^n > 0$, for some $n \geq 0$. That is, there is positive probability of reaching j from i in a finite number of steps. States i and j *communicate* if i is accessible from j and j is accessible from i .

Communication is an equivalence relation, which means that it satisfies the following three properties.

1. (*Reflexive*) Every state communicates with itself.
2. (*Symmetric*) If i communicates with j , then j communicates with i .
3. (*Transitive*) If i communicates with j , and j communicates with k , then i communicates with k .

Property 1 holds since $P_{ii}^0 = P(X_0 = i | X_0 = i) = 1$. Property 2 follows since the definition of communication is symmetric. For Property 3, assume that i communicates with j , and j communicates with k . Then, there exists $n \geq 0$ and $m \geq 0$ such that $P_{ij}^n > 0$ and $P_{jk}^m > 0$. Therefore,

$$P_{ik}^{n+m} = \sum_i P_{it}^n P_{tk}^m \geq P_{ij}^n P_{jk}^m > 0.$$

Thus, k is accessible from i . Similarly, i is accessible from k .

Since communication is an equivalence relation the state space can be partitioned into equivalence classes, called *communication classes*. That is, the state space can be divided into disjoint subsets, each of whose states communicate with each other but do not communicate with any states outside their class.

A modified transition graph is a useful tool for finding the communication classes of a Markov chain. Vertices of the graph are the states of the chain. A directed edge is drawn between i and j if $P_{ij} > 0$. For purposes of studying the communication relationship between states, it is not necessary to label the edges with probabilities.

■ **Example 3.11** Find the communication classes for the Markov chains with these transition matrices.

$$P = \begin{matrix} & \begin{matrix} a & b & c & d & e \end{matrix} \\ \begin{matrix} a \\ b \\ c \\ d \\ e \end{matrix} & \begin{pmatrix} 0 & 0 & 0 & 2/3 & 1/3 \\ 1/2 & 0 & 1/6 & 0 & 1/3 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 3/4 & 1/4 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix} \end{matrix}, \quad Q = \begin{matrix} & \begin{matrix} a & b & c & d & e & f \end{matrix} \\ \begin{matrix} a \\ b \\ c \\ d \\ e \\ f \end{matrix} & \begin{pmatrix} 1/6 & 1/3 & 0 & 0 & 1/2 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 3/4 & 1/4 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 4/5 & 0 & 0 & 1/5 & 0 & 0 \\ 0 & 0 & 1/2 & 0 & 1/2 & 0 \end{pmatrix} \end{matrix}.$$

Solution The transition graphs are shown in Figure 3.3. For the P -chain, the communication classes are $\{a, b, c, d\}$ and $\{e\}$. For the Q -chain, the communication classes are $\{a, d, e\}$, $\{b\}$, and $\{c, f\}$.

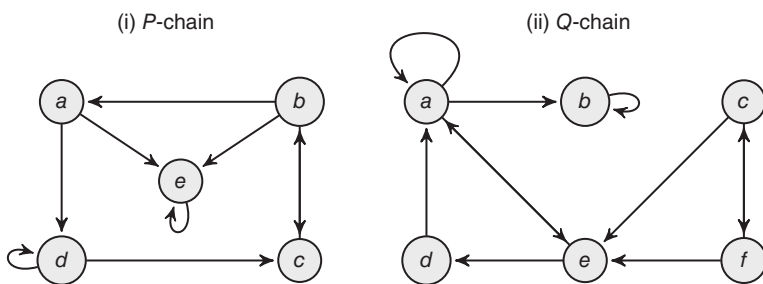


Figure 3.3 Transition graphs.

Most important is the case when the states of a Markov chain all communicate with each other.

Irreducibility

A Markov chain is called *irreducible* if it has exactly one communication class. That is, all states communicate with each other.

■ **Example 3.12** The Markov chain with transition matrix

$$P = \begin{matrix} & \begin{matrix} a & b & c & d & e & f \end{matrix} \\ \begin{matrix} a \\ b \\ c \\ d \\ e \\ f \end{matrix} & \begin{pmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 1/2 & 0 & 0 & 0 & 1/2 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 1/4 & 1/4 & 0 & 1/4 & 0 & 1/4 \\ 0 & 0 & 1 & 0 & 0 & 0 \end{pmatrix} \end{matrix}$$

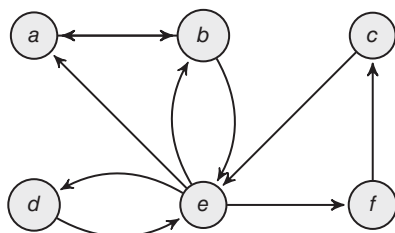


Figure 3.4 Transition graph for an irreducible Markov chain.

is irreducible, which can be seen by examining the transition graph in Figure 3.4. ■

Recurrence and Transience

Consider the transition graph in Figure 3.5. The communication classes are $\{a, b\}$ and $\{c\}$. From each state, consider the evolution of the chain started from that state and the probability that the chain eventually revisits that state.

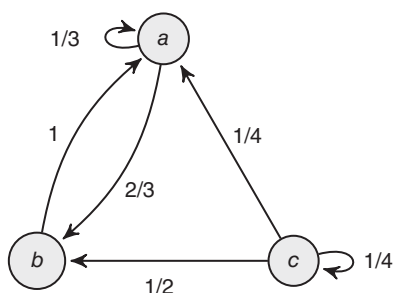


Figure 3.5

From a , the chain either returns to a in one step, or first moves to b and then returns to a on the second step. From a , the chain revisits a , with probability 1.

For the chain started in b , the chain first moves to a . It may continue to revisit a for many steps, but eventually it will return to b . This is because the probability that the chain stays at a forever is the probability that it continually transitions from a to a , which is equal to

$$\lim_{n \rightarrow \infty} (P_{aa})^n = \lim_{n \rightarrow \infty} \left(\frac{1}{3}\right)^n = 0.$$

Thus, from b , the chain revisits b , with probability 1.

For the chain started in c , the chain may revisit c for many successive steps. But with positive probability it will eventually move to either a or b . Once it does, it will never revisit c , as it is now *stuck* in the $\{a, b\}$ communication class. From c , there is positive probability that the chain started in c will never revisit c . In this case, that probability is $1 - 1/4 = 3/4$.

The states of a Markov chain, as this example illustrates, exhibit one of two contrasting behaviors. For the chain started in a given state, the chain either revisits that

state, with probability 1, or there is positive probability that the chain will never revisit that state.

Given a Markov chain $X_0, X_1, \dots$, let $T_j = \min\{n > 0 : X_n = j\}$ be the *first passage time* to state j . If $X_n \neq j$, for all $n > 0$, set $T_j = \infty$. Let

$$f_j = P(T_j < \infty | X_0 = j)$$

be the probability that the chain started in j eventually returns to j . For the three-state chain introduced in this section, $f_a = f_b = 1$, and $f_c = 1/4$. We classify the states j of a Markov chain according to whether or not $f_j = 1$.

Recurrent and Transient States

State j is said to be *recurrent* if the Markov chain started in j eventually revisits j . That is, $f_j = 1$.

State j is said to be *transient* if there is positive probability that the Markov chain started in j never returns to j . That is, $f_j < 1$.

Whether or not a state is eventually revisited is strongly related to how often that state is visited.

For the chain started in i , let

$$I_n = \begin{cases} 1, & \text{if } X_n = j, \\ 0, & \text{otherwise,} \end{cases}$$

for $n \geq 0$. Then, $\sum_{n=0}^{\infty} I_n$ is the number of visits to j . The expected number of visits to j is

$$E\left(\sum_{n=0}^{\infty} I_n\right) = \sum_{n=0}^{\infty} E(I_n) = \sum_{n=0}^{\infty} P(X_n = j | X_0 = i) = \sum_{n=0}^{\infty} P_{ij}^n,$$

where the infinite sum may possibly diverge to $+\infty$. For the chain started in i , the expected number of visits to j is the ij th entry of the matrix $\sum_{n=0}^{\infty} \mathbf{P}^n$.

Assume that j is recurrent. The chain started at j will eventually revisit j with probability 1. Once it hits j , the chain begins anew and behaves as if a new version of the chain started at j . We say the Markov chain *regenerates* itself. (This intuitive behavior is known as the *strong Markov property*. For a more formal treatment see Section 3.9.) From j , the chain will revisit j again, with probability 1, and so on. It follows that j will be visited infinitely many times, and

$$\sum_{n=0}^{\infty} P_{jj}^n = \infty.$$

On the other hand, assume that j is transient. Starting at j , the probability of eventually hitting j again is f_j , and the probability of never hitting j is $1 - f_j$. If the chain

hits j , the event that it will eventually revisit j is independent of past history. It follows that the sequence of successive visits to j behaves like an i.i.d. sequence of coin tosses where heads occurs if j is eventually hit and tails occurs if j is never hit again. The number of times that j is hit is the number of coin tosses until tails occurs, which has a geometric distribution with parameter $1 - f_j$. Thus, the expected number of visits to j is $1/(1 - f_j)$, and

$$\sum_{n=0}^{\infty} P_{jj}^n = \frac{1}{1 - f_j} < \infty.$$

In particular, a transient state will only be visited a finite number of times.

This leads to another characterization of recurrence and transience.

Recurrence, Transience

(i) State j is recurrent if and only if

$$\sum_{n=0}^{\infty} P_{jj}^n = \infty.$$

(ii) State j is transient if and only if

$$\sum_{n=0}^{\infty} P_{jj}^n < \infty.$$

Assume that j is recurrent and accessible from i . For the chain started in i there is positive probability of hitting j . And from j , the expected number of visits to j is infinite. It follows that the expected number of visits to j for the chain started in i is also infinite, and thus

$$\sum_{n=0}^{\infty} P_{ij}^n = \infty.$$

Assume that j is transient and accessible from i . By a similar argument the expected number of visits to j for the chain started in i is finite, and thus

$$\sum_{n=0}^{\infty} P_{ij}^n < \infty,$$

from which it follows that

$$\lim_{n \rightarrow \infty} P_{ij}^n = 0. \quad (3.5)$$

The long-term probability that a Markov chain eventually hits a transient state is 0.

Recurrence and transience are *class* properties of a Markov chain as described by the following theorem.

Recurrence and Transience are Class Properties

Theorem 3.3. *The states of a communication class are either all recurrent or all transient.*

Proof. Let i and j be states in the same communication class. Assume that i is recurrent. Since i and j communicate, there exists $r \geq 0$ and $s \geq 0$ such that $P_{ji}^r > 0$ and $P_{ij}^s > 0$. For $n \geq 0$,

$$P_{jj}^{r+n+s} = \sum_k \sum_l P_{jk}^r P_{kl}^n P_{lj}^s \geq P_{ji}^r P_{ii}^n P_{ij}^s.$$

Summing over n gives

$$\sum_{n=0}^{\infty} P_{jj}^{r+n+s} \geq P_{ji}^r \left(\sum_{n=0}^{\infty} P_{ii}^n \right) P_{ij}^s = \infty.$$

Since

$$\sum_{n=0}^{\infty} P_{jj}^n \geq \sum_{n=r+s}^{\infty} P_{jj}^n = \sum_{n=0}^{\infty} P_{jj}^{r+n+s},$$

it follows that $\sum_{n=0}^{\infty} P_{jj}^n$ diverges to infinity, and thus j is recurrent. Hence, if one state of a communication class is recurrent, all states in that class are recurrent.

On the other hand, if one state is transient, the other states must be transient. By contradiction, if the communication class contains a recurrent state then by what was just proven all the states are recurrent. ■

Corollary 3.4. *For a finite irreducible Markov chain, all states are recurrent.*

Proof. The states of an irreducible chain are either all recurrent or all transient. Assume that they are all transient. Then, state 1 will be visited for a finite amount of time, after which it is never hit again, similarly with state 2, and with all states. Since there are finitely many states, it follows that *none* of the states will be visited after some finite amount of time, which is not possible. ■

By Corollary 3.4, a finite Markov chain cannot have all transient states. This is not true for infinite chains, as the following classic example illustrates.

■ **Example 3.13 (Simple random walk)** A random walk on the integer line starts at 0 and moves left, with probability p , or right, with probability $1 - p$. For $0 < p < 1$, the process is an irreducible Markov chain, as every state is accessible from every other state. Is the chain recurrent or transient?

Solution Since the chain is irreducible, it suffices to examine one state. Choose state 0 and consider $\sum_{n=0}^{\infty} P_{00}^n$. Observe that from 0 the walk can only revisit 0 in an even number of steps. So $P_{00}^n = 0$, if n is odd. To move from 0 to 0 in exactly $2n$ steps requires that the walk moves n steps to the left and n steps to the right, in some order. Such a path of length $2n$ can be identified with a sequence of n Ls and n Rs. There are $\binom{2n}{n}$ such sequences. Each left move occurs with probability p and each right move occurs with probability $1 - p$. This gives

$$P_{00}^{2n} = \binom{2n}{n} p^n (1-p)^n.$$

The binomial coefficient $\binom{2n}{n}$ is estimated using *Stirling's approximation*

$$n! \approx n^n e^{-n} \sqrt{2\pi n}, \text{ for large } n.$$

The more precise statement is

$$\lim_{n \rightarrow \infty} \frac{n!}{n^n e^{-n} \sqrt{2\pi n}} = 1.$$

By Stirling's approximation, for large n ,

$$\binom{2n}{n} = \frac{(2n)!}{n!n!} \approx \frac{(2n)^{2n} e^{-2n} \sqrt{2\pi 2n}}{(n^n e^{-n} \sqrt{2\pi n})^2} = \frac{4^n}{\sqrt{\pi n}}.$$

Thus,

$$\sum_{n=0}^{\infty} P_{00}^{2n} = \sum_{n=0}^{\infty} \binom{2n}{n} p^n (1-p)^n \approx \sum_{n=1}^{\infty} \frac{(4p(1-p))^n}{\sqrt{\pi n}}.$$

Convergence of the infinite series depends upon p . We have

$$\sum_{n=0}^{\infty} P_{00}^{2n} \approx \begin{cases} \sum_{n=1}^{\infty} \frac{1}{\sqrt{\pi n}} = \infty, & \text{if } p = 1/2, \\ \sum_{n=1}^{\infty} \frac{\epsilon^n}{\sqrt{\pi n}} < \infty, & \text{if } p \neq 1/2, \end{cases}$$

where $\epsilon = 4p(1-p)$. If $p \neq 1/2$, then $0 < \epsilon < 1$, and $\epsilon^n \rightarrow 0$, as $n \rightarrow \infty$.

For $p = 1/2$, the random walk is recurrent. Each integer, no matter how large, is visited infinitely often. For $p \neq 1/2$, the walk is transient. With positive probability the walk will never return to its starting point.

Surprises await for random walk in higher dimensions. Simple symmetric random walk on the integer points in the plane $\mathbb{Z}^2$ moves left, right, up, or down, with probability $1/4$ each. The process has been called the *drunkard's walk*. As in one dimension, the walk is recurrent. The method of proof is similar. See Exercise 3.20. Letting 0 denote the origin in the plane, it can be shown that

$$\sum_{n=0}^{\infty} P_{00}^n \approx \sum_{n=1}^{\infty} \frac{1}{\pi n} = \infty.$$

Remarkably, in dimensions three and higher simple symmetric random walk is transient. This was first shown by George Pólya in 1921 and is known as *Pólya's Theorem*. It can be shown that in $\mathbb{Z}^3$,

$$\sum_{n=0}^{\infty} P_{00}^n \approx \sum_{n=1}^{\infty} \frac{1}{(\pi n)^{3/2}} < \infty.$$

The mathematician Shizuo Kakutani is quoted as explaining this result by saying, “A drunk man will find his way home, but a drunk bird may get lost forever.” ■

Canonical Decomposition

A set of states C is said to be *closed* if no state outside of C is accessible from any state in C . If C is closed, then $P_{ij} = 0$ for all $i \in C$ and $j \notin C$.

Closed Communication Class

Lemma 3.5. *A communication class is closed if it consists of all recurrent states. A finite communication class is closed only if it consist of all recurrent states.*

Proof. Let C be a communication class made up of recurrent states. Assume that C is not closed. Then, there exists states $i \in C$ and $j \notin C$ such that $P_{ij} > 0$. Since j is accessible from i , i is not accessible from j , otherwise j would be contained in C . Start the chain in i . With positive probability, the chain will hit j and then never hit i again. But this contradicts the assumption that i is recurrent.

On the other hand, assume that C is closed and finite. By the same argument given in the proof of Corollary 3.4 the states cannot all be transient. Hence, they are all recurrent. ■

The state space S of a finite Markov chain can be partitioned into transient and recurrent states as $S = T \cup R_1 \cup \cdots \cup R_m$, where T is the set of all transient states and the R_i are closed communication classes of recurrent states. This is called the *canonical decomposition*. The computation of many quantities associated with Markov chains can be simplified by this decomposition.

Given a canonical decomposition, the state space can be reordered so that the Markov transition matrix has the block matrix form

$$P = \begin{matrix} & \begin{matrix} T & R_1 & \cdots & R_m \end{matrix} \\ \begin{matrix} T \\ R_1 \\ \vdots \\ R_m \end{matrix} & \begin{pmatrix} * & * & \cdots & * \\ \mathbf{0} & P_1 & \cdots & \mathbf{0} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{0} & \mathbf{0} & \cdots & P_m \end{pmatrix} \end{matrix}.$$

Each submatrix $P_1, \dots, P_m$ is a square stochastic matrix corresponding to a closed recurrent communication class. By itself, each of these matrices is the matrix of an irreducible Markov chain with a restricted state space.

■ **Example 3.14** Consider the Markov chain with transition matrix

$$P = \begin{matrix} & \begin{matrix} a & b & c & d & e & f & g \end{matrix} \\ \begin{matrix} a \\ b \\ c \\ d \\ e \\ f \\ g \end{matrix} & \begin{pmatrix} 1/4 & 0 & 0 & 0 & 0 & 0 & 3/4 \\ 1/8 & 1/8 & 1/4 & 0 & 1/4 & 1/8 & 1/8 \\ 0 & 0 & 2/5 & 1/5 & 2/5 & 0 & 0 \\ 0 & 0 & 1/2 & 1/2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1/2 & 1/2 & 0 & 0 \\ 0 & 1/5 & 0 & 1/5 & 1/5 & 1/5 & 1/5 \\ 4/5 & 0 & 0 & 0 & 0 & 0 & 1/5 \end{pmatrix} \end{matrix}$$

described by the transition graph in Figure 3.6. Give the canonical decomposition.

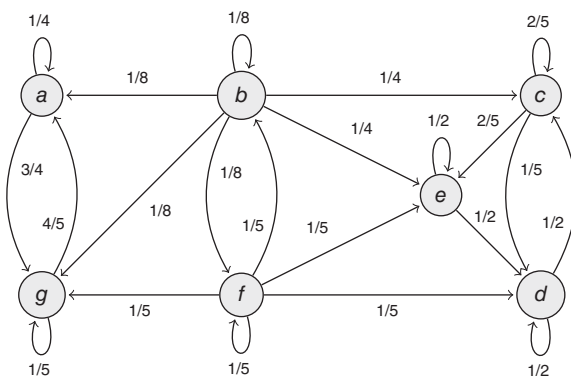


Figure 3.6

Solution States b and f are transient. The closed recurrent classes are $\{a, g\}$ and $\{c, d, e\}$. Reordering states gives the transition matrix in block matrix form

$$P = \begin{matrix} & \begin{matrix} b & f & a & g & c & d & e \end{matrix} \\ \begin{matrix} b \\ f \\ a \\ g \\ c \\ d \\ e \end{matrix} & \begin{pmatrix} 1/8 & 1/8 & 1/8 & 1/8 & 1/4 & 0 & 1/4 \\ 1/5 & 1/5 & 0 & 1/5 & 0 & 1/5 & 1/5 \\ 0 & 0 & 1/4 & 3/4 & 0 & 0 & 0 \\ 0 & 0 & 4/5 & 1/5 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 2/5 & 1/5 & 2/5 \\ 0 & 0 & 0 & 0 & 1/2 & 1/2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1/2 & 1/2 \end{pmatrix} \end{matrix}.$$

■

The canonical decomposition is useful for describing the long-term behavior of a Markov chain. The block matrix form facilitates taking matrix powers. For $n \geq 1$,

$$P^n = \begin{matrix} & T & R_1 & \cdots & R_m \\ \begin{matrix} T \\ R_1 \\ \vdots \\ R_m \end{matrix} & \begin{pmatrix} * & * & \cdots & * \\ \mathbf{0} & P_1^n & \cdots & \mathbf{0} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{0} & \mathbf{0} & \cdots & P_m^n \end{pmatrix} \end{matrix}.$$

Taking limits gives

$$\lim_{n \rightarrow \infty} P^n = \begin{matrix} & T & R_1 & \cdots & R_m \\ \begin{matrix} T \\ R_1 \\ \vdots \\ R_m \end{matrix} & \begin{pmatrix} \mathbf{0} & * & \cdots & * \\ \mathbf{0} & \lim_{n \rightarrow \infty} P_1^n & \cdots & \mathbf{0} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{0} & \mathbf{0} & \cdots & \lim_{n \rightarrow \infty} P_m^n \end{pmatrix} \end{matrix}.$$

Note that the entries of the columns corresponding to transient states are all 0 as a consequence of Equation (3.5).

The recurrent, closed communication classes $R_1, \dots, R_m$ behave like mini-irreducible Markov chains where all states communicate with each other. The asymptotic properties of the submatrices $P_1, \dots, P_m$ lead us to consider the properties of irreducible Markov chains, which is where our path now goes.

3.4 IRREDUCIBLE MARKOV CHAINS

The next theorem characterizes the stationary distribution π for finite irreducible Markov chains. It relates the stationary probability π_j to the expected number of steps between visits to j . Recall that $T_j = \min\{n > 0 : X_n = j\}$ is the first passage time to state j .

Limit Theorem for Finite Irreducible Markov Chains

Theorem 3.6. *Assume that $X_0, X_1, \dots$ is a finite irreducible Markov chain. For each state j , let $\mu_j = E(T_j | X_0 = j)$ be the expected return time to j . Then, μ_j is finite, and there exists a unique, positive stationary distribution π such that*

$$\pi_j = \frac{1}{\mu_j}, \text{ for all } j. \quad (3.6)$$

Furthermore, for all states i ,

$$\pi_j = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{m=0}^{n-1} P_{ij}^m. \quad (3.7)$$

The theorem is proved in Section 3.10.

Remark:

1. The fact that $\pi_j = 1/\mu_j$ is intuitive. If there is one visit to j every μ_j steps, then the proportion of visits to j is $1/\mu_j$.
2. The theorem does not assert that π is a limiting distribution. The convergence in Equation (3.7) is a weaker form of convergence than $\pi_j = \lim_{n \rightarrow \infty} P_{ij}^n$. We discuss in the next section that an additional assumption is needed for π to be a limiting distribution.
3. For finite irreducible Markov chains, all states are recurrent, and the expected return time $E(T_j|X_0 = j)$ is finite, for all j . However, for an infinite Markov chain if j is a recurrent state, even though the chain will eventually revisit j with probability 1, the expected number of steps between such visits need not be finite. The theorem can be extended to infinite irreducible Markov chains for which the expected return time $E(T_j|X_0 = j)$ is finite, for all j .

A recurrent state j is called *positive recurrent* if $E(T_j|X_0 = j) < \infty$, and *null recurrent* if $E(T_j|X_0 = j) = \infty$. Thus, the theorem holds for irreducible Markov chains for which all states are positive recurrent. See Exercise 3.27 for an example of a Markov chain in which all states are null recurrent.

In many applications, the expected time between visits to a given state is of particular importance.

■ **Example 3.15 (Earthquake recurrences)** The Chiayi–Tainan area of Taiwan was devastated by an earthquake on September 21, 1999. In Tsai (2002), Markov chains were used to study seismic activity in the region. A chain is constructed with states corresponding to Richter scale magnitudes of earthquake intensity. The state space is $\{M_2, M_3, M_4, M_5\}$, where M_k denotes an earthquake with a Richter level in the interval $[k, k + 1)$. A Markov transition matrix is estimated from historical data for the period 1973–1975:

$$P = \begin{matrix} & \begin{matrix} M_2 & M_3 & M_4 & M_5 \end{matrix} \\ \begin{matrix} M_2 \\ M_3 \\ M_4 \\ M_5 \end{matrix} & \begin{pmatrix} 0.785 & 0.194 & 0.018 & 0.003 \\ 0.615 & 0.334 & 0.048 & 0.003 \\ 0.578 & 0.353 & 0.069 & 0.000 \\ 0.909 & 0.000 & 0.091 & 0.000 \end{pmatrix} \end{matrix}.$$

The stationary distribution $\pi = (0.740, 0.230, 0.027, 0.003)$ is found with technology. Earthquakes of magnitude M_2 or greater tend to occur in this region about once every four months. If an M_5 earthquake occurs, investigators would like to know how long it will be before another M_5 earthquake.

The expected number of Markov chain transitions between M_5 earthquakes is $1/\pi_{M_5} = 1/0.003 = 333$. If earthquakes occur, on average, every four months, then according to the model it will take about $333 \times (4/12) = 111$ years before another M_5 earthquake. ■

■ **Example 3.16** For the frog-jumping random walk on an n -cycle, how many hops does it take, on average, for the frog to return to its starting lily pad?

Solution In the cycle graph, all vertices have the same degree. Hence, for simple random walk on the cycle, the stationary distribution is uniform on the set of vertices. Since $\pi_v = 1/n$, for all vertices v , it takes the frog, on average, $1/\pi_v = n$ hops to return to its starting pad. ■

First-Step Analysis

The expected return time $E(T_j|X_0 = j)$ is found by taking the reciprocal of the stationary probability π_j . Another approach is to condition on the first step of the chain and use the law of total expectation. This is called *first-step analysis*.

■ **Example 3.17** Consider a Markov chain with transition matrix

$$P = \begin{matrix} & \begin{matrix} a & b & c \end{matrix} \\ \begin{matrix} a \\ b \\ c \end{matrix} & \begin{pmatrix} 0 & 1 & 0 \\ 1/2 & 0 & 1/2 \\ 1/3 & 1/3 & 1/3 \end{pmatrix} \end{matrix}.$$

From state a , find the expected return time $E(T_a|X_0 = a)$ using first-step analysis.

Solution Let $e_x = E(T_a|X_0 = x)$, for $x = a, b, c$. Thus, e_a is the desired expected return time, and e_b and e_c are the expected *first passage* times to a for the chain started in b and c , respectively.

For the chain started in a , the next state is b , with probability 1. From b , the further evolution of the chain behaves as if the original chain started at b . Thus,

$$e_a = 1 + e_b.$$

From b , the chain either hits a , with probability $1/2$, or moves to c , where the chain behaves as if the original chain started at c . It follows that

$$e_b = \frac{1}{2} + \frac{1}{2}(1 + e_c).$$

Similarly, from c , we have

$$e_c = \frac{1}{3} + \frac{1}{3}(1 + e_b) + \frac{1}{3}(1 + e_c).$$

Solving the three equations gives

$$e_c = \frac{8}{3}, \quad e_b = \frac{7}{3}, \quad \text{and} \quad e_a = \frac{10}{3}.$$

The desired expected return time is $10/3$.

We leave it to the reader to verify that the stationary distribution is

$$\pi = \left(\frac{3}{10}, \frac{2}{5}, \frac{3}{10} \right).$$

The expected return time is simulated in the following R code. The chain started at a is run for 25 steps (long enough to return to a with very high probability), and the return time to a is found. The mean return time is estimated based on 10,000 trials.

R : Simulating an Expected Return Time

```
#returntime.R
> P
      a      b      c
a 0.00000 1.00000 0.00000
b 0.50000 0.00000 0.50000
c 0.33333 0.33333 0.33333
> init
[1] 1 0 0
> states
[1] "a" "b" "c"
> markov(init,P,25,states)
"a" "b" "c" "c" "c" "b" "a" "b" "a" "b" "a" "b" "c"
"b" "c" "a" "b" "c" "b" "c" "b" "a" "b" "a" "b" "a"

> trials <- 10000
> simlist <- numeric(trials)
> for (i in 1:trials) {
  path <- markov(init,P,25,states)
  # find the index of the 2nd occurrence of "a"
  # subtract 1 to account for time 0
  returntime <- which(path == "a")[2] - 1
  simlist[i] <- returntime }
# expected return time to state a
> mean(simlist)
[1] 3.3346
```

■

3.5 PERIODICITY

Finite irreducible Markov chains have unique, positive stationary distributions. Although they may not have limiting distributions, they have almost limiting behavior in the sense that for all states i and j , the partial averages $(1/n) \sum_{m=0}^{n-1} P_{ij}^m$ converge.

An example of a finite irreducible Markov chain with no limiting distribution is random walk on the n -cycle, when n is even. The graph is regular (all vertex degrees are the same) and the unique stationary distribution is uniform. But there is no limiting distribution since the chain flip-flops back and forth between even and odd states. The chain's position after n steps depends on the parity of the initial state.

It is precisely the finite irreducible Markov chains that do not exhibit this type of *periodic* behavior, which have limiting distributions.

For a Markov chain started in state i , consider the set of times when the chain can return to i . For the chain described by the graph in Figure 3.7a, from any state the set of possible return times is $\{2, 4, 6, 8, \dots\}$. The same is true for the chain in Figure 3.7b. The chain started from any state returns to that state in multiples of two steps. For the chain in Figure 3.7c, from each state the set of return times is $\{3, 6, 9, 12, \dots\}$. The chain started in a returns to a in multiples of three steps.

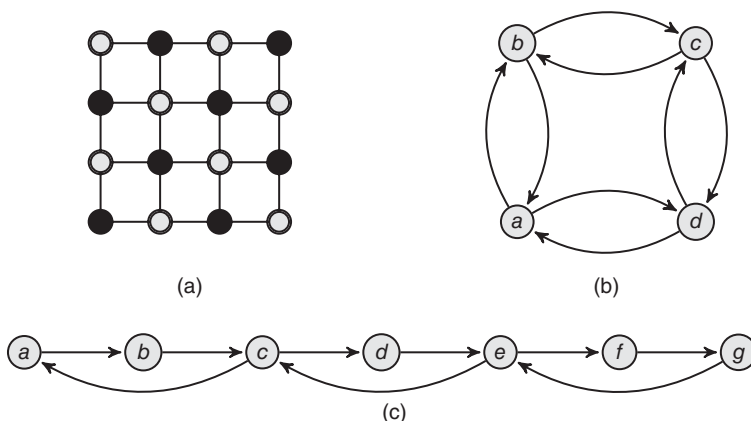


Figure 3.7 Periodic Markov chains.

The idea is formalized in the following definition. Recall that the *greatest common divisor* (*gcd*) of a set of positive integers is the largest integer that divides all the numbers of the set without a remainder.

Period

For a Markov chain with transition matrix P , the *period* of state i , denoted $d(i)$, is the greatest common divisor of the set of possible return times to i . That is,

$$d(i) = \gcd \{n > 0 : P_{ii}^n > 0\}.$$

If $d(i) = 1$, state i is said to be *aperiodic*. If the set of return times is empty, set $d(i) = +\infty$.

The definition of period gives that from state i , returns to i can only occur in multiples of $d(i)$ steps. And the period $d(i)$ is the largest such number with this property.

Periodicity, similar to recurrence and transience, is a class property.

Periodicity is a Class Property

Lemma 3.7. *The states of a communication class all have the same period.*

Proof. Let i and j be states in the same communication class with respective periods $d(i)$ and $d(j)$. Since i and j communicate, there exist positive integers r and s , such that $P_{ij}^r > 0$ and $P_{ji}^s > 0$. Then,

$$P_{ii}^{r+s} = \sum_k P_{ik}^r P_{ki}^s \geq P_{ij}^r P_{ji}^s > 0.$$

Thus $r + s$ is a possible return time for i , and hence $d(i)$ is a divisor of $r + s$. Assume that $P_{jj}^n > 0$ for some positive integer n . Then,

$$P_{ii}^{r+s+n} \geq P_{ij}^r P_{jj}^n P_{ji}^s > 0,$$

and thus $d(i)$ is a divisor of $r + s + n$. Since $d(i)$ divides both $r + s$ and $r + s + n$, it must also divide n . Thus, $d(i)$ is a common divisor of the set $\{n > 0 : P_{jj}^n > 0\}$. Since $d(j)$ is the largest such divisor, it follows that $d(i) \leq d(j)$. By the same argument with i and j reversed, we have that $d(j) \leq d(i)$. Hence, $d(i) = d(j)$. ■

■ **Example 3.18** Consider the transition graph in Figure 3.8. Identify the communication classes, their periods, and whether the class is recurrent or transient.

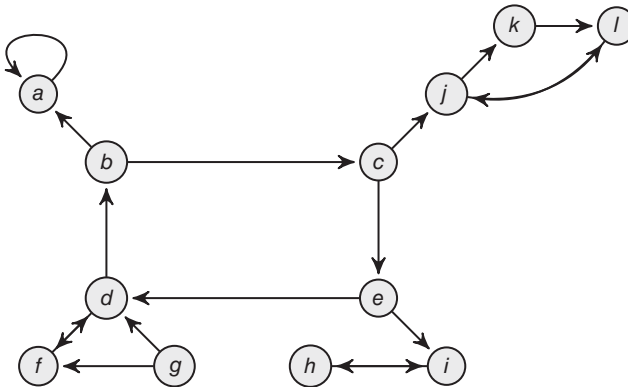


Figure 3.8

Solution Communication classes, with periods, are as follows:

- (i) $\{a\}$ is recurrent with period 1,
- (ii) $\{b, c, d, e, f\}$ is transient with period 2,

- (iii) $\{g\}$ is transient with period $+\infty$,
- (iv) $\{h, i\}$ is recurrent with period 2, and
- (v) $\{j, k, l\}$ is recurrent with period 1.

Observe that return times for state j can occur in multiples of three (first k , then l , then j) or in multiples of two (first l then j). Since two and three are relatively prime, their great common divisor is one. Hence, $d(j) = d(k) = d(l) = 1$. ■

From Lemma 3.7, it follows that all states in an irreducible Markov chain have the same period.

Periodic, Aperiodic Markov Chain

A Markov chain is called *periodic* if it is irreducible and all states have period greater than 1.

A Markov chain is called *aperiodic* if it is irreducible and all states have period equal to 1.

Note that any state i with the property that $P_{ii} > 0$ is necessarily aperiodic. Thus, a sufficient condition for an irreducible Markov chain to be aperiodic is that $P_{ii} > 0$ for some i . That is, at least one diagonal entry of the transition matrix is nonzero.

3.6 ERGODIC MARKOV CHAINS

A Markov chain is called *ergodic* if it is irreducible, aperiodic, and all states have finite expected return times. The latter is always true for finite chains. Thus, a finite Markov chain is ergodic if it is irreducible and aperiodic. It is precisely the class of ergodic Markov chains that have positive limiting distributions.

Fundamental Limit Theorem for Ergodic Markov Chains

Theorem 3.8. *Let $X_0, X_1, \dots$ be an ergodic Markov chain. There exists a unique, positive, stationary distribution π , which is the limiting distribution of the chain. That is,*

$$\pi_j = \lim_{n \rightarrow \infty} P_{ij}^n, \text{ for all } i, j.$$

Recall Theorem 3.2, which asserts the same limit result for Markov chains with regular transition matrices. The proof of the fundamental limit theorem for ergodic

Markov chains is given in Section 3.10, where it is also shown that a Markov chain is ergodic if and only if its transition matrix is regular.

■ **Example 3.19 (Modified Ehrenfest model)** In the Ehrenfest scheme of Example 3.7, at each discrete step one of N fleas is picked uniformly at random. The flea jumps from one dog to the other. Our dogs are named Cooper and Lisa. Let X_n be the number of fleas on Lisa after n jumps.

The Ehrenfest chain $X_0, X_1, \dots$ is irreducible and periodic with period 2. From state 0, the chain can only return to 0 in an even number of steps.

A modified Ehrenfest scheme picks a flea uniformly at random and then picks a dog uniformly at random for the flea to jump to. The transition probabilities are

$$P_{ij} = \begin{cases} i/(2N), & \text{if } j = i - 1, \\ (N - i)/(2N), & \text{if } j = i + 1, \\ 1/2, & \text{if } j = i, \\ 0, & \text{otherwise.} \end{cases}$$

Since $P_{ii} > 0$, the chain is aperiodic, and thus ergodic. The unique stationary distribution is binomial with parameters N and $1/2$, the same as in the regular Ehrenfest scheme. (We invite the reader to show this in Exercise 3.24.) By the fundamental limit theorem, the stationary distribution is the limiting distribution of the chain. The long-term stationary process can be simply described: for each of the N fleas, toss a fair coin. If heads, the flea jumps to Lisa, if tails it jumps to Cooper.

The modified Ehrenfest transition matrix for $N = 6$ fleas is

$$P = \begin{matrix} & \begin{matrix} 0 & 1 & 2 & 3 & 4 & 5 & 6 \end{matrix} \\ \begin{matrix} 0 \\ 1 \\ 2 \\ 3 \\ 4 \\ 5 \\ 6 \end{matrix} & \begin{pmatrix} 1/2 & 1/2 & 0 & 0 & 0 & 0 & 0 \\ 5/12 & 1/2 & 1/12 & 0 & 0 & 0 & 0 \\ 0 & 1/3 & 1/2 & 1/6 & 0 & 0 & 0 \\ 0 & 0 & 1/4 & 1/2 & 1/4 & 0 & 0 \\ 0 & 0 & 0 & 1/6 & 1/2 & 1/3 & 0 \\ 0 & 0 & 0 & 0 & 1/12 & 1/2 & 5/12 \\ 0 & 0 & 0 & 0 & 0 & 1/2 & 1/2 \end{pmatrix} \end{matrix},$$

with limiting distribution

$$\pi = \left(\frac{1}{64}, \frac{6}{64}, \frac{15}{64}, \frac{20}{64}, \frac{15}{64}, \frac{6}{64}, \frac{1}{64} \right).$$

■

■ **Example 3.20** Consider a Markov chain with transition matrix

$$P = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 & \cdots & n-1 & n \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \\ \vdots \\ n-1 \\ n \end{matrix} & \left(\begin{array}{ccccccc} 0 & 1 & 0 & 0 & \cdots & 0 & 0 \\ 1/2 & 0 & 1/2 & 0 & \cdots & 0 & 0 \\ 1/3 & 0 & 0 & 2/3 & \cdots & 0 & 0 \\ 1/4 & 0 & 0 & 0 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 1/(n-1) & 0 & 0 & 0 & \cdots & 0 & (n-2)/(n-1) \\ 1 & 0 & 0 & 0 & \cdots & 0 & 0 \end{array} \right) \end{matrix}.$$

The transition graph is shown in Figure 3.9. Find the limiting distribution.

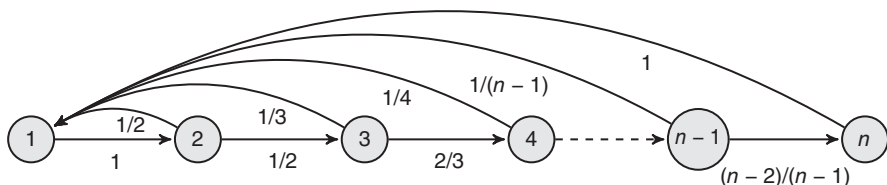


Figure 3.9

Solution State 1 is accessible from all states. Each state k is accessible from 1 by transitioning to 2, then 3, ..., to k . Thus, the chain is irreducible. It is also aperiodic for $n \geq 3$. For instance, one can reach state 1 from 1 in either two steps or three steps. Thus, $d(1) = 1$. The chain is ergodic, and the limiting distribution is gotten by finding the stationary distribution.

Let $\mathbf{x} = (x_1, x_2, \dots, x_n)$, with $x_1 = 1$. Solving

$$x_j = \sum_{i=1}^n x_i P_{ij} = x_{j-1} P_{j-1,j}, \text{ for } j = 2, \dots, n,$$

gives $x_2 = x_1 = 1$ and

$$\begin{aligned} x_j &= x_{j-1} \left(\frac{j-2}{j-1} \right) = x_{j-2} \left(\frac{j-3}{j-2} \right) \left(\frac{j-2}{j-1} \right) \\ &= x_{j-2} \left(\frac{j-3}{j-1} \right) = \cdots = x_2 \left(\frac{1}{j-1} \right) = \frac{1}{j-1}, \text{ for } j = 3, \dots, n. \end{aligned}$$

Hence,

$$\pi_j = \begin{cases} \frac{1}{c}, & \text{for } j = 1, \\ \frac{1}{c(j-1)}, & \text{for } j = 2, \dots, n, \end{cases}$$

where $c = 1 + \sum_{k=1}^{n-1} 1/k$.

An infinite version of this chain—see Exercise 3.27—does not have a stationary distribution. Although the chain is aperiodic and irreducible, it is not positive recurrent. That is, the expected return time between visits to the same state is infinite. ■

■ **Example 3.21 (PageRank)** Google's PageRank search algorithm is introduced in Chapter 1 in Example 1.1. The model is based on the *random surfer* model, which is a random walk on the *webgraph*. For this graph, each vertex represents an internet page. A directed edge connects i to j if there is a hypertext link from page i to page j . When the random surfer is at page i , they move to a new page by choosing from the available links on i uniformly at random.

Figure 3.10 shows a simplified network with seven pages. The network is described by the *network matrix*

$$N = \begin{matrix} & \begin{matrix} a & b & c & d & e & f & g \end{matrix} \\ \begin{matrix} a \\ b \\ c \\ d \\ e \\ f \\ g \end{matrix} & \begin{pmatrix} 0 & 0 & 0 & 0 & 1/2 & 1/2 & 0 \\ 1/3 & 0 & 1/3 & 0 & 0 & 1/3 & 0 \\ 0 & 0 & 0 & 1/2 & 0 & 1/2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 1/4 & 0 & 0 & 1/4 & 0 & 1/4 & 1/4 \\ 1/2 & 1/2 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix} \end{pmatrix}.$$

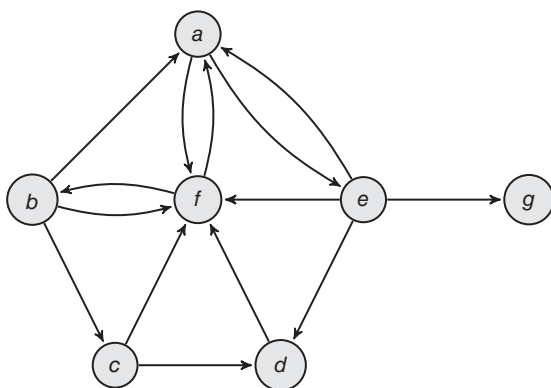


Figure 3.10

Note that N is not a stochastic matrix, as page g has no out-link. Row g consists of all 0s.

To insure that the walk reaches all pages in the network, the algorithm needs to account for (i) pages that have no out-links, called *dangling nodes*, and (ii) groups of pages that might result in the walk getting stuck in a subgraph. In the example network, node g is a dangling node.

Assume that the webgraph consists of k pages. In the PageRank algorithm, the fix for dangling nodes is to assume that when the random surfer reaches such a page they jump to a new page in the network uniformly at random. A new matrix $\mathbf{Q}$ is obtained where each row in the network matrix $\mathbf{N}$ corresponding to a dangling node is changed to one in which all entries are $1/k$. The matrix $\mathbf{Q}$ is a stochastic matrix.

For the problem of potentially getting stuck in small subgraphs of the webgraph, the solution proposed in the original paper by Brin and Page (1998) was to fix a *damping factor* $0 < p < 1$ for modifying the $\mathbf{Q}$ matrix. In their model, from a given page the random surfer, with probability $1 - p$, decides to not follow any links on the page and instead navigate to a new random page on the network. On the other hand, with probability p , they follow the links on the page as usual. This defines the PageRank transition matrix

$$\mathbf{P} = p\mathbf{Q} + (1 - p)\mathbf{A},$$

where $\mathbf{A}$ is a $k \times k$ matrix all of whose entries are $1/k$. The damping factor used by Google was originally set to $p = 0.85$.

With damping factor, the PageRank matrix $\mathbf{P}$ is stochastic, and the resulting random walk is aperiodic and irreducible. The PageRank of a page on the network is that page’s stationary probability.

Consider the sample network in Figure 3.10. See the following R code for the relevant calculations. The PageRank stationary distribution is

<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>	<i>g</i>
0.222	0.153	0.071	0.084	0.122	0.294	0.054

Assume that you are searching this network with the query “stochastic process,” and the search terms are found on pages a , c , e , and f . Ordered by their stationary probabilities, the PageRank algorithm would return the ordered pages f, a, e, c .

R: PageRank

```
# pagerank.R
> Q
      a      b      c      d      e      f      g
a 0.0000 0.0000 0.0000 0.0000 0.5000 0.5000 0.0000
b 0.3333 0.0000 0.3333 0.0000 0.0000 0.3333 0.0000
c 0.0000 0.0000 0.0000 0.5000 0.0000 0.5000 0.0000
d 0.0000 0.0000 0.0000 0.0000 0.0000 1.0000 0.0000
e 0.2500 0.0000 0.0000 0.2500 0.0000 0.2500 0.2500
f 0.5000 0.5000 0.0000 0.0000 0.0000 0.0000 0.0000
g 0.1429 0.1429 0.1429 0.1429 0.1429 0.1429 0.1429
> A <- matrix(rep(1/7,49),nrow=7)
```

```

> A
      a      b      c      d      e      f      g
a 0.1429 0.1429 0.1429 0.1429 0.1429 0.1429 0.1429
b 0.1429 0.1429 0.1429 0.1429 0.1429 0.1429 0.1429
c 0.1429 0.1429 0.1429 0.1429 0.1429 0.1429 0.1429
d 0.1429 0.1429 0.1429 0.1429 0.1429 0.1429 0.1429
e 0.1429 0.1429 0.1429 0.1429 0.1429 0.1429 0.1429
f 0.1429 0.1429 0.1429 0.1429 0.1429 0.1429 0.1429
g 0.1429 0.1429 0.1429 0.1429 0.1429 0.1429 0.1429
# Transition matrix with damping factor p=0.85
> P <- 0.85*Q + 0.15*A
> pr <- stationary(P)
> pr # Stationary probabilities
      a      b      c      d      e      f      g
0.2220 0.1527 0.0713 0.0843 0.1223 0.2935 0.0540

```

■

3.7 TIME REVERSIBILITY

Some Markov chains exhibit a directional bias in their evolution. Take, for instance, simple random walk on the integers, which moves from i to $i + 1$ with probability p , and from i to $i - 1$ with probability $1 - p$. If $p > 1/2$, the walk tends to move in the positive direction, mostly hitting ever larger integers. Similarly, if $p < 1/2$, over time the chain tends to hit ever smaller integers. However, if $p = 1/2$, the chain exhibits no directional bias.

The property of time reversibility can be explained intuitively as follows. If you were to take a movie of the Markov chain moving forward in time and then run the movie backwards, you could not tell the difference between the two.

Time Reversibility

An irreducible Markov chain with transition matrix P and stationary distribution π is *reversible*, or *time reversible*, if

$$\pi_i P_{ij} = \pi_j P_{ji}, \text{ for all } i, j. \quad (3.8)$$

Equations (3.8) are called the *detailed balance equations*. They say that for a chain in stationarity,

$$P(X_0 = i, X_1 = j) = P(X_0 = j, X_1 = i), \text{ for all } i, j.$$

That is, the frequency of transitions from i to j is equal to the frequency of transitions from j to i .

More generally (see Exercise 3.39), if a stationary Markov chain is reversible then

$$P(X_0 = i_0, X_1 = i_1, \dots, X_n = i_n) = P(X_0 = i_n, X_1 = i_{n-1}, \dots, X_n = i_0),$$

for all $i_0, i_1, \dots, i_n$.

■ **Example 3.22** A Markov chain has transition matrix

$$P = \begin{matrix} & \begin{matrix} 1 & 2 & 3 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \end{matrix} & \begin{pmatrix} 0 & 2/5 & 3/5 \\ 1/2 & 1/4 & 1/4 \\ 1/2 & 1/6 & 1/3 \end{pmatrix} \end{matrix}.$$

Determine if the chain is reversible.

Solution The chain is irreducible and aperiodic. The stationary distribution is $\pi = (1/3, 4/15, 2/5)$. We check the detailed balance equations

$$\pi_1 P_{12} = \left(\frac{1}{3}\right) \left(\frac{2}{5}\right) = \frac{2}{15} = \left(\frac{4}{15}\right) \left(\frac{1}{2}\right) = \pi_2 P_{21},$$

$$\pi_1 P_{13} = \left(\frac{1}{3}\right) \left(\frac{3}{5}\right) = \frac{1}{5} = \left(\frac{2}{5}\right) \left(\frac{1}{2}\right) = \pi_3 P_{31},$$

and

$$\pi_2 P_{23} = \left(\frac{4}{15}\right) \left(\frac{1}{4}\right) = \frac{1}{15} = \left(\frac{2}{5}\right) \left(\frac{1}{6}\right) = \pi_3 P_{32}.$$

Thus, the chain is time reversible. ■

If the stationary distribution of a Markov chain is uniform, it is apparent from Equation (3.8) that the chain is reversible if the transition matrix is symmetric.

■ **Example 3.23** Consider random walk on the n -cycle with transition matrix

$$P = \begin{cases} p, & \text{if } j \equiv i + 1 \pmod{n}, \\ 1 - p, & \text{if } j \equiv i - 1 \pmod{n}, \\ 0, & \text{otherwise.} \end{cases}$$

The stationary distribution is uniform. Hence, the random walk is time reversible for $p = 1/2$. For $p \neq 1/2$, directional bias of the walk will be apparent. For instance, if $p > 1/2$, the frequency of transitions from i to $i + 1$ is greater than the frequency of transitions from $i + 1$ to i . ■

■ **Example 3.24** Simple random walk on a graph is time reversible. If i and j are neighbors, then

$$\pi_i P_{ij} = \left(\frac{\deg(i)}{2e} \right) \left(\frac{1}{\deg(i)} \right) = \frac{1}{2e} = \left(\frac{\deg(j)}{2e} \right) \left(\frac{1}{\deg(j)} \right) = \pi_j P_{ji}.$$

If i and j are not neighbors, $P_{ij} = P_{ji} = 0$. ■

Reversible Markov Chains and Random Walk on Weighted Graphs

Random walk on a weighted graph is time reversible. In fact, every reversible Markov chain can be considered as a random walk on a weighted graph.

Given a reversible Markov chain with transition matrix P and stationary distribution π , construct a weighted graph on the state space by assigning edge weights $w(i, j) = \pi_i P_{ij}$. With these choice of weights, random walk on the weighted graph moves from i to j with probability

$$\frac{w(i, j)}{\sum_v w(i, v)} = \frac{\pi_i P_{ij}}{\sum_v \pi_i P_{iv}} = \frac{\pi_i P_{ij}}{\pi_i} = P_{ij}.$$

Conversely, given a weighted graph with edge weight function $w(i, j)$, the transition matrix of the corresponding Markov chain is obtained by letting

$$P_{ij} = \frac{w(i, j)}{\sum_v w(i, v)},$$

where the sum is over all neighbors of i . The stationary distribution is

$$\pi_i = \frac{\sum_y w(i, y)}{\sum_x \sum_y w(x, y)}.$$

One checks that

$$\begin{aligned} \pi_i P_{ij} &= \left(\frac{\sum_y w(i, y)}{\sum_x \sum_y w(x, y)} \right) \frac{w(i, j)}{\sum_y w(i, y)} \\ &= \frac{w(i, j)}{\sum_x \sum_y w(x, y)} \\ &= \left(\frac{\sum_y w(j, y)}{\sum_x \sum_y w(x, y)} \right) \frac{w(i, j)}{\sum_y w(j, y)} = \pi_j P_{ji}. \end{aligned}$$

Thus, the chain is time reversible.

■ **Example 3.25** A reversible Markov chain has transition matrix

$$P = \begin{matrix} & \begin{matrix} a & b & c & d \end{matrix} \\ \begin{matrix} a \\ b \\ c \\ d \end{matrix} & \begin{pmatrix} 0 & 4/5 & 1/5 & 0 \\ 4/6 & 1/6 & 1/6 & 0 \\ 1/4 & 1/4 & 0 & 1/2 \\ 0 & 0 & 2/3 & 1/3 \end{pmatrix} \end{matrix}. \quad (3.9)$$

Find the associated weighted graph.

Solution The stationary distribution is found to be $\pi = (5/18, 6/18, 4/18, 3/18)$. Arrange the quantities $\pi_i P_{ij}$ in a (nonstochastic) matrix

$$R = \begin{matrix} & \begin{matrix} a & b & c & d \end{matrix} \\ \begin{matrix} a \\ b \\ c \\ d \end{matrix} & \begin{pmatrix} 0 & 4/18 & 1/18 & 0 \\ 4/18 & 1/18 & 1/18 & 0 \\ 1/18 & 1/18 & 0 & 2/18 \\ 0 & 0 & 2/18 & 1/18 \end{pmatrix} \end{matrix},$$

where $R_{ij} = w(i, j) = \pi_i P_{ij}$. The matrix is symmetric. Multiplying the entries by 18 so that all weights are integers gives the weighted graph in Figure 3.11. ■

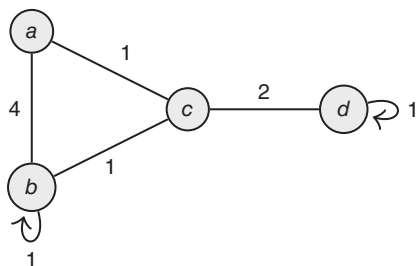


Figure 3.11 Weighted graph.

The next proposition highlights a key benefit of reversibility. It is often used to simplify computations for finding the stationary distribution of a Markov chain.

Proposition 3.9. *Let P be the transition matrix of a Markov chain. If $\mathbf{x}$ is a probability distribution which satisfies*

$$x_i P_{ij} = x_j P_{ji}, \text{ for all } i, j, \quad (3.10)$$

then $\mathbf{x}$ is the stationary distribution, and the Markov chain is reversible.

Proof. Assume that $\mathbf{x}$ satisfies the detailed balance equations. We have to show that $\mathbf{x} = \mathbf{xP}$. For all j ,

$$(xP)_j = \sum_i x_i P_{ij} = \sum_i x_j P_{ji} = x_j.$$

■

■ **Example 3.26 (Birth-and-death chain)** Birth-and-death chains were introduced in Example 2.9. From i , the process moves to either $i - 1$, i , or $i + 1$, with respective probabilities q_i , $1 - p_i - q_i$, and p_i . Show that a birth-and-death chain is time reversible and find the stationary distribution.

Solution To use Proposition 3.9, consider a vector $\mathbf{x}$ which satisfies the detailed balance equations. Then,

$$x_i P_{i,i+1} = x_{i+1} P_{i+1,i}, \text{ for } i = 0, 1, \dots$$

giving

$$x_i p_i = x_{i+1} q_{i+1}, \text{ for } i = 0, 1, \dots$$

Let $x_0 = 1$. Then, $x_1 = p_0/q_1$. Also, $x_2 = x_1 p_1/q_2 = (p_0 p_1)/(q_2 q_1)$. The general pattern is

$$x_k = \frac{p_0 p_1 \cdots p_{k-1}}{q_1 q_2 \cdots q_k} = \prod_{i=1}^k \frac{p_{i-1}}{q_i}, \text{ for } k = 0, 1, \dots$$

Normalizing gives the stationary distribution

$$\pi_k = \frac{x_k}{\sum_j x_j},$$

assuming that the infinite sum $\sum_j x_j$ converges. Thus, a necessary condition for a stationary distribution to exist, in which case the birth-and-death chain is time reversible, is that

$$\sum_j \prod_{i=1}^j \frac{p_{i-1}}{q_i} < \infty.$$

Birth-and-death chains encompass a large variety of special models. For instance, *random walk with a partially reflecting boundary* on $\{0, 1, \dots, N\}$ is achieved as a birth-and-death chain by letting $p_i = p$ and $q_i = q$, for $0 \leq i \leq N$, with $p + q < 1$. See Figure 3.12.

Minor modification to this derivation gives the stationary probabilities

$$\pi_k = \frac{x_k}{\sum_{i=0}^N x_i}, \text{ for } k = 0, 1, \dots, N,$$

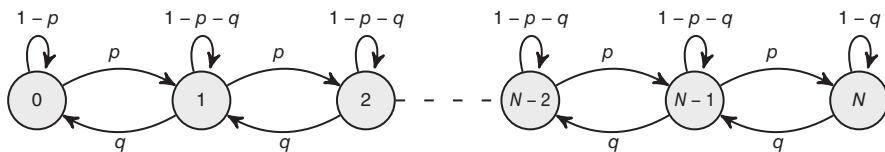


Figure 3.12 Random walk with partially reflecting boundaries.

where

$$x_k = \prod_{i=1}^k \left(\frac{p}{q} \right) = \left(\frac{p}{q} \right)^k, \text{ for } k = 0, 1, \dots, N.$$

For $p \neq q$, this gives

$$\pi_k = \left(1 - \frac{p}{q} \right) \left(\frac{p}{q} \right)^k / \left(1 - \left(\frac{p}{q} \right)^{N+1} \right), \text{ for } k = 0, 1, \dots, N.$$

For $p = q$, the stationary distribution is uniform. ■

3.8 ABSORBING CHAINS

Many popular board games can be modeled as Markov chains. The children's game *Chutes and Ladders* is based on an ancient Indian game called *Snakes and Ladders*. It is played on a 100-square board, as in Figure 3.13. Players each have a token and take turns rolling a six-sided die and moving their token by the corresponding number of squares. If a player lands on a *ladder*, they immediately move up the ladder to a higher-numbered square. If they move on a *chute*, or *snake*, they drop down to a lower-numbered square. The finishing square 100 must be reached by an exact roll of the die (or by landing on square 80 whose ladder climbs to the finish). The first player to land on square 100 wins.

The game is a Markov chain since the player's position only depends on their previous position and the roll of the die. The chain has 101 states as the game starts with all players off the board (state 0). For the Markov model, once the chain hits state 100 it stays at 100. That is, if P is the transition matrix, then $P_{100,100} = 1$.

Of particular interest is the average number of plays needed to reach the finish. The R script file **snakes.R** contains code for building the 101×101 transition matrix P . Here are the results from several simulations of the game.

R: Snakes and Ladders

```
> init <- c(1,rep(0,100)) # Start at square 0
> markov(init,P,150,0:100)
[1] 0 6 11 12 6 12 15 20 42 43 44 45 46 47 26
[16] 29 33 37 40 42 44 47 52 55 57 60 61 19 22 24
```

```

[31] 30 44 11 17 42 45 46 50 52 53 53 53 54 60 61
[46] 63 65 68 73 77 78 82 86 92 94 99 100
> markov(init,P,150,0:100)
[1] 0 14 18 24 30 34 40 44 45 67 70 74 100
> markov(init,P,150,0:100)
[1] 0 5 6 8 31 33 44 46 50 53 55 58 63 60 61
[16] 66 72 77 78 100
> markov(init,P,150,0:100)
[1] 0 2 5 8 11 14 20 26 31 37 39 40 42 47 53
[16] 58 63 60 60 65 69 72 76 82 86 24 25 29 31 35
[31] 40 42 26 31 34 35 39 43 45 11 13 19 24 26 32
[46] 35 38 39 42 44 26 84 88 90 96 100
> markov(init,P,150,0:100)
[1] 0 6 31 44 45 26 27 32 38 44 26 30 31 33 44
[16] 46 50 67 73 76 77 78 79 84 88 90 73 76 100
> markov(init,P,150,0:100)
[1] 0 3 31 34 40 42 47 26 30 32 38 44 47 53 59
[16] 19 23 25 31 44 26 29 31 33 37 41 42 26 31 32
[31] 38 43 44 50 52 58 19 24 30 31 35 41 44 46 67
[46] 68 91 97 99 99 99 99 99 99 99 99 99 100
> markov(init,P,150,0:100)
[1] 0 2 7 13 6 10 15 17 23 25 31 37 40 45 11
[16] 17 20 26 31 33 44 46 26 29 35 44 50 67 72 74
[31] 76 100

```

For these simulations, it took, respectively, 56, 12, 19, 55, 28, 57, and 31 steps to reach the winning square, for an average of 36.86 steps to win.

An exact analysis of the average time to win the game is given in this section, after establishing some theory.

In the Snakes and Ladders Markov chain every state, except 100, is transient. State 100 is recurrent. From 100, the chain stays at 100 forever. A state with this property is called *absorbing*.

Absorbing State, Absorbing Chain

State i is an *absorbing state* if $P_{ii} = 1$. A Markov chain is called an *absorbing chain* if it has at least one absorbing state.

Consider an absorbing Markov chain on k states for which t states are transient and $k - t$ states are absorbing. The states can be reordered, as in the canonical decomposition, with the transition matrix written in block matrix form

$$P = \left(\begin{array}{c|c} Q & R \\ \hline \mathbf{0} & I \end{array} \right) \quad (3.11)$$

where Q is a $t \times t$ matrix, R is a $t \times (k - t)$ matrix, $\mathbf{0}$ is a $(k - t) \times t$ matrix of 0s, and I is the $(k - t) \times (k - t)$ identity matrix.

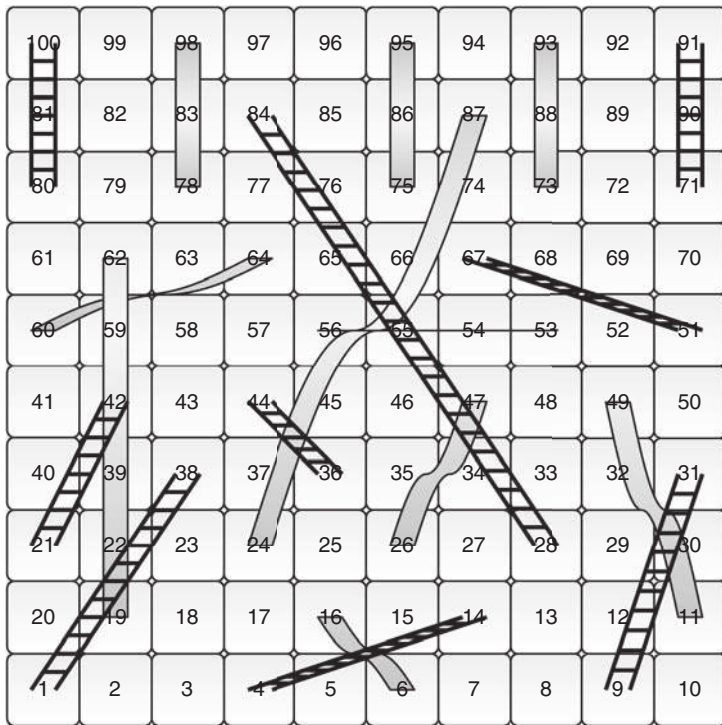


Figure 3.13 Children's Snakes and Ladders game. Board drawn using TikZ TeX package. *Source:* <http://tex.stackexchange.com/questions/85411/chutes-and-ladders/>. Reproduced with permission of Serge Ballif.

Computing powers of $\mathbf{P}$ is facilitated by the block matrix form. We have

$$\mathbf{P}^2 = \left(\begin{array}{c|c} \mathbf{Q} & \mathbf{R} \\ \hline \mathbf{0} & \mathbf{I} \end{array} \right) \left(\begin{array}{c|c} \mathbf{Q} & \mathbf{R} \\ \hline \mathbf{0} & \mathbf{I} \end{array} \right) = \left(\begin{array}{c|c} \mathbf{Q}^2 & (\mathbf{I} + \mathbf{Q})\mathbf{R} \\ \hline \mathbf{0} & \mathbf{I} \end{array} \right),$$

$$\mathbf{P}^3 = \left(\begin{array}{c|c} \mathbf{Q}^3 & (\mathbf{I} + \mathbf{Q} + \mathbf{Q}^2)\mathbf{R} \\ \hline \mathbf{0} & \mathbf{I} \end{array} \right),$$

and, in general,

$$\mathbf{P}^n = \left(\begin{array}{c|c} \mathbf{Q}^n & (\mathbf{I} + \mathbf{Q} + \cdots + \mathbf{Q}^{n-1})\mathbf{R} \\ \hline \mathbf{0} & \mathbf{I} \end{array} \right), \text{ for } n \geq 1. \quad (3.12)$$

To find the limiting matrix $\lim_{n \rightarrow \infty} \mathbf{P}^n$, we make use of the following lemma from linear algebra.

Lemma 3.10. *Let A be a square matrix with the property that $A^n \rightarrow \mathbf{0}$, as $n \rightarrow \infty$. Then,*

$$\sum_{n=0}^{\infty} A^n = (I - A)^{-1}.$$

The lemma gives the matrix analog of the sum of a geometric series of real numbers. That is,

$$\sum_{n=0}^{\infty} r^n = (1 - r)^{-1},$$

if $r^n \rightarrow 0$, as $n \rightarrow \infty$.

Proof. For fixed n ,

$$\begin{aligned} (I - A)(I + A + A^2 + \cdots + A^n) &= I + A + A^2 + \cdots + A^n \\ &\quad - (A + A^2 + \cdots + A^n + A^{n+1}) \\ &= I - A^{n+1}. \end{aligned}$$

If $I - A$ is invertible, then

$$(I + A + \cdots + A^n) = (I - A)^{-1}(I - A^{n+1}).$$

Taking limits, as $n \rightarrow \infty$, gives

$$\sum_{n=0}^{\infty} A^n = (I - A)^{-1},$$

since $A^{n+1} \rightarrow \mathbf{0}$, as $n \rightarrow \infty$.

To show that $I - A$ is invertible, consider the linear system $(I - A)x = \mathbf{0}$. Invertibility of $I - A$ is equivalent to the fact that the only solution to this system is $x = \mathbf{0}$. We have that $\mathbf{0} = (I - A)x = x - Ax$. That is, $x = Ax$. Iterating gives

$$x = Ax = A(Ax) = A^2x = \cdots = A^n x, \text{ for all } n \geq 1.$$

Taking limits on both sides of the equation gives

$$x = \lim_{n \rightarrow \infty} A^n x = \mathbf{0}. \quad \blacksquare$$

To apply Lemma 3.10 to Equation (3.12), observe that $Q^n \rightarrow \mathbf{0}$, as $n \rightarrow \infty$. The matrix Q is indexed by transient states. If i and j are transient,

$$\lim_{n \rightarrow \infty} Q_{ij}^n = \lim_{n \rightarrow \infty} P_{ij}^n = 0,$$

as the long-term probability that a Markov chain hits a transient state is 0. Taking limits in Equation (3.12) gives

$$\begin{aligned}\lim_{n \rightarrow \infty} P^n &= \lim_{n \rightarrow \infty} \left(\begin{array}{c|c} Q^n & (I + Q + \cdots + Q^{n-1})R \\ \hline \mathbf{0} & I \end{array} \right) \\ &= \left(\begin{array}{c|c} \lim_{n \rightarrow \infty} Q^n & \lim_{n \rightarrow \infty} (I + Q + \cdots + Q^{n-1})R \\ \hline \mathbf{0} & I \end{array} \right) \\ &= \left(\begin{array}{c|c} \mathbf{0} & (I - Q)^{-1}R \\ \hline \mathbf{0} & I \end{array} \right).\end{aligned}$$

Consider the interpretation of the limiting submatrix $(I - Q)^{-1}R$. The matrix is indexed by transient rows and absorbing columns. The ij th entry is the long-term probability that the chain started in transient state i is absorbed in state j . If the Markov chain has only one absorbing state, this submatrix will be a $(k - 1)$ -element column vector of 1s.

Example 3.27 (Gambler's ruin) A gambler starts with \$2 and plays a game where the chance of winning each round is 60%. The gambler either wins or loses \$1 on each round. The game stops when the gambler either gains \$5 or goes bust. Find the probability that the gambler is eventually ruined.

Solution The game is an absorbing Markov chain with absorbing states 0 and 5. The transition matrix in canonical form is

$$P = \begin{array}{c} \begin{array}{cccccc} & 1 & 2 & 3 & 4 & 0 & 5 \\ \begin{array}{c} 1 \\ 2 \\ 3 \\ 4 \\ 0 \\ 5 \end{array} & \begin{pmatrix} 0 & 0.6 & 0 & 0 & 0.4 & 0 \\ 0.4 & 0 & 0.6 & 0 & 0 & 0 \\ 0 & 0.4 & 0 & 0.6 & 0 & 0 \\ 0 & 0 & 0.4 & 0 & 0 & 0.6 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix} \end{array},\end{array}$$

with

$$Q = \begin{array}{c} \begin{array}{cccc} & 1 & 2 & 3 & 4 \\ \begin{array}{c} 1 \\ 2 \\ 3 \\ 4 \end{array} & \begin{pmatrix} 0 & 0.6 & 0 & 0 \\ 0.4 & 0 & 0.6 & 0 \\ 0 & 0.4 & 0 & 0.6 \\ 0 & 0 & 0.4 & 0 \end{pmatrix} \end{array} \quad \text{and} \quad R = \begin{array}{c} \begin{array}{cc} & 0 & 5 \\ \begin{array}{c} 1 \\ 2 \\ 3 \\ 4 \end{array} & \begin{pmatrix} 0.4 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0.6 \end{pmatrix} \end{array}.\end{array}$$

This gives

$$\begin{aligned}
 (I - Q)^{-1}R &= \begin{pmatrix} 1 & -0.6 & 0 & 0 \\ -0.4 & 1 & -0.6 & 0 \\ 0 & -0.4 & 1 & -0.6 \\ 0 & 0 & -0.4 & 1 \end{pmatrix}^{-1} \begin{pmatrix} 0.4 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0.6 \end{pmatrix} \\
 &= \begin{matrix} & 0 & 5 \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{pmatrix} 0.616 & 0.384 \\ 0.360 & 0.640 \\ 0.190 & 0.810 \\ 0.076 & 0.924 \end{pmatrix} \end{matrix}.
 \end{aligned}$$

If the gambler starts with \$2, the probability of their eventual ruin is 0.36.

R: Gambler's Ruin

Commands for working with matrices in R are explained in Appendix E. The R command `solve(mat)` computes the matrix inverse of `mat`. The command `diag(k)` generates a $k \times k$ identity matrix.

```

> P
      1      2      3      4      0      5
1 0.0 0.6 0.0 0.0 0.4 0.0
2 0.4 0.0 0.6 0.0 0.0 0.0
3 0.0 0.4 0.0 0.6 0.0 0.0
4 0.0 0.0 0.4 0.0 0.0 0.6
0 0.0 0.0 0.0 0.0 0.0 1.0
5 0.0 0.0 0.0 0.0 0.0 0.0 1.0
> Q <- P[1:4,1:4]
> R <- P[1:4,5:6]
> Q
      1      2      3      4
1 0.0 0.6 0.0 0.0
2 0.4 0.0 0.6 0.0
3 0.0 0.4 0.0 0.6
4 0.0 0.0 0.4 0.0
> R
      0      5
1 0.4 0.0
2 0.0 0.0
3 0.0 0.0
4 0.0 0.6

```

```
> solve(diag(4)-Q) %*% R
      0      5
1 0.616114 0.38389
2 0.360190 0.63981
3 0.189573 0.81043
4 0.075829 0.92417
```

■

For absorbing Markov chains, the matrix $(I - Q)^{-1}$ is called the *fundamental matrix*. Its importance is highlighted by the next theorem. Recall that if i is transient, then for the chain started in i , the expected number of visits to i is finite.

Expected Number of Visits to Transient States

Theorem 3.11. *Consider an absorbing Markov chain with t transient states. Let F be a $t \times t$ matrix indexed by transient states, where F_{ij} is the expected number of visits to j given that the chain starts in i . Then,*

$$F = (I - Q)^{-1}.$$

Two proofs are given. The first uses the method of first-step analysis.

Proof #1. Let T be the set of transient states. Assume that $i, j \in T$. Consider the chain started in i . On the first step, the chain moves to some state k . If k is an absorbing state, then the chain will never visit j , unless $j = i$, in which case it has visited j one time. If k is a transient state, then the expected number of visits to j is F_{kj} , if $j \neq i$, and $1 + F_{ki}$, if $j = i$. This gives

$$\begin{aligned} F_{ij} &= \begin{cases} \sum_{k \in T} P_{ik}(1 + F_{ki}) + \sum_{k \notin T} P_{ik}, & \text{if } j = i, \\ \sum_{k \in T} P_{ik}F_{kj}, & \text{if } j \neq i, \end{cases} \\ &= \begin{cases} 1 + \sum_{k \in T} Q_{ik}F_{ki}, & \text{if } j = i, \\ \sum_{k \in T} Q_{ik}F_{kj}, & \text{if } j \neq i, \end{cases} \\ &= \delta_{ij} + \sum_{k \in T} Q_{ik}F_{kj}, \end{aligned} \tag{3.13}$$

where $\delta_{ij} = 1$, if $i = j$, and 0, otherwise. The second equality is because if i and k are transient states, then $P_{ik} = Q_{ik}$.

In matrix terms, Equation (3.13) says that $F = I + QF$, or $(I - Q)F = I$. It follows that $I - Q$ is invertible and $(I - Q)^{-1} = F$. ■

Proof #2. For the chain started in i , define indicator variables

$$I_n = \begin{cases} 1, & \text{if } X_n = j, \\ 0, & \text{otherwise,} \end{cases}$$

for $n = 0, 1, \dots$. Then, $\sum_{n=0}^{\infty} I_n$ is the number of visits to j . The expected number of visits is

$$\begin{aligned} F_{ij} &= E\left(\sum_{n=0}^{\infty} I_n\right) = \sum_{n=0}^{\infty} E(I_n) \\ &= \sum_{n=0}^{\infty} P(X_n = j | X_0 = i) \\ &= \sum_{n=0}^{\infty} P_{ij}^n = \sum_{n=0}^{\infty} Q_{ij}^n = \left(\sum_{n=0}^{\infty} Q^n\right)_{ij} = (I - Q)^{-1}_{ij}, \end{aligned}$$

as a consequence of Lemma 3.10. ■

Expected Time to Absorption

For an absorbing Markov chain started in transient state i , let a_i be the expected absorption time, the expected number of steps to reach some absorbing state. The number of transitions from i to an absorbing state is simply the sum of the number of transitions from i to each of the transient states until eventual absorption. The expected number of steps from i to transient state j is F_{ij} . It follows that

$$a_i = \sum_{k \in T} F_{ik}.$$

In vector form, $\mathbf{a} = \mathbf{F}\mathbf{1}$, where $\mathbf{1}$ is the column vector of all 1s. That is, the expected absorption times are the row sums of the fundamental matrix.

We summarize results for absorbing Markov chains.

Absorbing Markov Chains

For an absorbing Markov chain with all states either transient or absorbing, let $\mathbf{F} = (\mathbf{I} - \mathbf{Q})^{-1}$.

1. (*Absorption probability*) The probability that from transient state i the chain is absorbed in state j is $(\mathbf{F}\mathbf{R})_{ij}$.
2. (*Absorption time*) The expected number of steps from transient state i until the chain is absorbed in some absorbing state is $(\mathbf{F}\mathbf{1})_i$.

■ **Example 3.28 (Snakes and Ladders)** We find the expected absorption time to square 100 in Snakes and Ladders. See the **snakes.R** script file. The transition matrix is a 101×101 matrix stored in the variable **P**.

R: Snakes and Ladders

```
> Q <- P[1:100,1:100]
> F <- solve(diag(100)-Q)
> a <- F %*% rep(1,100) # Expected absorption times
> round(t(a),2)
```

	0	1	2	3	4	5	6	7	8	9
39.60	40.25	40.07	39.57	39.84	39.67	39.46	39.23	38.98	39.05	
	10	11	12	13	14	15	16	17	18	19
39.11	38.64	38.22	37.84	37.50	37.02	36.08	35.82	35.66	35.55	
	20	21	22	23	24	25	26	27	28	29
35.50	35.50	33.82	34.31	34.66	34.94	35.15	35.30	37.52	37.25	
	30	31	32	33	34	35	36	37	38	39
36.77	36.60	36.41	36.19	35.93	35.64	35.68	35.55	35.31	34.83	
	40	41	42	43	44	45	46	47	48	49
34.37	33.93	34.10	34.75	33.87	31.95	31.61	31.27	30.20	28.47	
	50	51	52	53	54	55	56	57	58	59
28.58	29.88	29.61	29.22	28.75	28.24	29.29	28.28	27.94	26.87	
	60	61	62	63	64	65	66	67	68	69
25.95	25.16	22.57	22.19	21.16	20.42	20.43	20.44	20.03	19.85	
	70	71	72	73	74	75	76	77	78	79
19.81	20.50	20.50	20.52	17.59	18.74	19.56	20.12	20.47	20.64	
	80	81	82	83	84	85	86	87	88	89
24.12	25.62	24.48	23.53	22.56	21.62	20.90	18.54	17.63	17.79	
	90	91	92	93	94	95	96	97	98	99
16.79	15.94	16.59	14.17	12.14	10.82	10.82	10.82	6.00	6.00	

The desired expectation is a_0 . At the start of the game, it takes, on average, 39.60 moves to reach the ending square. ■

■ **Example 3.29 (Graduation)** Recall the graduation Markov chain of Example 2.19. Students at a 4-year college either drop out, repeat a year, or move on to the next year. The chain is an absorbing chain with graduating and dropping out as absorbing states. Relevant matrices are

$$P = \begin{matrix} & \begin{matrix} \text{Fr} & \text{So} & \text{Jr} & \text{Sr} & \text{Drop} & \text{Grad} \end{matrix} \\ \begin{matrix} \text{Fr} \\ \text{So} \\ \text{Jr} \\ \text{Se} \\ \text{Drop} \\ \text{Grad} \end{matrix} & \begin{pmatrix} 0.03 & 0.91 & 0 & 0 & 0.06 & 0 \\ 0 & 0.03 & 0.91 & 0 & 0.06 & 0 \\ 0 & 0 & 0.03 & 0.93 & 0.04 & 0 \\ 0 & 0 & 0 & 0.03 & 0.04 & 0.93 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix} \end{matrix},$$

$$Q = \begin{matrix} & \begin{matrix} \text{Fr} & \text{So} & \text{Jr} & \text{Sr} \end{matrix} \\ \begin{matrix} \text{Fr} \\ \text{So} \\ \text{Jr} \\ \text{Se} \end{matrix} & \begin{pmatrix} 0.03 & 0.91 & 0 & 0 \\ 0 & 0.03 & 0.91 & 0 \\ 0 & 0 & 0.03 & 0.93 \\ 0 & 0 & 0 & 0.03 \end{pmatrix} \end{matrix}, \quad \text{and} \quad R = \begin{matrix} & \begin{matrix} \text{Drop} & \text{Grad} \end{matrix} \\ \begin{matrix} \text{Fr} \\ \text{So} \\ \text{Jr} \\ \text{Se} \end{matrix} & \begin{pmatrix} 0.06 & 0 \\ 0.06 & 0 \\ 0.04 & 0 \\ 0.04 & 0.93 \end{pmatrix} \end{matrix}.$$

This gives

$$(I - Q)^{-1}R = \begin{pmatrix} 0.97 & -0.91 & 0 & 0 \\ 0 & 0.97 & -0.91 & 0 \\ 0 & 0 & 0.97 & -0.93 \\ 0 & 0 & 0 & 0.97 \end{pmatrix}^{-1} \begin{pmatrix} 0.06 & 0 \\ 0.06 & 0 \\ 0.04 & 0 \\ 0.04 & 0.93 \end{pmatrix}$$

$$= \begin{matrix} & \begin{matrix} \text{Drop} & \text{Grad} \end{matrix} \\ \begin{matrix} \text{Fr} \\ \text{So} \\ \text{Jr} \\ \text{Se} \end{matrix} & \begin{pmatrix} 0.191 & 0.809 \\ 0.138 & 0.862 \\ 0.081 & 0.919 \\ 0.041 & 0.959 \end{pmatrix} \end{matrix}.$$

For a student who starts as a first-year, the probability of eventually graduating is 0.809. ■

R: Graduation

```
> P
      Fr    So    Jr    Se Drop Grad
Fr  0.03 0.91 0.00 0.00 0.06 0.00
So  0.00 0.03 0.91 0.00 0.06 0.00
Jr  0.00 0.00 0.03 0.93 0.04 0.00
Se  0.00 0.00 0.00 0.03 0.04 0.93
Drop 0.00 0.00 0.00 0.00 1.00 0.00
Grad 0.00 0.00 0.00 0.00 0.00 1.00
> Q <- P[1:4,1:4]
> R <- P[1:4,5:6]
> Absorb <- solve(diag(4) - Q) %*% R
> Absorb
      Drop    Grad
Fr 0.190975 0.80902
So 0.137633 0.86237
Jr 0.080774 0.91923
Se 0.041237 0.95876
```

Expected Hitting Times for Irreducible Chains

For an irreducible Markov chain, first hitting times can be analyzed as absorption times for a suitably modified chain. In particular, assume that P is the transition matrix

of an irreducible Markov chain. To find the expected time until state i is first hit, consider a new chain in which i is an absorbing state. The transition matrix $\tilde{P}$ for the new chain is gotten by zeroing out the i th row of the P matrix and setting $\tilde{P}_{ii} = 1$. The resulting Q matrix is obtained from $\tilde{P}$ by deleting the i th row and the i th column of $\tilde{P}$. The time that the original P -chain first hits i is equal to the time that the modified $\tilde{P}$ -chain is absorbed in i .

■ **Example 3.30** Consider random walk on the weighted graph in Figure 3.14. Starting from each vertex in the graph, find the expected number of steps until the walk first hits f .

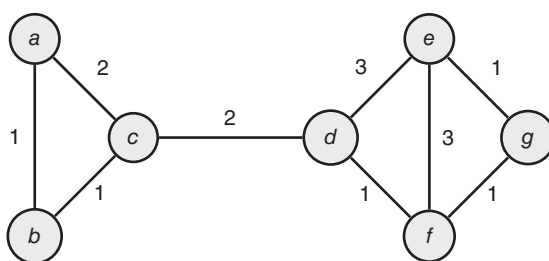


Figure 3.14

Solution The transition matrix is

$$P = \begin{matrix} & \begin{matrix} a & b & c & d & e & f & g \end{matrix} \\ \begin{matrix} a \\ b \\ c \\ d \\ e \\ f \\ g \end{matrix} & \begin{pmatrix} 0 & 1/3 & 2/3 & 0 & 0 & 0 & 0 \\ 1/2 & 0 & 1/2 & 0 & 0 & 0 & 0 \\ 2/5 & 1/5 & 0 & 2/5 & 0 & 0 & 0 \\ 0 & 0 & 1/3 & 0 & 1/2 & 1/6 & 0 \\ 0 & 0 & 0 & 3/7 & 0 & 3/7 & 1/7 \\ 0 & 0 & 0 & 1/5 & 3/5 & 0 & 1/5 \\ 0 & 0 & 0 & 0 & 1/2 & 1/2 & 0 \end{pmatrix} \end{matrix}.$$

By making f an absorbing state, the resulting Q matrix is

$$Q = \begin{matrix} & \begin{matrix} a & b & c & d & e & g \end{matrix} \\ \begin{matrix} a \\ b \\ c \\ d \\ e \\ g \end{matrix} & \begin{pmatrix} 0 & 1/3 & 2/3 & 0 & 0 & 0 \\ 1/2 & 0 & 1/2 & 0 & 0 & 0 \\ 2/5 & 1/5 & 0 & 2/5 & 0 & 0 \\ 0 & 0 & 1/3 & 0 & 1/2 & 0 \\ 0 & 0 & 0 & 3/7 & 0 & 1/7 \\ 0 & 0 & 0 & 0 & 1/2 & 0 \end{pmatrix} \end{matrix},$$

which gives

$$F = (I - Q)^{-1} = \begin{matrix} & \begin{matrix} a & b & c & d & e & g \end{matrix} \\ \begin{matrix} a \\ b \\ c \\ d \\ e \\ g \end{matrix} & \begin{pmatrix} 3.847 & 2.165 & 4.412 & 2.294 & 1.235 & 0.176 \\ 3.247 & 2.965 & 4.412 & 2.294 & 1.235 & 0.176 \\ 2.647 & 1.765 & 4.412 & 2.294 & 1.235 & 0.176 \\ 1.147 & 0.765 & 1.912 & 2.29 & 1.235 & 0.176 \\ 0.529 & 0.353 & 0.882 & 1.059 & 1.647 & 0.235 \\ 0.265 & 0.176 & 0.441 & 0.529 & 0.824 & 1.118 \end{pmatrix} \end{matrix}.$$

Row sums are

$$\begin{matrix} a \\ b \\ c \\ d \\ e \\ g \end{matrix} \begin{pmatrix} 14.129 \\ 14.329 \\ 12.529 \\ 7.529 \\ 4.706 \\ 3.353 \end{pmatrix},$$

which gives the expected numbers of steps, starting from each vertex in the graph, to first hit f . ■

■ **Example 3.31** A coin is flipped repeatedly until three heads in a row appear. What is the expected number of flips needed?

Solution One approach is to condition on the first coin flip and use the law of total expectation. Here is a Markov chain solution.

Let X_n be the most recent run of heads gotten after the n th coin flip. States are depicted as $\{\emptyset, H, HH, HHH\}$, with HHH an absorbing state and $\emptyset$ representing the initial state. Note that if tails is ever flipped then the run of heads starts over again. Thus, $\emptyset$ also represents having gotten tails on the last flip. The transition matrix is

$$P = \begin{matrix} & \begin{matrix} \emptyset & H & HH & HHH \end{matrix} \\ \begin{matrix} \emptyset \\ H \\ HH \\ HHH \end{matrix} & \begin{pmatrix} 1/2 & 1/2 & 0 & 0 \\ 1/2 & 0 & 1/2 & 0 \\ 1/2 & 0 & 0 & 1/2 \\ 0 & 0 & 0 & 1 \end{pmatrix} \end{matrix}.$$

This gives

$$I - Q = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} - \begin{pmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{2} & 0 & \frac{1}{2} \\ \frac{1}{2} & 0 & 0 \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & -\frac{1}{2} & 0 \\ -\frac{1}{2} & 1 & -\frac{1}{2} \\ -\frac{1}{2} & 0 & 1 \end{pmatrix}$$

and

$$F = (I - Q)^{-1} = \begin{pmatrix} 8 & 4 & 2 \\ 6 & 4 & 2 \\ 4 & 2 & 2 \end{pmatrix},$$

with expected absorption times

$$a = F\mathbf{1} = \begin{matrix} & \text{HHH} \\ \emptyset & \\ \text{H} & \\ \text{HH} & \end{matrix} \begin{pmatrix} 14 \\ 12 \\ 8 \end{pmatrix}.$$

It takes on average 14 flips to get three heads in a row. ■

Patterns in Sequences

The last example illustrates a general method for problems involving the occurrence of patterns in random trials. Assume that the elements of a set S are repeatedly sampled. A *pattern* is a sequence $(p_1, p_2, \dots, p_n)$, such that each of the p_i is an element of S . In the last example, $S = \{H, T\}$, and the desired pattern is (H,H,H).

For $k = 1, \dots, n$, let $s_k = (p_1, \dots, p_k)$ be the subsequence consisting of the first k elements of the pattern. A Markov chain is constructed with state space $\{\emptyset, s_1, \dots, s_n\}$, where s_n , the desired pattern, is an absorbing state. The absorption time for the Markov chain is equal to the time until the pattern first appears in repeated sampling from S .

■ **Example 3.32** A biased coin comes up heads, with probability $2/3$, and tails, with probability $1/3$. The coin is repeatedly flipped. How many flips are needed, on average, until the pattern HTHTH first appears?

Solution An absorbing Markov chain is constructed with transition matrix

$$P = \begin{matrix} & \emptyset & H & HT & HTH & HTHT & HTHTH \\ \emptyset & \\ H & \\ HT & \\ HTH & \\ HTHT & \\ HTHTH & \end{matrix} \begin{pmatrix} 1/3 & 2/3 & 0 & 0 & 0 & 0 \\ 0 & 2/3 & 1/3 & 0 & 0 & 0 \\ 1/3 & 0 & 0 & 2/3 & 0 & 0 \\ 0 & 2/3 & 0 & 0 & 1/3 & 0 \\ 1/3 & 0 & 0 & 0 & 0 & 2/3 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}.$$

The fundamental matrix is

$$F = (I - Q)^{-1} = \begin{pmatrix} 2/3 & -2/3 & 0 & 0 & 0 \\ 0 & 1/3 & -1/3 & 0 & 0 \\ -1/3 & 0 & 1 & -2/3 & 0 \\ 0 & -2/3 & 0 & 1 & -1/3 \\ -1/3 & 0 & 0 & 0 & 1 \end{pmatrix}^{-1}$$

$$= \begin{pmatrix} 45/8 & 81/4 & 27/4 & 9/2 & 3/2 \\ 33/8 & 81/4 & 27/4 & 9/2 & 3/2 \\ 33/8 & 69/4 & 27/4 & 9/2 & 3/2 \\ 27/8 & 63/4 & 21/4 & 9/2 & 3/2 \\ 15/8 & 27/4 & 9/4 & 3/2 & 3/2 \end{pmatrix}.$$

The sum of the first row of the fundamental matrix is

$$\frac{45}{8} + \frac{81}{4} + \frac{27}{4} + \frac{9}{2} + \frac{3}{2} = \frac{309}{8} = 38.625.$$

It takes, on average, 38.625 flips before HTHTH first appears. ■

R : Simulation of Number of Flips Needed For HTHTH

```
# pattern.R
# P(Heads) = 2/3, P(Tails) = 1/3
> trials <- 100000
> simlist <- numeric(trials)
> for (i in 1:trials) {
+   pattern <- c(1,0,1,0,1) # 1:Heads, 0:Tails
+   state <- sample(c(0,1),5,prob=c(1/3,2/3),replace=T)
+   k <- 5
+   while (!prod(state==pattern))
+     { flip <- sample(c(0,1),1,prob=c(1/3,2/3))
+       state <- c(tail(state,4),flip)
+       k <- k + 1 }
+   simlist[i] <- k }
> mean(simlist)
[1] 38.67718
# exact expectation is 38.625
```

In the last example, successive trials (e.g., coin flips) are independent and identically distributed. However, this is not necessary. In the next example, the trials themselves form a Markov chain.

■ **Example 3.33** Assume that successive occurrences of DNA nucleotides on a chromosome are modeled by a Markov chain with transition matrix

$$\tilde{P} = \begin{matrix} & \begin{matrix} a & c & g & t \end{matrix} \\ \begin{matrix} a \\ c \\ g \\ t \end{matrix} & \begin{pmatrix} 0.3 & 0.2 & 0.3 & 0.2 \\ 0.4 & 0.3 & 0.1 & 0.2 \\ 0.25 & 0.15 & 0.4 & 0.2 \\ 0.2 & 0.2 & 0.3 & 0.3 \end{pmatrix} \end{matrix}.$$

Consider searching sequentially across the chromosome until the pattern *accgc* first appears. On average, how many steps (e.g., successive nucleotides), are needed?

Solution An absorbing Markov chain is built on $\{\emptyset, a, ac, acc, accg, accgc\}$, with transition matrix

$$P = \begin{matrix} & \begin{matrix} \emptyset & a & ac & acc & accg & accgc \end{matrix} \\ \begin{matrix} \emptyset \\ a \\ ac \\ acc \\ accg \\ accgc \end{matrix} & \begin{pmatrix} 0.7 & 0.3 & 0 & 0 & 0 & 0 \\ 0.5 & 0.3 & 0.2 & 0 & 0 & 0 \\ 0.3 & 0.4 & 0 & 0.3 & 0 & 0 \\ 0.5 & 0.4 & 0 & 0 & 0.1 & 0 \\ 0.6 & 0.25 & 0 & 0 & 0 & 0.15 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix} \end{matrix}.$$

To understand the matrix entries, observe, for instance, that from state acc , the chain moves to (i) $accg$, with probability $\tilde{P}_{cg} = 0.1$, or to (ii) a , with probability $\tilde{P}_{ca} = 0.4$, or to (iii) $\emptyset$, with the complementary probability 0.5. We have

$$\begin{aligned} F = (I - Q)^{-1} &= \begin{pmatrix} 0.3 & -0.3 & 0 & 0 & 0 \\ -0.5 & 0.7 & -0.2 & 0 & 0 \\ -0.3 & -0.4 & 1 & -0.3 & 0 \\ -0.5 & -0.4 & 0 & 0.9 & 0 \\ -0.6 & -0.25 & 0 & 0 & 1 \end{pmatrix}^{-1} \\ &= \begin{pmatrix} 2201.85 & 1111.11 & 222.22 & 66.67 & 6.67 \\ 2198.52 & 1111.11 & 222.22 & 66.67 & 6.67 \\ 2190.19 & 1106.11 & 222.22 & 66.67 & 6.67 \\ 2167.41 & 1094.44 & 218.89 & 66.67 & 6.67 \\ 1870.74 & 944.444 & 188.89 & 56.67 & 6.67 \end{pmatrix}. \end{aligned}$$

The sum of the first row is

$$2201.85 + 1111.11 + 222.22 + 66.67 + 6.67 = 3608.52,$$

which gives the average number of nucleotides needed to reach the desired pattern. ■

3.9 REGENERATION AND THE STRONG MARKOV PROPERTY*

A Markov chain is sometimes observed from a fixed time $n > 0$ into the future. Assume that $X_0, X_1, \dots$ is a Markov chain with transition matrix P . Then, the process started at $n > 0$, $X_n, X_{n+1}, X_{n+2}, \dots$ is also a Markov chain with transition matrix P . This is a consequence of the Markov property, which says that given the present, past, and future are independent.

The *strong Markov property* asserts that for certain types of random times called *stopping times*, the Markov property holds. If S is a stopping time, then the sequence $X_S, X_{S+1}, X_{S+2}, \dots$ is a Markov chain. Given that the present time is a stopping time, past and future are independent.

An integer-valued random variable S is a stopping time for a Markov chain if, for each s , the event $\{S = s\}$ can be determined from $X_0, \dots, X_s$. In other words, if the outcomes $X_0, \dots, X_s$ are known, then it can be determined whether or not $\{S = s\}$ occurs.

An important example of a stopping time is the *first hitting time* random variable

$$T_i = \min\{n \geq 0 : X_n = i\},$$

which is the first time that a Markov chain hits state i . For instance, consider the weather Markov chain. Let r denote rain. Then, T_r is the first day that it rains. For any day t , If we are given the succession of weather states up to time t , $X_0, \dots, X_t$, then it can be determined whether or not the first day that it rained was on day t . This shows that T_r is a stopping time. By the strong Markov property, the sequence $X_{T_r}, X_{T_r+1}, \dots$ is a Markov chain.

A closely related stopping time for a Markov chain started at i , is the *first return time*

$$T_i^+ = \min\{n \geq 1 : X_n = i\}.$$

The strong Markov property says that the chain started at T_i^+ looks the same as the chain started at i . We say that at time T_i^+ the Markov chain *regenerates itself* and probabilistically starts anew.

More generally, given a nonempty subset of states $A \subseteq S$, the first time the chain hits a state in A

$$T_A = \min\{n \geq 0 : X_n \in A\}$$

is a stopping time.

A random time that is not a stopping time is the *last* visit to state i . Knowing whether or not the last visit to i occurs at time s cannot be determined from just $X_0, \dots, X_s$. It requires knowledge of the entire Markov sequence $X_0, X_1, \dots$

Strong Markov Property

Let $X_0, X_1, \dots$ be a Markov chain with transition matrix $\mathbf{P}$. Let S be a stopping time. Then, $X_S, X_{S+1}, \dots$ is a Markov chain with transition matrix $\mathbf{P}$.

Proof. For states $i, j, i_0, i_1, \dots$, consider

$$\begin{aligned} &P(X_{S+1} = j, X_S = i, X_u = i_u, 0 \leq u < S) \\ &= \sum_s P(S = s, X_{s+1} = j, X_s = i, X_u = i_u, 0 \leq u < s) \\ &= \sum_s P(S = s | X_{s+1} = j, X_s = i, X_u = i_u, 0 \leq u < s) \\ &\quad \times P(X_{s+1} = j | X_s = i, X_u = i_u, 0 \leq u < s) \\ &\quad \times P(X_s = i, X_u = i_u, 0 \leq u < s) \end{aligned}$$

$$\begin{aligned}
&= \sum_s P(S = s | X_s = i, X_u = i_u, 0 \leq u < s) \\
&\quad \times P_{ij} P(X_s = i, X_u = i_u, 0 \leq u < s) \\
&= P_{ij} P(X_S = i, X_u = i_u, 0 \leq u < S).
\end{aligned}$$

The third equality is by conditional probability. The fourth equality is because (i) S is a stopping time and the event $\{S = s\}$ is determined by $X_0, \dots, X_s$, and (ii) by the Markov property. It follows that

$$\begin{aligned}
P(X_{S+1} = j | X_S = i, X_u = i_u, 0 \leq u < S) \\
&= \frac{P(X_{S+1} = j, X_S = i, X_u = i_u, 0 \leq u < S)}{P(X_S = i, X_u = i_u, 0 \leq u < S)} \\
&= P_{ij}.
\end{aligned}$$

■

3.10 PROOFS OF LIMIT THEOREMS*

In this section, we prove the main limit theorems from this chapter. Each proof is given after restating the corresponding theorem.

Limit Theorem for Regular Markov Chains

Theorem 3.2. *A Markov chain whose transition matrix $\mathbf{P}$ is regular has a limiting distribution, which is the unique, positive, stationary distribution of the chain.*

Proof of Theorem 3.2. This is a direct consequence of two forthcoming results: Proposition 3.13, which gives that regular Markov chains are ergodic, and Theorem 3.8, the fundamental limit theorem for ergodic Markov chains. ■

Finite Irreducible Markov Chains

Theorem 3.6. *Assume that $X_0, X_1, \dots$ is a finite irreducible Markov chain. For each state j , let $\mu_j = E(T_j | X_0 = j)$ be the expected return time to j . Then, μ_j is finite, and there exists a unique, positive stationary distribution π such that*

$$\pi_j = \frac{1}{\mu_j}, \text{ for all } j. \quad (3.14)$$

Furthermore, for all i and j ,

$$\pi_j = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{m=0}^{n-1} P_{ij}^m. \quad (3.15)$$

Proof of Theorem 3.6. Let $X_0, X_1, \dots$ be an irreducible Markov chain with state space $\{1, \dots, k\}$. Given states i, j , consider the chain started in i . Since the chain is irreducible and all states are recurrent, the chain will visit j infinitely often.

Let Y_1 be the time the chain first hits j . Since j is recurrent, the chain will return to j infinitely often. For $n \geq 2$, let Y_n be the number of steps from the $(n-1)$ st visit to j to the n th visit to j . By the strong Markov property, each time the chain visits j it probabilistically restarts itself independently of past history. Thus, $Y_1, Y_2, \dots$ is an i.i.d. sequence with common mean $\mu_j = E(T_j | X_0 = j)$.

Assume that $\mu_j < \infty$. By the strong law of large numbers,

$$\lim_{n \rightarrow \infty} \frac{Y_1 + \dots + Y_n}{n} = \mu_j, \text{ with probability 1.} \quad (3.16)$$

Define indicator variables

$$I_m = \begin{cases} 1, & \text{if } X_m = j, \\ 0, & \text{otherwise,} \end{cases}$$

for $m = 0, 1, \dots$. Then, $\sum_{m=0}^{n-1} I_m$ is the number of visits to j in the first n steps of the chain. The long-term expected proportion of visits to j is

$$\lim_{n \rightarrow \infty} E \left(\frac{1}{n} \sum_{m=0}^{n-1} I_m \right) = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{m=0}^{n-1} E(I_m) = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{m=0}^{n-1} P_{ij}^m.$$

Since there are n visits to j by time $Y_1 + \dots + Y_n$, for large n ,

$$\frac{1}{n} \sum_{m=0}^{n-1} I_m \approx \frac{n}{Y_1 + \dots + Y_n},$$

giving that

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{m=0}^{n-1} P_{ij}^m = \lim_{n \rightarrow \infty} \frac{n}{Y_1 + \dots + Y_n} = \frac{1}{\mu_j}, \text{ with probability 1.} \quad (3.17)$$

Let $\pi_j = 1/\mu_j$, for all j . Then, $\boldsymbol{\pi} = (\pi_1, \dots, \pi_k)$ is the desired stationary distribution by the following four properties.

1. Since the μ_j are positive and finite, $\boldsymbol{\pi}$ is positive.
2. Summing the entries of $\boldsymbol{\pi}$ gives

$$\sum_{j=1}^k \pi_j = \sum_{j=1}^k \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{m=0}^{n-1} P_{ij}^m = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{m=0}^{n-1} \sum_{j=1}^k P_{ij}^m = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{m=0}^{n-1} 1 = 1.$$

Thus, $\boldsymbol{\pi}$ is a probability distribution.

3. To show π is a stationary distribution, we need to show that

$$\sum_{i=1}^k \frac{1}{\mu_i} P_{ij} = \frac{1}{\mu_j}.$$

We have that

$$\begin{aligned} \sum_{i=1}^k \frac{1}{\mu_i} P_{ij} &= \sum_{i=1}^k \left(\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{m=0}^{n-1} P_{ii}^m \right) P_{ij} \\ &= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{m=0}^{n-1} \sum_{i=1}^k P_{ii}^m P_{ij} \\ &= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{m=0}^{n-1} P_{ij}^{m+1} \\ &= \lim_{n \rightarrow \infty} \left(\frac{n+1}{n} \right) \left(\frac{1}{n+1} \right) \left(\sum_{m=0}^n P_{ij}^m - P_{ij}^0 \right) \\ &= \left(\lim_{n \rightarrow \infty} \frac{n+1}{n} \right) \left(\lim_{n \rightarrow \infty} \frac{1}{n+1} \sum_{m=0}^n P_{ij}^m \right) - \lim_{n \rightarrow \infty} \frac{P_{ij}^0}{n+1} \\ &= \frac{1}{\mu_j}. \end{aligned}$$

4. For uniqueness, assume that $\pi = \pi P$ is a stationary distribution. Then, $\pi = \pi P^n$, for all n , and thus $\pi = \lim_{n \rightarrow \infty} \pi P^n$. Pointwise,

$$\begin{aligned} \pi_j &= \lim_{n \rightarrow \infty} \sum_{i=1}^k \pi_i P_{ij}^n = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{m=0}^{n-1} \sum_{i=1}^k \pi_i P_{ij}^m \\ &= \sum_{i=1}^k \pi_i \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{m=0}^{n-1} P_{ij}^m = \sum_{i=1}^k \pi_i \frac{1}{\mu_j} = \frac{1}{\mu_j}. \end{aligned}$$

For the second equality, we use the fact that if a sequence converges to a limit, then the sequence of partial averages also converges to that limit. We have shown that if π is a stationary distribution, then necessarily $\pi_j = 1/\mu_j$.

To finish the proof of Theorem 3.6, it remains to show that for a finite Markov chain the expected return time μ_j is finite, for all j . A recurrent state j with finite expected return time is called positive recurrent. If the expected return time is infinite, the state is called null recurrent.

To show that the states of a finite irreducible Markov chain are all positive recurrent, we show that a finite irreducible Markov chain contains at least one positive recurrent state. The result will follow as a consequence of the following lemma.

Positive and Null Recurrence are Class Properties

Lemma 3.12. *All the states in a recurrent communication class are either positive recurrent or null recurrent.*

Proof of Lemma. Assume that i is a positive recurrent state. Let j be another state in the same communication class as i . Since both states communicate, there exist positive integers r and s such that $P_{ji}^r > 0$ and $P_{ij}^s > 0$. Thus,

$$\begin{aligned} \frac{1}{\mu_j} &= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{m=0}^{n-1} P_{jj}^m \\ &\geq \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{m=r+s}^{n-1} P_{ji}^r P_{ii}^{m-r-s} P_{ij}^s \\ &= \lim_{n \rightarrow \infty} \left(\frac{n-r-s}{n} \right) P_{ji}^r \left(\frac{1}{n-r-s} \sum_{m=r+s}^{n-1} P_{ii}^{m-r-s} \right) P_{ij}^s \\ &= P_{ji}^r \left(\frac{1}{\mu_i} \right) P_{ij}^s > 0. \end{aligned}$$

Hence, $\mu_j < \infty$ and j is positive recurrent. Having shown that positive recurrence is a class property, it follows that null recurrence is a class property. For if the communication class of a null recurrent state contains a positive recurrent state, then all states in the class are positive recurrent, leading to a contradiction. ■

A finite irreducible Markov chain is recurrent. We show that at least one state must be positive recurrent. If not, then all states are null recurrent, and all expected return times are infinite. See Equation (3.16). Since the Y_i are non-negative, this equation still holds with $\mu_j = +\infty$. And by Equation (3.17),

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{m=1}^n P_{ij}^m = 0, \text{ for all } i, j.$$

Sum over j to obtain

$$\begin{aligned} 0 &= \sum_{j=1}^k \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{m=1}^n P_{ij}^m = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{j=1}^k \sum_{m=1}^n P_{ij}^m \\ &= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{m=1}^n \sum_{j=1}^k P_{ij}^m = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{m=1}^n 1 = 1, \end{aligned}$$

a contradiction. Thus, a finite irreducible Markov chain contains at least one positive recurrent state. By Lemma 3.12, all states are positive recurrent. And Theorem 3.6 is proved. ■

Remark: The theorem holds for infinite irreducible chains that are positive recurrent. For infinite irreducible chains that are null recurrent, no stationary distribution exists.

Fundamental Limit Theorem for Ergodic Markov Chains

Theorem 3.8. *Let $X_0, X_1, \dots$ be an ergodic Markov chain. There exists a unique, positive, stationary distribution π , which is the limiting distribution of the chain. That is,*

$$\pi_j = \lim_{n \rightarrow \infty} P_{ij}^n, \text{ for all } i, j.$$

Proof of Theorem 3.8. Two proofs will be given. One is probabilistic, based on an elegant technique known as *coupling*. The other relies on linear algebra and the eigenstructure of the transition matrix.

Both proofs are a consequence of the following proposition, which says that finite ergodic Markov chains are precisely those chains that have regular transition matrices. Recall that a square matrix is regular if some power of the matrix has all positive entries.

Ergodic Chains and Regular Matrices

Proposition 3.13. *Assume that P is the transition matrix of a finite Markov chain. The Markov chain is ergodic if and only if P is regular.*

The proof of this proposition relies on the following lemma.

Lemma 3.14. *If i is an aperiodic state, there exists a positive integer N such that $P_{ii}^n > 0$ for all $n \geq N$.*

Proof of Lemma. The following is based on Hoel et al. (1986) and uses results from number theory.

Assume that i is an aperiodic state. Let $T = \{n > 0 : P_{ii}^n > 0\}$. By definition, $\gcd(T) = 1$. The set T is closed under addition, since if $m, n \in T$, then

$$P_{ii}^{m+n} = \sum_k P_{ik}^m P_{ki}^n \geq P_{ii}^m P_{ii}^n > 0.$$

That is, $m + n \in T$.

We claim that T contains two consecutive integers. If not, then there exists $k, m \in T$ such that $k \geq 2$, $k + m \in T$, and any two integers in T differ by at least k . Furthermore, since $\gcd(T) = 1$, there is an $n \in T$ such that k is not a divisor of n . Write $n = qk + r$, where $q \geq 0$ and $0 < r < k$. Since T is closed under addition, $(q + 1)(m + k) \in T$ and $n + (q + 1)m \in T$. Their difference is

$$(q + 1)(m + k) - n - (q + 1)m = k + qk - n = k - r > 0.$$

Thus, we have found two elements of T whose difference is positive and smaller than k , giving a contradiction.

Hence, T contains consecutive integers, say m and $m + 1$. Let $N = m^2$. We show that $n \in T$, for all $n \geq N$, which establishes the lemma. For $n \geq N$, write $n - N = qm + r$, for $q \geq 0$ and $0 \leq r < m$. Then,

$$n = m^2 + qm + r = (m - r + q)m + r(m + 1) \in T. \quad \blacksquare$$

Proof of Proposition 3.13. Assume that $\mathbf{P}$ is the transition matrix of a finite ergodic chain. Since the chain is irreducible, for states i and j there exists $m \geq 0$, such that $P_{ij}^m > 0$. The number $m = m(i, j)$ depends on i and j . Let $M^* = \max_{i,j} m(i, j)$. We can take the maximum since the chain is finite.

Since the chain is aperiodic, by Lemma 3.14, there exists $N > 0$ such that $P_{ii}^n > 0$, for all $n \geq N$. The number $N = N(i)$ depends on i . Let $N^* = \max_i N(i)$. Then, N^* does not depend on i , and for all $n \geq N^*$, $P_{ii}^n > 0$, for all i .

Let $X = M^* + N^*$. We claim that $\mathbf{P}^X$ is positive. For states i and j ,

$$P_{ij}^X = P_{ij}^{(X-m(i,j))+m(i,j)} = \sum_{t=1}^k P_{it}^{X-m(i,j)} P_{tj}^{m(i,j)} \geq P_{ii}^{X-m(i,j)} P_{ij}^{m(i,j)} > 0.$$

The last inequality is because (i) $P_{ij}^{m(i,j)} > 0$, and (ii) $P_{ii}^{X-m(i,j)} > 0$, since

$$X - m(i, j) \geq X - M^* = N^*.$$

Thus, $\mathbf{P}$ is regular.

Conversely, assume that $\mathbf{P}$ is regular. Then, $\mathbf{P}^n > 0$ for some positive integer N . Thus, all states communicate and the chain is irreducible. It suffices to show that the chain is aperiodic. Stochastic matrices have the property that if $\mathbf{P}^n$ is positive then $\mathbf{P}^{N+m}$ is positive for all $m \geq 0$, a property we leave for the reader to prove. (See Exercise 3.33.) For any state i , the set of possible return times to i includes $N, N + 1, \dots$, and $\gcd\{N, N + 1, \dots\} = 1$. That is, i is aperiodic. $\blacksquare$

Coupling Proof of Fundamental Limit Theorem

The method of proof is based on *coupling*, a probabilistic technique first introduced by the German mathematician Wolfgang Doeblin in the 1930s.

Here is a bird's-eye view. Let $X_0, X_1, \dots$ be an ergodic Markov chain on S with transition matrix $\mathbf{P}$. Since the chain is irreducible, it has a unique stationary distribution π . We need to show that for all i and j ,

$$\lim_{n \rightarrow \infty} P(X_n = j | X_0 = i) = \pi_j.$$

Consider a second chain $Y_0, Y_1, \dots$ with the same transition matrix $\mathbf{P}$, but with initial distribution π . That is, the Y chain is a stationary chain. For all $n \geq 0$, the distribution of Y_n is π .

The X and Y chains are run independently of each other. Eventually, at some finite time T , both chains will hit the same state. The chains are then *coupled* so that $X_n = Y_n$, for $n \geq T$. Since the Y chain is in stationarity, from time T onwards the X chain is also in stationarity, from which follows the result.

To elaborate, let $T = \min\{n \geq 0 : X_n = Y_n\}$ be the first time that the X and Y chains hit the same state. Define a new process by letting

$$Z_n = \begin{cases} X_n, & \text{if } n < T, \\ Y_n, & \text{if } n \geq T, \end{cases}$$

for $n \geq 0$. Then, $Z_0, Z_1, \dots$ is a Markov chain with the same transition matrix and initial distribution as the X chain.

Consider the bivariate process $(Z_0, Y_0), (Z_1, Y_1), \dots$. The bivariate process is a Markov chain on the state space $\mathcal{S} \times \mathcal{S}$ with transition matrix $\tilde{P}$ defined by

$$\tilde{P}_{(i,k),(j,l)} = P_{ij}P_{kl}.$$

The bivariate process represents a *coupling* of the original X chain with the stationary Y chain. The chains are coupled in such a way so that once both chains hit the same state then from that time onward the chains march forward in lockstep. See Example 3.34 for an illustration of the construction.

We show that the bivariate Markov chain is ergodic. Since the X chain is ergodic, P is regular, and there exists some $N > 0$ such that $P^N > 0$. For this choice of N , and for all i, j, k, l ,

$$\tilde{P}_{(i,k),(j,l)}^N = P_{ij}^N P_{kl}^N > 0.$$

Thus, $\tilde{P}$ is regular, and by Proposition 3.13 the bivariate chain is ergodic. From any state, the bivariate chain reaches any other state in finite time, with probability 1. In particular, it eventually hits a state of the form (j, j) . The event that $(Z_n, Y_n) = (j, j)$ for some j implies that the two chains have coupled by time n . It follows that T , the first time the two chains meet, is finite with probability 1. Hence,

$$\lim_{n \rightarrow \infty} P(T > n) = \lim_{n \rightarrow \infty} 1 - P(T \leq n) = 1 - P(T < \infty) = 0.$$

Consider

$$P(Y_n = j) = P(Y_n = j, T \leq n) + P(Y_n = j, T > n).$$

Taking limits, as $n \rightarrow \infty$, the left-hand side converges to π_j , and the rightmost term converges to 0. Thus, $P(Y_n = j, T \leq n) \rightarrow \pi_j$, as $n \rightarrow \infty$.

For any initial distribution,

$$\begin{aligned} P(X_n = j) &= P(Z_n = j) \\ &= P(Z_n = j, T \leq n) + P(Z_n = j, T > n) \\ &= P(Z_n = j | T \leq n)P(T \leq n) + P(Z_n = j, T > n) \\ &= P(Y_n = j | T \leq n)P(T \leq n) + P(Z_n = j, T > n) \\ &= P(Y_n = j, T \leq n) + P(Z_n = j, T > n). \end{aligned}$$

Taking limits gives

$$\lim_{n \rightarrow \infty} P(X_n = j) = \lim_{n \rightarrow \infty} P(Y_n = j, T \leq n) + \lim_{n \rightarrow \infty} P(Z_n = j, T > n) = \pi_j,$$

which completes the proof. ■

■ **Example 3.34** The coupling construction is illustrated on a two-state chain with state space $\mathcal{S} = \{a, b\}$ and transition matrix

$$P = \begin{matrix} & \begin{matrix} a & b \end{matrix} \\ \begin{matrix} a \\ b \end{matrix} & \begin{pmatrix} 9/10 & 1/10 \\ 1/5 & 4/5 \end{pmatrix} \end{matrix}.$$

The chain is ergodic with stationary distribution $\pi = (2/3, 1/3)$.

For the following simulation, the X chain was started at a , the Y chain in π . Both chains were run independently for 12 steps. The chains coupled at time $T = 8$.

Chain	0	1	2	3	4	5	6	7	8	9	10	11	12
X	a	a	a	a	a	b	b	b	a	a	b	a	a
Y	b	b	b	b	b	a	a	a	a	a	b	b	b
Z	a	a	a	a	a	b	b	b	a	a	b	b	b
(Z, Y)	ab	ab	ab	ab	ab	ba	ba	ba	aa	aa	bb	bb	bb

The transition matrix $\tilde{P}$ of the bivariate process is

$$\tilde{P} = \begin{matrix} & \begin{matrix} aa & ab & ba & bb \end{matrix} \\ \begin{matrix} aa \\ ab \\ ba \\ bb \end{matrix} & \begin{pmatrix} (9/10)^2 & (9/10)(1/10) & (1/10)(9/10) & (1/10)^2 \\ (9/10)(1/5) & (9/10)(4/5) & (1/10)(1/5) & (1/10)(4/5) \\ (1/5)(9/10) & (1/5)(1/10) & (4/5)(9/10) & (4/5)(1/10) \\ (1/5)^2 & (1/5)(4/5) & (4/5)(1/5) & (4/5)^2 \end{pmatrix} \end{matrix}$$

$$= \begin{matrix} & \begin{matrix} aa & ab & ba & bb \end{matrix} \\ \begin{matrix} aa \\ ab \\ ba \\ bb \end{matrix} & \begin{pmatrix} 81/100 & 9/100 & 9/100 & 1/100 \\ 9/50 & 36/50 & 1/50 & 4/50 \\ 9/50 & 1/50 & 36/50 & 4/50 \\ 1/25 & 4/25 & 4/25 & 16/25 \end{pmatrix} \end{matrix}.$$
■

Linear Algebra Proof of Fundamental Limit Theorem

Asymptotic properties of Markov chains are related to the eigenstructure of the transition matrix.

Eigenvalues of a Stochastic Matrix

Lemma 3.15. *A stochastic matrix $\mathbf{P}$ has an eigenvalue $\lambda^* = 1$. All other eigenvalues λ of $\mathbf{P}$ are such that $|\lambda| \leq 1$.*

If $\mathbf{P}$ is a regular matrix, then the inequality is strict. That is, $|\lambda| < 1$ for all $\lambda \neq \lambda^$.*

Proof. Let $\mathbf{P}$ be a $k \times k$ stochastic matrix. Since the rows of $\mathbf{P}$ sum to 1, we have that $\mathbf{P}\mathbf{1} = \mathbf{1}$, where $\mathbf{1}$ is a column vector of all 1s. Thus, $\lambda^* = 1$ is an eigenvalue of $\mathbf{P}$. Let λ be any other eigenvalue of $\mathbf{P}$ with corresponding eigenvector $\mathbf{z}$. Let $|z_m| = \max_{1 \leq i \leq k} |z_i|$. That is, z_m is the component of $\mathbf{z}$ of maximum absolute value. Then,

$$|\lambda||z_m| = |\lambda z_m| = |(Pz)_m| = \left| \sum_{i=1}^k P_{mi} z_i \right| \leq \sum_{i=1}^k P_{mi} |z_i| \leq |z_m| \sum_{i=1}^k P_{mi} = |z_m|.$$

Thus, $|\lambda| \leq 1$.

Assume that $\mathbf{P}$ is regular. Then, $\mathbf{P}^N > 0$, for some $N > 0$. Since $\mathbf{P}^N$ is a stochastic matrix, the first part of the lemma holds for $\mathbf{P}^N$. If λ is an eigenvalue of $\mathbf{P}$, then λ^N is an eigenvalue of $\mathbf{P}^N$. Let $\mathbf{x}$ be the corresponding eigenvector, with $|x_m| = \max_{1 \leq i \leq k} |x_i|$. Then,

$$|\lambda|^N |x_m| = |(P^N x)_m| = \left| \sum_{i=1}^k P_{mi}^N x_i \right| \leq \sum_{i=1}^k P_{mi}^N |x_i| \leq |x_m| \sum_{i=1}^k P_{mi}^N = |x_m|.$$

Since the entries of $\mathbf{P}^N$ are all positive, the last inequality is an equality only if $|x_1| = |x_2| = \dots = |x_k|$. And the first inequality is an equality only if $x_1 = \dots = x_k$. But the constant vector whose components are all the same is an eigenvector associated with the eigenvalue 1. Hence, if $\lambda \neq 1$, one of the inequalities is strict. Thus, $|\lambda|^N < 1$, and the result follows. ■

The fundamental limit theorem for ergodic Markov chains is a consequence of the Perron–Frobenius theorem for positive matrices.

Perron–Frobenius Theorem

Theorem 3.16. *Let $\mathbf{M}$ be a $k \times k$ positive matrix. Then, the following statements hold.*

1. *There is a positive real number λ^* which is an eigenvalue of $\mathbf{M}$. For all other eigenvalues λ of $\mathbf{M}$, $|\lambda| < \lambda^*$. The eigenvalue λ^* is called the Perron–Frobenius eigenvalue.*
2. *The eigenspace of eigenvectors associated with λ^* is one-dimensional.*

3. There exists a positive right eigenvector $\mathbf{v}$ associated with λ^* , and a positive left eigenvector $\mathbf{w}$ associated with λ^* . Furthermore,

$$\lim_{n \rightarrow \infty} \frac{1}{(\lambda^*)^n} \mathbf{M}^n = \mathbf{v} \mathbf{w}^T,$$

where the eigenvectors are normalized so that $\mathbf{w}^T \mathbf{v} = 1$.

The proof of the Perron–Frobenius theorem can be found in many advanced linear algebra textbooks, including Horn and Johnson (1990).

For an ergodic Markov chain, the transition matrix $\mathbf{P}$ is regular and $\mathbf{P}^N$ is a positive matrix for some integer N . The Perron–Frobenius theorem applies. The Perron–Frobenius eigenvalue of $\mathbf{P}^N$ is $\lambda^* = 1$, with associated right eigenvector $\mathbf{v} = \mathbf{1}$, and associated left eigenvector $\mathbf{w}$.

If $\lambda^* = 1$ is an eigenvalue of $\mathbf{P}^N$, then $(\lambda^*)^{1/N} = 1$ is an eigenvalue of $\mathbf{P}$, with associated right and left eigenvectors $\mathbf{v}$ and $\mathbf{w}$, respectively. Normalizing $\mathbf{w}$ so that its components sum to 1 gives the unique, positive stationary distribution $\boldsymbol{\pi}$, which is the limiting distribution of the chain. The limiting matrix $\mathbf{v} \mathbf{w}^T$ is a stochastic matrix all of whose rows are equal to $\mathbf{w}^T$. ■

EXERCISES

- 3.1 Consider a Markov chain with transition matrix

$$\mathbf{P} = \begin{pmatrix} 1/2 & 1/4 & 0 & 1/4 \\ 0 & 1/2 & 1/2 & 0 \\ 1/4 & 1/4 & 1/2 & 0 \\ 0 & 1/4 & 1/2 & 1/4 \end{pmatrix}.$$

Find the stationary distribution. Do not use technology.

- 3.2 A stochastic matrix is called *doubly stochastic* if its rows and columns sum to 1. Show that a Markov chain whose transition matrix is doubly stochastic has a stationary distribution, which is uniform on the state space.

- 3.3 Determine which of the following matrices are regular.

$$\mathbf{P} = \begin{pmatrix} 0.4 & 0.6 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{pmatrix}, \quad \mathbf{Q} = \begin{pmatrix} 0 & 1 \\ p & 1-p \end{pmatrix}, \quad \mathbf{R} = \begin{pmatrix} 0 & 1 & 0 \\ 0.25 & 0.5 & 0.25 \\ 1 & 0 & 0 \end{pmatrix}.$$

- 3.4 Consider a Markov chain with transition matrix

$$\mathbf{P} = \begin{pmatrix} 1-a & a & 0 \\ 0 & 1-b & b \\ c & 0 & 1-c \end{pmatrix},$$

where $0 < a, b, c < 1$. Find the stationary distribution.

3.5 A Markov chain has transition matrix

$$P = \begin{pmatrix} 0 & 1/4 & 0 & 0 & 3/4 \\ 3/4 & 0 & 0 & 0 & 1/4 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 1/4 & 3/4 & 0 & 0 & 0 \end{pmatrix}.$$

- Describe the set of stationary distributions for the chain.
- Use technology to find $\lim_{n \rightarrow \infty} P^n$. Explain the long-term behavior of the chain.
- Explain why the chain does not have a limiting distribution, and why this does not contradict the existence of a limiting matrix as shown in (b).

3.6 Consider a Markov chain with transition matrix

$$P = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 & 5 & \dots \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \\ \vdots \end{matrix} & \begin{pmatrix} 1/2 & 1/2 & 0 & 0 & 0 & \dots \\ 2/3 & 0 & 1/3 & 0 & 0 & \dots \\ 3/4 & 0 & 0 & 1/4 & 0 & \dots \\ 4/5 & 0 & 0 & 0 & 1/5 & \dots \\ 5/6 & 0 & 0 & 0 & 0 & \dots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix} \end{matrix},$$

defined by

$$P_{ij} = \begin{cases} i/(i+1), & \text{if } j = 1, \\ 1/(i+1), & \text{if } j = i+1, \\ 0, & \text{otherwise.} \end{cases}$$

- Does the chain have a stationary distribution? If yes, exhibit the distribution. If no, explain why.
- Classify the states of the chain.
- Repeat part (a) with the row entries of P switched. That is, let

$$P_{ij} = \begin{cases} 1/(i+1), & \text{if } j = 1, \\ i/(i+1), & \text{if } j = i+1, \\ 0, & \text{otherwise.} \end{cases}$$

3.7 A Markov chain has n states. If the chain is at state k , a coin is flipped, whose heads probability is p . If the coin lands heads, the chain stays at k . If the coin lands tails, the chain moves to a different state uniformly at random. Exhibit the transition matrix and find the stationary distribution.

3.8 Let

$$P_1 = \begin{pmatrix} 1/4 & 3/4 \\ 1/2 & 1/2 \end{pmatrix} \quad \text{and} \quad P_2 = \begin{pmatrix} 1/5 & 4/5 \\ 4/5 & 1/5 \end{pmatrix}.$$

Consider a Markov chain on four states whose transition matrix is given by the block matrix

$$P = \begin{pmatrix} P_1 & \mathbf{0} \\ \mathbf{0} & P_2 \end{pmatrix}.$$

- (a) Does the Markov chain have a unique stationary distribution? If so, find it.
- (b) Does $\lim_{n \rightarrow \infty} P^n$ exist? If so, find it.
- (c) Does the Markov chain have a limiting distribution? If so, find it.

3.9 Let P be a stochastic matrix.

- (a) If P is regular, is P^2 regular?
- (b) If P is the transition matrix of an irreducible Markov chain, is P^2 the transition matrix of an irreducible Markov chain?

3.10 A Markov chain has transition matrix P and limiting distribution π . Further assume that π is the initial distribution of the chain. That is, the chain is in stationarity. Find the following:

- (a) $\lim_{n \rightarrow \infty} P(X_n = j | X_{n-1} = i)$
- (b) $\lim_{n \rightarrow \infty} P(X_n = j | X_0 = i)$
- (c) $\lim_{n \rightarrow \infty} P(X_{n+1} = k, X_n = j | X_0 = i)$
- (d) $\lim_{n \rightarrow \infty} P(X_0 = j | X_n = i)$

3.11 Consider a simple symmetric random walk on $\{0, 1, \dots, k\}$ with reflecting boundaries. If the walk is at state 0, it moves to 1 on the next step. If the walk is at k , it moves to $k - 1$ on the next step. Otherwise, the walk moves left or right, with probability $1/2$.

- (a) Find the stationary distribution.
- (b) For $k = 1,000$, if the walk starts at 0, how many steps will it take, on average, for the walk to return to 0?

3.12 A Markov chain has transition matrix

$$P = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 1/4 & 0 & 3/4 \\ 1/2 & 0 & 1/2 & 0 \\ 0 & 3/4 & 0 & 1/4 \end{pmatrix}.$$

Find the set of all stationary distributions.

3.13 Find the communication classes of a Markov chain with transition matrix

$$P = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 & 5 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{matrix} & \begin{pmatrix} 1/2 & 0 & 0 & 0 & 1/2 \\ 1/3 & 1/2 & 1/6 & 0 & 0 \\ 0 & 1/4 & 0 & 1/2 & 1/4 \\ 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 0 \end{pmatrix} \end{matrix}.$$

Rewrite the transition matrix in canonical form.

3.14 The California Air Resources Board warns the public when smog levels are above certain thresholds. Days when the board issues warnings are called *episode* days. Lin (1981) models the daily sequence of episode and nonepisode days as a Markov chain with transition matrix

$$P = \begin{matrix} & \begin{matrix} \text{Nonepisode} & \text{Episode} \end{matrix} \\ \begin{matrix} \text{Nonepisode} \\ \text{Episode} \end{matrix} & \begin{pmatrix} 0.77 & 0.23 \\ 0.24 & 0.76 \end{pmatrix} \end{matrix}.$$

- (a) What is the long-term probability that a given day will be an episode day?
 - (b) Over a year's time about how many days are expected to be episode days?
 - (c) In the long-term, what is the average number of days that will transpire between episode days?
- 3.15** On a chessboard a single random knight performs a simple random walk. From any square, the knight chooses from among its permissible moves with equal probability. If the knight starts on a corner, how long, on average, will it take to return to that corner?
- 3.16** As in the previous exercise, find the expected return time from a corner square for the following chess pieces: (i) queen, (ii) rook, (iii) king, (iv) bishop. Order the pieces by which pieces return quickest.
- 3.17** Consider a Markov chain with transition matrix

$$P = \begin{pmatrix} 1/2 & 1/2 \\ 0 & 1 \end{pmatrix}.$$

Obtain a closed form expression for P^n . Exhibit the matrix $\sum_{n=0}^{\infty} P^n$ (some entries may be $+\infty$). Explain what this shows about the recurrence and transience of the states.

- 3.18** Use first-step analysis to find the expected return time to state b for the Markov chain with transition matrix

$$P = \begin{matrix} & \begin{matrix} a & b & c \end{matrix} \\ \begin{matrix} a \\ b \\ c \end{matrix} & \begin{pmatrix} 1/2 & 1/2 & 0 \\ 1/4 & 0 & 3/4 \\ 1/2 & 1/2 & 0 \end{pmatrix} \end{matrix}.$$

- 3.19** Consider random walk on the graph in Figure 3.15. Use first-step analysis to find the expected time to hit d for the walk started in a . (*Hint: By exploiting symmetries in the graph, the solution can be found by solving a 3×3 linear system.*)

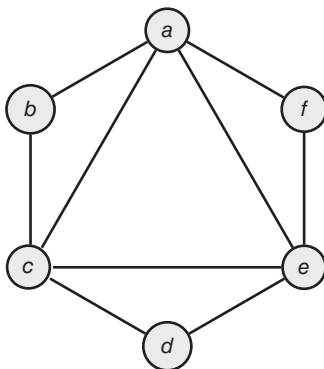


Figure 3.15

- 3.20** Show that simple symmetric random walk on $\mathbb{Z}^2$, that is, on the integer points in the plane, is recurrent. As in the one-dimensional case, consider the origin.
- 3.21** Show that simple symmetric random walk on $\mathbb{Z}^3$ is transient. As in the one-dimensional case, consider the origin and show

$$\begin{aligned} P_{00}^{2n} &= \frac{1}{6^{2n}} \sum_{0 \leq j+k \leq n} \frac{(2n)!}{j!j!k!k!(n-j-k)!(n-j-k)!}, \\ &\leq \frac{1}{2^{2n}} \binom{2n}{n} \left(\frac{1}{3^n} \frac{n!}{(n/3)!(n/3)!(n/3)!} \right). \end{aligned}$$

Then, use Stirling's approximation.

- 3.22** Consider the general two-state chain

$$P = \begin{matrix} & \begin{matrix} 1 & 2 \end{matrix} \\ \begin{matrix} 1 \\ 2 \end{matrix} & \begin{pmatrix} 1-p & p \\ q & 1-q \end{pmatrix} \end{matrix},$$

where p and q are not both 0. Let T be the first return time to state 1, for the chain started in 1.

- Show that $P(T \geq n) = p(1 - q)^{n-2}$, for $n \geq 2$.
- Find $E(T)$ and verify that $E(T) = 1/\pi_1$, where π is the stationary distribution of the chain.

3.23 Consider a k -state Markov chain with transition matrix

$$P = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & \cdots & k-2 & k-1 & k \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ \vdots \\ k-2 \\ k-1 \\ k \end{matrix} & \begin{pmatrix} 1/k & 1/k & 1/k & \cdots & 1/k & 1/k & 1/k \\ 1 & 0 & 0 & \cdots & 0 & 0 & 0 \\ 0 & 1 & 0 & \cdots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 0 & 0 & 0 \\ 0 & 0 & 0 & \cdots & 1 & 0 & 0 \\ 0 & 0 & 0 & \cdots & 0 & 1 & 0 \end{pmatrix} \end{matrix}.$$

Show that the chain is ergodic and find the limiting distribution.

- 3.24** Show that the stationary distribution for the modified Ehrenfest chain of Example 3.19 is binomial with parameters N and $1/2$.
- 3.25** Read about the Bernoulli–Laplace model of diffusion in Exercise 2.12.
 - Find the stationary distribution for the cases $k = 2$ and $k = 3$.
 - For general k , show that $\pi_j = \binom{k}{j}^2 / \binom{2k}{k}$, for $j = 0, 1, \dots, k$, satisfies the equations for the stationary distribution and is thus the unique limiting distribution of the chain.
- 3.26** Assume that $(p_1, \dots, p_k)$ is a probability vector. Let P be a $k \times k$ transition matrix defined by

$$P_{ij} = \begin{cases} p_j, & \text{if } i = 1, \dots, k-1, \\ 0, & \text{if } i = k, j < k, \\ 1, & \text{if } i = k, j = k. \end{cases}$$

Describe all the stationary distributions for P .

- 3.27** Sinclair (2005). Consider the infinite Markov chain on the non-negative integers described by Figure 3.16.
 - Show that the chain is irreducible and aperiodic.
 - Show that the chain is recurrent by computing the first return time to 0 for the chain started at 0.
 - Show that the chain is null recurrent.

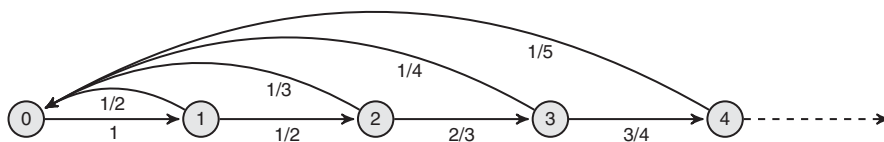


Figure 3.16

3.28 Consider a Markov chain with transition matrix

$$P = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \\ 6 \\ 7 \end{matrix} & \begin{pmatrix} 1/3 & 1/3 & 1/3 & 0 & 0 & 0 & 0 \\ 2/3 & 0 & 1/3 & 0 & 0 & 0 & 0 \\ 0 & 2/3 & 1/3 & 0 & 0 & 0 & 0 \\ 0 & 1/4 & 0 & 1/2 & 0 & 1/4 & 0 \\ 0 & 0 & 1/4 & 0 & 1/4 & 0 & 1/2 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 1/8 & 1/8 & 1/8 & 1/8 & 1/4 & 1/4 \end{pmatrix} \end{matrix}.$$

Identify the communication classes. Classify the states as recurrent or transient.

For all i and j , determine $\lim_{n \rightarrow \infty} P_{ij}^n$ without using technology.

3.29 Consider a Markov chain with transition matrix

$$P = \begin{matrix} & \begin{matrix} a & b & c & d & e & f & g \end{matrix} \\ \begin{matrix} a \\ b \\ c \\ d \\ e \\ f \\ g \end{matrix} & \begin{pmatrix} 0.6 & 0.2 & 0 & 0 & 0.2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0.5 & 0.5 \\ 0 & 0 & 0.3 & 0.7 & 0 & 0 & 0 \\ 0 & 0.3 & 0.4 & 0.3 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0.1 & 0 & 0 & 0 & 0 & 0.9 \\ 0 & 0.2 & 0 & 0 & 0 & 0.8 & 0 \end{pmatrix} \end{matrix}.$$

Identify the communication classes. Classify the states as recurrent or transient, and determine the period of each state.

3.30 A graph is *bipartite* if the vertex set can be colored with two colors black and white such that every edge in the graph joins a black vertex and a white vertex. See Figure 3.7(a) for an example of a bipartite graph. Show that for simple random walk on a connected graph, the walk is periodic if and only if the graph is bipartite.

3.31 For the network graph in Figure 3.17, find the PageRank for the nodes of the network using a damping factor of $p = 0.90$. See Example 3.21.

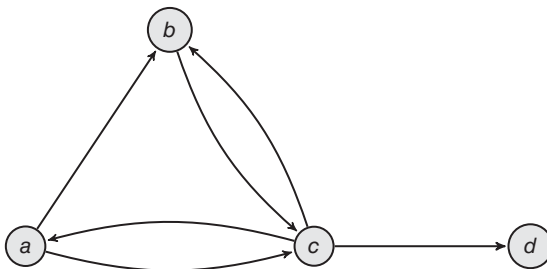


Figure 3.17

3.32 Let $X_0, X_1, \dots$ be an ergodic Markov chain with transition matrix $\mathbf{P}$ and stationary distribution π . Define the bivariate process $Z_n = (X_n, X_{n-1})$, for $n \geq 1$, with $Z_0 = (X_0, X_0)$.

- Give an intuitive explanation for why $Z_0, Z_1, \dots$ is a Markov chain.
- Determine the transition probabilities in terms of $\mathbf{P}$. That is, find

$$P(Z_n = (i, j) | Z_{n-1} = (s, t)).$$

- Find the limiting distribution.

3.33 Assume that $\mathbf{P}$ is a stochastic matrix. Show that if $\mathbf{P}^N$ is positive, then $\mathbf{P}^{N+m}$ is positive for all $m \geq 0$.

3.34 Let $\mathbf{P}$ be the transition matrix of an irreducible, but not necessarily ergodic, Markov chain. For $0 < p < 1$, let

$$\tilde{\mathbf{P}} = p\mathbf{P} + (1 - p)\mathbf{I},$$

where $\mathbf{I}$ is the identity matrix. Show that $\tilde{\mathbf{P}}$ is a stochastic matrix for an ergodic Markov chain with the same stationary distribution as $\mathbf{P}$. Give an intuitive description for how the $\tilde{\mathbf{P}}$ chain evolves compared to the $\mathbf{P}$ -chain.

3.35 Let $\mathbf{Q}$ be a $k \times k$ stochastic matrix. Let $\mathbf{A}$ be a $k \times k$ matrix each of whose entries is $1/k$. For $0 < p < 1$, let

$$\mathbf{P} = p\mathbf{Q} + (1 - p)\mathbf{A}.$$

Show that $\mathbf{P}$ is the transition matrix for an ergodic Markov chain.

3.36 Let $X_0, X_1, \dots$ be an ergodic Markov chain on $\{1, \dots, k\}$ with stationary distribution π . Assume that the chain is in stationarity.

- Find $\text{Cov}(X_m, X_{m+n})$.
- Find $\lim_{n \rightarrow \infty} \text{Cov}(X_m, X_{m+n})$.

3.37 Show that all two-state Markov chains, except for the trivial chain whose transition matrix is the identity matrix, are time reversible.

3.38 You throw five dice and set aside those dice that are sixes. Throw the remaining dice and again set aside the sixes. Continue until you get all sixes.

(a) Exhibit the transition matrix for the associated Markov chain, where X_n is the number of sixes after n throws. See also Exercise 2.11.

(b) How many turns does it take, on average, before you get all sixes?

3.39 Show that if $X_0, X_1, \dots$ is reversible, then for the chain in stationarity

$$P(X_0 = i_0, X_1 = i_1, \dots, X_n = i_n) = P(X_n = i_0, X_{n-1} = i_1, \dots, X_0 = i_n),$$

for all $i_0, i_1, \dots, i_n$.

3.40 Consider a *biased random walk* on the n -cycle, which moves one direction with probability p and the other direction with probability $1 - p$. Determine whether the walk is time reversible.

3.41 Show that the Markov chain with transition matrix

$$P = \begin{matrix} & \begin{matrix} a & b & c & d \end{matrix} \\ \begin{matrix} a \\ b \\ c \\ d \end{matrix} & \begin{pmatrix} 1/6 & 1/6 & 0 & 2/3 \\ 1/5 & 2/5 & 2/5 & 0 \\ 0 & 1/3 & 1/6 & 1/2 \\ 4/9 & 0 & 1/3 & 2/9 \end{pmatrix} \end{matrix}$$

is reversible. The chain can be described by a random walk on a weighted graph. Exhibit the graph such that all the weights are integers.

3.42 Consider random walk on $\{0, 1, 2, \dots\}$ with one reflecting boundary. If the walk is at 0, it moves to 1 on the next step. Otherwise, it moves left, with probability p , or right, with probability $1 - p$. For what values of p is the chain reversible? For such p , find the stationary distribution.

3.43 A Markov chain has transition matrix

$$P = \begin{pmatrix} 1/2 & 1/4 & 0 & 1/4 \\ p & 0 & 1-p & 0 \\ 0 & 1/4 & 1/2 & 1/4 \\ q & 0 & 1-q & 0 \end{pmatrix}.$$

(a) For what values of p and q is the chain ergodic?

(b) For what values of p and q is the chain reversible?

3.44 Markov chains are used to model nucleotide substitutions and mutations in DNA sequences. Kimura gives the following transition matrix for such a model.

$$P = \begin{matrix} & \begin{matrix} a & g & c & t \end{matrix} \\ \begin{matrix} a \\ g \\ c \\ t \end{matrix} & \begin{pmatrix} 1-p-2r & p & r & r \\ p & 1-p-2r & r & r \\ q & q & 1-p-2q & p \\ q & q & p & 1-p-2q \end{pmatrix} \end{matrix}.$$

Find a vector $\mathbf{x}$ that satisfies the detailed-balance equations. Show that the chain is reversible and find the stationary distribution. Confirm your result for the case $p = 0.1$, $q = 0.2$, and $r = 0.3$.

3.45 If $\mathbf{P}$ is the transition matrix of a reversible Markov chain, show that $\mathbf{P}^2$ is, too. Conclude that $\mathbf{P}^n$ is the transition matrix of a reversible Markov chain for all $n \geq 1$.

3.46 Given a Markov chain with transition matrix $\mathbf{P}$ and stationary distribution $\boldsymbol{\pi}$, the *time reversal* is a Markov chain with transition matrix $\tilde{\mathbf{P}}$ defined by

$$\tilde{P}_{ij} = \frac{\pi_j P_{ji}}{\pi_i}, \text{ for all } i, j.$$

(a) Show that a Markov chain with transition matrix $\mathbf{P}$ is reversible if and only if $\mathbf{P} = \tilde{\mathbf{P}}$.

(b) Show that the time reversal Markov chain has the same stationary distribution as the original chain.

3.47 Consider a Markov chain with transition matrix

$$\mathbf{P} = \begin{matrix} & \begin{matrix} 1 & 2 & 3 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \end{matrix} & \begin{pmatrix} 1/3 & 0 & 2/3 \\ 1/2 & 1/2 & 0 \\ 1/6 & 1/3 & 1/2 \end{pmatrix} \end{matrix}.$$

Find the transition matrix of the time reversal chain (see Exercise 3.46).

3.48 Consider a Markov chain with transition matrix

$$\mathbf{P} = \begin{matrix} & \begin{matrix} a & b & c \end{matrix} \\ \begin{matrix} a \\ b \\ c \end{matrix} & \begin{pmatrix} 1-\alpha & \alpha & 0 \\ 0 & 1-\beta & \beta \\ \gamma & 0 & 1-\gamma \end{pmatrix} \end{matrix},$$

where $0 < \alpha, \beta, \gamma < 1$. Find the transition matrix of the time reversal chain (see Exercise 3.46).

3.49 Consider an absorbing chain with t transient and $k - t$ absorbing states. For transient state i and absorbing state j , let B_{ij} denote the probability starting at i that the chain is absorbed in j . Let $\mathbf{B}$ be the resulting $t \times (k - t)$ matrix. By first-step analysis show that $\mathbf{B} = (\mathbf{I} - \mathbf{Q})^{-1}\mathbf{R}$.

3.50 Consider the following method for shuffling a deck of cards. Pick two cards from the deck uniformly at random and then switch their positions. If the same two cards are chosen, the deck does not change. This is called the *random transpositions* shuffle.

(a) Argue that the chain is ergodic and the stationary distribution is uniform.

- (b) Exhibit the 6×6 transition matrix for a three-card deck.
- (c) How many shuffles does it take on average to reverse the original order of the deck of cards?
- 3.51** A deck of k cards is shuffled by the *top-to-random* method: the top card is placed in a uniformly random position in the deck. (After one shuffle, the top card stays where it is with probability $1/k$.) Assume that the top card of the deck is the ace of hearts. Consider a Markov chain where X_n is the position of the ace of hearts after n top-to-random shuffles, with $X_0 = 1$. The state space is $\{1, \dots, k\}$. Assume that $k = 6$.
- (a) Exhibit the transition matrix and find the expected number of shuffles for the ace of hearts to return to the top of the deck.
- (b) Find the expected number of shuffles for the bottom card to reach the top of the deck.
- 3.52** The board for a modified Snakes and Ladder game is shown in Figure 3.18. The game is played with a tetrahedron (four-faced) die.
- (a) Find the expected length of the game.
- (b) Assume that the player is on square 6. Find the probability that they will find themselves on square 3 before finishing the game.

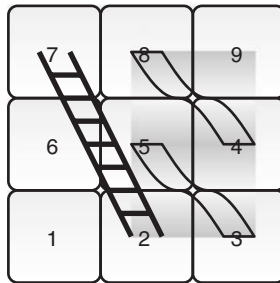


Figure 3.18

- 3.53** When an NFL football game ends in a tie, under *sudden-death* overtime the two teams play at most 15 extra minutes and the team that scores first wins the game. A Markov chain analysis of sudden-death is given in Jones (2004). Assuming two teams A and B are evenly matched, a four-state absorbing Markov chain is given with states PA : team A gains possession, PB : team B gains possession, A : A wins, and B : B wins. The transition matrix is

$$P = \begin{matrix} & \begin{matrix} PA & PB & A & B \end{matrix} \\ \begin{matrix} PA \\ PB \\ A \\ B \end{matrix} & \begin{pmatrix} 0 & 1-p & p & 0 \\ 1-p & 0 & 0 & p \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \end{matrix},$$

where p is the probability that a team scores when it has the ball. Which team first receives the ball in overtime is decided by a coin flip.

- (a) If team A receives the ball in overtime, find the probability that A wins.
- (b) An alternate overtime procedure is the *first-to-six rule*, where the first time to score six points in overtime wins the game. Consider two evenly matched teams. Let α be the probability that a team scores a touchdown (six points). Let β be the probability that a team scores a field goal (three points). Assume for simplicity that touchdowns and field goals are the only way points can be scored. Develop a 10-state Markov chain model for overtime play.
- (c) For the 2002 regular NFL season, there were 6,049 possessions, 1,270 touchdowns, and 737 field goals. Using these data compare the probability that A wins the game for each of the two overtime procedures.

- 3.54** A mouse is placed in the maze in Figure 3.19 starting in box A. A piece of cheese is put in box I. From each room the mouse moves to an adjacent room through an open door, choosing from the available doors with equal probability.
- (a) How many rooms, on average, will the mouse visit before it finds the cheese?
 - (b) How many times, on average, will the mouse visit room A before it finds the cheese?

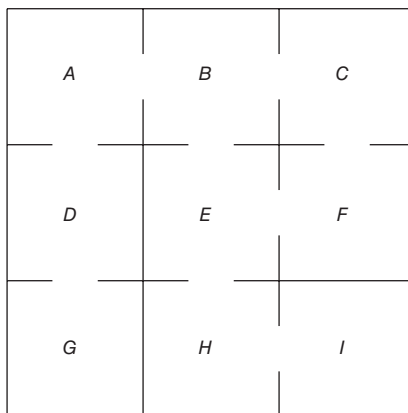


Figure 3.19 Mouse in a maze.

- 3.55** In a sequence of fair coin flips, how many flips, on average, are required to first see the pattern H-H-T-H?
- 3.56** A biased coin has heads probability $1/3$ and tails probability $2/3$. If the coin is tossed repeatedly, find the expected number of flips required until the pattern H-T-T-H-H appears.
- 3.57** In repeated coin flips, consider the set of all three-element patterns:

$$\{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}.$$

Which patterns take the longest time, on average, to appear in repeated sampling? Which take the shortest?

3.58 A sequence of 0s and 1s is generated by a Markov chain with transition matrix

$$P = \begin{matrix} & \begin{matrix} 0 & 1 \end{matrix} \\ \begin{matrix} 0 \\ 1 \end{matrix} & \begin{pmatrix} 1/4 & 3/4 \\ 3/4 & 1/4 \end{pmatrix} \end{matrix}.$$

The first element of the sequence is decided by a fair coin flip. On average, how many steps are required for the pattern 0-0-1-1 to first appear?

3.59 Consider random walk on the weighted graph in Figure 3.20.

- If the walk starts in a , find the expected number of steps to return to a .
- If the walk starts in a , find the expected number of steps to first hit b .
- If the walk starts in a , find the probability that the walk hits b before c .

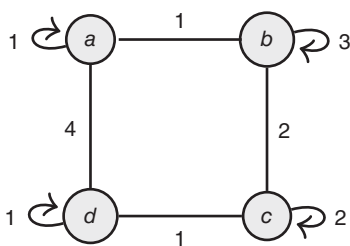


Figure 3.20

- 3.60** For a Markov chain started in state i , let T denote the *fifth time* the chain visits state i . Is T a stopping time? Explain.
- 3.61** Consider the weather Markov chain $X_0, X_1, \dots$ of Example 2.3. Let T be the first time that it rains for 40 days in a row. Is $X_T, X_{T+1}, X_{T+2}, \dots$ a Markov chain? Explain.
- 3.62** Let S be a random variable that is constant, with probability 1, where that constant is some positive integer. Show that S is a stopping time. Conclude that the Markov property follows from the strong Markov property.
- 3.63** R : Hourly wind speeds in a northwestern region of Turkey are modeled by a Markov chain in Sahin and Sen (2001). Seven wind speed levels are the states

of the chain. The transition matrix is

$$P = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \\ 6 \\ 7 \end{matrix} & \begin{pmatrix} 0.756 & 0.113 & 0.129 & 0.002 & 0 & 0 & 0 \\ 0.174 & 0.821 & 0.004 & 0.001 & 0 & 0 & 0 \\ 0.141 & 0.001 & 0.776 & 0.082 & 0 & 0 & 0 \\ 0.003 & 0 & 0.192 & 0.753 & 0.052 & 0 & 0 \\ 0 & 0 & 0.002 & 0.227 & 0.735 & 0.036 & 0 \\ 0 & 0 & 0 & 0.007 & 0.367 & 0.604 & 0.022 \\ 0 & 0 & 0 & 0 & 0.053 & 0.158 & 0.789 \end{pmatrix} \end{pmatrix}.$$

- (a) Find the limiting distribution by (i) taking high matrix powers, and (ii) using the `stationary` command in the **utilities.R** file. How often does the highest wind speed occur? How often does the lowest speed occur?
 - (b) Simulate the chain for 100,000 steps and estimate the proportion of times that the chain visits each state.
- 3.64** R: The evolution of forest ecosystems in the United States and Canada is studied in Strigul et al. (2012) using Markov chains. Five-year changes in the state of the forest soil are modeled with a 12-state Markov chain. The transition matrix can be found in the R script file **forest.R**. About how many years does it take for the ecosystem to move from state 1 to state 12?
- 3.65** R: Simulate the gambler's ruin problem for a gambler who starts with \$15 and quits when he reaches \$50 or goes bust. Use your code to simulate the probability of eventual ruin and compare to the exact probability.
- 3.66** R: Simulate the expected hitting time for the random walk on the hexagon in Exercise 3.19.
- 3.67** R: Simulate the dice game of Exercise 3.38. Verify numerically the theoretical expectation for the number of throws needed to get all sixes.
- 3.68** R : Write a function `reversal(mat)`, whose input is the transition matrix of an irreducible Markov chain and whose output is the transition matrix of the reversal chain.
- 3.69** R : Make up your own board game which can be modeled as a Markov chain. Ask interesting questions and answer them by simulation and/or an exact analysis.

4

BRANCHING PROCESSES

Every moment dies a man, every moment one is born.

—Alfred Tennyson, *The Vision of Sin*

Every moment dies a man, every moment one and one-sixteenth is born.

—Mathematician Charles Babbage in a letter to Alfred Tennyson
suggesting a change “in your otherwise beautiful poem.”

4.1 INTRODUCTION

Branching processes are a class of stochastic processes that model the growth of populations. They are widely used in biology and epidemiology to study the spread of infectious diseases and epidemics. Applications include nuclear chain reactions and the spread of computer software viruses. Their original motivation was to study the extinction of family surnames, an issue of concern to the Victorian aristocracy in 19th century Britain.

In 1873, the British statistician Sir Francois Galton posed the following question in the *Educational Times*.

Problem 4001: A large nation, of whom we will only concern ourselves with adult males, N in number, and who each bear separate surnames colonize a district. Their law of population is such that, in each generation, a_0 percent of the adult males have no male children who reach adult life; a_1 have one such male child; a_2 have two; and so on up to

a_5 who have five. Find (1) what proportion of their surnames will have become extinct after r generations; and (2) how many instances there will be of the surname being held by m persons.

The Reverend Henry William Watson replied with a solution. The study of branching processes grew out of Watson and Galton's collaboration. Their results were independently discovered by the French statistician Irénée-Jules Bienaymé. The basic branching process model is sometimes called a Bienaymé–Galton–Watson process.

We use the imagery of populations, generations, children, and offspring. Assume that we have a population of individuals, each of which independently produces a random number of children according to a probability distribution $\mathbf{a} = (a_0, a_1, a_2, \dots)$. That is, an individual gives birth to k children with probability a_k , for $k \geq 0$, independent of other individuals. Call $\mathbf{a}$ the *offspring distribution*.

The population grows or declines from generation to generation. Let Z_n be the size (e.g., number of individuals) of the n th generation, for $n \geq 0$. Assume $Z_0 = 1$. That is, the population starts with one individual. The sequence $Z_0, Z_1, \dots$ is a *branching process*. See Figure 4.1 for a realization of such a process through three generations.

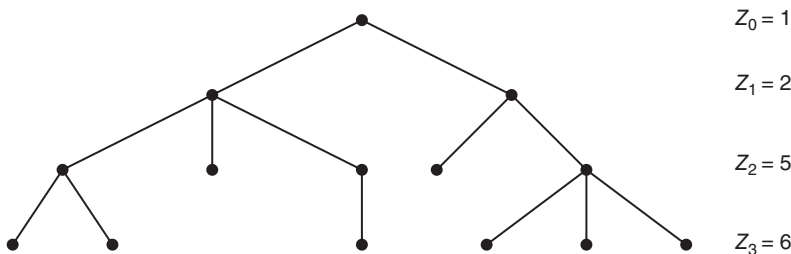


Figure 4.1 Branching process.

A branching process is a Markov chain since the size of a generation only depends on the size of the previous generation and the number of their offspring. If Z_n is given, then the size of the next generation Z_{n+1} is independent of $Z_0, \dots, Z_{n-1}$.

Assume $0 < a_0 < 1$. If $a_0 = 0$, then the population only grows and 0 is not in the state space. If $a_0 = 1$, then $Z_n = 0$, for all $n \geq 1$. We also assume that there is positive probability that an individual gives birth to more than one offspring, that is, $a_0 + a_1 < 1$.

Galton's first question, "What proportion of their surnames will have become extinct after r generations?" leads one to examine the recurrence and transience properties of the Markov chain.

It should be clear that 0 is an absorbing state. If a generation has no individuals, there will be no offspring. Under the initial assumptions, all other states of a branching process are transient.

Lemma 4.1. *In a branching process, all nonzero states are transient.*

Proof. If $Z_n = 0$, say the process has become *extinct* by generation n . Consider the probability that a population of size $i > 0$ goes extinct in one generation, that is, $P(Z_{n+1} = 0 | Z_n = i)$. If a generation has i individuals and the next generation has none, then each individual produced zero offspring, which occurs with probability $(a_0)^i$, by independence.

To show i is transient, we need to show that f_i , the probability of eventually hitting i for the chain started in i , is less than one. If the chain starts with i individuals, then the event that the chain eventually hits i is $\{Z_n = i \text{ for some } n \geq 1\} \subseteq \{Z_1 > 0\}$. Hence,

$$\begin{aligned} f_i &= P(Z_n = i \text{ for some } n \geq 1 | Z_0 = i) \\ &\leq P(Z_1 > 0 | Z_0 = i) \\ &= 1 - P(Z_1 = 0 | Z_0 = i) \\ &= 1 - (a_0)^i < 1, \end{aligned}$$

since $a_0 > 0$. ■

Since all nonzero states are transient and the chain has infinite state space, there are two possibilities for the long-term evolution of the process: either it gets absorbed in state 0, that is, the population eventually goes extinct, or the population grows without bound.

4.2 MEAN GENERATION SIZE

In a branching process, the size of the n th generation is the sum of the total offspring of the individuals of the previous generation. That is,

$$Z_n = \sum_{i=1}^{Z_{n-1}} X_i, \quad (4.1)$$

where X_i denotes the number of children born to the i th person in the $(n-1)$ th generation. Because of the independence assumption, $X_1, X_2, \dots$ is an i.i.d. sequence with common distribution $\mathbf{a}$. Furthermore, Z_{n-1} is independent of the X_i .

Equation (4.1) represents Z_n as a random sum of i.i.d. random variables. Results for such random sums can be applied to find the moments of Z_n .

Let $\mu = \sum_{k=0}^{\infty} k a_k$ be the mean of the offspring distribution. To find the mean of the size of the n th generation $E(Z_n)$, condition on Z_{n-1} . By the law of total expectation,

$$\begin{aligned} E(Z_n) &= \sum_{k=0}^{\infty} E(Z_n | Z_{n-1} = k) P(Z_{n-1} = k) \\ &= \sum_{k=0}^{\infty} E\left(\sum_{i=1}^{Z_{n-1}} X_i \mid Z_{n-1} = k\right) P(Z_{n-1} = k) \end{aligned}$$

$$\begin{aligned}
 &= \sum_{k=0}^{\infty} E \left(\sum_{i=1}^k X_i \middle| Z_{n-1} = k \right) P(Z_{n-1} = k) \\
 &= \sum_{k=0}^{\infty} E \left(\sum_{i=1}^k X_i \right) P(Z_{n-1} = k) \\
 &= \sum_{k=0}^{\infty} k \mu P(Z_{n-1} = k) = \mu E(Z_{n-1}),
 \end{aligned}$$

where the fourth equality is because the X_i are independent of Z_{n-1} . Iterating the resulting recurrence relation gives

$$E(Z_n) = \mu E(Z_{n-1}) = \mu^2 E(Z_{n-2}) = \cdots = \mu^n E(Z_0) = \mu^n, \text{ for } n \geq 0,$$

since $Z_0 = 1$.

Three Cases

For the long-term expected generation size,

$$\lim_{n \rightarrow \infty} E(Z_n) = \lim_{n \rightarrow \infty} \mu^n = \begin{cases} 0, & \text{if } \mu < 1, \\ 1, & \text{if } \mu = 1, \\ \infty, & \text{if } \mu > 1. \end{cases}$$

A branching process is said to be *subcritical* if $\mu < 1$, *critical* if $\mu = 1$, and *supercritical* if $\mu > 1$. For a subcritical branching process, mean generation size declines exponentially to zero. For a supercritical process, mean generation size exhibits long-term exponential growth. The limits suggest three possible regimes depending on μ : long-term extinction, stability, and boundless growth. However, behavior of the *mean* generation size does not tell the whole story.

Insight into the evolution of a branching process is gained by simulation. We simulated 10 generations $Z_0, \dots, Z_{10}$ of a branching process with Poisson offspring distribution, choosing three values for the Poisson mean parameter corresponding to three types of branching process: $\mu = 0.75$ (subcritical), $\mu = 1$ (critical), and $\mu = 1.5$ (supercritical). Each process was simulated five times. See the R script file **branching.R**. Results are shown in Table 4.1.

For $\mu = 0.75$, all simulated paths result in eventual extinction. Furthermore, the extinction occurs fairly rapidly.

When $\mu = 1$, all but one of the simulations in Table 4.1 become extinct by the 10th generation.

TABLE 4.1 Simulations of a Branching Process for Three Choices of μ

μ	Z_0	Z_1	Z_2	Z_3	Z_4	Z_5	Z_6	Z_7	Z_8	Z_9	Z_{10}
0.75	1	2	3	3	2	1	1	0	0	0	0
0.75	1	0	0	0	0	0	0	0	0	0	0
0.75	1	2	0	0	0	0	0	0	0	0	0
0.75	1	0	0	0	0	0	0	0	0	0	0
0.75	1	3	3	1	3	1	0	0	0	0	0
1	1	1	1	0	0	0	0	0	0	0	0
1	1	2	0	0	0	0	0	0	0	0	0
1	3	6	6	5	6	7	8	8	8	6	5
1	1	3	4	1	2	1	0	0	0	0	0
1	1	2	1	1	2	1	2	1	0	0	0
1.5	1	2	3	10	22	41	93	173	375	763	1,597
1.5	1	1	1	1	2	4	7	9	11	19	29
1.5	1	4	5	18	34	68	127	246	521	1,011	2,065
1.5	1	1	2	0	0	0	0	0	0	0	0
1.5	1	2	5	3	2	6	9	17	18	13	19

Indeed, for a general branching process, in the subcritical and critical cases ($\mu \leq 1$), the population becomes extinct with probability 1.

In the supercritical case $\mu = 1.5$, most simulations in Table 4.1 seem to grow without bound. However, one realization goes extinct. We will see that in the general supercritical case, the probability that the population eventually dies out is less than one, but typically greater than zero.

Extinction in the Subcritical Case

Assume that $Z_0, Z_1, \dots$ is a subcritical branching process. Let $E_n = \{Z_n = 0\}$ be the event that the population is extinct by generation n , for $n \geq 1$. Let E be the event that the population is ultimately extinct. Then,

$$E = \{Z_n = 0, \text{ for some } n \geq 1\} = \bigcup_{n=1}^{\infty} E_n,$$

and $E_1 \subseteq E_2 \subseteq \dots$. It follows that the probability that the population eventually goes extinct is

$$P(E) = P\left(\bigcup_{n=1}^{\infty} E_n\right) = \lim_{n \rightarrow \infty} P(E_n) = \lim_{n \rightarrow \infty} P(Z_n = 0). \quad (4.2)$$

The probability that the population is extinct by generation n is

$$\begin{aligned}
 P(Z_n = 0) &= 1 - P(Z_n \geq 1) \\
 &= 1 - \sum_{k=1}^{\infty} P(Z_n = k) \\
 &\geq 1 - \sum_{k=1}^{\infty} kP(Z_n = k) \\
 &= 1 - E(Z_n) = 1 - \mu^n.
 \end{aligned}$$

Taking limits gives

$$P(E) = \lim_{n \rightarrow \infty} P(Z_n = 0) \geq \lim_{n \rightarrow \infty} 1 - \mu^n = 1,$$

since $\mu < 1$. Thus, $P(E) = 1$. With probability 1, a subcritical branching process eventually goes extinct.

■ **Example 4.1** Subcritical branching processes have been used to model the spread of infections and disease in highly vaccinated populations. Farrington and Grant (1999) cite several examples, including the spread of measles and mumps, the outbreak of typhoidal salmonellae reported in Scotland in 1967–1990, and outbreaks of human monkeypox virus in past decades. Becker (1974) finds evidence of subcriticality in European smallpox data from 1950 to 1970.

Often the goal of these studies is to use data on observed outbreaks to estimate the unknown mean offspring parameter μ as well as the number of generations of spread until extinction. ■

Variance of Generation Size

To explore the process of extinction in the critical and supercritical cases ($\mu \geq 1$), we first consider the variance of the size of the n th generation $\text{Var}(Z_n)$. Let σ^2 denote the variance of the offspring distribution. By the law of total variance,

$$\text{Var}(Z_n) = \text{Var}(E(Z_n|Z_{n-1})) + E(\text{Var}(Z_n|Z_{n-1})).$$

We have shown that

$$E(Z_n|Z_{n-1} = k) = E\left(\sum_{i=1}^k X_i\right) = \sum_{i=1}^k E(X_i) = \mu k,$$

which gives $E(Z_n|Z_{n-1}) = \mu Z_{n-1}$. Similarly, $\text{Var}(Z_n|Z_{n-1}) = \sigma^2 Z_{n-1}$, since

$$\text{Var}(Z_n|Z_{n-1} = k) = \text{Var}\left(\sum_{i=1}^k X_i\right) = \sum_{i=1}^k \text{Var}(X_i) = \sigma^2 k,$$

using the independence of the X_i . Applying the law of total variance,

$$\begin{aligned} \text{Var}(Z_n) &= \text{Var}(\mu Z_{n-1}) + E(\sigma^2 Z_{n-1}) \\ &= \mu^2 \text{Var}(Z_{n-1}) + \sigma^2 \mu^{n-1}, \text{ for } n \geq 1. \end{aligned} \quad (4.3)$$

With $\text{Var}(Z_0) = 0$, Equation (4.3) yields

$$\begin{aligned} \text{Var}(Z_1) &= \mu^2 \text{Var}(Z_0) + \sigma^2 = \sigma^2, \\ \text{Var}(Z_2) &= \mu^2 \text{Var}(Z_1) + \sigma^2 \mu = \sigma^2 \mu(1 + \mu), \text{ and} \\ \text{Var}(Z_3) &= \mu^2 \text{Var}(Z_2) + \sigma^2 \mu^2 = \sigma^2 \mu^2(1 + \mu + \mu^2). \end{aligned}$$

The general pattern, proved by induction on n , gives

$$\text{Var}(Z_n) = \sigma^2 \mu^{n-1} \sum_{k=0}^{n-1} \mu^k = \begin{cases} n\sigma^2, & \text{if } \mu = 1, \\ \sigma^2 \mu^{n-1}(\mu^n - 1)/(\mu - 1), & \text{if } \mu \neq 1. \end{cases}$$

In the subcritical case, both the mean and variance of generation size tend to 0.

In the critical case, the mean size of every generation is one, but the variance is a linearly growing function of n .

In the supercritical case, the variance grows exponentially large. The potentially large difference between the mean μ^n and variance suggests that in some cases both extinction and boundless growth are possible outcomes.

To explore the issue more carefully, we will find the probability of ultimate extinction when $\mu \geq 1$. First, however, new tools are needed.

4.3 PROBABILITY GENERATING FUNCTIONS

For a discrete random variable X taking values in $\{0, 1, \dots\}$, the *probability generating function of X* is the function

$$\begin{aligned} G(s) &= E(s^X) = \sum_{k=0}^{\infty} s^k P(X = k) \\ &= P(X = 0) + sP(X = 1) + s^2P(X = 2) + \dots \end{aligned}$$

The function is a power series whose coefficients are probabilities. Observe that $G(1) = 1$. The series converges absolutely for $|s| \leq 1$. To emphasize the underlying random variable X , we may write $G(s) = G_X(s)$.

The generating function represents the distribution of a discrete random variable as a power series. If two power series are equal, then they have the same coefficients. Hence, if two discrete random variables X and Y have the same probability generating function, that is, $G_X(s) = G_Y(s)$ for all s , then X and Y have the same distribution.

■ **Example 4.2** Let X be uniformly distributed on $\{0, 1, 2\}$. Find the probability generating function of X .

Solution

$$G(s) = E(s^X) = \frac{1}{3} + s\left(\frac{1}{3}\right) + s^2\left(\frac{1}{3}\right) = \frac{1}{3}(1 + s + s^2).$$

■

■ **Example 4.3** Assume that X has a geometric distribution with parameter p . Find the probability generating function of X .

Solution

$$G(s) = E(s^X) = \sum_{k=1}^{\infty} s^k p(1-p)^{k-1} = sp \sum_{k=1}^{\infty} (s(1-p))^{k-1} = \frac{sp}{1-s(1-p)},$$

for $|s| < 1$.

■

Probabilities for X can be obtained from the generating function by successive differentiation. We have that

$$G(0) = P(X = 0),$$

$$G'(0) = \sum_{k=1}^{\infty} k s^{k-1} P(X = k) \Big|_{s=0} = P(X = 1),$$

$$G''(0) = \sum_{k=2}^{\infty} k(k-1) s^{k-2} P(X = k) \Big|_{s=0} = 2P(X = 2),$$

and so on. In general,

$$G^{(j)}(0) = \sum_{k=j}^{\infty} k(k-1) \cdots (k-j+1) s^{k-j} P(X = k) \Big|_{s=0} = j! P(X = j),$$

and thus

$$P(X = j) = \frac{G^{(j)}(0)}{j!}, \quad \text{for } j = 0, 1, \dots,$$

where $G^{(j)}$ denotes the j th derivative of G .

■ **Example 4.4** A random variable X has probability generating function

$$G(s) = (1 - p + sp)^n.$$

Find the distribution of X .

Solution We have $P(X = 0) = G(0) = (1 - p)^n$. For $1 \leq j \leq n$, the j th derivative of G is

$$G^{(j)}(s) = n(n-1) \cdots (n-j+1)p^j(1-p+sp)^{n-j},$$

which gives

$$\begin{aligned} P(X = j) &= \frac{G^{(j)}(0)}{j!} \\ &= \frac{n(n-1) \cdots (n-j+1)}{j!} p^j (1-p)^{n-j} \\ &= \binom{n}{j} p^j (1-p)^{n-j}. \end{aligned}$$

For $j > n$, $G^{(j)}(0) = 0$, and thus $P(X = j) = 0$. We see that X has a binomial distribution with parameters n and p . ■

Sums of Independent Random Variables

Generating functions are useful tools for working with sums of independent random variables. Assume that $X_1, \dots, X_n$ are independent. Let $Z = X_1 + \cdots + X_n$. The probability generating function of Z is

$$\begin{aligned} G_Z(s) &= E(s^Z) = E(s^{X_1 + \cdots + X_n}) \\ &= E\left(\prod_{k=1}^n s^{X_k}\right) = \prod_{k=1}^n E(s^{X_k}) \\ &= G_{X_1}(s) \cdots G_{X_n}(s), \end{aligned}$$

where the fourth equality is by independence. The generating function of an independent sum is the product of the individual generating functions. If the X_i are also identically distributed, then

$$G_Z(s) = G_{X_1}(s) \cdots G_{X_n}(s) = [G_X(s)]^n,$$

where X is a random variable with the same distribution as the X_i .

■ **Example 4.5** In Example 4.4, it is shown that the generating function of a binomial random variable with parameters n and p is $G(s) = (1 - p + ps)^n$. Here is a derivation using the fact that a sum of i.i.d. Bernoulli random variables has a binomial distribution.

Solution Let $X_1, \dots, X_n$ be an i.i.d. sequence of Bernoulli random variables with parameter p . The common generating function of the X_i is

$$G(s) = E(s^{X_i}) = s^0 P(X_i = 0) + s^1 P(X_i = 1) = (1 - p) + sp.$$

The sum $Z = X_1 + \dots + X_n$ has a binomial distribution with parameters n and p . The probability generating function of Z is thus

$$G_Z(s) = [G(s)]^n = (1 - p + ps)^n. \quad \blacksquare$$

Moments

The probability generating function of X can be used to find the mean, variance, and higher moments of X . Observe that

$$G'(1) = E(Xs^{X-1}) \Big|_{s=1} = E(X).$$

Also,

$$G''(1) = E(X(X-1)s^{X-2}) \Big|_{s=1} = E(X(X-1)) = E(X^2) - E(X),$$

which gives

$$\begin{aligned} \text{Var}(X) &= E(X^2) - E(X)^2 = (E(X^2) - E(X)) + E(X) - E(X)^2 \\ &= G''(1) + G'(1) - G'(1)^2. \end{aligned}$$

Example 4.6 For a geometric random variable with parameter p , the generating function is

$$G(s) = \frac{sp}{1 - s(1 - p)},$$

as shown in Example 4.3. Use the generating function to find the mean and variance of the geometric distribution.

Solution For the mean,

$$G'(s) = \frac{p}{(1 - s(1 - p))^2},$$

which gives $E(X) = G'(1) = 1/p$. For the variance,

$$G''(s) = \frac{2p(1 - p)}{(1 - s(1 - p))^3}, \quad \text{and} \quad G''(1) = \frac{2(1 - p)p}{p^2}.$$

This gives

$$\text{Var}(X) = G''(1) + G'(1) - G'(1)^2 = \frac{2(1 - p)p}{p^2} + \frac{1}{p} - \frac{1}{p^2} = \frac{1 - p}{p^2}.$$

We summarize some key properties of probability generating functions. ■

Properties of Probability Generating Function

1. Let $G(s) = E(s^X)$ be the probability generating function of a discrete random variable X . Then,
 - (a) $G(1) = 1$,
 - (b) $P(X = k) = G^{(k)}(0)/k!$, for $k \geq 0$,
 - (c) $E(X) = G'(1)$,
 - (d) $\text{Var}(X) = G''(1) + G'(1) - G'(1)^2$.
2. If X and Y are random variables such that $G_X(s) = G_Y(s)$ for all s , then X and Y have the same distribution.
3. If X and Y are independent, then $G_{X+Y}(s) = G_X(s)G_Y(s)$.

4.4 EXTINCTION IS FOREVER

Probability generating functions are especially useful for analyzing branching processes. We use them to find the probability that a branching process eventually goes extinct.

For $n \geq 0$, let

$$G_n(s) = \sum_{k=0}^{\infty} s^k P(Z_n = k)$$

be the generating function of the n th generation size Z_n . Let

$$G(s) = \sum_{k=0}^{\infty} s^k a_k$$

be the generating function of the offspring distribution. We have

$$G_n(s) = E(s^{Z_n}) = E\left(s^{\sum_{k=1}^{Z_{n-1}} X_k}\right) = E\left(E\left(s^{\sum_{k=1}^{Z_{n-1}} X_k} \mid Z_{n-1}\right)\right),$$

where the last equality is by the law of total expectation. From the independence of Z_{n-1} and the X_k ,

$$\begin{aligned} E\left(s^{\sum_{k=1}^{Z_{n-1}} X_k} \mid Z_{n-1} = z\right) &= E\left(s^{\sum_{k=1}^z X_k} \mid Z_{n-1} = z\right) \\ &= E\left(s^{\sum_{k=1}^z X_k}\right) = E\left(\prod_{k=1}^z s^{X_k}\right) \\ &= \prod_{k=1}^z E(s^{X_k}) = [G(s)]^z, \end{aligned}$$

for all z . This gives

$$E \left(s^{\sum_{k=1}^{Z_{n-1}} X_k} | Z_{n-1} \right) = [G(s)]^{Z_{n-1}}.$$

Taking expectations,

$$G_n(s) = E \left(G(s)^{Z_{n-1}} \right) = G_{n-1}(G(s)), \quad \text{for } n \geq 1.$$

The probability generating function of Z_n is the composition of the generating function of Z_{n-1} and the generating function of the offspring distribution.

Observe that $G_0(s) = s$, and $G_1(s) = G_0(G(s)) = G(s)$. From the latter we see that the distribution of Z_1 is the offspring distribution $\mathbf{a}$.

Continuing,

$$G_2(s) = G_1(G(s)) = G(G(s)) = G(G_1(s)),$$

and

$$G_3(s) = G_2(G(s)) = G(G(G(s))) = G(G_2(s)).$$

In general,

$$G_n(s) = G_{n-1}(G(s)) = \underbrace{G(\cdots G(G(s)) \cdots)}_{n\text{-fold}} = G(G_{n-1}(s)). \quad (4.4)$$

The generating function of Z_n is the n -fold composition of the offspring distribution generating function.

Equation (4.4) is typically not useful for computing the actual distribution of Z_n . For an arbitrary offspring distribution, the distribution of Z_n will be complicated with no tractable closed-form expression. However, the equation is central to the proof of the following theorem, which characterizes the extinction probability for a branching process.

Extinction Probability

Theorem 4.2. *Given a branching process, let G be the probability generating function of the offspring distribution. Then, the probability of eventual extinction is the smallest positive root of the equation $s = G(s)$.*

If $\mu \leq 1$, that is, in the subcritical and critical cases, the extinction probability is equal to 1.

Remark: We have already shown that in the subcritical $\mu < 1$ case, the population goes extinct with probability 1. The theorem gives that this is also true for $\mu = 1$, even though for each generation the expected generation size is $E(Z_n) = \mu^n = 1$. For the supercritical case $\mu > 1$, the expected generation size Z_n grows without bound.

However, the theorem gives that even in this case there is positive probability of eventual extinction.

Before proving Theorem 4.2, we offer some examples of its use. Let e denote the probability of eventual extinction.

■ **Example 4.7** Find the extinction probability for a branching process with offspring distribution $\mathbf{a} = (1/6, 1/2, 1/3)$.

Solution The mean of the offspring distribution is

$$\mu = 0(1/6) + 1(1/2) + 2(1/3) = 7/6 > 1,$$

so this is the supercritical case. The offspring generating function is

$$G(s) = \frac{1}{6} + \frac{s}{2} + \frac{s^2}{3}.$$

Solving

$$s = G(s) = \frac{1}{6} + \frac{s}{2} + \frac{s^2}{3}$$

gives the quadratic equation $s^2/3 - s/2 + 1/6 = 0$, with roots $s = 1$ and $s = 1/2$. The smallest positive root is the probability of eventual extinction $e = 1/2$. ■

■ **Example 4.8** A branching process has offspring distribution

$$a_k = (1 - p)^k p, \quad \text{for } k = 0, 1, \dots$$

Find the extinction probability in the supercritical case.

Solution The offspring distribution is a variant of the geometric distribution. The generating function is

$$G(s) = \sum_{k=0}^{\infty} s^k (1-p)^k p = p \sum_{k=0}^{\infty} (s(1-p))^k = \frac{p}{1 - s(1-p)}, \quad \text{for } |s(1-p)| < 1.$$

The mean of the offspring distribution is

$$\mu = G'(1) = \left. \frac{p(1-p)}{(1 - s(1-p))^2} \right|_{s=1} = \frac{1-p}{p}.$$

The supercritical case $\mu > 1$ corresponds to $p < 1/2$.

To find the extinction probability, solve

$$s = G(s) = \frac{p}{1 - s(1-p)},$$

which gives the quadratic equation

$$(1-p)s^2 - s + p = 0,$$

with roots

$$s = \frac{1 \pm \sqrt{1 - (4(1-p)p)}}{2(1-p)} = \frac{1 \pm (1-2p)}{2(1-p)}.$$

The roots are 1 and $p/(1-p)$. For $0 < p < 1/2$, the smaller root is $p/(1-p) = 1/\mu$, that is, $e = 1/\mu$. ■

We explore the extinction probability result in Example 4.8 with the use of simulation. Let $p = 1/4$. We simulated a branching process with offspring distribution $a_k = (3/4)^k(1/4)$, for $k \geq 0$. The process is supercritical with $\mu = 3$. Results are collected in Table 4.2. Four of the 12 runs went extinct by time $n = 10$. The exact extinction probability is $e = 1/3$. (The fact that 4 out of 12 is exactly $1/3$ is, we assure the reader, pure coincidence.)

R: Simulating the Extinction Probability

The branching process is simulated 10,000 times, keeping track of the number of times the process goes extinct by the 10th generation. See the file **branching.R**.

```
> branch(10, 1/4)
 1   4   6  16  71  205  569  1559 4588 13726 40800

> trials <- 10000
> simlist <- replicate(trials, branch(10, 1/4)[11])
# Estimate of extinction probability
> sum(simlist==0)/trials
[1] 0.332
```

The function `branch(n, p)` simulates n steps of a branching process whose offspring distribution is geometric with parameter p . The `replicate` command repeats the simulation 10,000 times, storing the outcome of Z_{10} for each trial in the vector `simlist`. The proportion of 0s in `simlist` estimates the extinction probability e .

Our conclusions are slightly biased since we assume that if extinction takes place it will occur by time $n = 10$. Of course, extinction could occur later. However, it appears from the simulations that if extinction occurs it happens very rapidly.

TABLE 4.2 Simulation of a Supercritical Branching Process, with $\mu = 3$. Four of the 12 runs go extinct by the 10th generation.

Z_0	Z_1	Z_2	Z_3	Z_4	Z_5	Z_6	Z_7	Z_8	Z_9	Z_{10}
1	4	25	97	394	1160	3475	10685	31885	95757	287130
1	1	1	3	7	11	47	165	515	1525	4689
1	10	37	115	350	1124	3455	10073	29896	88863	267386
1	1	1	2	2	3	0	0	0	0	0
1	0	0	0	0	0	0	0	0	0	0
1	8	31	71	248	779	2282	6864	19895	59196	178171
1	0	0	0	0	0	0	0	0	0	0
1	4	7	13	34	106	380	1123	3385	10200	30090
1	16	49	163	447	1284	3794	11592	34626	104390	312704
1	1	1	5	16	51	155	559	1730	5378	15647
1	1	3	31	79	267	883	2637	8043	23970	71841
1	0	0	0	0	0	0	0	0	0	0

■ **Example 4.9 (Lotka's estimate of the extinction probability)** One of the earliest applications of branching processes is contained in the work of Alfred Lotka, considered the father of demographic analysis, who estimated the probability that a male line of descent would ultimately become extinct. Based on the 1920 census data, Lotka fitted the distribution of male offspring to a *zero-adjusted* geometric distribution of the form

$$a_0 = 0.48235 \quad \text{and} \quad a_k = (0.2126)(0.5893)^{k-1}, \quad \text{for } k \geq 1.$$

The generating function of the offspring distribution is

$$G(s) = 0.48235 + 0.2126 \sum_{k=1}^{\infty} (0.5893)^{k-1} s^k = 0.48235 + \frac{(0.2126)s}{1 - (0.5893)s}.$$

Lotka found the extinction probability as the numerical solution to $G(s) = s$, giving the value $e = 0.819$.

The mean of the male offspring distribution is $\mu = 1.26$. It is interesting that despite a mean number of children (sons and daughters) per individual of about 2.5, the probability of extinction of family surnames is over 80% See Lotka (1931) and Hull (2001). ■

■ **Example 4.10** A *worm* is a self-replicating computer virus, which exploits computer network security vulnerabilities to spread itself. The *Love Letter* was a famous worm, which attacked tens of millions of Windows computers in 2000. It was spread as an attachment to an email message with the subject line ILOVEYOU. When users clicked on the attachment the worm automatically downloaded onto their machines.

Sellke et al. (2008) modeled the spread of computer worms as a branching process. The worms they analyzed are spread by randomly scanning from the 2^{32} current IP addresses to find a vulnerable host. Let V denote the total number of vulnerable hosts. Then, $p = V/2^{32}$ is the probability of finding a vulnerable host in one scan. If a worm scans at most M hosts, then infected hosts represent the individuals of a branching process whose offspring distribution is binomial with parameters M and p . Since the mean of the offspring distribution is Mp , it follows that the spread of the worm will eventually die out, with probability 1, if $Mp \leq 1$, or $M \leq 1/p$.

Typically, M is large and p is small and thus the offspring binomial distribution is well approximated by a Poisson distribution with parameter $\lambda = Mp$. The total number of infected hosts T before the worm eventually dies out is the *total progeny* of the branching process. See Exercises 4.24 and 4.25, where the mean and variance of the total progeny of a branching process are derived. If the worm starts out with I infected hosts then the mean and variance of the total number of infected hosts before the virus dies out is

$$E(T) = \frac{I}{1 - \lambda} \quad \text{and} \quad \text{Var}(T) = \frac{I}{(1 - \lambda)^3}.$$

■

Proof of Extinction Probability Theorem 4.2

Proof. Let $e_n = P(Z_n = 0)$ denote the probability that the population goes extinct by generation n . We have

$$\begin{aligned} e_n &= P(Z_n = 0) = G_n(0) = G(G_{n-1}(0)) \\ &= G(P(Z_{n-1} = 0)) = G(e_{n-1}), \end{aligned} \tag{4.5}$$

for $n \geq 1$. From Equation (4.2), $e_n \rightarrow e$, as $n \rightarrow \infty$. Taking limits on both sides of Equation (4.5), as $n \rightarrow \infty$, and using the fact that the probability generating function is continuous, gives $e = G(e)$. Thus, e is a root of the equation $s = G(s)$.

Let x be a positive solution of $s = G(s)$. We need to show that $e \leq x$. Since $G(s) = \sum_k s^k P(X = k)$ is an increasing function on $(0, 1]$, and $0 < x$,

$$e_1 = P(Z_1 = 0) = G_1(0) = G(0) \leq G(x) = x.$$

By induction, assuming $e_k \leq x$, for all $k < n$,

$$e_n = P(Z_n = 0) = G_n(0) = G(G_{n-1}(0)) = G(e_{n-1}) \leq G(x) = x.$$

Taking limits as $n \rightarrow \infty$, gives $e \leq x$. This proves the first part of the theorem.

The remainder of the theorem is essentially revealed by Figure 4.2. Consider the intersection of the graph of $y = G(s)$ with the line $y = s$ on the interval $[0, 1]$. Observe

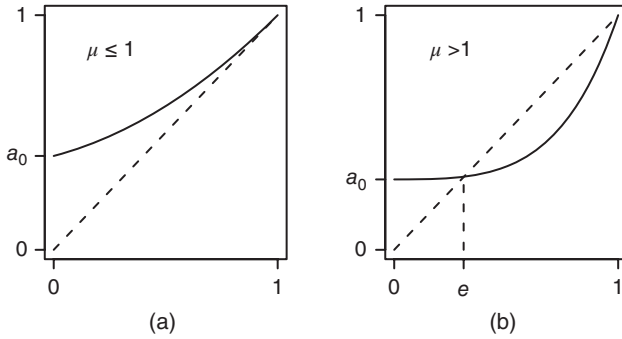


Figure 4.2 Graph of $G(s)$.

that $G(0) = a_0$ and $G(1) = 1$. Furthermore, the continuous and differentiable function G is convex (concave up) as

$$G''(s) = \sum_{k=2}^{\infty} k(k-1)s^{k-2}P(X = k) > 0.$$

It follows that the graph of $y = G(s)$ can intersect the line $y = s$ at either one or two points.

What distinguishes the two cases is the derivative of $G(s)$ at $s = 1$. Recall that $G'(1) = \mu$.

(i) If $\mu = G'(1) \leq 1$, we are in the setting of Figure 4.2(b). Since

$$G'(s) = \sum_{k=1}^{\infty} ks^{k-1}P(X = k) > 0$$

is a strictly increasing function of s , we have $G'(s) < G'(1) = 1$, for $0 < s < 1$. Let $h(s) = s - G(s)$. Then, $h'(s) = 1 - G'(s) > 0$, for $0 < s < 1$. Since h is increasing and $h(1) = 0$, it follows that $h(s) < 0$, for $0 < s < 1$. That is, $s < G(s)$. Hence, the graph of $G(s)$ lies above the line $y = s$, for $0 < s < 1$, and $s = 1$ is the only point of intersection. Hence, the extinction probability is $e = 1$.

(ii) If $\mu = G'(1) > 1$, we are in the setting of Figure 4.2(b). Here,

$$h(0) = 0 - G(0) = -a_0 < 0.$$

Also, $h'(1) = 1 - G'(1) = 1 - \mu < 0$, thus $h(s)$ is decreasing at $s = 1$. Since $h(1) = 0$, there is some $0 < t < 1$ such that $h(t) > 0$. It follows that there is a number e between 0 and 1 such that $h(e) = 0$. That is, $e = G(e)$. This is the desired extinction probability. ■

EXERCISES

- 4.1** Consider a branching process with offspring distribution $\mathbf{a} = (a, b, c)$, where $a + b + c = 1$. Let $\mathbf{P}$ be the Markov transition matrix. Exhibit the first three rows of $\mathbf{P}$. That is, find P_{ij} for $i = 0, 1, 2$ and $j = 0, 1, \dots$
- 4.2** Find the probability generating function of a Poisson random variable with parameter λ . Use the pgf to find the mean and variance of the Poisson distribution.
- 4.3** Let $X \sim \text{Poisson}(\lambda)$ and $Y \sim \text{Poisson}(\mu)$. Assume that X and Y are independent. Use probability generating functions to find the distribution of $X + Y$.
- 4.4** If X is a negative binomial distribution with parameters r and p , then X can be written as the sum of r i.i.d. geometric random variables with parameter p . Use this fact to find the pgf of X . Then, use the pgf to find the mean and variance of the negative binomial distribution.
- 4.5** The k th factorial moment of a random variable X is

$$E(X(X-1)\cdots(X-k+1)) = E\left(\frac{X!}{(X-k)!}\right), \quad \text{for } k \geq 0.$$

- (a) Given the probability generating function G of X , show how to find the k th factorial moment of X .
- (b) Find the k th factorial moment of a binomial random variable with parameters n and p .
- 4.6** Let $X_1, X_2, \dots$ be a sequence of i.i.d. Bernoulli random variables with parameter p . Let N be a Poisson random variable with parameter λ , which is independent of the X_i .
- (a) Find the probability generating function of $Z = \sum_{i=1}^N X_i$.
- (b) Use (a) to identify the probability distribution of Z .
- 4.7** Give the probability generating function for an offspring distribution in which an individual either dies, with probability $1 - p$, or gives birth to three children, with probability p . Also find the mean and variance of the number of children in the fourth generation.
- 4.8** If X is a discrete random variable with generating function G . Show that

$$P(X \text{ is even}) = \frac{1 + G(-1)}{2}.$$

- 4.9** Let $Z_0, Z_1, \dots$ be a branching process whose offspring distribution mean is μ . Let $Y_n = Z_n / \mu^n$, for $n \neq 0$. Show that $E(Y_{n+1} | Y_n) = Y_n$.
- 4.10** Show by induction that for $\mu \neq 1$,

$$\text{Var}(Z_n) = \sigma^2 \mu^{n-1} \frac{\mu^n - 1}{\mu - 1}.$$

- 4.11** Use the generating function representation of Z_n in Equation (4.4) to find $E(Z_n)$.
- 4.12** A branching process has offspring distribution $\mathbf{a} = (1/4, 1/4, 1/2)$. Find the following:
- μ .
 - $G(s)$.
 - The extinction probability.
 - $G_2(s)$.
 - $P(Z_2 = 0)$.
- 4.13** Use numerical methods to find the extinction probability for a branching process with Poisson offspring distribution with parameter $\lambda = 1.5$.
- 4.14** A branching process has offspring distribution with $a_0 = p$, $a_1 = 1 - p - q$, and $a_2 = q$. For what values of p and q is the process supercritical? In the supercritical case, find the extinction probability.
- 4.15** Assume that the offspring distribution is uniform on $\{0, 1, 2, 3, 4\}$. Find the extinction probability.
- 4.16** Consider a branching process where $Z_0 = k$. That is, the process starts with k individuals. Let $G(s)$ be the probability generating function of the offspring distribution. Let $G_n(s)$ be the probability generating function of Z_n for $n = 0, 1, \dots$
- Find the probability generating function $G_1(s)$ in terms of $G(s)$.
 - True or False: $G_{n+1}(s) = G_n(G(s))$, for $n = 1, 2, \dots$
 - True or False: $G_{n+1}(s) = G(G_n(s))$, for $n = 1, 2, \dots$
- 4.17** For $0 < p < 1$, let $\mathbf{a} = (1 - p, 0, p)$ be the offspring distribution of a branching process. Each individual in the population can have either two or no offspring. Assume that the process starts with two individuals.
- Find the extinction probability.
 - Write down the general term P_{ij} for the Markov transition matrix of the branching process.
- 4.18** Consider a branching process with offspring distribution
- $$\mathbf{a} = (p^2, 2p(1 - p), (1 - p)^2), \quad \text{for } 0 < p < 1.$$
- The offspring distribution is binomial with parameters 2 and $1 - p$. Find the extinction probability.
- 4.19** Let $T = \min\{n : Z_n = 0\}$ be the time of extinction for a branching process. Show that $P(T = n) = G_n(0) - G_{n-1}(0)$, for $n \geq 1$.
- 4.20** Consider the offspring distribution defined by $a_k = (1/2)^{k+1}$, for $k \geq 0$.
- Find the extinction probability.
 - Show by induction that

$$G_n(s) = \frac{n - (n - 1)s}{n + 1 - ns}.$$

- See Exercise 4.19. Find the distribution of the time of extinction.

- 4.21** The *linear fractional case* is one of the few branching process examples in which the generating function $G_n(s)$ can be explicitly computed. For $0 < p < 1$, let

$$a_0 = \frac{1 - c - p}{1 - p}, \quad a_k = cp^{k-1}, \quad \text{for } k = 1, 2, \dots,$$

where $0 < c < 1 - p$ is a parameter. The offspring distribution is a geometric distribution rescaled at 0.

- (a) Find μ , the mean of the offspring distribution.
 (b) Assume that $\mu = 1$. Show, by induction, that

$$G_n(s) = \frac{np - (np + p - 1)s}{1 - p + np - nps}.$$

- (c) For $\mu > 1$, Athreya and Ney (1972) show

$$G_n(s) = \frac{(\mu^n e - 1)s + e(1 - \mu^n)}{(\mu^n - 1)s + e - \mu^n},$$

where $e = (1 - c - p)/(p(1 - p))$ is the extinction probability. See Example 4.9. Observe that Lotka's model falls in the linear fractional case. For Lotka's data, find the probability that a male line of descent goes extinct by the third generation.

- 4.22** *Linear fractional case, continued.* A rumor-spreading process evolves as follows. At time 0, one person has heard a rumor. At each discrete unit of time every person who has heard the rumor decides how many people to tell according to the following mechanism. Each person flips a fair coin. If heads, they tell no one. If tails, they proceed to roll a fair die until 5 appears. The number of rolls needed determines how many people they will tell the rumor.
- (a) After four *generations*, how many people, on average, have heard the rumor?
 (b) Find the probability that the rumor-spreading process will stop after four generations.
 (c) Find the probability that the rumor-spreading process will eventually stop.
- 4.23** Let $\mathbf{a}$ be an offspring distribution with generating function G . Let X be a random variable with distribution $\mathbf{a}$. Let Z be a random variable whose distribution is that of X conditional on $X > 0$. That is, $P(Z = k) = P(X = k | X > 0)$. Find the generating function of Z in terms of G .
- 4.24** Let $T_n = Z_0 + Z_1 + \dots + Z_n$ be the total number of individuals up through generation n . Let $T = \lim_{n \rightarrow \infty} T_n$ be the *total progeny* of the branching process. Find $E(T)$ for the subcritical, critical, and supercritical cases.
- 4.25** *Total progeny, continued.* Let $\phi_n(s) = E(s^{T_n})$ be the probability generating function of T_n , as defined in Exercise 4.24.

- (a) Show that ϕ_n satisfies the recurrence relation

$$\phi_n(s) = sG(\phi_{n-1}(s)), \quad \text{for } n = 1, 2, \dots,$$

where $G(s)$ is the pgf of the offspring distribution. Hint: Condition on Z_1 and use Exercise 4.16(a).

- (b) From (a), argue that

$$\phi(s) = sG(\phi(s)),$$

where $\phi(s)$ is the pgf of the total progeny T .

- (c) Use (b) to find the mean of T in the subcritical case.

4.26 In a lottery game, three winning numbers are chosen uniformly at random from $\{1, \dots, 100\}$, sampling without replacement. Lottery tickets cost \$1 and allow a player to pick three numbers. If a player matches the three winning numbers they win the jackpot prize of \$1,000. For matching exactly two numbers, they win \$15. For matching exactly one number they win \$3.

- (a) Find the distribution of net winnings for a random lottery ticket. Show that the expected value of the game is -70.8 cents.
- (b) *Parlaying* bets in a lottery game occurs when the winnings on a lottery ticket are used to buy tickets for future games. Hoppe (2007) analyzes the effect of parlaying bets on several lottery games. Assume that if a player matches either one or two numbers they parlay their bets, buying respectively 3 or 15 tickets for the next game. The number of tickets obtained by parlaying can be considered a branching process. Find the mean of the offspring distribution and show that the process is subcritical.
- (c) See Exercise 4.19. Let T denote the duration of the process, that is, the length of the parlay. Find $P(T = k)$, for $k = 1, \dots, 4$.
- (d) Hoppe shows that the probability that a single parlayed ticket will ultimately win the jackpot is approximately $p/(1 - m)$, where p is the probability that a single ticket wins the jackpot, and m is the mean of the offspring distribution of the associated branching process. Find this probability and show that the parlaying strategy increases the probability that a ticket will ultimately win the jackpot by slightly over 40%.

4.27 Consider a branching process whose offspring distribution is Bernoulli with parameter p .

- (a) Find the probability generating function for the n th generation size Z_n . Describe the distribution of Z_n .
- (b) For $p = 0.9$, find the extinction probability and the expectation of total progeny.

4.28 In a *branching process with immigration*, a random number of immigrants W_n is independently added to the population at the n th generation.

- (a) Let H_n be the probability generating function of W_n . If G_n is the generating function of the size of the n th generation, show that

$$G_n(s) = G_{n-1}(G(s))H_n(s).$$

- (b) Assume that the offspring distribution is Bernoulli with parameter p , and the immigration distribution is Poisson with parameter λ . Find the generating function $G_n(s)$, and show that

$$\lim_{n \rightarrow \infty} G_n(s) = e^{-\lambda(1-s)/(1-p)}.$$

What can you conclude about the limiting distribution of generation size?

- 4.29** R: Examine the proof of Theorem 4.2 and observe that

$$e_n = G(e_{n-1}), \quad \text{for } n \geq 1, \quad (4.6)$$

where $e_n = P(Z_n = 0)$ is the probability that the population goes extinct by generation n . Since $e_n \rightarrow e$, as $n \rightarrow \infty$, Equation (4.6) is the basis for a numerical, recursive method to approximate the extinction probability in the supercritical case. To find e :

1. Initialize with $e_0 \in (0, 1)$.
2. Successively compute $e_n = G(e_{n-1})$, for $n \geq 1$.
3. Set $e = e_n$, for large n .

Convergence can be shown to be exponentially fast, so that n can often be taken to be relatively small (e.g., $n \approx 10$ -20). Use this numerical method to find the extinction probability for the following cases.

- (a) $a_0 = 0.8, a_4 = 0.1, a_9 = 0.1$.
- (b) Offspring distribution is uniform on $\{0, 1, \dots, 10\}$.
- (c) $a_0 = 0.6, a_3 = 0.2, a_6 = 0.1, a_{12} = 0.1$.

- 4.30** R: Simulate the branching process in Exercise 4.12. Use your simulation to estimate the extinction probability e .
- 4.31** R: Simulating a branching process whose offspring distribution is uniformly distributed on $\{0, 1, 2, 3, 4\}$.
- (a) Use your simulation to estimate the probability that the process goes extinct by the third generation. Compare with the exact result obtained by numerical methods.
 - (b) See Exercise 4.15. Use your simulation to estimate the extinction probability e . Assume that if the process goes extinct it will do so by the 10th generation with high probability.
- 4.32** R: Simulate the branching process with immigration in Exercise 4.28(b), with $p = 3/4$ and $\lambda = 1.2$. Illustrate the limit result in Exercise 4.28(c) with $n = 100$.

- 4.33** R: Simulate the total progeny for a branching process whose offspring distribution is Poisson with parameter $\lambda = 0.60$. Estimate the mean and variance of the total progeny distribution.
- 4.34** R: Based on the numerical algorithm in Exercise 4.29, write an R function `extinct(offspring)` to find the extinction probability for any branching process with a finite offspring distribution.